
TTT4295 - ACOUSTIC SIGNAL PROCESSING

PETER SVENSSON
Prel. version 16 Sep. 2024

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Preface

This compendium, for the course "Acoustic Signal Processing" (ASP) is based on a collection of the lecture notes that have been developed over time for a previous course "Audio technology" (AT). For this first version for the ASP course, finished in September 2024, some chapters, on wave fundamentals, microphones, and loudspeakers, have been condensed, compared to the original compendium for the AT course.

The reader will have to excuse the inconsistent styles of illustrations.

If you find some mistakes, please send a message to Peter Svensson (peter.svensson@ntnu.no).

Signal processing basics and measurement techniques

1.1 Introduction: Signals and systems

Two very central concepts in signal processing are *signals* and *systems*. The block diagram in Figure 1.1 represents a so-called single-input-single-output system. Such a system will generate an output signal, or response signal, $y(t)$, which is completely determined by its input signal, or excitation signal, $x(t)$, according to some rule.

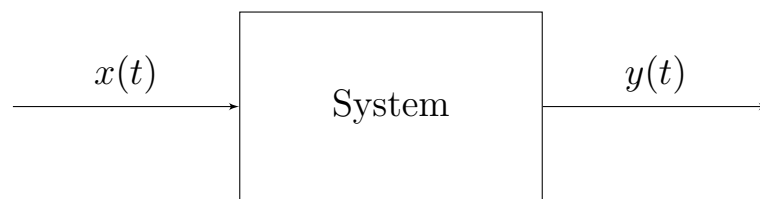


Figure 1.1: A single-input, single-output system. Such a system has a well-defined relationship between the input signal, or excitation signal, to the system, $x(t)$, and the output signal, or response signal, $y(t)$.

Some examples of systems in acoustics could be:

- A loudspeaker. The electrical voltage sent to the loudspeaker would be the input signal, $x(t) = e_{\text{LSP}}(t)$. The output signal, $y(t) = p_{\text{LSP},1\text{m}}(t)$, could be the sound pressure at 1 m distance, in front of the loudspeaker.
- A microphone. The sound pressure in the sound wave at the location of the microphone membrane *before the microphone is placed there, that is, in the free-field (FF)*, is commonly viewed as the input signal, $x(t) = p_{\text{FF}}(t)$. The electrical voltage that comes out of the microphone amplifier is then the output signal or response signal, $y(t) = e_{\text{MIC}}(t)$.
- A filter. An electronic filter (equalizer, bandpass-filter, delay unit, reverberation unit etc) takes an electronic signal as input signal, and gives a processed electronic signal as output signal. A special case is an amplifier which is supposed to just amplify the signal, equally much for all frequencies.

- A room. When acoustic measurements are carried out in a room, usually a loudspeaker is used as sound source, and a microphone as a measurement sensor. Therefore the electrical voltage sent to the loudspeaker would be the input signal, $x(t) = e_{\text{LSP}}(t)$, and the electrical voltage generated by the microphone amplifier would be the output signal, $y(t) = e_{\text{MIC}}(t)$. We would actually want to have the free-field sound pressure at the microphone as the response signal of the room, but a measurement microphone is often very close to ideal, that is, $e_{\text{MIC}}(t) \propto p_{\text{FF}}(t)$, so the microphone voltage is quite a good "proxy" for the sound pressure.
- A mass-spring system, with the input signal being the applied external force, $x(t) = f_{\text{EXT}}(t)$, and the vibration velocity of the mass being the output signal, $y(t) = u_{\text{MASS}}(t)$.

Some examples of signals in acoustics could be:

- The sound pressure, $p(t)$, in one point in space.
- The electrical voltage, $e(t)$, to or from an amplifier. For a measurement microphone, we typically assume that $e_{\text{MIC}}(t) \propto p_{\text{FF}}(t)$ across a certain frequency range (as mentioned above for the room example).

Sometimes, our interest is to measure the *properties of a system*. Other times, we are interested in measuring and analyzing *a signal* in itself. However, our only possibility to find out the properties of a system is to analyze the excitation and response signals^a.

^aIn some cases, where one can make some assumptions about the system and the excitation signal, it can be possible to estimate the properties of the system only by analyzing the response signal. This is often called *blind estimation*.

We start by looking at signals.

1.1.1 Signals

Signals can be divided in various ways, e.g. *signals of finite duration* vs. *signals of infinite duration*.

Some examples of finite duration signals are:

- A transient signal, like a gun-shot, a hand-clap, etc.
- A measurement signal such as a shorter or longer frequency-sweep signal.
- A speech sound, a single syllable, or a word, or a sentence.

Examples of infinite duration signals are:

- Random noise.
- A periodic signal, which could be a sinusoidal signal, a music sound, a specially constructed measurement signal etc

Real-world signals might contain a mix of these different categories.

A special signal: a sinusoidal, and its complex representation - the $j\omega$ -method

Thanks to Fourier analysis, we can analyze a situation for the special signal which is a favorite in signal processing: a single sinusoidal*

$$x(t) = A \cos(\omega t + \varphi) \quad (1.1)$$

with the angular frequency $\omega = 2\pi f$, where f is the frequency. A is the amplitude of the sinusoidal and φ is the phase. The solution for a signal with a single frequency is useful since all signals could be expressed as

*We choose a cosine instead of a sine because the following discussion will be a tiny bit easier if we do.

a sum (or distribution) of sinusoidals. Now the mathematical operations tend to be a little bit tedious with sinusoidals, and it was realized in the 1800s, first for electrical circuits and then for acoustics, that the mathematical operations would be much simpler by introducing complex signals.

Thus, we introduce a "companion signal" to $x(t)$ which is a sine-version of $x(t)$:

$$x_{\text{companion}}(t) = A \sin(\omega t + \varphi) \quad (1.2)$$

We will let our signal $x(t)$ "drag along" this companion signal, but instead of always having to write this:

$$\begin{cases} x(t) = A \cos(\omega t + \varphi) \\ x_{\text{companion}}(t) = A \sin(\omega t + \varphi) \end{cases}$$

we use the beautiful relationship called Euler's formula*, which lets us "store" both of those two signals in a single complex signal, which we call $\underline{x}(t)$:

$$\begin{aligned} \underline{x}(t) &= x(t) + jx_{\text{companion}}(t) = A [\cos(\omega t + \varphi) + j \sin(\omega t + \varphi)] \\ &= A e^{j(\omega t + \varphi)} = A e^{j\varphi} e^{j\omega t} = \underline{A} e^{j\omega t} \end{aligned}$$

where we "packed" both the A (amplitude) and φ (phase) from Eq. (7.2.1) into a *complex* amplitude $\underline{A}$. This complex amplitude is sometimes called the "phasor".

A remarkable fact is that we can now use $\underline{x}(t)$ instead of $x(t)$ in our derivations and various calculations without worrying about what effects this companion signal might have on our calculations. Things will automatically be handled correctly, such that at any stage in our calculate, we can take the real part of the complex signal, and get our real-world signal $x(t)$ back.

At the price of having to work with signals with complex values instead of the real-world signals with only real values, the math will be significantly easier. We try to demonstrate that this complex-signal approach makes it easier through a numerical example.

Example: we have two sinusoidals with the same frequency but different amplitudes and phase values (in radians):

$$\begin{aligned} x_1(t) &= 7 \cdot \cos(\omega t + 0.5), \\ x_2(t) &= 3 \cdot \cos(\omega t - 2), \end{aligned}$$

If we add these together, we get a single sinusoidal with the same frequency but a new amplitude and phase. What is this new amplitude?

Old method We use this trigonometrical relation: $\cos(A + B) = \cos A \cdot \cos B - \sin A \cdot \sin B$ to get:

$$\begin{aligned} x_1(t) &= 7 \cdot \cos(\omega t) \cdot \cos(0.5) - 7 \cdot \sin(\omega t) \cdot \sin(0.5) \\ &= 6.143 \cos(\omega t) - 3.356 \sin(\omega t), \\ x_2(t) &= 3 \cdot \cos(\omega t) \cdot \cos(2) + 3 \cdot \sin(\omega t) \cdot \sin(2) \\ &= -1.248 \cos(\omega t) + 2.728 \sin(\omega t) \end{aligned}$$

Then we can add the sine and cosine terms separately:

$$x_{\text{new}}(t) = x_1(t) + x_2(t) = 4.895 \cos(\omega t) - 0.628 \sin(\omega t) \quad (1.3)$$

*Euler's formula: $e^{j\alpha} = \cos \alpha + j \sin \alpha$.

As the last step, we realize that thanks to the trigonometric relation above, it should be possible to write $x_{\text{new}}(t)$ as $A \cdot \cos(\omega t + \varphi)$. Thus we apply the trig. relation to $x_{\text{new}}(t)$:

$$x_{\text{new}}(t) = A \cdot \cos(\omega t) \cos(\varphi) - A \cdot \sin(\omega t) \sin(\varphi) \quad (1.4)$$

and we can find the sought A and φ by making the expressions in Eqs. (1.3) and (1.4) equal:

$$\begin{cases} A \cdot \cos(\varphi) = 4.895 \\ -A \cdot \sin(\varphi) = -0.628 \end{cases}$$

First we divide the second equation by the first and get

$$\tan(\varphi) = \frac{0.628}{4.895} = 0.128 \Rightarrow \varphi = \arctan(0.128) = 0.127$$

Then we insert this value for φ in the first equation and get A :

$$A = \frac{4.895}{\cos(0.127)} = 4.93$$

and this is the sought amplitude.

New method We write the signals x_1 and x_2 in their complex versions:

$$\underline{x}_1(t) = 7 \cdot e^{j(\omega t + 0.5)} = 7e^{j0.5}e^{j\omega t},$$

$$\underline{x}_2(t) = 3 \cdot e^{j(\omega t - 2)} = 3e^{-j2}e^{j\omega t}$$

We add them together, and can then break out the factor $e^{j\omega t}$:

$$\underline{x}_{\text{new}}(t) = [7e^{j0.5} + 3e^{-j2}]e^{j\omega t} = \underline{A} \cdot e^{j\omega t}$$

To find the value of $\underline{A}$ (on the polar form), we have to convert the two complex amplitudes to the real + imaginary form, sum them, and convert back to the polar form:

$$\begin{aligned} \underline{A} &= 7e^{j0.5} + 3e^{-j2} = 7\cos(0.5) + j7\sin(0.5) + 3\cos(-2) + j3\sin(-2) \\ &= 6.143 + j3.356 - 1.248 - j2.728 = 4.895 + j0.628 \\ &= \sqrt{4.895^2 + 0.628^2}e^{j \tan^{-1} \frac{0.628}{4.895}} = 4.935e^{j0.129} \end{aligned}$$

and 4.94 is the sought amplitude. The small difference from the old method is caused by rounding errors for the last decimal. We also recognize the phase shift, $\varphi = 0.129$ to have the same (very similar) value with the two methods.

It has to be said that the complex method is particularly easy to use when you use some calculator, or software, which can handle complex numbers directly, like MatlabTM.

You might wonder: what if our real-world signal is a sinusoidal and not a cosine??? That is no problem, because a sinusoidal can be written as a cosine. Remember that $\sin \alpha = \cos(\frac{\pi}{2} - \alpha)$ (and vice versa). Thus:

$$x(t) = A \cdot \sin(\omega t + \phi) = A \cdot \cos\left(\frac{\pi}{2} - (\omega t + \phi)\right)$$

Also, since $\cos(-\alpha) = \cos(\alpha)$,

$$\begin{aligned} x(t) &= A \cdot \cos\left(\omega t + \phi - \frac{\pi}{2}\right) = A \cdot \text{Re}\left[e^{j(\phi - \frac{\pi}{2})}e^{j\omega t}\right] \\ &\Rightarrow \underline{x}(t) = \underline{A}e^{j\omega t}, \quad \text{where } \underline{A} = A \cdot e^{j(\phi - \frac{\pi}{2})} \end{aligned}$$

You might also wonder: what if our real-world signal is not a (co-)sinusoidal but music, noise, speech, etc? Remember the Fourier analysis idea: all signals can be viewed as a sum of sinusoids. (many sinusoids might be needed, but still, it will be a sum of sinusoids).

To round this off, we can repeat that the real-world sinusoidal signal can always be retrieved by taking the real part of the complex signal, $\underline{x}(t)$. However, most of the time we don't need to retrieve the sinusoidal - we know quite well what a sinusoidal looks like. Most of the time, we are primarily interested in the *amplitude of the sinusoidal* and then we can inspect *the magnitude of the complex signal amplitude*, $|\underline{A}|$ - that magnitude is exactly the amplitude of the real-world signal.

In addition to the magnitude of the complex signal amplitude, we are often interested in the phase of the complex signal amplitude - but the phase is mostly relevant when we have two or more signal components which are added because the resulting interference is critically affected by the phase difference between the signals.

1.1.2 Systems

If a system like in Fig. 1.1, is *linear and time-invariant* (LTI), it can be characterized by its transfer function, or *frequency response* function. The frequency response function of the system is directly connected to the use of single-frequency signals:

The frequency response tells, for each frequency, how much the amplitude of the signal is changed/scaled/-multiplied by the system, *and* how much the phase of the signal is changed by the system.

When we use the complex-signal approach, then the input and output signals in Fig. 1.1 can be written as

$$\begin{cases} \underline{x}(t) = \underline{X}(\omega)e^{j\omega t} \\ \underline{y}(t) = \underline{Y}(\omega)e^{j\omega t} \end{cases}$$

where we write the complex signal amplitudes as functions of ω to indicate that a signal might have different (complex) amplitudes at different frequencies. With the complex signals, the frequency response function can be expressed as a complex number, $\underline{F}(\omega)$:

$$\underline{F}(\omega) = \frac{\underline{Y}(\omega)}{\underline{X}(\omega)}$$

Then:

The magnitude of $\underline{F}(\omega)$, $|\underline{F}(\omega)|$, will tell how much the amplitude of the input sinusoidal $x(t)$ is changed (scaled, multiplied) by the system.

The phase of $\underline{F}(\omega)$, which can be denoted as $\angle \underline{F}(\omega)$ or $\arg(\underline{F}(\omega))$, tells how many radians the system changes the phase of the signal, $x(t)$, for each (angular) frequency ω .

1.1.3 Signals vs Systems

We can summarize some important differences between the complex numbers that represent the signals and the complex numbers that represent the system:

- The complex amplitude of a signal is a constant complex number, $\underline{X}(\omega)$, which is always assumed to be multiplied by $e^{j\omega t}$ and it will then give a sinusoidal complex function of time, $\underline{x}(t)$. We might be interested in:

- The signal's amplitude, $\hat{x} = |\underline{X}(\omega)|$
- The signal's rms-value, $x_{\text{rms}} = \frac{1}{\sqrt{2}}|\underline{X}(\omega)|$
- The real-world signal's time-dependence

$$x(t) = \text{Re}[\underline{x}(t)] = \text{Re}[\underline{X}(\omega)e^{j\omega t}]$$

- The complex value of a system's frequency response function, $\underline{F}(\omega)$, is a constant complex number for each frequency. *This complex number does not represent a function of time!!* We might be interested in:
 - The system's frequency response magnitude, $|\underline{F}(\omega)|$
 - The system's frequency response phase shift, $\angle \underline{F}(\omega)$
- For the special system example of the so-called *radiation impedance* of a vibrating system*, we might be interested in (we denote the frequency response function with $\underline{Z}(\omega)$ instead of $\underline{F}(\omega)$ for this special system example)
 - The real part of the system's frequency response, $\text{Re}[\underline{Z}(\omega)]$, and
 - the imaginary part of the system's frequency response, $\text{Im}[\underline{Z}(\omega)]$

The last example is special; it is quite rare that we are interested in the real and imaginary parts of a system's frequency response.

As a last note, we can observe that in practically all textbooks, there is no underline (or other symbol) to mark that values might have a complex value. From now on, the underline notation will not be used.

*How can the radiation impedance be viewed as a system??? Well, we can view the vibration velocity of a vibrating surface as an input signal, and the reaction force that acts on the vibrating surface as an output signal, or response signal, and then the radiation impedance becomes the frequency response function of this system.

1.2 Analysis of sound signals

The sound pressure signal is our primary signal of interest, and it can be viewed as the input signal to our hearing. A good-quality microphone is close to an ideal transducer in many (but not all) applications: the frequency response is close to flat over a wide frequency range, and the signal-to-noise ratio can be high. Therefore, when the electrical voltage signal from a microphone amplifier is digitized by an audio interface, analysis software on a computer directly gets a digital version of this signal of interest.

1.2.1 The RMS-value revisited

The rms-value is the most important quantity in noise measurements since the loudness we perceive is largely determined by the rms-value of a sound pressure signal.

In acoustics, it is more common to express a sound's intensity/strength by the sound pressure level, in dB, rather than an rms-value in Pa. But, there is a well-defined relationship between the two, see Eqs. (1.10) and (1.11), and we will move back and forth between the rms-values, in Pa, and the level values, in dB.

Later, we will see more details on this, e.g., that the frequency response, and the dynamic range, of our hearing is making things a bit more complicated*. It is well known from previous courses how to calculate the rms-value of signals, but we will see one interesting aspect related to the human hearing. So, we repeat some basics first: The rms-value, squared, is defined as:

$$p_{\text{rms}}^2 = \frac{1}{T} \int_0^T p_{\text{real}}^2(t) dt \quad (1.5)$$

p_{rms}^2 is the mean value of the square of $p_{\text{real}}(t)$ and it corresponds to the mean power or mean intensity of the signal. As mentioned in Table 1.3 (in the Appendix), often p_{rms} is just written p , but in this compendium we try to write out p_{rms} , to make a clear distinction from p , which is also used for the complex amplitude for a time-harmonic signal.

For one harmonic signal of (angular) frequency ω , using the real notation from Eq. (??), we get

$$p_{\text{rms}}^2 = \frac{1}{T} \int_0^T \hat{p}^2 \cos^2(\omega t + \varphi) dt = \frac{\hat{p}^2}{T} \int_0^T \frac{1}{2} [1 + \cos(2\omega t + 2\varphi)] dt = \frac{\hat{p}^2}{2}$$

(the integral over the cos-term is zero because the integration time, T , should be one period of the periodic signal). We can also use the complex notation, but we must then include a factor of 1/2 in the definition

$$p_{\text{rms}}^2 = \frac{1}{2T} \int_0^T |p_{\text{complex}}(t)|^2 dt = \frac{1}{2T} \int_0^T |A \cdot e^{j\omega t}|^2 dt = \frac{|A|^2}{2} = \frac{\hat{p}^2}{2}$$

For the sum of two signals with the same frequency, which could be caused by a microphone picking up two signals from two different sources (but with the same frequency)[†], or a direct sound and a single reflection, as illustrated in Fig. 1.2. (Remember that A_1 and A_2 are two complex constants)

$$\begin{aligned} p_{\text{total}}(t) &= A_1 e^{j\omega t} + A_2 e^{j\omega t} = (|A_1| e^{j\varphi_1} + |A_2| e^{j\varphi_2}) e^{j\omega t} \\ &= e^{j\varphi_1} (|A_1| + |A_2| e^{j(\varphi_2 - \varphi_1)}) e^{j\omega t} = A_{\text{total}} e^{j\omega t} \\ \Rightarrow p_{\text{rms}}^2 &= \frac{|A_{\text{total}}|^2}{2} = \frac{1}{2} \left| |A_1| + |A_2| e^{j(\varphi_2 - \varphi_1)} \right|^2 \\ &= \dots = \frac{|A_1|^2}{2} + \frac{|A_2|^2}{2} + |A_1| |A_2| \cos(\varphi_2 - \varphi_1) \end{aligned}$$

*As one example, the perceived loudness is not proportional to p_{rms} , but approximately to $p_{\text{rms}}^{0.4}$

[†]The sound pressure picked up is a simple sum of the two individual sound waves if we have linear conditions. One of the properties of a linear system is that superposition is valid, which we use here.



Figure 1.2: The superposition of two harmonic signals of the same angular frequency, ω , can for instance occur if two loudspeakers emit sound waves of the same (angular) frequency - or if there is a single source with a direct sound and one reflected sound wave.

Depending on the phase difference between the two components, we get positive or negative interference, and the maximum and minimum rms-value squared, respectively, will be:

$$p_{\text{rms,max}}^2 = \frac{1}{2} (|A_1| + |A_2|)^2, \text{ if } \varphi_2 = \varphi_1, \quad (1.6)$$

$$p_{\text{rms,min}}^2 = \frac{1}{2} (|A_1| - |A_2|)^2, \text{ if } \varphi_2 = \varphi_1 \pm \pi \quad (1.7)$$

We need to remember this: when two signals of the same frequency are summed, *interference* will occur and the resulting signal will have an amplitude which can be anything between the extreme values in Eqs. (1.6) and (1.7).

An important application which uses these expressions is *active noise control* (ANC), the idea behind which is to cancel out the sound from a (primary) noise source by having another (secondary) source radiate the same frequency and same amplitude but with reversed phase (i.e., let $\varphi_2 = \varphi_1 \pm \pi$). If done perfectly, it could lead to $p_{\text{total}} = 0$!

You can try this sound example demonstrating Active noise control (ANC) yourself: put two loudspeakers face-to-face (quite close to each other). Play a low-frequency pure tone, e.g., 100 Hz, in these three different ways:

1. Send a signal to one loudspeaker but not the other.
2. Now send the same signal, with the same polarity, to both speakers. The resulting sound pressure level should be 6 dB higher than in 1. (WHY? Because the rms-value will be twice the value of the single-speaker rms-value, and $20 \lg(2) = 6.02$ dB.)
3. Finally, send the same signal to both speakers, but with the opposite polarity to one of them. Note how quiet it gets! This is a form of Active Noise Control.

Why can this demonstration be called ANC? View lsp 1 as a noise source and lsp 2 as an ANC source. By selecting the right amplitude and phase (should differ by 180 degrees), for each frequency that the noise source emits, the total sound pressure gets much lower (if not perfectly zero). Please note that the loudspeakers will vibrate with practically the same vibration amplitudes in case 2 and 3, and still the total sound wave can differ so much in amplitude!

Now we look at two signals with different frequencies. This could be the same two loudspeakers as above, but now radiating two different frequencies, as illustrated in Fig. 1.3. We denote $\omega_2 = \omega_1 + \Delta\omega$, and get the total signal as

$$p_{\text{total}}(t) = A_1 e^{j\omega_1 t} + A_2 e^{j\omega_2 t} = |A_1| e^{j\varphi_1} e^{j\omega_1 t} + |A_2| e^{j\varphi_2} e^{j(\omega_1 + \Delta\omega)t}$$



Figure 1.3: The superposition of two harmonic signals of two different angular frequencies, ω_1 and ω_2 .

$$= e^{j\varphi_1} \left(|A_1| + |A_2| e^{j(\varphi_2 - \varphi_1 + \Delta\omega t)} \right) e^{j\omega_1 t}$$

So, it seems as if we can view this as a signal of (angular) frequency ω_1 , and a time-varying amplitude. Then we can get the rms-value via the complex notation as before:

$$\begin{aligned} p_{\text{total,rms}}^2 &= \frac{1}{2T} \int_0^T |p_{\text{complex}}(t)|^2 dt = \frac{1}{2T} \int_0^T \left| |A_1| + |A_2| e^{j(\varphi_2 - \varphi_1 + \Delta\omega t)} \right|^2 dt \\ &= \frac{|A_1|^2}{2} + \frac{|A_2|^2}{2} + \frac{1}{T} \int_0^T |A_1| |A_2| \cos(\varphi_2 - \varphi_1 + \Delta\omega t) dt \\ &= p_{1,\text{rms}}^2 + p_{2,\text{rms}}^2 + 2p_{1,\text{rms}}p_{2,\text{rms}} \frac{1}{T} \int_0^T \cos(\varphi_2 - \varphi_1 + \Delta\omega t) dt \end{aligned}$$

There are some interesting consequences of the integral term:

1. If the integration time, T , is chosen as a multiple of the period time of the time-variation caused by $\Delta\omega$, then the integral is zero, and

$$p_{\text{total,rms}}^2 = p_{1,\text{rms}}^2 + p_{2,\text{rms}}^2$$

This is a very important fact: we can sum rms²-values for signal components of different frequencies! The total sound pressure level will be

$$L_{p,\text{total}} = 10 \cdot \lg \frac{p_{1,\text{rms}}^2 + p_{2,\text{rms}}^2}{p_0^2} = 10 \cdot \lg \left(10^{L_{p,1}/10} + 10^{L_{p,2}/10} \right)$$

where $L_{p,i} = 10 \cdot \lg(p_{i,\text{rms}}^2/p_0^2)$. This summation is often called *uncorrelated addition*.

2. However, if $\Delta\omega$ is very small, then the total rms²-value, *computed over a short time interval T* , will vary up and down. As an example: so-called "beats" are heard if two sinusoids of almost the same frequency are played simultaneously.

We can make the same observation as above if we recall this trigonometric identity:

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

This identity can be written on a form which looks useful:

$$\begin{aligned} \cos(\omega_1 t) + \cos(\omega_2 t) &= 2 \cos \frac{(\omega_1 + \omega_2)t}{2} \cos \frac{(\omega_1 - \omega_2)t}{2} \\ &= 2 \cos(\omega_{\text{average}} t) \cos \frac{\Delta\omega t}{2} \end{aligned}$$

Interpretation: we can understand the sum of two sinusoids of different frequencies as:

1. Two independent sinusoidals (left side of the equation), or
2. One sinusoidal, with the average frequency and a time-varying amplitude (right side of the equation)

The functioning of our hearing affects what we hear when two sinusoidals are present in a signal:

- If two sinusoidals that are far apart in frequency (large $\Delta\omega$) are played at the same time, we hear two tones, each with constant amplitude.
- If two sinusoidals with a small $\Delta\omega$ are played at the same time, then we hear one tone with a fluctuating amplitude (“beating”).
- In a transition area we hear one tone with a constant amplitude but a non-pure, “rough” character.

Our hearing has a set of band-pass filters, which splits up the incoming signal into different frequency bands. These bands are called “auditory filters”, or “critical bands”, and they are quite similar to 1/6-th octave-band filters, except for the lowest audible frequencies, where the bandwidth is a bit wider than that.

One consequence of such a splitting up the signal into frequency range components is that when two tonal components end up in the same auditory channel, then they will interfere and be perceived as one sound with a fluctuating or rough amplitude. When two tonal components end up in different auditory channels, then they will not interfere, and they will be perceived as two separate sounds.

1.2.2 Computing the RMS-value in practice

Eq. (1.5) gave the definition of the RMS-value (squared) for a continuous-time signal, $p_{\text{real}}(t)$, using an integral. In practice, signal processing is implemented for *digital signals*, i.e., discrete-time signals, sampled with a certain sampling frequency, f_S . Then the signal, $p(n)$, will be known at discrete time samples, n , and an estimate of the rms-value will be computed as

$$\hat{p}_{\text{rms}}^2 = \frac{1}{N} \sum_{n=0}^{N-1} p_{\text{real}}^2(n) \quad (1.8)$$

for a signal of N samples length. We realize that even if our signal is a sinusoidal with an amplitude p_{peak} and an exactly known rms-value: $p_{\text{rms}} = p_{\text{peak}}/\sqrt{2}$, our estimate, $\hat{p}_{\text{rms}}^2$, will depend on how many samples, N , of the signal that are used for computing the estimate. Fig. 1.4 (a) illustrates that in some cases (the solid blue curve with angular frequency ω_0), the number of samples and the frequency of the signal makes the samples perfectly distributed, and the rms-estimate will be exactly correct. In most cases, however, as is also illustrated in Fig. 1.4 (a) (the dashed red curve with angular frequency $1,05\omega_0$), the sampling will not be perfectly distributed and the rms-estimate will deviate more or less from the correct value.

The rms-estimate will vary by tiny amounts depending on exactly how the samples are distributed along the sinusoidal. In particular, the “abrupt” start and end of the sampling will contribute to this variation, and this variation can be reduced by applying a window which suppresses the signal at the ends by known amounts. Using a so-called window function, the squared rms-estimate would be:

$$\hat{p}_{\text{rms,windowed}}^2 = C_{\text{correction},w}^2 \frac{1}{N} \sum_{n=0}^{N-1} [p_{\text{real}}(n)w(n)]^2 \quad (1.9)$$

where $w(n)$ is a window function and $C_{\text{correction},w}^2$ is the correction factor, which compensates for the systematically removed energy of the signal. The correction factor is simply

$$C_{\text{correction},w}^2 = \frac{1}{\frac{1}{N} \sum_{n=0}^{N-1} w^2(n)} \xrightarrow{N \rightarrow \infty} \frac{1}{\frac{1}{T} \int_0^T w^2(t) dt}$$

where the final value, for $N \rightarrow \infty$ gives the theoretical correction factor. A common window function is the Hann window, which is a "raised cosine" function. A Hann window of length N samples is given by

$$w_{\text{Hann}}(n) = 0.5 \left[1 - \cos \left(2\pi \frac{n}{N-1} \right) \right], \quad n \in [0, N-1]$$

with the correction factor

$$C_{\text{correction,Hann}} = \sqrt{\frac{8}{3}} \sqrt{\frac{N}{N-1}} \xrightarrow{N \rightarrow \infty} 1.633...$$

Fig. 1.4 (b) illustrates the Hann window, and the effect on a signal the window has. The effect of the number of samples on the RMS-value estimate, for unwindowed and Hann-windowed signals is further illustrated in Fig. 1.5. These curves are for one particular sinusoidal, with one arbitrary phase shift, φ :

$$p_{\text{example,sinusoidal}}(n) = \sin \left(\frac{2\pi f n}{f_s} + \varphi \right)$$

Different values of φ would give different details for these curves but some observations are general:

- With a Hann window (or other window), the RMS-estimate has much less variability (when N varies) than without a window.
- With a Hann window, the error will be smaller than 1% if the number of samples cover at least 1.5 periods of the sinusoidal, and it is substantially larger without a window.
- Without a window, the RMS-estimate becomes exactly correct when the number of samples is an integer times a half period.

These observations are, by the way, also true for signals with two or more sinusoids.

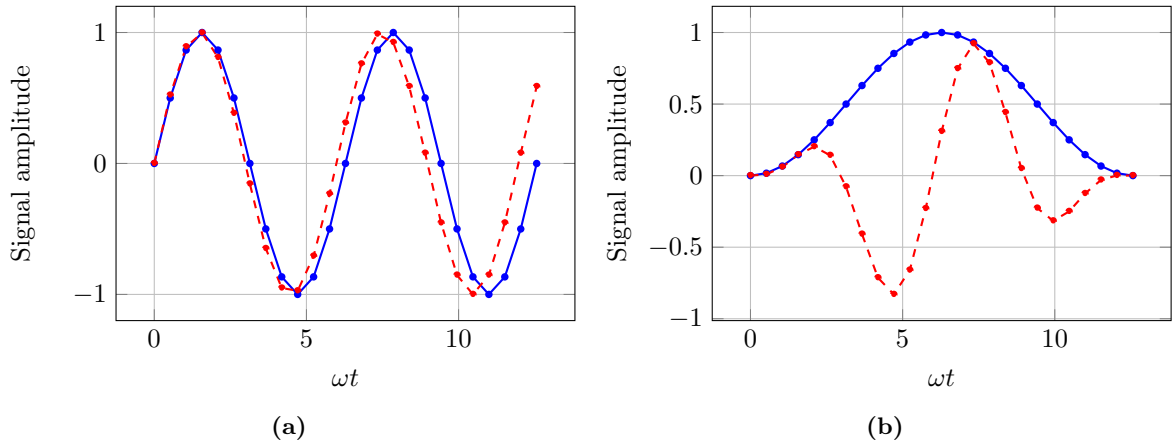


Figure 1.4: (a) Two sampled sinusoids, the solid blue curve has an angular frequency ω_0 such that the frequency is the sampling frequency divided by an integer ($2\pi f_s / \omega_0 = \text{an integer}$), and the dashed red curve has an angular frequency which is 5% higher. (b) A Hann window, solid blue curve, and the sinusoidal from (a) (dashed red curve in (a)) multiplied with the Hann window (dashed red curve).

1.2.3 The sound pressure level (SPL)

In acoustics, the RMS-value, as discussed in the previous section, is usually expressed as a sound pressure level, in dB, relative to a reference value.

The sound pressure level, or SPL, L_p , is by ISO [1] defined as^a

$$L_p \equiv 10 \lg \frac{p_{\text{rms}}^2}{p_0^2} \text{ dB} \quad (1.10)$$

where p_{rms} is the root-mean-square sound pressure and p_0 is the reference sound pressure. For sound in air and other gases, the reference sound pressure is given by $20 \mu\text{Pa}$. The definition above is equivalent with

$$L_p = 20 \lg \frac{p_{\text{rms}}}{p_0} \text{ dB} \quad (1.11)$$

The reference value, $p_0 = 20 \mu\text{Pa}$, is chosen so that the weakest audible sounds get a value around 0 dB. At some frequencies, and for some frequencies, a person might very well hear sounds that are weaker than $20 \mu\text{Pa}$, which means that a sound pressure level of, e.g., -5 dB is audible. The negative value has no special importance more than signaling that the rms-value of the signal is smaller than the fixed reference value.

Eqs. (1.10) and (1.11) above also tell us how to get the rms-value from a dB-value:

$$p_{\text{rms}} = p_0 10^{\frac{L_p}{20}}, \quad p_{\text{rms}}^2 = p_0^2 10^{\frac{L_p}{10}} \quad (1.12)$$

^aNote that lg is the preferred symbol for the base-10 logarithm, according to ISO.

Regulations on the maximum allowed noise levels in dwellings, workplaces etc. are based on the sound pressure level, L_p . As the frequency response of our hearing is not flat, in measurements of noise levels, one normally applies a so-called A-weighting filter, which reflects our hearing's frequency-dependent sensitivity (at low sound levels; there is also a C-weighting for high sound levels, see more below). When one measures with this A-weighting filter one gets the

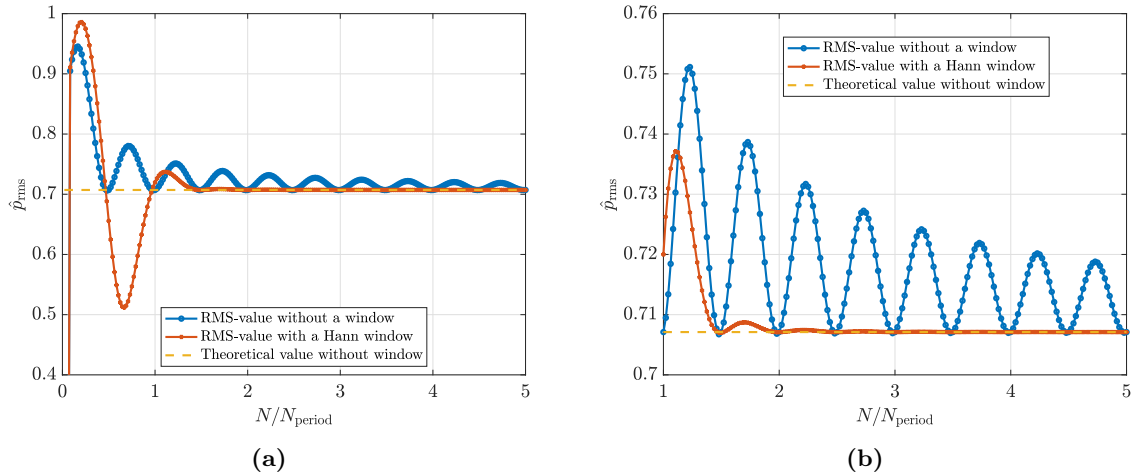


Figure 1.5: (a) The RMS-value for a single sinusoid, computed without (blue curve) and with a Hann window (red curve), as function of how many samples N are used for the signal, relative to the number of samples per sinusoidal period. (b) Zoomed-in version of (a).

Frequency-weighted sound pressure level. Using A-weighting, it is denoted L_A , and is expressed in dB [2]:

$$L_A = 10 \lg \frac{p_{A\text{-weighted,rms}}^2}{p_0^2} \text{ dB},$$

where $p_{A\text{-weighted,rms}}$ is the rms-value of the A-weighted (filtered) sound pressure signal.

Note that the A-weighted sound pressure level should *not* be expressed in dBA, even though it is not uncommon to see the dBA used.

The background for the A-weighting filter is the so-called isophone curves, or equal- loudness-level contours, [3], which show which sinusoids are perceived as equally loud, see Fig. 1.6. The so-called *loudness level*, denoted L_N , has the unit phon, and is the parameter marked for each curve. One very important observation is that the curves are not parallel and therefore, the human hearing can (unfortunately) not be approximated with a single frequency response, which also means that the hearing is a non-linear system.

One can see in Fig. 1.6 that our hearing seems to be most sensitive around 3-3.5 kHz, and least sensitive for low frequencies. The maximum sensitivity is due to the amplification caused by our ear canal (a quarter-wave resonance). These curves are averages for a large normal- hearing population. Individual values can deviate substantially from these curves.

We can also see in Fig. 1.6 that the "Threshold of hearing"-curve goes below 0 dB for a small frequency range. Thus, most normal-hearing people can hear sounds with a negative level in that frequency range, but as explained above, after Eq. (1.11), a negative dB-value simply indicates that the rms-value of the signal is smaller than the reference value.

As one example: the signals with

$$L_p = 66 \text{ dB at } 63 \text{ Hz},$$

$$L_p = 30 \text{ dB at } 1 \text{ kHz},$$

$$L_p = 26 \text{ dB at } 4 \text{ kHz}$$

all follow the 30-phon curve and as a consequence, are perceived as equally loud.

Another example: assume that we are listening to music over loudspeakers with one harmonic tone with a fundamental frequency of 250 Hz. Assume that the harmonics happen to have sound pressure levels such that they all give a loudness level, L_N , of 70 phon, for every single harmonic component. Now, we turn down the sound pressure level by 30 dB (by turning down the gain of the amplifier, which will reduce all frequency components with the same scale value. By inspecting the diagram in Fig. 1.6, we can find the original sound pressure levels which gave 70 phon, and the resulting loudness levels after the sound pressure levels were reduced by 30 dB, see Table 1.1. We can then see that when we reduce the volume, it will be perceived as a weakening of the bass, since the resulting loudness level is lower for the 250 Hz-frequency.

The standardized weighting-curves (A-, C-, and Z-), [2], are simply filters that are meant to approximate the human hearing at low levels (A), and high levels (C). Previously there was also a B-weighting curve for medium levels but it is now obsolete (that is, not part of the IEC standard anymore). The Z-weighting is simply a flat curve, that is, no weighting.

One might have expected that the B-weighting would be used in many normal noise situations, but the A-weighted level has been shown to work well as a measure for both noise disturbance and risk of hearing impairments and is used in practically all regulations. In fact, the B-weighting was never used in any standards and it was ultimately removed as a result. Sometimes there are complementary requirements about C-weighted levels and/or peak-values in addition to equivalent sound levels in requirements.

f	250	500	1k	[Hz]
$L_{N,\text{original}}$	70	70	70	[phon]
$L_{p,\text{original}}$	76	71	70	[dB]
$L_{p,\text{reduced}}$	46	41	40	[dB]
$L_{N,\text{reduced}}$	35	38	40	[phon]
ΔL_N	35	32	30	[phon]

Table 1.1: Numerical example which illustrates the change in loudness level, L_N , when the sound pressure level, L_p , is lowered by 30 dB. The last row shows that when the sound pressure level is turned down, for instance when the volume control is turned down on an amplifier, a reduction in bass will be perceived.

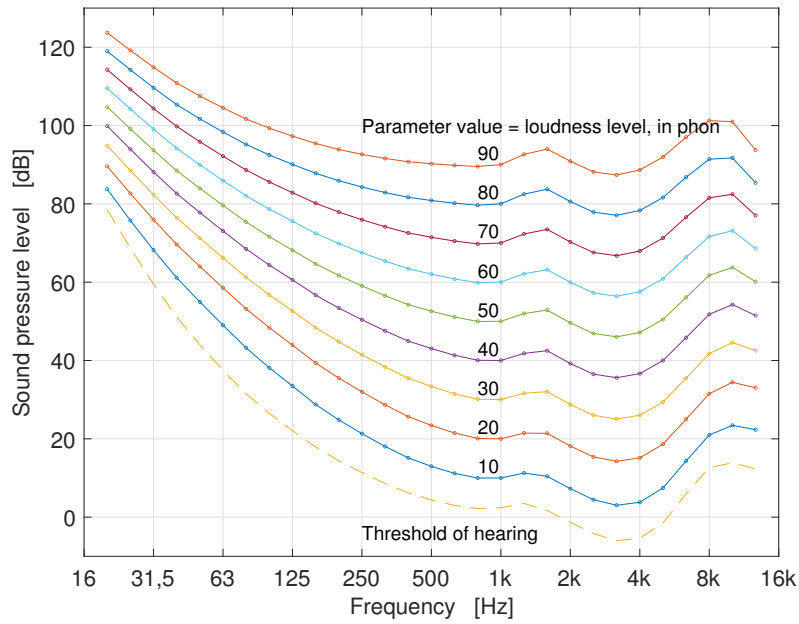


Figure 1.6: The equal loudness contours (the diagram was made using the Matlab function iso226.m (see mathworks.com) by Jeff Walkers).

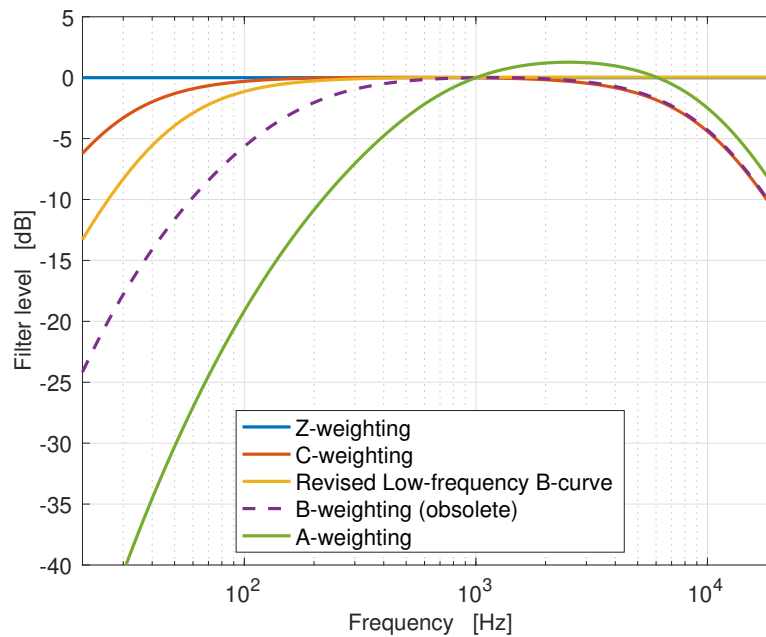


Figure 1.7: The standardized frequency-weighting curves for sound levels.

The international standard IEC 61672 specifies that all sound level meters (which is the most common acoustic measurement instrument) must include an A-weighting filter. The C-weighting filter, must also be included for the highest grade of sound level meters, so-called “precision sound level meters” (Class 1). The Z-weighting (flat frequency response) is optional.

Two classes are defined in the standard: Class 1 has smaller tolerance for error than Class 2. Some standards accept measurements with a Class 2 sound level meter, whereas others require Class 1. This classification involves the quality of the microphone, and also requires that a calibration can be made. Therefore, smartphone apps can typically not fulfill such classification (typically, the built-in microphone quality is varying, and can not be calibrated). However, some smartphone apps and microphones can actually perform well as sound level meters, see further below.

For some audio technology applications, the A-weighted level is not accurate enough for measuring the perceived level:

- In broadcasting, the perceived level by listeners should stay more or less constant across speech programs, different types of music, commercials etc. Therefore, special measurement quantities have been developed for this application, so-called Loudness monitoring, [4]. This involves another frequency-weighting: the revised low-frequency B-curve, also shown in Fig. 1.7.
- In listening tests for evaluating sound reproduction equipment, it is very important that two sound reproductions that are to be compared should be set to as similar perceived level as possible - otherwise the louder will practically always be perceived as better! More on this in a later chapter on hearing.
- The ITU standard for the broadcasting level monitoring, with the revised low- frequency B-curve, has been shown in listening tests to give excellent correlation with perceived loudness (at typical sound reproduction listening levels), see Fig. 1.8, [5]. The perceived loudness (“Subjective gain”) was evaluated for twelve different proposed loudness metrics. The measure based on this “revised low-frequency B-curve” performed best of all twelve. Interestingly, unweighted equivalent level performed almost as well,

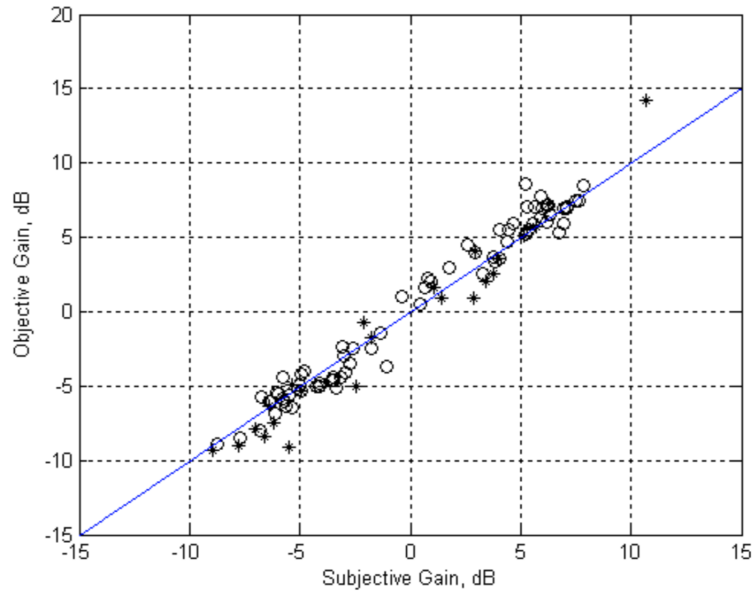


Figure 17: Meter M [$Leq(RLB)$].

Figure 1.8: Results from perception experiments on loudness monitoring, from [5]. "Subjective gain" is the loudness rated by test persons, and "objective gain" is the value calculated by the "Revised low-frequency B-curve".

A-weighted equivalent level performed second worst of all twelve, and B-weighted equivalent level was around the middle of all twelve. The other competing algorithms were commercial (hidden) algorithms.

The frequency-weighting discussed above has a direct analogy with the perception of light. The light intensity that hits a surface (so-called irradiance) has the unit W/m^2 . A weighting curve, or filter, that represents the human vision has been defined in the same way as for hearing. Interestingly, the light intensity, after weighting with such a weighting curve, has got its own quantity, illuminance, with its own unit, lux. Also, a light source's radiated light power (so-called radiant flux) is measured in watts, but after applying the same weighting curve, one gets the quantity luminous flux with the unit lumen.

The weighting curve is called the luminosity function, and was standardized in 1931 by the International Commission on Illumination (CIE). In a similar way as hearing has different frequency responses for different intensities, there are two different weighting curves for vision. The so-called photopic function applies to typical light levels whereas the so-called scotopic function is valid for low light intensities. To complete this (sidetrack) topic, the quantity luminous intensity, with the unit candela, is luminous flux per solid angle. A corresponding physical (or radiometric) quantity is radiant intensity, with the unit W/sr . There is no corresponding acoustic quantity which expresses directional radiation strength similarly. The unit candela is one of the seven fundamental units in the SI system.

1.2.4 Time-weighting of the sound pressure level (SPL)

Sound signals usually fluctuate over time, and there are two standardized ways to handle the time-dependence when measuring the sound pressure level, [2]:

1. Time-weighted sound level, denoted $L_F(t)$ or $L_S(t)$ depending on the choice of time constant: "Fast" or "Slow". The time-weighted sound level changes with time, and is therefore marked as a function of time.

2. Time-averaged sound level, or Equivalent continuous sound level, denoted $L_{\text{eq},T}$ where T is the averaging time interval. This level is a single value, i.e., not a function of time.

Both of these are expressed in dB. Below, we look at the second one first.

The equivalent continuous sound level

The equivalent continuous sound level is simply given by averaging $p^2(t)$ (possibly after it has been filtered with one of the standardized filters in the previous subsection) over a time period:

$$L_{\text{eq},T} \equiv 10 \lg \frac{\frac{1}{T} \int_0^T p^2(t) dt}{p_0^2} \text{ dB} \quad (1.13)$$

or

$$L_{\text{Aeq},T} \equiv 10 \lg \frac{\frac{1}{T} \int_0^T p_A^2(t) dt}{p_0^2} \text{ dB} \quad (1.14)$$

where T is the time period over which the averaging of $p^2(t)$ takes place. Either the unweighted, $p(t)$, or the A-weighted, $p_A(t)$, sound pressure signal is to be used, and consequently gives unweighted or A-weighted sound level.

For workplaces the time $T = 8$ hours is usually used. Not all sound level meters have the possibility to calculate an equivalent continuous sound level; they are called integrating- averaging sound level meters. They include a start and a stop function, in some form.

If we look at the expressions in Eqs. (1.13) and (1.14), and compare with Eq. (1.5), we realize that the expressions in the numerators of Eqs. (1.13) and (1.14) are just the RMS-values squared. Furthermore, by checking the expressions in Eqs. (1.8) and (1.9), we realize that "The equivalent continuous sound level" is apparently based on an *unwindowed* RMS-value estimate. But, wasn't it argued above that using a window always gives a smaller uncertainty for the RMS-value? It was indeed, but for this particular use, two arguments speak for not using a window:

1. The equivalent continuous sound level is usually measured over a long time period T , e.g., a work day of 8 hours. Then the uncertainty for the unwindowed rms-value is very small: imagine Fig. 1.5 (a) for very large values of N/N_{period} .
2. Sound level meters which measure the equivalent continuous sound level have a start and a stop button, and then a symmetrical window, like a Hann window, can not be applied without storing the entire signal.

The time-weighted sound level

The most common type of sound level meter is calculating and displaying the time-weighted sound level, which is based on a low-pass filtered version of $p^2(t)$, using one of two time constants, denoted by F or S. For each of these, the unweighted or the A-weighted sound level can be computed. For the F time constant, one can use the unfiltered sound pressure signal, $p(t)$,

$$L_F(t) \equiv 10 \lg \frac{\frac{1}{\tau_F} \int_{-\infty}^t p^2(\xi) e^{-(t-\xi)/\tau_F} d\xi}{p_0^2} \quad (1.15)$$

or one can use the A-weighted sound pressure signal, $p_A(t)$,

$$L_{\text{AF}}(t) \equiv 10 \lg \frac{\frac{1}{\tau_F} \int_{-\infty}^t p_A^2(\xi) e^{-(t-\xi)/\tau_F} d\xi}{p_0^2} \quad (1.16)$$

This integral simply expresses a convolution of $p^2(t)$ and an exponential weighting function. Such an exponential weighting function is the impulse response of a first-order low-pass filter. The two standardized time constants are:

F: $\tau_F = 125$ ms Levels are denoted L_{AF} or L_{CF} , for A- and C-weighting, respectively.

S: $\tau_S = 1$ s Levels are denoted L_{AS} or L_{CS} , for A- and C-weighting, respectively

According to IEC 61672-1, as a minimum, a sound level meter must have the “F” time-weighting.

Again, we compare with the previous expressions for the RMS-value squared, and realize that for this “Time-weighted sound level”, a window is actually used for computing the RMS-estimate and the window is an exponentially shaped window. An exponential window is asymmetrical, giving the weight 1 to the latest value of $p^2(t)$ and then an exponentially decreasing amplitude towards earlier times. Without going into the details here, we claim that the multiplication of $p^2(t)$ with an exponentially shaped window can be implemented very efficiently as a convolution, or filtering, with a first-order low-pass filter, as mentioned above. In addition, such an implementation automatically gives continuously updated values of $\hat{p}_{\text{rms}}^2(t)$.

The maximum time-weighted sound level

The maximum time-weighted sound level, within a stated time interval, is denoted $L_{AF\text{max}}$ or $L_{CF\text{max}}$, or $L_{AS\text{max}}$ or $L_{CS\text{max}}$. This is simply the maximum reading of, e.g., $L_{AF}(t)$ during the interval.

The peak sound level

The peak sound level is the highest sound level, calculated from the *instantaneous* $p^2(t)$ (or $p_A^2(t)$) during a stated time interval.

Note the difference between those two levels which both describe the highest value in two different forms: the maximum time-weighted sound level always includes the F or S time-weighting (low-pass filtering), which leads to a lower value than the peak sound level. These two levels require a “hold” function on the sound level meter.

1.2.5 Sound exposure

A final quantity to mention is the sound exposure, E , which is the time integral, over a specified time interval, of the squared sound pressure signal, unweighted or weighted. The quantity has the unit Pa^2s .

$$E_T \equiv \int_{t_1}^{t_2} p^2(t) dt$$

$$E_{A,T} \equiv \int_{t_1}^{t_2} p_A^2(t) dt$$

where $T = t_2 - t_1$ is the time interval over which the sound exposure level is computed. The sound exposure can also be expressed in dB, which is the sound exposure level, with the reference value $400 \cdot 10^{-12} \text{ Pa}^2\text{s}$. It can use unweighted or A-weighted (or C-weighted) sound pressure signals:

$$L_{E,T} \equiv 10 \lg \frac{\int_{t_1}^{t_2} p^2(t) dt}{p_0^2 T_0}$$

$$L_{AE,T} \equiv 10 \lg \frac{\int_{t_1}^{t_2} p_A^2(t) dt}{p_0^2 T_0}$$

T_0 is a reference value, which is 1 s.

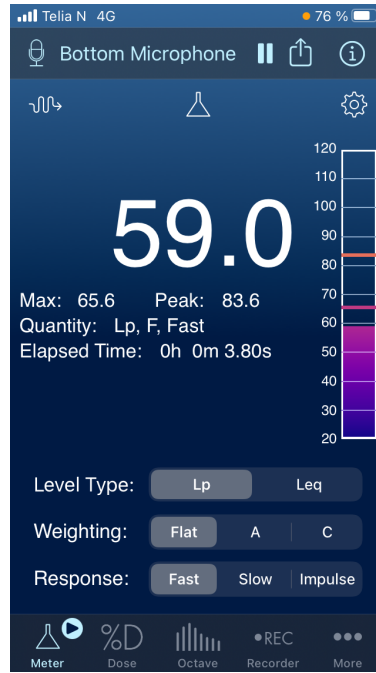


Figure 1.9: Screenshot from the smartphone app "SoundMeter X" from Faber Acoustical.

1.2.6 Example of smartphone sound level meter app

Fig. 1.9 shows a screenshot from a smartphone with the "SoundMeter X" app from Faber Acoustical [6].

We can see:

"Level type": "Lp" and "Leq", corresponds to Time-weighted sound level and Equivalent continuous sound level, respectively.

"Weighting": "Flat" (corresponding to "Z"), "A" and "C"

"Response": "Fast" (= "F"), "Slow" (= "S"), and "Impulse" (which is not standardized).

We can also see that Max and Peak values are displayed.

There has been much interest in sound level meter apps for smartphone apps, and several papers have evaluated a number of apps and microphones, e.g [7], [8].

1.2.7 Spectral analysis - Octave bands

A spectral analysis of sound is often required. Standardized band-pass filters with constant relative bandwidths, so-called *octave band filters* can then be applied [9]. See the illustration in Fig. 1.10.

The centre frequencies, or mid-band frequencies, of octave bands $f_{m,ob}$, are given by

$$f_{m,ob} = f_r 10^{3x/10}$$

where f_r is the reference frequency, which is 1000 Hz, and x is an integer, positive or negative (...,-2,-1,0,1,2,...):

$$x = -2 \Rightarrow f_{m,ob} = 251.19... \text{ Hz} \approx 250 \text{ Hz}$$

$$x = -1 \Rightarrow f_{m,ob} = 501.19... \text{ Hz} \approx 500 \text{ Hz}$$

$$x = 0 \Rightarrow f_{m,ob} = 1000 \text{ Hz}$$

etc

Note that an octave-band filter is usually called the “500 Hz-filter” etc even if the exact mid-band frequency is actually 501.19.. Hz etc. The frequency 500 Hz is called the ”nominal mid-band frequency” and it is used as a label, or name, for the band since it is more practical to use than the exact frequencies.

Furthermore, for octave bands, the lower and upper cutoff-frequencies, f_1 and f_2 , and the bandwidths, Δf , are:

$$\begin{aligned} f_{1,\text{ob}} &= f_{\text{m,ob}} 10^{-1.5/10} \approx 0.708 f_{\text{m,ob}} \\ f_{2,\text{ob}} &= f_{\text{m,ob}} 10^{1.5/10} \approx 1.413 f_{\text{m,ob}} \\ \Delta f_{\text{ob}} &= f_{\text{m,ob}} \left(10^{1.5/10} - 10^{-1.5/10} \right) \approx 0.705 f_{\text{m,ob}} \end{aligned}$$

Fractional-octave band analysis

There are alternatives to octave-band analysis with higher frequency resolution, so-called fractional-octave-band filters. For instance, 1/3-octave bands are used in quite many standards for various acoustic measurements. In addition, 1/6-octave bands are very similar to the so-called ”Equivalent Rectangular Bandwidth” (ERB) filters, that are often used to model the auditory system. For a $1/b$ -th octave band, the mid-frequencies, lower and higher cutoff frequencies, and bandwidths, are given by

$$\begin{aligned} f_{\text{m},1/b} &= f_{\text{r}} 10^{3x/(10b)} \\ f_{1,1/b} &= f_{\text{m},1/b} 10^{-3/(20b)} \\ f_{2,1/b} &= f_{\text{m},1/b} 10^{3/(20b)} \\ \Delta f_{2,1/b} &= f_{\text{m},1/b} \left(10^{3/(20b)} - 10^{-3/(20b)} \right) \end{aligned}$$

where, again, x is a positive or negative integer, and f_{r} is 1000 Hz. We can note that sets of fractional-octave-bands for which b differ by a factor of 3, every third midband frequency will coincide. Thus the 1/3-octave band midband frequencies are as follows, and the ones with underscore are also octave band midband frequencies:

$$\begin{aligned} f_{\text{m},1/3} &\approx \dots, \underline{63}, 80, 100, \underline{125}, 160, 200, \underline{250}, 315, 400, \underline{500}, \\ &630, 800, \underline{1000}, 1250, \dots \end{aligned}$$

As mentioned above, note that we use the approximate values, or nominal midband frequencies, 250, 315, etc since they are practical *labels* for the bands. Note also that for series where b is a multiple of 3, the values repeat for each decade - so if we memorize the values between 100 and 1000, we automatically know them for

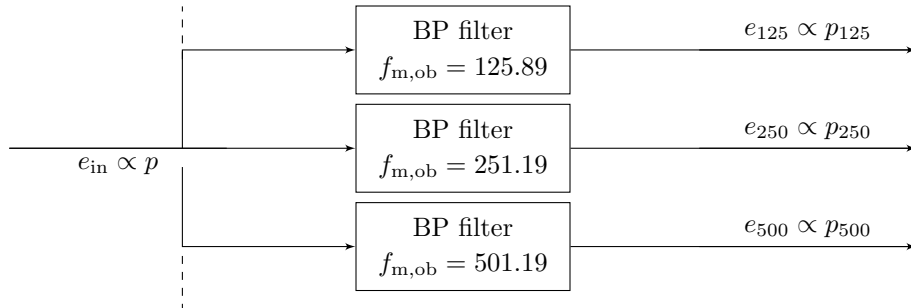


Figure 1.10: Octave band analysis is done by a number of bandpass filters connected in parallel.

lower and higher decades too.

The bandpass filters in Fig. 1.10 are constructed such that the energy of a signal is decomposed into the various octave bands. This decomposition means that *the energy of the input signal, the broadband signal, should be the same as the sum of the energies of the octave band signals!* Thus, if we assume that a signal has no spectral content below and above the octave bands of 125 Hz and 4 kHz, respectively, then

$$p_{\text{broadband,rms}}^2 = p_{\text{octave 125,rms}}^2 + p_{\text{octave 250,rms}}^2 + \dots + p_{\text{octave 4k,rms}}^2 \quad (1.17)$$

The energy decomposition also requires that the bandpass filters must have unity passband gain, that is, 0 dB passband gain. A unity passband gain implies that if a signal contains only a single sinusoidal, with a certain amplitude, then the output signal in one of the (fractional-)octave bands in Fig. 1.10 should (ideally) be identical to this input signal: a sinusoidal with the same amplitude.

Numerical example 1:

Let's assume that a measurement instrument is giving us sound pressure levels in octave bands as in the table below. What is the sound pressure level for the broadband signal?

Octave band:	125	250	500	1k	2k	4k	[Hz]
$L_{p,\text{octave}}:$	45	42	40	35	34	33	[dB] re. 20 μPa

We can use Eq. (1.17) to find the broadband signal, but we apparently need to convert the octave band levels to p_{rms}^2 -values for each octave band. After summing, we also need to convert the broadband p_{rms}^2 -value to a sound pressure level in dB. Using the definitions in Eqs. (1.10) and (1.12), we can derive the wanted value:

$$\begin{aligned} L_{p,\text{broadband}} &= 10 \lg \frac{p_{\text{broadband,rms}}^2}{p_0^2} = 10 \lg \frac{p_{\text{octave 125,rms}}^2 + \dots + p_{\text{octave 4k,rms}}^2}{p_0^2} \\ &= 10 \lg \frac{p_0^2 \cdot 10^{\frac{L_{p,\text{oct.125}}}{10}} + \dots + p_0^2 \cdot 10^{\frac{L_{p,\text{oct.4k}}}{10}}}{p_0^2} = 10 \lg \left(10^{\frac{L_{p,\text{oct.125}}}{10}} + \dots + 10^{\frac{L_{p,\text{oct.4k}}}{10}} \right) \\ &= 10 \lg \left(10^{\frac{45}{10}} + \dots + 10^{\frac{33}{10}} \right) = 48,1 \text{ dB re. } 20 \mu\text{Pa} \end{aligned} \quad (1.18)$$

Numerical example 2:

A number of sound pressure levels have been measured in a room, to find a representative average level. But, in most situations, the correct average level is given by *the mean of the p_{rms}^2 -values*. As one example, we have measured these sound pressure levels:

Position, i :	1	2	3	4	5
$L_{p,i}:$	45	47	44	44	46
					[dB] re. 20 μPa

The average level is then

$$\begin{aligned} L_{p,\text{average}} &= 10 \lg \frac{p_{\text{rms,average}}^2}{p_0^2} = 10 \lg \frac{\frac{1}{5} (p_{1,\text{rms}}^2 + \dots + p_{5,\text{rms}}^2)}{p_0^2} \\ &= 10 \lg \frac{\frac{1}{5} \left(p_0^2 10^{\frac{L_{p,1}}{10}} + \dots + p_0^2 10^{\frac{L_{p,5}}{10}} \right)}{p_0^2} = 10 \lg \left[\frac{1}{5} \left(10^{\frac{L_{p,1}}{10}} + \dots + 10^{\frac{L_{p,5}}{10}} \right) \right] \end{aligned} \quad (1.19)$$

$$= 10 \lg \left[\frac{1}{5} \left(10^{\frac{45}{10}} + \dots + 10^{\frac{46}{10}} \right) \right] = 45,4 \text{ dB re. } 20 \mu\text{Pa}$$

Note the important difference between the expressions in Eqs. (1.18) and (1.19) for :

- *Summing* a number of sound pressure levels, such as octave band values, Eq. (1.18), or
- *Averaging* a number of sound pressure levels, such as measurements in different positions, or at different times, Eq. (1.19).

When one has carried out a number of rms-squared summations, starting from dB-values, one usually observes that when two levels, a larger L_1 and a smaller L_2 , are to be summed, *and they differ by X dB* - then the result will always be Y dB larger than L_1 , *regardless of the value L_1 !*

A numerical example:

If we have two sound levels, $L_1 = 60$ dB and $L_2 = 57$ dB, then their sum (p^2 -based addition) will be 61.8 dB.

If $L_1 = 42$ dB and $L_2 = 39$ dB, then the sum will be 43.8 dB, etc.

Thus if one learns a few such relationships (a difference of 3 dB always means the sum is 1.8 dB than the larger of the two levels, etc), then one can quite easily make rough estimates of level summations.

Hint:

You could easily generate a diagram, using e.g. Matlab, with "Difference in level. [dB]" on the x-axis, and "Addition to highest level. [dB]" on the y-axis. According to the numerical example above, the curve in that diagram should pass through (-3 dB, +1.8 dB).

1.2.8 FFT-based spectral analysis

In the previous subsection, spectral analysis was done by splitting up the signal into band-pass filtered versions, and then the rms-value could be calculated for each such band. This analysis can be done for analog, continuous-time signals, $p(t)$, using analog/electronic bandpass filters, followed by "RMS-detecting" electronic circuits. Usually the analysis is of course done using digital signals, $p(n)$, and digital signal processing for the bandpass filtering and RMS calculations.

A very useful alternative is to use Fourier analysis for doing a spectral analysis of a signal. Then we basically have to use digital signals, $p(n)$, and digital signal processing in the form of the Discrete Fourier Transform (DFT), usually in the form of the very efficient Fast Fourier Transform (FFT)*.

Since the FFT can be computed so efficiently, it is possible to get very high-resolution spectral analysis done by the FFT-based approach†.

The DTFT, DFT, and FFT

We assume that we have a time-domain signal $x(n)$ of length N samples. We can then analyze the presence in the signal of a component with *any* angular frequency ω by carrying out this sum, which is the *Discrete-Time Fourier Transform*, or DTFT:

$$X(\omega) = \sum_{n=0}^{N-1} x(n) e^{-j\omega \frac{n}{f_s}} \quad (1.20)$$

*A very fascinating description of an analog instrument which carried out Fourier analysis in the 1800s can be viewed in the Veritasium channel on Youtube: <https://www.youtube.com/watch?v=IgF3OX8nT0w>.

†It is possible to compute a high-resolution spectral analysis using band-pass filters, but the implementation of many bandpass filters with a sharp filter characteristic is very costly.

where, again, f_S is the sampling frequency, and X is a complex number where the magnitude and phase tells us the amplitude and phase shift of the sinusoidal which the signal contains. The DTFT gives us the value X for one angular frequency ω , but we must be a bit careful: if we compute the sum in Eq. (1.20) for a number of samples, N , then in general, N will not be an integer number of periods for the angular frequency ω . As a result, the value $X(\omega)$ will vary (somewhat) for different signal lengths N , and we might need to study the convergence of $X(\omega)$ as N is increased.

This can be expanded into the *Discrete Fourier Transform*, or DFT, which gives the values $X(\omega_k)$ for a set of frequencies, $\omega_k, k \in [0, N_{\text{DFT}} - 1]$, where N_{DFT} is the same as the length N samples of the time-domain signal $p(n)$:

$$X(\omega_k) = \sum_{n=0}^{N_{\text{DFT}}-1} x(n) e^{-j\omega_k \frac{n}{f_S}}$$

and the frequencies are regularly distributed, $\omega_k = 2\pi k f_S / N_{\text{DFT}}$, so

$$X(\omega_k) = \sum_{n=0}^{N_{\text{DFT}}-1} x(n) e^{-j \frac{2\pi n k}{N_{\text{DFT}}}} \quad (1.21)$$

The time-domain (TD) signal can be recovered exactly (apart from numerical/computation noise) from the frequency-domain (FD) coefficients through the inverse DFT:

$$x(n) = \frac{1}{N_{\text{DFT}}} \sum_{k=0}^{N_{\text{DFT}}-1} X(\omega_k) e^{j \frac{2\pi n k}{N_{\text{DFT}}}} \quad (1.22)$$

We can observe the very strong similarity between the DFT, Eq. (1.21), and the inverse DFT, Eq. (1.22). The scale factor, $1/N_{\text{DFT}}$, differs between implementations, and between text books. The scale factors shown here correspond to the Matlab implementation, but below we will also find it useful with an alternative scaling scheme. With the alternative scaling scheme the DFT is

$$X'(\omega_k) = \frac{1}{N_{\text{DFT}}} \sum_{n=0}^{N_{\text{DFT}}-1} x(n) e^{-j \frac{2\pi n k}{N_{\text{DFT}}}} \quad (1.23)$$

and the TD signal is recovered by

$$x(n) = \sum_{k=0}^{N_{\text{DFT}}-1} X'(\omega_k) e^{j \frac{2\pi n k}{N_{\text{DFT}}}} \quad (1.24)$$

Some important observations can be made about the DFT:

1. The DFT is assuming that the signal $x(n)$ is periodical. *This is a very useful observation to keep in mind, which can explain many further observations that can be made!* This assumption of periodicity also means that there is no uncertainty of the value of $X(\omega_k)$ as was described for the DTFT above!
2. The frequency-domain (FD) coefficients, $X(\omega_k)$, are also periodical, that is,

$$X(\omega_k + m \cdot 2\pi f_S) = X(\omega_k)$$

where $m = [\pm 1, \pm 2, \pm 3, \dots]$. In particular, that means that $X(2\pi f_S) = X(0)$.

3. For a real-valued signal $x(n)$, the FD coefficients are "symmetrically complex-conjugated" around 0 rad/s:

$$X(-\omega_k) = X(\omega_k)^*$$

4. For a real-valued signal $x(n)$, the FD coefficients for 0 rad/s and $\omega_k = 2\pi f_S/2$ must be real-valued (the latter exists only if N_{DFT} is even).

5. Items 3 and 4 lead to the following:

- (a) The TD signal has N_{DFT} different/independent real-valued numbers
- (b) The FD representation has N_{DFT} *complex* values, which seems like twice as much information as the TD signal, *but* the FD representation has redundancy and exactly N_{DFT} independent values: two real values (item 4), and $N_{\text{DFT}}/2 - 1$ complex values (item 5).

6. Item 5 leads to the possibility to save some storage and processing: once a DFT has been computed, it is possible to throw away the redundant values and keep only the FD coefficients number for $k = [0, N_{\text{DFT}}/2]$ ($N_{\text{DFT}}/2 + 1$ values) and do all subsequent FD processing for this reduced set.

7. The DC-component, that is, $X(0)$, is simply the sum of all TD sample values:

$$X(0) = \sum_{n=0}^{N_{\text{DFT}}-1} x(n)$$

8. If N_{DFT} has a value which is 2^M , where M is an integer, then the faster FFT algorithm can be used. Curiously, the Matlab function `fft` computes the DFT in general, and computes the FFT if the input vector happens to have the length which fit with the FFT requirement.

Power spectrum/power spectral density, PSD

In previous sections, we discussed the squared rms-value for a signal, and the squared rms-value is often said to represent the *power* of a signal, even if the square of a signal (that is, the real-world signal which we process in its digital form) does not have the unit of watts. However, the power in some physical system is proportional to the square of a real-world signal, and therefore we often use the term power to describe the square of a TD signal. Similarly, the square of the magnitudes of the FD coefficients that result from a DFT represent the power of the signal. A very important equality relationship exists between the power expressions in the TD and the FD: Parseval's formula, which can be written as follows for digital signals and the DFT (which, again, assumes that $x(n)$ is periodical):

$$x_{\text{RMS}}^2 = \frac{1}{N_{\text{DFT}}} \sum_{n=0}^{N_{\text{DFT}}-1} x^2(n) = \frac{1}{N_{\text{DFT}}^2} \sum_{k=0}^{N_{\text{DFT}}-1} |X(\omega_k)|^2 = \sum_{k=0}^{N_{\text{DFT}}-1} |X'(\omega_k)|^2 \quad (1.25)$$

Please note that for Eq. (1.25) to hold, one must sum the energy of *all* frequency "bins". Thus, if one discards the redundant, mirrored complex-conjugated ones (as suggested earlier), one has to compensate for that discarded energy.

The coefficients $|X'(\omega_k)|^2$ are called the *Power Spectral Density* (coefficients).

Spectral leakage and the use of windowing

We will now demonstrate a common effect of the DFT, the so-called "spectral leakage" effect. First, we generate a sinusoidal with a frequency which is the sampling frequency divided by some integer. Such a very special case implies that if we plot two "frames" of N_{DFT} samples right after each other, the two frames connect perfectly and the repeated signal looks like the perfect sinusoidal, as illustrated in Fig. 1.11 (a). The DFT, using the $X'(\omega_k)$ form in Eq. (1.23), then has a non-zero value only for exactly the frequency of that sinusoidal, see Fig. 1.11 (b). Note that there are two lines because of the mirroring effect of the DFT when the TD signal is real-valued. We can easily check by hand if Eq. (1.25) holds. The RMS-value squared for the TD signal is easy to calculate since the sinusoidal is sampled perfectly: $x_{\text{RMS}}^2 = 0,5$. In Fig. 1.11(b) we see that $X'(\omega_k) = 0,5$, so

the squared value (the PSD coefficient) is $|X'(\omega_k)|^2 = 0,25$. Since there are identical values for two frequency bins, the sum will be 0.5.

Now we construct a sinusoidal with a slightly different frequency, and plot the same quantities in Fig. 1.12. When the single "frame" of N_{DFT} samples is repeated in Fig. 1.12 (a), the two connected signals create a little jump, and it doesn't anymore look like a perfect sinusoidal. The DFT assumes that the periodical signal has exactly that little jump, *and that little jump implies that there must be other frequency components!* In Fig. 1.12 (b), the DFT does indeed have non-zero values for many frequencies, with a peak *in the neighborhood* of the single frequency which we started with.

We check the Parseval relationship again. The RMS value squared of the TD signal will not be exactly 0.5 this time since the samples are not perfectly distributed along the sinusoidal. The RMS value squared happens to be 0,479895 for this specific case, and if we sum all the PSD-coefficients, that is, all the $|X'(\omega_k)|^2$ -values, we get 0,479895 as well.

Now we apply a Hann window to the TD-signal before computing the DFT, exactly like in the section on the computation of RMS-values in practice. For the same frequencies as in Figs. 1.11 and 1.12, we get the results in Figs. 1.13 and 1.14. Note that the TD signals have been scaled by a Hann-window correcting scale factor which gives an (almost) correct RMS-value. A main observation is that the Hann window gives much more similar spectral leakage effects between the two example frequencies, see Figs. 1.13 (b) and 1.14 (b).

These examples demonstrate the so-called *spectral leakage* effect: even if the underlying signal contains a single frequency, its energy is distributed across many frequencies, and that effect is called "leaking". The amount of leaking is strongly affected by the use of a window, and different windows have different spectral leakage characteristics.

It should, however, be realized that windows are used for *stationary signals*, in order to suppress the abrupt steps at the start and/or end of the analysis window. Often, we analyze transient signals or impulse responses, and they should usually have died out before the end of the analysis window. For transient signals, we do not (usually) want to use windows since a window modifies the spectrum, or frequency response, in unknown ways^a.

^aThe window modifies the spectrum of a stationary signal as well, but in known ways!

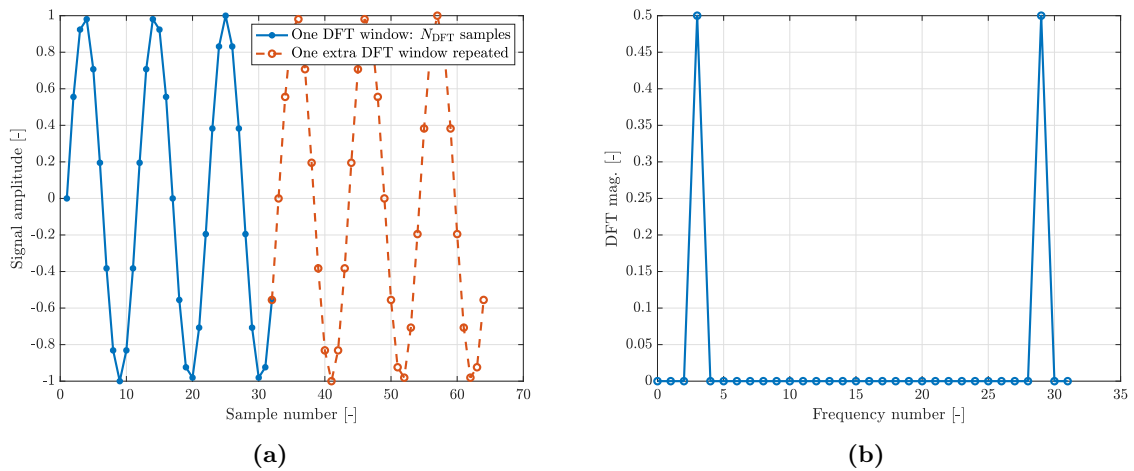


Figure 1.11: (a) A sinusoidal with a frequency selected to be the sampling frequency divided by an integer is sampled by a DFT-window of 32 samples. A repetition is also plotted, illustrating that the repetition connects perfectly. Thus, the periodicity which is assumed by the DFT creates an ideal sinusoidal. (b) The DFT coefficient magnitudes (based on Eq. (1.23)) of the signal in (a). Note the 32 frequency "bins" and the mirroring effect.

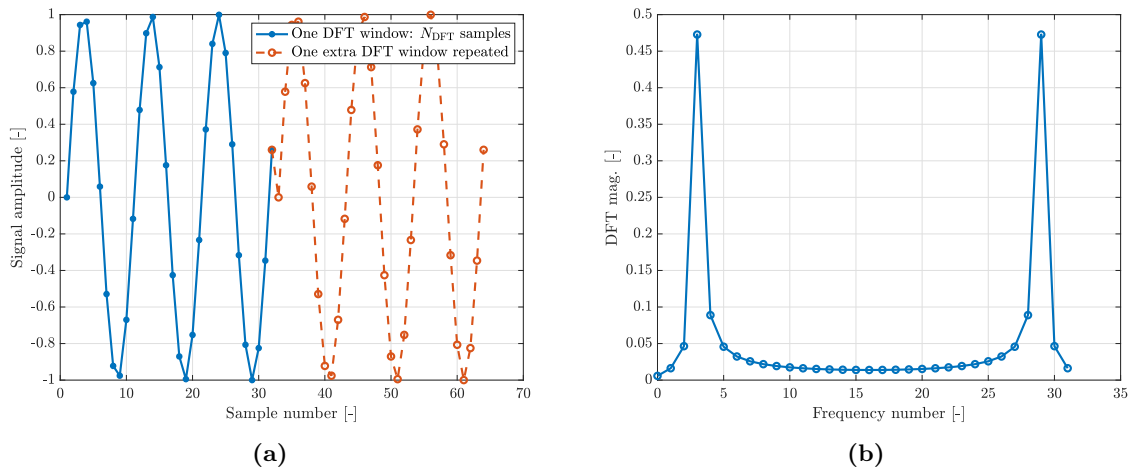


Figure 1.12: (a) A sinusoidal with a frequency selected to *not* be the sampling frequency divided by an integer, sampled by a DFT-window of 32 samples. A repetition is also plotted, illustrating that the repetition creates a jump. Thus, the periodicity which is assumed by the DFT creates a range of frequency components. (b) The DFT coefficient magnitudes (based on Eq. (1.23)) of the signal in (a). Note the 32 frequency "bins", the mirroring effect, and the spectral leakage.

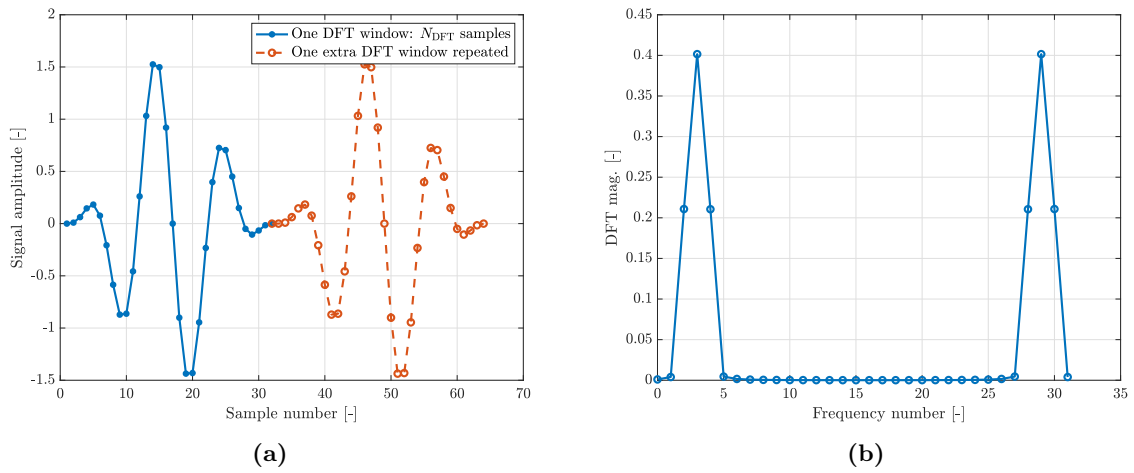


Figure 1.13: (a) The same sinusoidal as in Fig. 1.11 (a), but now with a Hann window applied. (b) The DFT coefficient magnitudes (based on Eq. (1.23)) of the signal in (a). Note that the ideal spectrum in Fig. 1.11 (b) is now replaced by a not-quite-ideal spectrum.

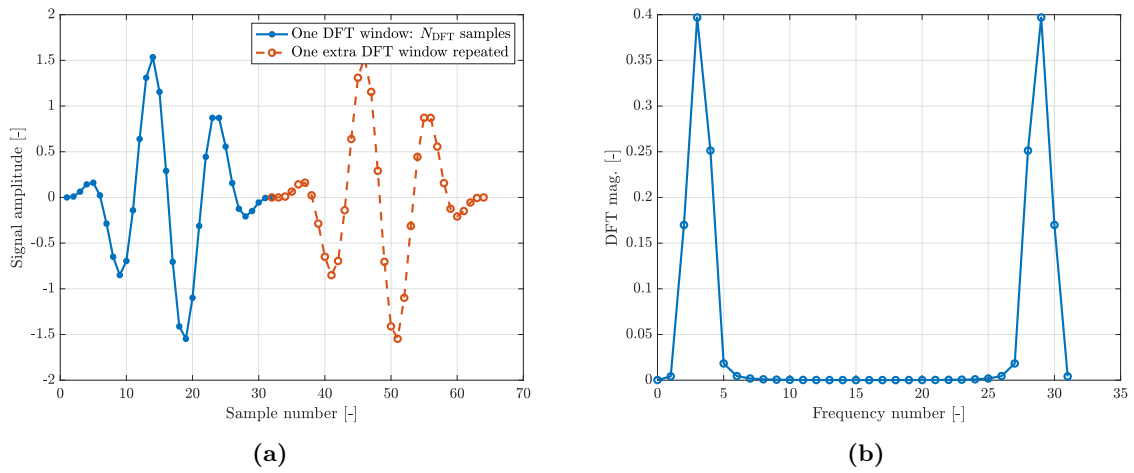


Figure 1.14: (a) The same sinusoidal as in Fig. 1.12 (a), but now with a Hann window applied. (b) The DFT coefficient magnitudes (based on Eq. (1.23)) of the signal in (a). Note that the spectrum has much less spectral leakage than the one in Fig. 1.12 (b).

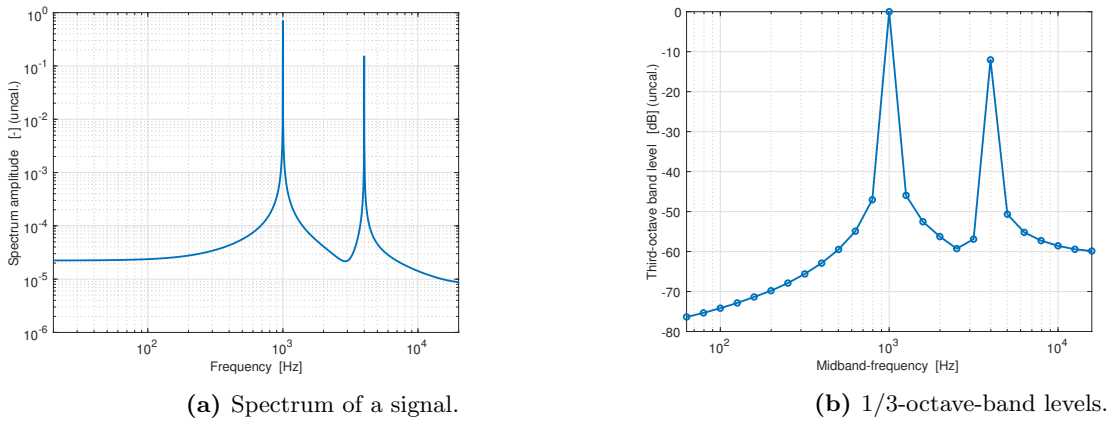


Figure 1.15: (a) Spectrum and (b) 1/3-octave band levels of a synthetic signal with two sinusoids, at 1 kHz and 4 kHz. The spectrum in (a) seems to show that there are not only two pure tones, but this is an effect of taking an FFT of a signal with frequency components that are not exact multiples of the frequency resolution.

FFT-based fractional-octave-band analysis

The IEC standard referred to above gives detailed requirements for the bandpass filters that should be used in (fractional-)octave band analysis as illustrated in Fig. 1.10. However, it can often be more practical to use FFT-based analysis, even though it is not so straightforward to evaluate if such analysis fulfills the requirements for fractional-octave bands or not.

Assume that we have computed a spectrum of a time-signal with an FFT, and stored in e.g., `pspectrumvec`, and an associated frequency vector (with resolution $\Delta f = f_S/n_{\text{FFT}}$) is stored in `fvec`. Then, the level in a single fractional-octave-band, with frequency limits f_1 and f_2 is computed as:

```
% Lazy version:
ivec = find(fvec > f1 & fvec < f2);
% More efficient version:
df = fs/nfft;
ivec = [round(f1/df)+1:round(f2/df)+1];
Lband = 10*log10( sum( abs(pspectrumvec(ivec)).^2 ));
```

After taking care of the scaling issues (e.g., related to FFT-size used), this way to compute octave-band levels can be very efficient. One example is shown in Fig. 1.15, where (a) shows the spectrum of a synthetic signal with a 1 kHz component and a 4 kHz component (and the amplitude of the latter was 0.25 times the amplitude of the former). In Fig. 1.15 (b), the corresponding 1/3 octave-band levels have been computed from the spectrum in Fig. 1.15 (a). The levels are expressed with some arbitrary reference value. In the processing, a sampling frequency of 44.1 kHz was used, and a signal length of 65536 samples, so that an FFT size of 65536 was used to compute the spectrum. As a sidenote, we could acknowledge that in the special case that the frequency components are multiples of $\Delta f = f_S/n_{\text{FFT}}$, then the spectrum in Fig. 1.15 (a) would have had single peaks at precisely those frequency components, and the amplitude for other frequencies would be around the computation noise floor, which is around 10^{-15} . In other words, the non-zero amplitudes to the side of the expected frequency components are consequences of the fact that the frequency components in the signal don't happen to be multiples of $\Delta f = f_S/n_{\text{FFT}} = 44100/65536 \approx 0.672913$ Hz. This is the spectral leakage effect discussed in the section on the PSD.

1.3 LTI systems

A linear, time-invariant (LTI) system with one input signal and one output signal is illustrated in Fig. 1.16(a). Such a system can be characterized by its *impulse response* (IR), $h(n)$ (in the discrete-time/digital domain), or, equivalently, its *transfer function* (TF)/*frequency response* function, $H(\omega)$. The input-output relationship for the discrete-time signals is given by the convolution

$$y(n) = x(n) * h(n) = \sum_{k=0}^{\infty} h(k)x(n-k) \quad (1.26)$$

where we have assumed that the impulse response is causal (by starting the sum at $k = 0$ rather than at $k = -\infty$). In practice, a measured impulse response is usually known only for a finite length, a finite impulse response, even if the underlying real-world impulse response might have an infinite extent. For a finite impulse response of length N samples, the sum in Eq. (1.26) ends at $N - 1$ rather than ∞ . The frequency-domain relationship is

$$Y(\omega) = H(\omega)X(\omega) \quad (1.27)$$

where X, Y, H are the Discrete-time Fourier transforms (DTFT) of x, y, h , respectively, see also Eq. (1.20),

$$H(\omega) = \mathcal{F}[h(n)] = \sum_{n=0}^{\infty} h(n)e^{-j\omega \frac{n}{f_s}} \quad (1.28)$$

etc for x and y . The computation of the DTFT is straightforward if the signals $x(n)$ and $y(n)$ are of finite length, or periodical. For general (non-periodical) signals, it is a bit more intricate to compute the DTFT, but we move onto the DFT since the DFT is what is used in practice.

The DFT gives us values only for the discrete frequencies $f_k = k\Delta f = k\frac{1}{T} = k\frac{f_s}{n_{\text{DFT}}}$ where k are integers, $k \in [0, n_{\text{DFT}} - 1]$, f_s is the sampling frequency, and $n_{\text{DFT}} = N$ is the number of time samples in one period, T , of a time signal *which is assumed to be periodic*. See also Eq. (1.21). Eq. (1.27) can then be expressed as

$$Y(\omega_k) = H(\omega_k)X(\omega_k) \quad (1.29)$$

where X, Y, H now are the Discrete Fourier transforms (DFTs) of x_{periodic} , y_{periodic} , and h_{periodic} , respectively:

$$\begin{aligned} X(\omega_k) &= \text{DFT}[x_{\text{periodic}}(n)] = \sum_{n=0}^{N-1} x_{\text{periodic}}(n)e^{-j\omega_k \frac{n}{f_s}} \\ &= \sum_{n=0}^{N-1} x_{\text{periodic}}(n)e^{-jk\frac{2\pi f_s}{N}\frac{n}{f_s}} = \sum_{n=0}^{N-1} x_{\text{periodic}}(n)e^{-j\frac{2\pi n}{N}k} \end{aligned} \quad (1.30)$$

etc for y_{periodic} and h_{periodic} . The subscript "periodic" is written as a reminder that the Discrete Fourier Transform *assumes* that the time signals x and y are periodic. The DFT in Eq. (1.30) is computed from a

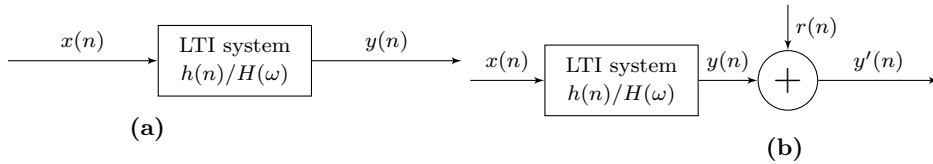


Figure 1.16: (a) A single-input, single-output LTI system. (b) In a measurement situation, the system in (a) will have some random disturbance signal, r , added to the output signal.

sequence of $N_{\text{DFT}} = N$ time samples of x or y , and therefore one such set of DFT-values, $X(\omega_k)$, "doesn't know anything about" (isn't affected by) the time signal outside that window in time, or frame.

For the IR/TF pair, $h(n)/H(\omega_k)$, the DFT periodicity assumption takes a special role, as described in Appendix 1.5.1 - if a system has an impulse response which is longer than the DFT size n_{DFT} , then the impulse response is *periodicized*, or wrapped around itself for certain measurement methods.

A common task is to find the impulse response and/or transfer function/frequency response function of such an LTI system and then we encounter the situation in Fig. 1.16(b) - we can not observe the ideal output signal $y(n)$, but one which has some disturbance signal, $r(n)$, added to it, and we can call the observable output signal $y'(n)$. Usually we assume that $r(n)$ is a random signal, which is independent of $x(n)$.

In some fields, the term "Transfer function" is used for $H(s)$, the Laplace transform of the continuous-time impulse response, $h(t)$, and the function $H(\omega)$ or $H(j\omega)$ is then called "Frequency response function". In acoustical engineering, we find that the terms Transfer function and Frequency response function might both be used for $H(\omega)$.

1.3.1 Is the system analog or digital?

The "system" that we are interested in often exists in the physical world, and then the system should rather have been defined with an analog, continuous-time impulse response, $h(t)$, rather than its digital, discrete-time equivalence, $h(n)$. When our focus is on the real-world system, then it makes more sense to use the continuous-time descriptors. However, this section deals with the *measurement* of the real-world systems, and our measurement systems are exclusively working in the digital domain. Therefore, all our measurement algorithms must work on digital signals, and we use the discrete-time representations. We could illustrate the measurement system as in Fig. 1.17.

1.3.2 Methods for measuring LTI systems

In measurement situations the terms "input signal" and "output signal" become ambiguous. The signal x that is used to excite a system (which can be a loudspeaker, a room, etc) is an input signal when viewed from the system to be identified, *but acts as an output signal when viewed from the measurement system* that generates and sends out the excitation signal. The same situation is true for the response signal from the system to be identified, $y(n)$.

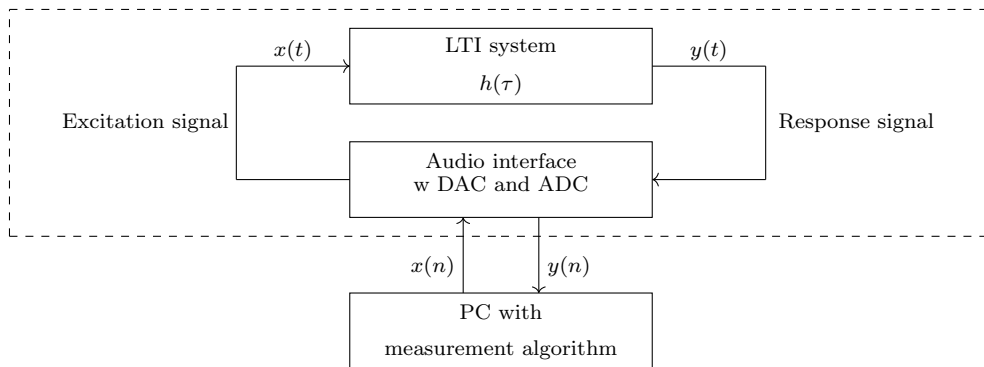


Figure 1.17: A real-world LTI system has analog signals $x(t)$ and $y(t)$ as input and output signals, respectively, but the measurement algorithm generates a digital excitation signal $x(n)$ and receives a digital response signal $y(n)$. From the measurement algorithm's perspective, everything inside the dashed box will belong to the system-to-be-measured.

Therefore, from now on, we will* call the signal x the *excitation signal* or the *stimulus signal*. The signal y will then be called the *response signal*, see Fig. 1.17.

The transfer function or impulse response of an LTI system can be measured in different ways. Two distinctly different situations can occur*:

1. We *can* choose the system excitation signal, $x(n)$.
2. We *can not* choose the system excitation signal, $x(n)$ – we can only observe/measure it.

Furthermore, the type of excitation signal can be

1. Finite-length. A common example is the so-called chirp signal.
2. Periodic, with a Dirac pulse as (periodic) autocorrelation function. This is true for some specially constructed signals, such as so-called Maximum-Length Sequences (MLS). For so-called pseudo-random noise sequences (a sequence of N samples that form a noise-like sequence, which is repeated periodically), the periodic autocorrelation function is *close to* a Dirac pulse.
3. General periodic signals that are not of type 2. As an example, a sinusoidal signal has a sine function as autocorrelation function.
4. A random signal, such as white or pink noise.

Some comments to this measurement task:

- When we try to measure a system's impulse response, $h(n)$, then we are *assuming* that the system is an LTI system. Further below, we will comment non-linearities and time-variance issues briefly.
- In electroacoustical applications, the system-to-be-measured typically involves a loudspeaker, a room (in a broad sense; it could be, e.g., a car compartment or an anechoic room), and a microphone, as illustrated in Fig. 1.18. It might be that we are interested in only one of those three components, but we will have to live with having all three components included. Alternatively, we can try to remove some of them by making separate measurements and try to remove their influence.
- In all measurement situations, the audio interface will be included in the system-to-be-measured. Audio interfaces can usually have very good electronic properties when it comes to frequency response flatness,

*That is, we will try to systematically use the terms stimulus/excitation signal, and response signal, respectively.

*In both situations here, we can observe/measure both x and y . There is also a technique called *blind system identification*, where one has access only to the response signal, y .

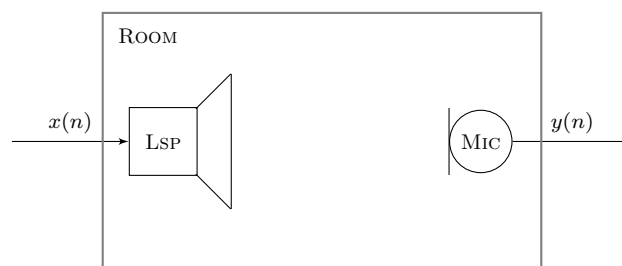


Figure 1.18: Illustration of a common electroacoustical measurement situation, where the actual component of interest might be the loudspeaker *or* the room *or* the microphone, but out of necessity, all three components will be present in the system that is being measured. Discrete-time excitation and response signals are indicated, which means that digital-to-analog (DAC) and analog-to-digital (ADC) converters, together with amplifiers, are considered as part of the system to be measured.

linearity, signal-to-noise ratio. In some measurement situations, it is impossible to use the same audio interface for the ADC and DAC (e.g., when cables can not be used between the excitation end and response end of the system to be measured, or when a recording of an excitation signal is used). Then a critical issue might occur: when the DAC and ADC have different sampling frequencies (which they will always have to larger or smaller degrees), *the system to be measured will effectively behave like a time-variant system*.

A number of different methods exist to determine the TF or IR of a system, as listed in Table 1.2. They are all described in separate subsections in the following.

Table 1.2: Methods for determining $h(n)$ and/or $H(\omega)$

Method	Measured quantity	Excitation signal
Stepped sine	$ H(\omega) $	Sinusoidals
Direct measurement of IR	$h(n)$	Pulse signal
Time-domain deconvolution	$h(n)$	A signal which has an explicitly known inverse signal
DFT-based deconvolution	$H(\omega_k)$	Any finite-length signal, or truly periodical signal, with broadband energy content, e.g., a chirp
Cross-correlation based system identification	$h_{\text{periodic}}(n)$	A signal with a (periodic) Dirac-pulse auto-correlation function e.g., a Maximum-Length Sequence (MLS)
Cross- and auto-spectra system estimation	$H(\omega_k)$	Any random signal with broadband energy content
Adaptive filter based system estimation (LMS)	$h(n)$	Any signal (but the system properties will only be determined for the frequencies of the excitation signal)

1.3.3 The classical method: stepped sine or swept sine, for determining $|H(\omega)|$

A mostly outdated method is to excite a system with a sinusoidal and simply measure the rms-values (or peak values) of the excitation and response signals. The magnitude of the transfer function, $|H(\omega)|$, can then be established as

$$|H(\omega)| = \frac{y_{\text{rms}}}{x_{\text{rms}}} = \frac{y_{\text{peak}}}{x_{\text{peak}}}$$

where $x(t)$ is a sinusoidal: $x(t) = x_{\text{peak}}e^{j\omega t}$ (which enforces that also $y(t)$ is on this form.)* One has to step through the number of sinusoidals that one is interested in. This method only requires instruments that can measure the rms-value (or peak-value, as sinusoidals are used), such as a sound level meter, or an oscilloscope, but it is slow and doesn't provide us with the phase function of $H(\omega)$ †. The impulse response can not be found from such a measurement. Furthermore, this technique is not able to reduce noise through averaging or impulse response truncation, or to reduce the influence of reflections through impulse response windowing, see section 1.4. For some instrumentation, it might be possible to use a slowly swept sinusoidal, rather than a stepped one. This classical method was the standard way to measure the frequency response of a loudspeaker into the 1980s.

*Here it makes some sense to specify that signals are functions of continuous time since this measurement method can be used with analog instruments - even though most instruments are digital.

†With an oscilloscope, or a special phase meter, the phase shift between x and y could be observed as well.

1.3.4 Direct measurement of the impulse response, $h(n)$, with a pulse excitation signal

A very special case of finite-length excitation signal is a pulse. We could generate a signal, $x(n)$, which is zero everywhere except for a single sample, where it has the maximum possible amplitude. Then, the response signal $y(n)$ will directly give us the impulse response, without the need for any processing. This approach was introduced in the 1970s, [10], where an analog rectangular-window signal was generated and fed to a loudspeaker. This technique is, however, not very efficient since we get a very low signal-to-noise ratio: a single-sample pulse (or a very short rectangular-window signal) contains very little signal energy. The signal-to-noise ratio can be improved substantially by using averaging as described in Section 1.4.3.

1.3.5 Deconvolution-based determination of $h(n)$

If the excitation signal, $x(n)$, is not the ideal pulse as in Section 1.3.4 above, then the response signal does not any longer equal the impulse response, and the influence of the excitation signal must be removed. This can be done by *deconvolution* = convolving the response signal with the inverse of the excitation signal. This deconvolution is a time-domain operation which is possible for some kinds of excitation signals. The process requires the existence of an inverse signal, $x^{-1}(n)$, which is such that

$$x(n) * x^{-1}(n) = p(n)$$

where $p(n)$ is a delayed pulse, that is, a signal which is zero everywhere except for sample $n_{\text{modelling delay}}$ where $p = 1$. If such an inverse signal can be found, then the time-shifted impulse response can be found by convolving the response signal with this inverse signal,

$$h(n - n_{\text{modelling delay}}) = y(n) * x^{-1}(n) \quad (1.31)$$

One example of an excitation signal which has an inverse can be described by the signal's z -transform

$$\begin{aligned} x(n) &= [1, a, a^2, a^3, \dots] \text{ for } n = [0, 1, 2, 3, \dots] \Rightarrow \\ X(z) &= 1 + az^{-1} + a^2z^{-2} + \dots = \frac{1}{1 - az^{-1}} \end{aligned} \quad (1.32)$$

where $a < 1$. The signal $x(n)$ is the impulse response of a first-order low-pass filter, and a perfect inverse of this signal is found via the z -transform,

$$\begin{aligned} X^{-1}(z) &= 1 - az^{-1} \Rightarrow \\ x^{-1}(n) &= [1, -a, 0, 0, \dots] \text{ for } n = [0, 1, 2, 3, \dots] \end{aligned} \quad (1.33)$$

No modelling delay is needed for this signal/inverse pair - the convolution of the two signals will give a pulse of amplitude 1 at $n = 0$.

We get a possibly non-intuitive effect here. Imagine that we construct the time signal, or rather discrete-time sequence, in Eq. (1.32). It is obviously a signal of infinite length, even if the amplitude dies out exponentially. If we convolve this signal with the filter impulse response represented by Eq. (1.33), which has an impulse response of length two samples, the result is a perfect Dirac pulse.

So: we feed a signal of infinite length into a filter, and out comes a single pulse!

As remarkable as that might seem, it corresponds to the (perhaps) more intuitive frequency-domain observation that if we send a signal with a certain spectrum into a filter with a frequency response which is the inverse of the input signal's spectrum, the signal which comes out should have a flat spectrum. An important detail is that the phase functions must be inverted as well, for the perfect pulse to result.

The pulse excitation technique for measuring the impulse response of loudspeakers that was described in Section 1.3.4 was further developed by the use of *pre-emphasis*, meaning pre-filtering the pulse by boosting the

low frequencies, and then use a compensating *de-emphasis* on the response signal. Such filtering has been done by analog filters, see [11]. The matching pre-emphasis/de-emphasis filter pairs have been used for a long time in various applications. In the analog sound reproduction domain, vinyl LPs have always used a standardized pre-/de-emphasis filter pair, a so-called RIAA-filter. Those two filters should together, i.e., when they are connected in series, create a flat frequency response.

1.3.6 DFT-based deconvolution gives $H(\omega_k)$ for finite-length, or periodical, excitation signals

Most finite-length excitation signals *do not* have inverse signals that are practical or possible to use for time-domain deconvolution, but, the DFT can be used for *frequency-domain* deconvolution. Thus, the complex-valued transfer function can be determined for a set of discrete frequencies, ω_k , via Eq. (1.29), which leads to

$$H(\omega_k) = \frac{Y(\omega_k)}{X(\omega_k)} = \frac{\text{DFT}[y(n)]}{\text{DFT}[x(n)]} \quad (1.34)$$

which is valid:

- if x is of finite length (with a length smaller than $T = n_{\text{DFT}} \cdot f_S$), and y has significant amplitude only within the same time length,

or

- if x , and therefore y , are periodical (with a period which equals $T = n_{\text{DFT}} \cdot f_S$).

Most often one prefers to use the FFT, which is a special case of DFT (If $n_{\text{DFT}} = 128, 256, 512, \dots$ the efficient FFT-algorithm can be used*). We can note that the magnitude of the complex $H(\omega)$ should be identical to the one measured with the stepped sine method in Section 1.3.3 (if the noise is negligible, and all the sound paths are included in both methods). We can choose an arbitrary excitation signal $x(n)$ as long as it has energy for all frequencies (because the division in Eq. (1.34) will cause trouble otherwise). The periodicized impulse response (see Appendix 1.5.1 for an explanation of this) can subsequently be computed,

$$h_{\text{periodic}}(n) = \text{DFT}^{-1}[H(\omega_k)] \quad (1.35)$$

The DFT size needs to be large enough that both finite-length signals, x and y , have died out (below the observable amplitude range, that is, below the amplitude of the disturbance signal indicated in Fig. 1.16(b)). If the signal y has died out, then the resulting periodicized impulse response, h_{periodic} , should have no detectable wrap-around effect of the type illustrated in Fig. 1.36, and we could as well describe the impulse response as $h(n)$ rather than $h_{\text{periodic}}(n)$. It can be noted that the response signal will typically be much longer than the excitation signal since the response signal is the convolution of the excitation signal and the impulse response.

A small summary of the previous subsection, and the expressions above, is in place:

- If we have an excitation signal, $x(n)$, for which we know an inverse signal, $x^{-1}(n)$, explicitly (which we usually do not), then we can find a system's impulse response with Eq. (1.31)^a.
- If we don't know the inverse to the excitation signal explicitly, then we can do the deconvolution using Eqs. (1.34) and (1.35).

We should also remember that we can usually use zero-padding and a large enough FFT-size that the periodicized impulse response, $h_{\text{periodic}}(n)$, has no detectable difference from the true impulse response,

*In Matlab, the function `fft` calculates the DFT or the FFT, depending on the length of the input signal and the optional input parameter which specifies the DFT/FFT size.

$h(n)$.

^aSomewhat ironically, we might want to carry out the convolution in Eq. (1.31) using FFT anyways, since the FFT-based convolution is much faster than the time-domain convolution, for long time signals.

Special case of finite-length excitation signal - the exponential chirp

A popular choice of finite-length excitation signal is a so-called exponential chirp signal, which is a sinusoidal with an exponentially increasing frequency. This signal is sometimes called "log-sweep", or "exponential sine sweep". A chirp can be very short and almost pulse-like, or very long, like a slowly varying sinusoidal. Its spectrum will be smooth, without any clear zeros. Expressed as a continuous-time signal, this expression can be used to generate a chirp which sweeps from f_1 to f_2 in T seconds,

$$x_{\text{chirp}}(t) = \sin \left[\frac{2\pi f_1 T}{R} \left(e^{\frac{Rt}{T}} - 1 \right) \right] \quad (1.36)$$

where $R = \ln \frac{f_2}{f_1}$. The more useful discrete-time version is

$$x_{\text{chirp}}(n) = \sin \left[\frac{2\pi f_1 T}{R} \left(e^{\frac{Rn}{Tf_s}} - 1 \right) \right] \quad (1.37)$$

Fig. 1.19 shows an example of a very short chirp signal.

This exponential sine sweep signal has the peculiar, but extremely useful, effect that any non-linearities of the system will appear as separate components in the resulting $h(n)$, which can be analyzed separately, and thus give estimates on the amount of non-linearities in the system that is measured, [12]. These non-linear, or harmonic-distortion, components appear at the end of a measured impulse response and become more separable the longer the sweep is.

Figs. 1.20 to 1.23 illustrate an imagined measurement situation. The process leads to an estimate, $h_{\text{periodic}}(n)$, of the impulse response, $h(n)$, which is longer than the original impulse response (128 samples rather than 50), but the first 50 samples are identical within the computational accuracy.

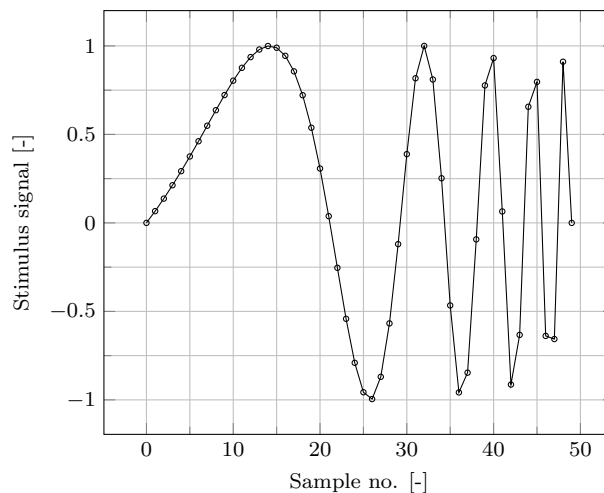


Figure 1.19: A 50 sample long exponential chirp.

For the chirp signal in Eq. (1.36), an inverse filter, $x^{-1}(t)$, is available such that $x(t) * x^{-1}(t) = \delta(t)$, [13]:

$$x^{-1}(t) = \frac{f_1}{RT} e^{-\frac{Rt}{T}} x_{\text{chirp}}(-t)$$

that is, the inverse signal is a time-reversed version of the excitation signal, with an additional shaping of the amplitude (the amplitude stays constant for the chirp itself), by the factor $e^{-\frac{Rt}{T}}$.

1.3.7 Cross-correlation based estimate of $h_{\text{periodic}}(n)$

If the excitation signal $x(n)$ is a signal whose (periodic) auto-correlation function is a Dirac-pulse, then the IR can be found as a *periodic cross-correlation between the excitation and the response signals*. Some special measurement signals possess *almost* this property, such as Maximum-Length Sequences (MLS), as further explored below, and such measurement signals can give a very accurate estimate, $\hat{h}_{\text{periodic}}$, of the IR. Ideally, the periodic Dirac-pulse property means that

$$r_{xx,\text{periodic}}(n) = \begin{cases} 1, n = 0(+m \cdot n_{\text{period}}) \\ 0, n \neq 0(+m \cdot n_{\text{period}}) \end{cases} \quad (1.38)$$

where $m \in [0, \pm 1, \pm 2, \dots]$, but this is rarely achieved, and instead of 0, the value of $|r_{xx}(n)|$ should be "small enough" for $n \neq 0$. As will be commented further below, the MLS has a small *constant* value for $r_{xx}(n)$, $n \neq 0$ and that is particularly useful for estimating impulse responses.

If the stimulus signal has this Dirac pulse property, then the impulse response can be estimated* as the periodic cross-correlation signal, as given by Eq. (1.49):

$$\hat{h}_{\text{periodic}}(n) = r_{xy,\text{periodic}}(n) = \frac{1}{N} \sum_{k=0}^{N-1} x_{\text{periodic}}(k) y_{\text{periodic}}(k+n) \quad (1.39)$$

In contrast to the DFT-based measurement method, we get the impulse response as the primary output, and can find the TF using a DFT. The periodic cross-correlation in Eq. (1.39) can be expressed as a circular convolution

*If the stimulus signal has the ideal Dirac autocorrelation, then the estimate is exact, apart from noise/disturbance signals.

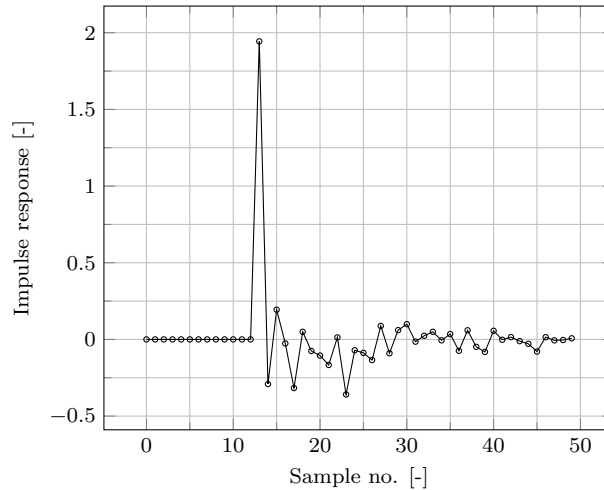


Figure 1.20: An example impulse response, $h(n)$, with a length of 50 samples.

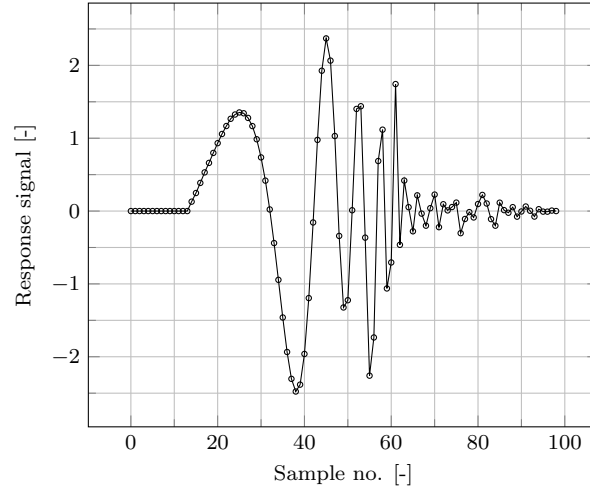


Figure 1.21: The response signal from the system, $y(n)$, which is the convolution of the stimulus signal in Fig. 1.19 with the impulse response in Fig. 1.20. The length is 99 samples.

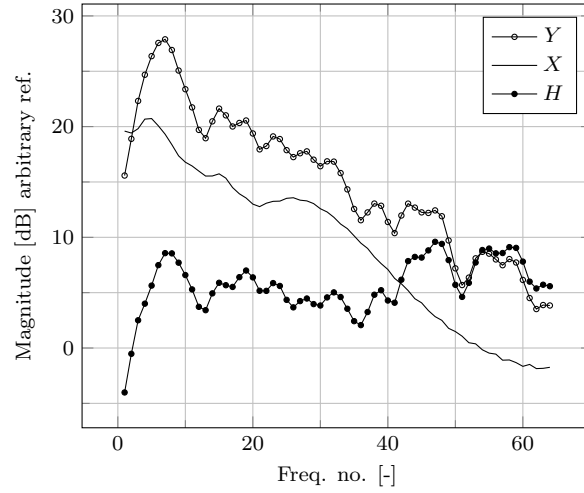


Figure 1.22: The frequency-domain functions X , Y and H , based on a 128-point DFT of the signals in Figs. 1.19 (X) and 1.21 (Y). The transfer function H was computed with Eq. (1.34).

between the response signal, $y_{\text{periodic}}(n)$, and the time-reversed/mirrored version of the (periodic) excitation signal, $x_{\text{periodic}}(n)$, as given in Eq. (1.5.3):

$$\hat{h}_{\text{periodic}}(n) = \frac{1}{N} x_{\text{periodic}}^{\text{mirrored}}(n) \otimes y_{\text{periodic}}(n) \quad (1.40)$$

This convolution (with the extra scaling factor in front of the sum) of the response signal, $y_{\text{periodic}}(n)$ and a time-reversed version of the excitation signal, $x_{\text{periodic}}^{\text{mirrored}}(n)$ is simply a filtering, and this time-reversed version of the excitation signal is sometimes called a *matched filter*. The response signal, $y_{\text{periodic}}(n)$, is filtered with (convolved with) this matched filter*.

*Matched filters are used for many different types of signals used in communication applications, in order to enhance the detection of signals in noisy transmission channels. For such applications, the perfect Dirac pulse property in Eq. (1.38) is not necessary; just an "impulse-like enough" autocorrelation function, [14].

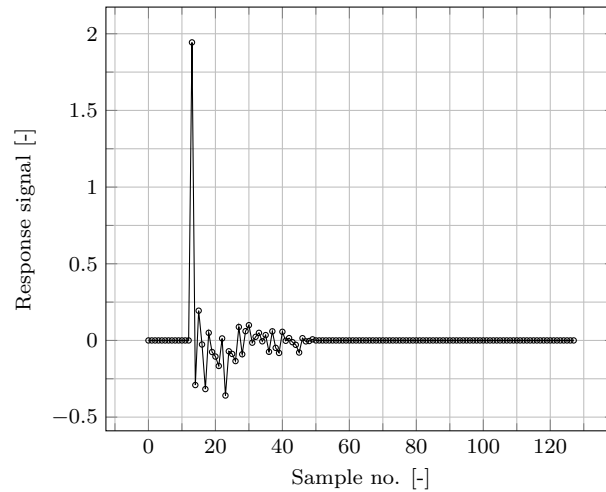


Figure 1.23: The resulting impulse response, computed with Eq. (1.35). The first 50 samples are identical (within the "computation noise") with the impulse response in Fig. 1.20.

Eqs. (1.39) and (1.40) describe time-domain correlation and convolution, respectively, but this processing can also be done more efficiently using DFT, see Eq. (1.5.4), which implies that

$$\begin{aligned}\hat{h}_{\text{periodic}}(n) &= r_{xy,\text{periodic}}(n) \\ &= \frac{1}{N} \text{DFT}^{-1} \left[(\text{DFT}[x_{\text{periodic}}(n)])^* \cdot \text{DFT}[y_{\text{periodic}}(n)] \right]\end{aligned}\quad (1.41)$$

In the same way as for time-domain vs. DFT-based convolution, the DFT-based periodic cross-correlation is faster than the time-domain version (especially if the faster FFT-algorithm is used), and the advantage for the DFT-based approach gets larger and larger the bigger the DFT size is. For the special MLS signals (see the following section), however, there is one special way to compute the periodical cross-correlation, and that is to use the so-called Fast Hadamard Transform, [15], which we don't describe any further here. It suffices to say that the Fast Hadamard Transform is even faster than the FFT version of the DFT approach in Eq. (1.41).

A special stimulus signal for cross-correlation: The Maximum-Length Sequence (MLS)

As argued above, the DFT-based method to find the impulse response, Eqs. (1.34) and (1.35), is usually much faster than the time-domain cross-correlation, Eq. (1.40). There is, however, a very interesting special case of stimulus signal, $x(n)$, for which the cross-correlation in Eq. (1.40) can actually be done faster than the DFT-based eq. (1.34). This special case is a so-called Maximum-length sequence (MLS), which is a special sequence of values 0 and 1 (that usually gets mapped to values -1 and 1). This efficiency is caused by the so-called Fast Hadamard Transform, which performs the cross-correlation particularly efficiently. An MLS, when low-pass filtered, sounds like random white noise, but it is a so-called pseudo-random noise: a deterministic signal that has the properties of a random noise signal. The periodic autocorrelation function does have Dirac pulses at $n = 0 + m \cdot n_{\text{period}}$, [16]. The MLS has "run out of fashion" in room acoustics measurements because it has turned out to be very sensitive to time-variance in the system to be measured, [17].

An MLS is remarkably easy to generate, either in hardware, or in software. It can be implemented as a sequence of shift registers holding the 0 or 1, with the values clocked forward + feedback connectors in exactly the right selected positions. With these connectors chosen correctly, an N -unit shift register will give a signal which is periodical with period $n_{\text{period}} = 2^N - 1$. An example is shown in Fig. 1.24 for $N = 4$. The summation should be a modulo-2 addition ($0+0 = 0$; $0+1 = 1+0 = 1$; $1+1 = 0$). There is also a Matlab function, `mls`, which gives an mls of the desired length. The Matlab function scales and shifts the signal so the values are not 0 and 1.

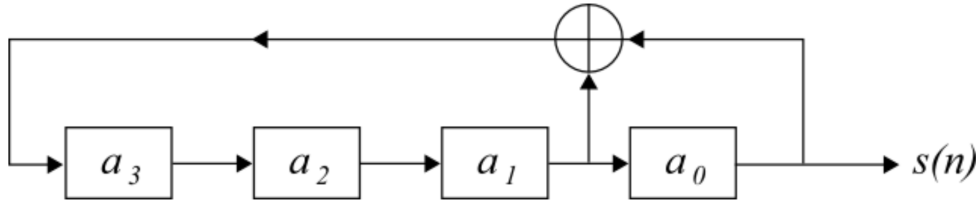


Figure 1.24: A 4-cell shift register which generates a Maximum-Length Sequence (MLS), $s(n)$, of period $2^4 - 1 = 15$. Illustration from www.wikipedia.com: "Maximum length sequence".

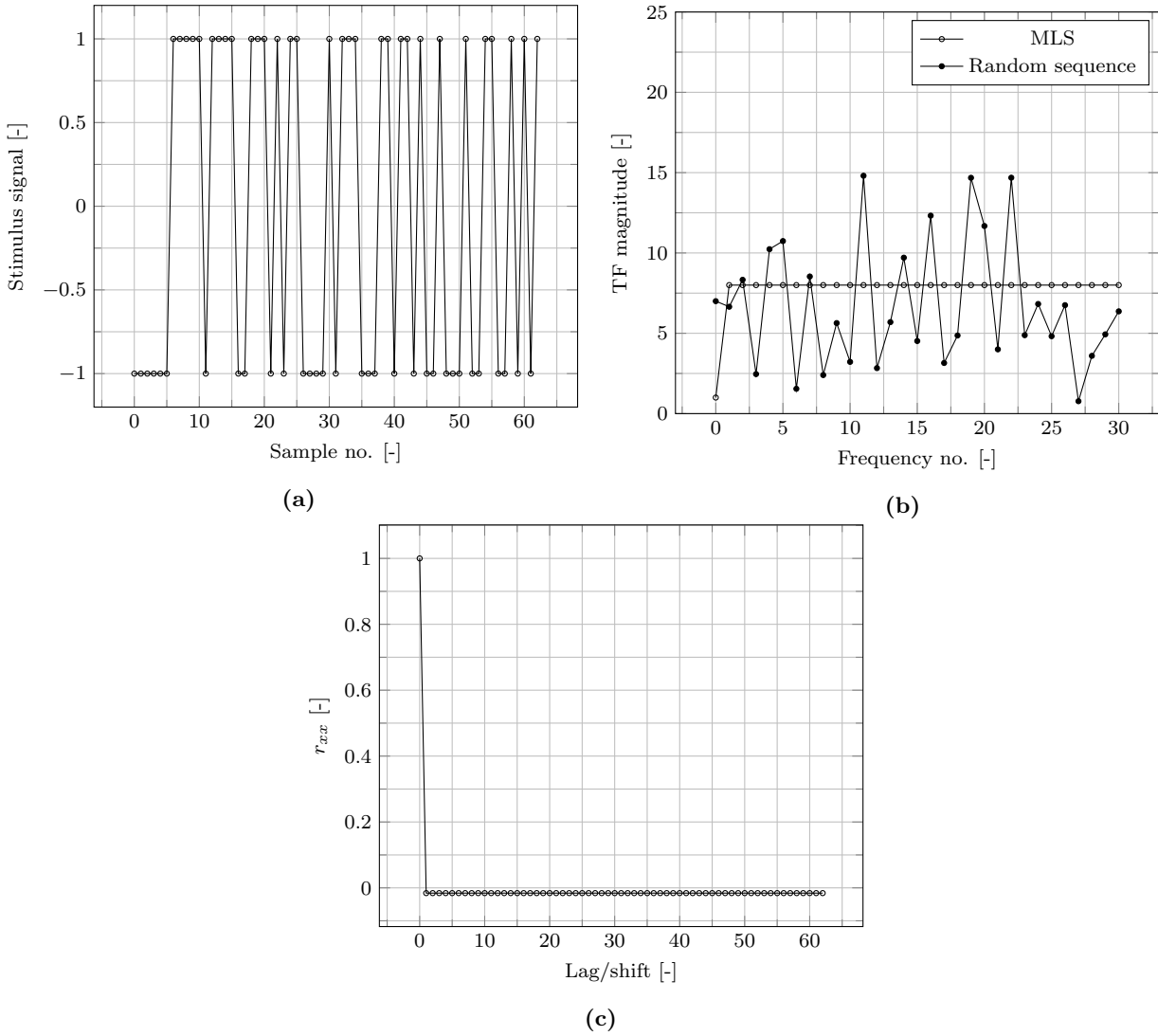


Figure 1.25: (a) An MLS of length/period 63 samples, and (b) its spectrum, together with the spectrum of a *random* binary sequence. Note that only half the spectra are shown because of the symmetry around half the sampling frequency caused by the real-valued signals. (c) The periodical autocorrelation signal for the MLS in (a).

Fig. 1.25 (a) shows an example of an MLS signal, with a period length $2^6 - 1 = 63$ samples. Note that the values have been scaled and shifted to -1 and 1, rather than 0 and 1. Fig. 1.25 (b) shows the spectrum

of the MLS in Fig 1.25 (a), together with the spectrum of a random binary sequence. Note that the spectrum of the MLS is perfectly flat, except for the DC component. The DC component reflects the fact that the autocorrelation function of the MLS had the little remaining constant residual value between the desired pulse values. The periodic autocorrelation function is computed with Eq. (1.48), and the result is shown in Fig. 1.25(c).

Now we can inspect how a measurement situation would work with the MLS. Let's construct a sequence of 3 periods of the 63-sample MLS in Fig. 1.25 to excite a system with a 50-sample long IR as in Fig. 1.20. The result will be a response signal of 238 samples length, as shown in Fig. 1.26 (a). Whenever a periodic signal is initiated, or stopped, there will be build-up period and a decay period before and after the stationary, periodic conditions have been established. Therefore, the first frame of data points should not be used, but the second and third frames are identical (except for additional disturbance signals). By selecting one of these frames of 63 samples, it is possible to find the impulse response in several different ways:

1. To use DFT-based deconvolution, Eqs. (1.34) and (1.35).
2. To use time-domain periodical cross correlation, Eq. (1.39), (which can be called filtering with a matched filter) or the equivalent DFT-based version, Eq. (1.41).
3. To use time-domain periodical cross correlation using the Fast Hadamard Transform, [15]. This is not described any further here but is a very fast method to carry out the time-domain periodical cross correlation.

Fig. 1.26(b) shows the impulse response computed with methods 1 and 2 above.

1.3.8 Estimating $H(\omega)$ for random excitation signals using cross and autospectra

In some measurement situations, the excitation signal, $x(n)$, is outside our control but can be measured. Typically, x then has a component of random noise and possibly some tonal components as well.

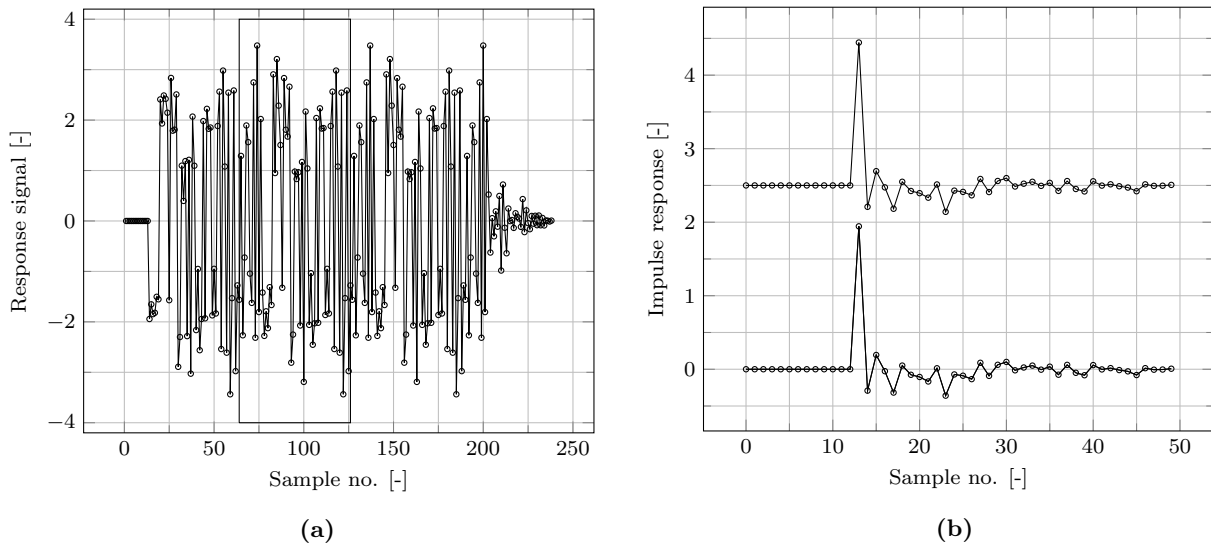


Figure 1.26: (a) The response signal, $y(n)$, when 3 periods of the 63-sample MLS in Fig. 1.25(a) are used as excitation signal to the system with a 50-sample IR as in Fig. 1.20. The box indicates frame number 2, samples 64-126, which is used as one frame of the stationary state. (b) The resulting impulse response computed with DFT-based deconvolution (lower curve), and periodical cross-correlation (upper curve). The upper impulse response has been shifted vertically for readability.

An estimate of the transfer function $H(\omega_k)$ can be found as

$$\hat{H}(\omega_k) = \frac{\hat{S}_{xy}(\omega_k)}{\hat{S}_{xx}(\omega_k)} \quad (1.42)$$

where $\hat{S}_{xx}(\omega_k)$ is the estimate of the autospectrum (power spectral density) of signal $x(n)$, based on averages of short-term Fourier transforms of windows of samples of the signal x . Likewise, the quantity $\hat{S}_{xy}(\omega)$ is the estimate of the crossspectrum of signals x and y . The situation can be illustrated as in Figs. 1.27 and 1.28. With this approach, one has to accept that the response signal in any given frame, say frame 2, will have a component which will be the decay of the excitation signal in the previous frame, no. 1. This part of the response signal, in frame 2, is unrelated to the excitation signal in frame 2, and thus appears as a disturbance signal. Likewise, part of the decay caused by the excitation signal in frame 2 will end up in frame 3. This "across-frames leakage" implies that the estimation of $H(\omega_k)$ will always have a limited accuracy. Usually, the so-called *coherence function* is inspected to find out how large part of the response signal has a constant relationship to the excitation signal. The coherence function tells both about the amount of across-frames leakage and the amount of external disturbance signal. It can be noted that methods to reduce the influence of the external disturbance signal will not necessarily reduce the influence of the across-frame leakage: boosting the excitation signal amplitude helps with respect to the disturbance signal but does not affect the across-frame leakage. On the other hand, increasing the period length will reduce the effect of across-frame leakage, but not directly suppress the disturbance signal.*

Periodical excitation signals, as described in the previous sections, do not have any of this across-frame leakage, and thus give more accurate estimates of the transfer function than the expression in Eq. (1.42). When periodical signals have broadband character, but with (pseudo-)randomized phase so that the signal seems like a noise signal, it is sometimes called a *pseudo-noise* signal.

*Indirectly, a longer measurement period can suppress the disturbance signal by using impulse response truncation techniques as described in Section 1.4.4.

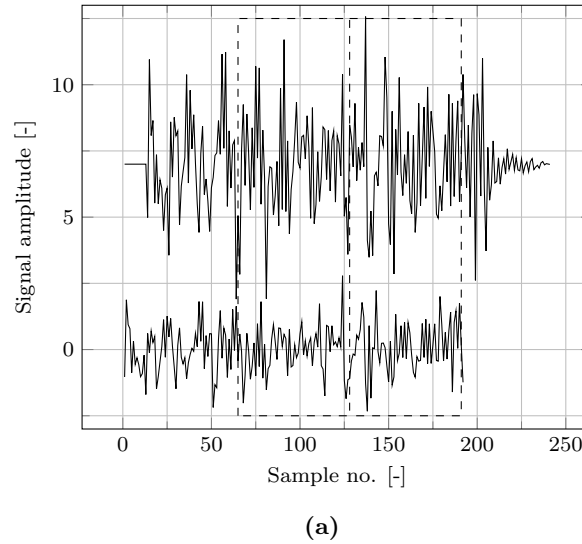


Figure 1.27: A random excitation signal, $x(n)$, of length 3*64 samples (lower curve), and the resulting response signal, $y(n)$, (upper curve) after being convolved with the 50-sample impulse response in Fig. 1.20. The dashed boxes indicate 64-sample frames number 2 and 3, which can be used for the cross-spectra estimates.

1.3.9 Adaptive filter based estimation of $h(n)$ for random excitation - the LMS method

The previous section described how an estimate of the transfer function, $H(\omega_k)$, can be estimated from observations of the excitation signal, x_{periodic} and the response signal, y_{periodic} , even if one can not choose what x_{periodic} should be. It is also possible to estimate the impulse response, $h(n)$, directly from these observations, using the least-mean squares (LMS) approach. It works directly on time signals and is remarkably simple in a time-iterative implementation. Furthermore, the LMS technique automatically leads to that an impulse response estimate can be updated continuously, and is therefore very useful in adaptive-filter applications [18].

Referring to Fig. 1.29 the LMS algorithm gives us an estimate, $\hat{h}(n)$, of the impulse response $h(n)$, and can update that estimate for each new time step n . We write the impulse response estimate, at time sample n , as a vector of length N samples, $\hat{\mathbf{h}}_n$. This estimate can then be updated using the last error sample $e(n)$, and the last N samples of the input signal, $\mathbf{x}_n$:

$$\hat{\mathbf{h}}_{n+1} = \hat{\mathbf{h}}_n - \alpha e(n) \mathbf{x}_n^{\text{mirrored}} \quad (1.43)$$

where $e(n)$ is a single value: the latest error, defined as

$$e(n) = \hat{y}(n) - y(n)$$

and α is a step size. A large value of α gives a faster convergence, but should not exceed a certain value since that would give an unstable algorithm. There are several interesting aspects with this algorithm:

- The algorithm in Eq. (1.43) is very simple to implement.
- The step-by-step updating of the estimates implies that we automatically get an *adaptive* filter which adjusts itself to changes in the system.
- The algorithm can be used while the filter is in use.

We can inspect how this algorithm works by looking at an example. Let's assume that we have a known impulse response of just two samples length,

$$\mathbf{h}_{\text{true}} = [h_{\text{true}}(0), h_{\text{true}}(1)]$$

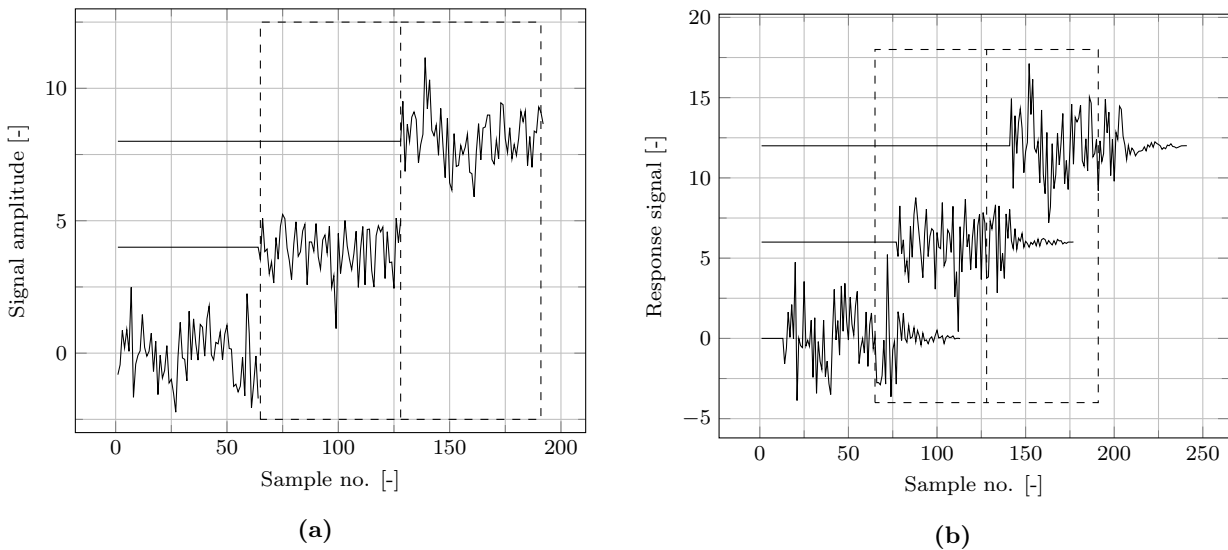


Figure 1.28: (a) The random excitation signal in Fig. 1.27 is split up into three frames of 64 samples each. (b) The response signal in Fig. 1.27 is split up into three *partly overlapping* frames that are caused by the three stimulus frames in (a). The three curves are separated vertically for readability. The dashed boxes indicate frames 2 and 3, of 64 samples each.

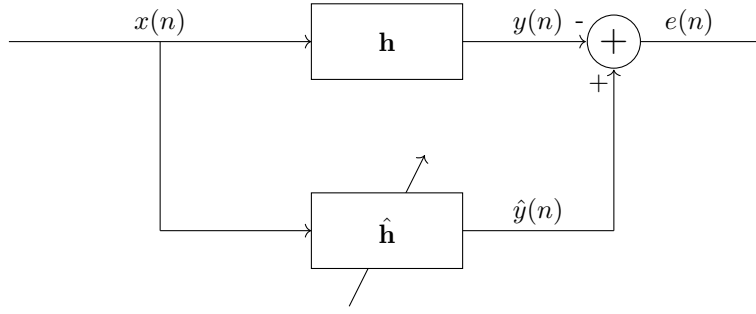


Figure 1.29: The LMS algorithm finds an estimate, $\hat{\mathbf{h}}$, of an impulse response to be identified, $\mathbf{h}$. The LMS algorithm needs access to the input signal, $x(n)$, and the error signal, $e(n)$, which is defined as the difference between the IR model output $\hat{y}(n)$ and the true output signal, $y(n)$.

We can then write out an expression for the error signal $e(n)$ since the convolution with a two-coefficient FIR filter is easy. Thus

$$e(n) = \hat{y}(n) - y(n) = \left[x(n)\hat{h}(0)_n + x(n-1)\hat{h}(1)_{n-1} \right] - \left[x(n)h_{\text{true}}(0) + x(n-1)h_{\text{true}}(1) \right]$$

where we have taken into account that the filter coefficient estimates, $\hat{h}(k)_n$, might get updated for each time step n . Now, the expression in Eq. (1.43) means that the two filter coefficients get updated according to

$$\begin{cases} \hat{h}(0)_{n+1} = \hat{h}(0)_n - \alpha(n)e(n)x(n) \\ \hat{h}(1)_{n+1} = \hat{h}(1)_n - \alpha(n)e(n)x(n-1) \end{cases}$$

So, the two coefficients will get updated as

$$\begin{aligned} \hat{h}(0)_{n+1} &= \hat{h}(0)_n - \alpha(n)x^2(n) \left[\hat{h}(0)_n - h_{\text{true}}(0) \right] \\ &\quad - \alpha(n)x(n)x(n-1) \left[\hat{h}(1)_{n-1} - h_{\text{true}}(1) \right], \\ \hat{h}(1)_{n+1} &= \hat{h}(1)_n - \alpha(n)x(n)x(n-1) \left[\hat{h}(0)_n - h_{\text{true}}(0) \right] \\ &\quad - \alpha(n)x^2(n-1) \left[\hat{h}(1)_{n-1} - h_{\text{true}}(1) \right] \end{aligned}$$

Here we can notice something important: if x is a random signal, then the expected values, $E[\cdot]$, are

$$E[x(n)x(n-1)] = 0, \quad E[x^2(n)] > 0$$

so the update rules are, in the mean,

$$\begin{aligned} \hat{h}(0)_{n+1} &\rightarrow \hat{h}(0)_n - \alpha(n)x^2(n) \left[\hat{h}(0)_n - h_{\text{true}}(0) \right] \\ \hat{h}(1)_{n+1} &\rightarrow \hat{h}(1)_n - \alpha(n)x^2(n-1) \left[\hat{h}(1)_{n-1} - h_{\text{true}}(1) \right] \end{aligned}$$

We can observe that if the latest estimate, $\hat{h}(0)_n$, is smaller than the true value, $h_{\text{true}}(0)$, then the update rule will make $\hat{h}(0)$ *increase*. Likewise, if $\hat{h}(0)_n$ is larger than the true value, $h_{\text{true}}(0)$, then the update rule will make $\hat{h}(0)$ *decrease*. Thus, the individual coefficients $\hat{h}$ will always move towards the true values.

This basic form of LMS has been developed through various modifications, e.g. the normalized LMS (N-LMS), the filtered-X LMS, and more. It has found extensive use in Active Noise Control (ANC) applications, [19] and also for the equalization of loudspeakers, [20].

1.4 Further processing techniques for LTI system measurements

1.4.1 One or two measurement channels for the DFT-based method?

The very common DFT-based method for determining $H(\omega_k)$, described in section 1.3.6, can be done with a single-channel or a two-channel measurement system, as illustrated in Fig. 1.30. The algorithm for determining $H(\omega_k)$ needs to know the two digital signals $x(n)$ and $y(n)$, but since the algorithm generates $x(n)$, this signal is already known, and therefore a single-channel measurement system as in Fig. 1.30(a) can work. However, the blocks marked "DAC" and "ADC" in Fig. 1.30 are not ideal and therefore they can be represented by transfer functions. If we use a somewhat improper way to mix digital-to-analog signals, we can represent the DAC by TF'_{DAC} , which has a digital input signal, $x(n)$, and an analog output signal, $x(t)$.

What kind of imperfections could the transfer functions $\text{TF}'_{\text{DAC/ADC}}$ have? Four factors can be identified:

1. The DAC and ADC typically introduce a constant delay, which is usually not known, but which can be measured easily enough (by first replacing the "System" by a straight wire from the DAC to the ADC. If one measures this measurement system TF, it has to be taken into account for every measurement.
2. Depending on the type of computer, operating system, sound card driver software, and measurement software, the time-bases for the two signals (= number sequences) x and y , might or might not be perfectly synchronized. Any constant extra time-delay will be included in factor 1 above, but time-varying extra delays might ruin the possibility to measure transfer functions.
3. The analog electronics parts of the sound card typically have software and/or hardware gains/volume controls. Such volume controls will make the estimate $\hat{H}(\omega_k)_{\text{single-channel}}$ change in magnitude as one changes the gain setting, whereas the estimate $\hat{H}(\omega_k)_{\text{two-channel}}$ is unaffected.
4. The analog electronics parts of the sound card typically have close-to-ideal frequency responses - but they are only close to ideal.

The single-channel approach will determine $H(\omega_k) = \text{TF}_{\text{system}}$ as follows (see Eq. (1.34)),

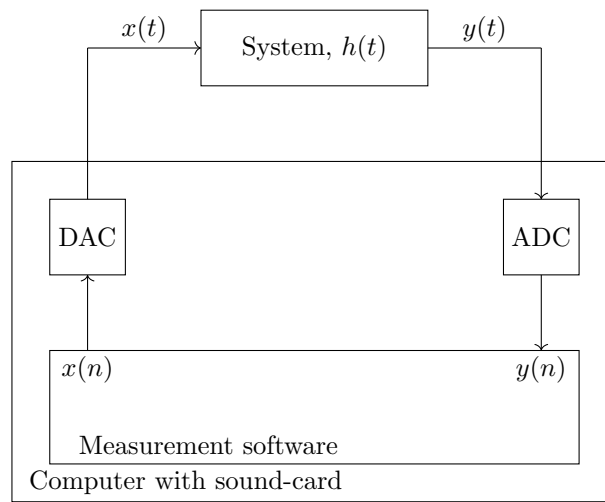
$$\begin{aligned}\hat{H}(\omega_k)_{\text{single-channel}} &= \frac{\text{DFT}[y]}{\text{DFT}[x]} = \frac{\text{DFT}[x] \cdot \text{TF}'_{\text{DAC}} \cdot \text{TF}_{\text{system}} \cdot \text{TF}'_{\text{ADC}}}{\text{DFT}[x]} \\ &= \text{TF}'_{\text{DAC}} \cdot \text{TF}_{\text{system}} \cdot \text{TF}'_{\text{ADC}}\end{aligned}\quad (1.44)$$

On the other hand, with the two-channel approach as in Fig. 1.30(b), we get

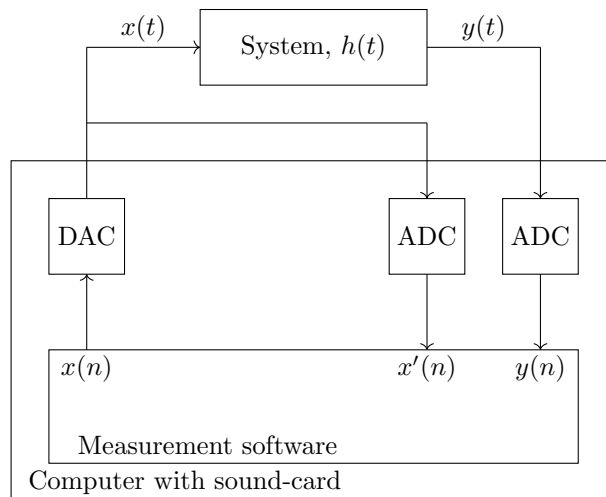
$$\begin{aligned}\hat{H}(\omega_k)_{\text{two-channel}} &= \frac{\text{DFT}[y]}{\text{DFT}[x']} = \frac{\text{DFT}[x] \cdot \text{TF}'_{\text{DAC}} \cdot \text{TF}_{\text{system}} \cdot \text{TF}'_{\text{ADC}}}{\text{DFT}[x] \cdot \text{TF}'_{\text{DAC}} \cdot \text{TF}'_{\text{ADC}}} \\ &= \text{TF}_{\text{system}}\end{aligned}\quad (1.45)$$

where we have ignored that different channels of ADC or DAC will have ever so slightly different transfer functions, primarily of type 4 in the list above. It might be possible to set different gains for different DAC/ADC channels, and then factor 3 in the list above also becomes relevant. Finally, some simpler multi-channel ADCs might use multiplexing, which means that the different channels are not measured at the same instant, but one-after-another, which would introduce an extra constant delay.

The two-channel approach is preferable.



(a)



(b)

Figure 1.30: Set-up for the measurement of a system's impulse response or transfer function, using (a) One measurement channel, or (b) Two measurement channels. DAC = digital-to-analog converter, ADC = analog-to-digital converter.

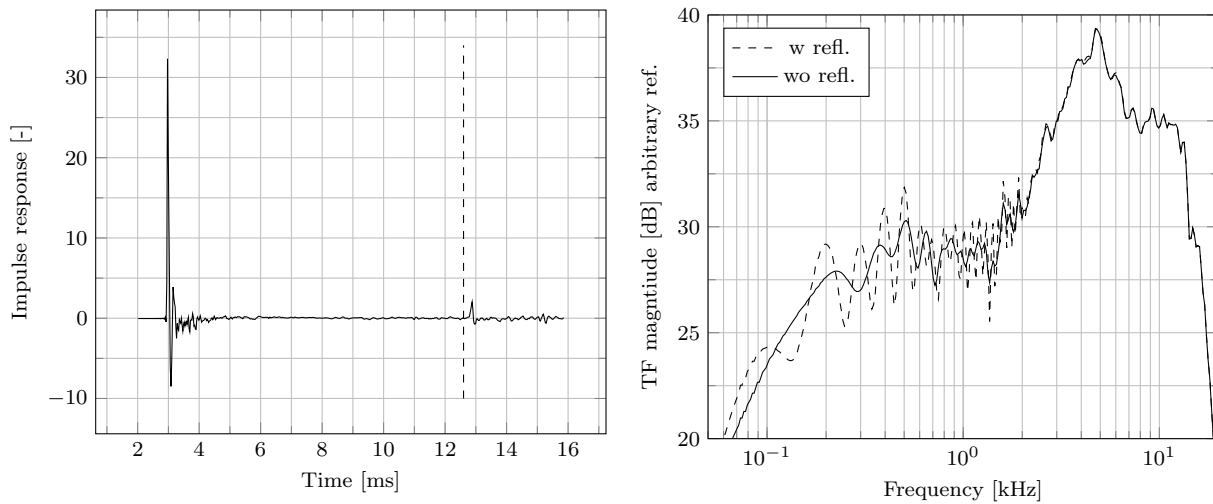


Figure 1.31: (a) A recorded time response for an excitation of a loudspeaker with a very short rectangular pulse (after averaging), indicating an unwanted reflection after the time at the vertical dashed line. (b) The Fourier transform of the impulse response which is truncated at 16 ms and thus includes the reflection (dashed line), or is truncated at 12.6 ms and thus excludes the reflection (solid line).

1.4.2 Windowing the impulse response

A powerful post-processing of measured impulse responses is to use *windowing*. There are two reasons to do this:

- Create a quasi-anechoic impulse response from a measurement in a reflective room. Unwanted reflections can be removed through windowing, as in Fig. 1.31(a). One can thus simulate an anechoic room, at the price of getting the *shortened* impulse response, and a shortened impulse response will not give the correct low-frequency response, see subsection 1.4.2. When the frequency response is computed via Eq. (1.30), we can see substantial effects of removing the unwanted reflection in Fig. 1.31(b).*
- Suppress noise in a room impulse response (or any impulse response). As illustrated in Fig. 1.35, a room impulse response has useful information during the entire IR decay, but after a certain time, the measured IR will be completely dominated by stationary background noise. By keeping the IR only up to the "knee-point", where the more or less exponential IR decay (= useful information), crosses into the stationary background noise, the amount of noise in the measured IR is minimized, [21].

It can also be realized that the DFT-based approach gives the transfer function $H(\omega_k)$ in Eq. (1.34) which involves the effect of all the reflections in the system. Yet, after transforming to the (periodic) impulse response according to Eq. (1.35), windowing the IR as in Fig. 1.31, and transforming back to the transfer function according to Eq. (1.30), then $H(\omega_k)$ has the effects of the reflections removed!

The influence on the frequency response of cutting a measured impulse response

The frequency response (or transfer function) of a loudspeaker is ideally measured in anechoic conditions, but when we use the impulse response measurement technique, it is possible to create an "anechoic time window"

*One realizes that to create as long as possible quasi-anechoic window, we need to make sure the first substantial reflection (wall, floor, ceiling) comes as late as possible after the direct sound. It might be helpful to visualize an ellipsoid, with the source and microphone at the two focal points. An ellipsoid is created by tying a string to the two focal points, and then the surface of the ellipsoid is created by grabbing one point after another along the string, and sweep it around in space. The goal is then to create an ellipsoid such that the length of that string, *minus the distance between the two focal points*, is as large as possible.

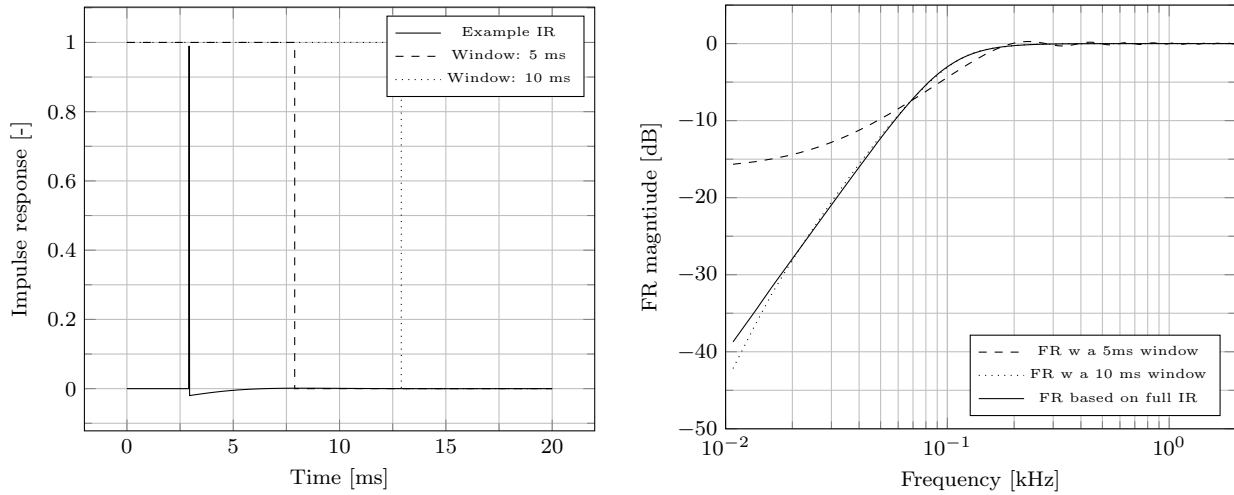


Figure 1.32: (a) The impulse response of an idealized loudspeaker: a second-order high-pass filter, with a time delay. Also shown are two time windows that simulate a 5ms and a 10ms anechoic time window. (b) The frequency responses that correspond to the impulse response in (a), without and with the two time windows.

as described above. The length of the anechoic time window is easy to determine:

$$t_{\text{anechoic time window}} = \frac{r_{\text{reflection}} - r_{\text{direct}}}{c}$$

where c is the speed of sound, r_{direct} is the distance from the source to the microphone, and $r_{\text{reflection}}$ is the path length from the source (loudspeaker) via the closest reflecting surface, to the microphone. The measured impulse response is then cut (that is, a rectangular window is applied) just before the reflection, and the crucial question is: *What is the influence on the response measurement of a cutting of the impulse response?*

We can study this by synthesizing an ideal loudspeaker response: a 2nd-order highpass-filter at 100 Hz, and a 3 ms delay, which simulates that we have measured at a distance of 1 m. Then we apply (with a bit of an arbitrary choice) a rectangular window with length 5 ms and 10 ms, after the direct sound, respectively. The results are shown in Fig. 1.32. It might hard to imagine how the time windows can have such a significant effect, since it seems like the loudspeaker impulse response has died out well before the windows end. But, Fig. ?? shows a zoomed-in version of the impulse response in Fig. 1.32 (a). One can realize, from the zoomed-in version, that the decay of the impulse response has a low amplitude but that the time window can have a strong effect. Especially the 5ms-window seems to cut the strongest oscillation in two.

The cutting of the high-pass filter's impulse response decay can cause significant distortion in the measurement of the low-frequency response.

1.4.3 Suppressing disturbance signals by averaging measured signals

As a general rule in measurements, we should aim for a high enough, or even as high as possible, signal-to-noise ratio in our measurements. By making a few measurements with various gain settings, we could find out what settings that will give us a high SNR. Generally, increasing the gain in the various places in the measurement chain: gain for the DAC, power amplifier, microphone amplifier, gain for the ADC, would increase the SNR - as long as one does not increase the non-linearities, which will eventually happen.

An efficient way to suppress the effect of external noise/disturbance signals for a measured impulse response or transfer function is to use *averaging*. By creating a repetition of excitation signals (pulses, chirps, MLS) the response signal can be averaged directly as time signals. Equivalently, is any further linear processing is done

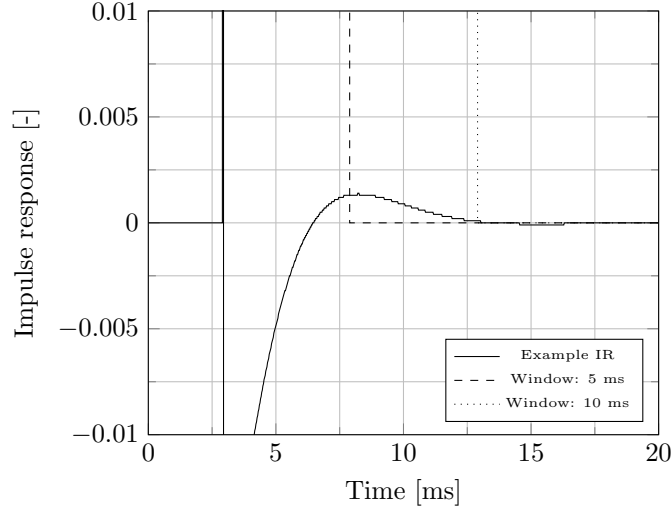


Figure 1.33: Zoomed-in version of the diagram in Fig. 1.32 (a).

to estimate the impulse response or transfer function, these estimates can be averaged. Note, however, that averaging the response signals directly is computationally much cheaper! The impulse response of the system will stay constant across repetition periods (as long as the system is indeed time-invariant), but the random components of the disturbance signal, r in Fig. 1.16(b), will vary from time frame to time frame and the resulting noise will gradually decrease. This can be expressed as follows. If we have measured two estimates, $\hat{h}_1(n)$ and $\hat{h}_2(n)$ of a true IR, $h(n)$, then these two estimates will also contain two independent samples of disturbance signals:

$$\begin{aligned}\hat{h}_1(n) &= h(n) + r_1(n), \\ \hat{h}_2(n) &= h(n) + r_2(n),\end{aligned}$$

and the average of these two will be

$$\hat{h}_{2 \text{ av.}} = h(n) + \frac{1}{2} [r_1(n) + r_2(n)] = h(n) + r_{2 \text{ av.}}(n)$$

The rms-value of the average of N independent samples of a random signal will be reduced as

$$r_{\text{rms}, N \text{ av.}} = \frac{r_{\text{rms}, \text{single frame}}}{\sqrt{N}}$$

which corresponds to a reduction by 3 dB for $N = 2$. Fig. 1.34 shows a "Single-shot response" and a "Signal-averaged response", to use the terms from Ref. [10]. We could note that the "signal-averaged response" can be described as $h_{\text{periodic}}(n)$ whereas the "single-shot response" could be described as $h(n)$. However, as mentioned earlier, if the measurement/analysis time period is long enough, then there is no detectable difference between the periodicized impulse response $h_{\text{periodic}}(n)$ and $h(n)$.

As pointed out at the start of this text, we assume that our system under study is an LTI system. In reality, all systems will have various degrees of time-variance and non-linearities, and this might be something that affects the quality of our IR and TF estimates. One simple example is the inevitable time-variance which is caused by temperature changes in the air since that leads to a change in the speed of sound. As a sidenote, mechanical/vibrating systems might display much less time-variance effects over short measurement periods than systems that involve sound propagation in air.

The presence of time-variance is fortunately quite easy to detect. Using a similar model as above, where the effect of averaging was inspected, we can imagine a situation where we measure two IR estimates, that are based on *two slightly different true IRs* (because of time-variance):

$$\begin{aligned}\hat{h}_1(n) &= h_{\text{true},1}(n) + r_1(n), \\ \hat{h}_2(n) &= h_{\text{true},2}(n) + r_2(n),\end{aligned}$$

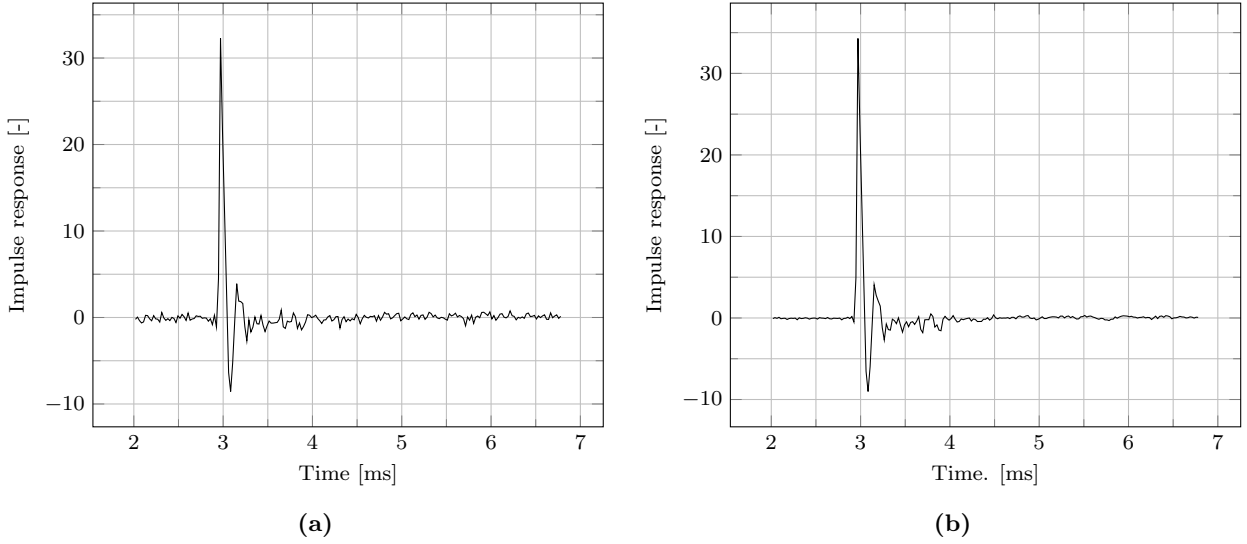


Figure 1.34: (a) A single frame of recorded time response for an excitation of a loudspeaker with a very short rectangular pulse and (b) the result after averaging 16 such recorded frames, a "Signal-averaged response", as it was called in Ref. [10].

If we use $h_{\text{true},1}(n)$ as a reference, we can express the difference between the two true IRs as $d(n) = h_{\text{true},2}(n) - h_{\text{true},1}(n)$. Then, if we take the *difference between the two estimates*,

$$\hat{\Delta}(n) = \hat{h}_2(n) - \hat{h}_1(n) = d(n) + r_2(n) - r_1(n) = d(n) + r_{\text{sum of } 2}(n)$$

If there is no time-variance, $\hat{\Delta}(n)$ will contain only the effects of external noise, and if the external noise is truly random and stationary, then

$$r_{\text{rms, sum of } N} = \sqrt{N} \cdot r_{\text{rms, single frame}}$$

For $N = 2$, it implies that $r_{\text{sum of } 2}$ will be 3 dB higher than $r_{\text{single frame}}$. Then, if $\hat{\Delta}$ has a level which is *more than 3 dB higher* than $r_{\text{single frame}}$, then there might be significant time-variance. A changing temperature leads to a characteristic increase of the difference signal with +6 dB per octave at high frequencies. The $r_{\text{single frame}}$ can be estimated with a single IR/TF measurement without excitation signal, as described above.

1.4.4 Suppressing disturbance signals by using a longer measurement period

An alternative to using averaging for suppressing disturbance signals is to use a longer measurement period T , which can be used for chirp signals (by using a longer chirp) or periodic signals, but not for pulses. The impulse response is concentrated in time towards the time zero, since all impulse responses for passive systems must have a decaying amplitude. The impulse response's distribution over time is independent of the measurement period, but random disturbance signals will have their energy distributed evenly across the measurement period thanks to the processing involved. As a result, the signal-to-noise ratio for the temporal range where the impulse response is located will get higher and higher, the longer the measurement period is. For truly random disturbance signals, the averaging technique and the longer-measurement-period technique will give the same signal-to-noise ratio for a certain total measurement period $T_{\text{total}} = T_{\text{long single measurement period}}$ or $T_{\text{total}} = n_{\text{averages}} \cdot T_{\text{short measurement period}}$. However, using a single long chirp turns out to be less sensitive to time-variance than averaging.

1.4.5 Quantifying the disturbance signal

We usually want to quantify the disturbance signal, commonly in terms of a signal-to-noise ratio (SNR). For an impulse response measurement, a first glimpse is easy to get by plotting the measured impulse response on a logarithmic amplitude axis (vs. a linear time-axis). Fig. 1.35 shows an example of a measured impulse

response in a room, and the two heavy solid lines indicate the exponential decay of the room, and the constant background noise in the room, respectively. The dashed line shows the peak IR value, the vertical distance between the peak IR value and the constant background noise, the direct-sound-to-background noise ratio, is informative but not immediately useful as quantification, because:

- The *total* signal energy, i.e., the useful part of the measurement, is distributed exponentially over time, and the peak value does not give the total amount of useful energy; we would need to integrate over time to get a measure. Having said that, for some measurements it is only the direct sound which is analyzed and then this direct-sound-to-background noise ratio is more informative than the total-signal-energy-to-background noise ratio.
- The broadband time-domain result does not give any information about the frequency-dependence.

Both of these weaknesses can be improved upon by carrying out two measurements:

1. The IR or TF is measured, as intended.
2. Another IR or TF measurement is carried out *without any stimulus signal*

By carrying out exactly the same processing steps, and with all measurement amplifiers set to the same gains, for those two measurements, a transformation to transfer functions will then give two curves: the interesting frequency response of the part of the system that we want to analyze, and the spectrum of the background noise, respectively. Since the spectrum of the background noise is usually random in character, it might be useful to smooth the spectrum into frequency band levels. This way, the frequency-dependent signal-to-noise ratio will be visible, and we know that it will be the relevant noise measure that is shown: the noise in the analyzed impulse response window.

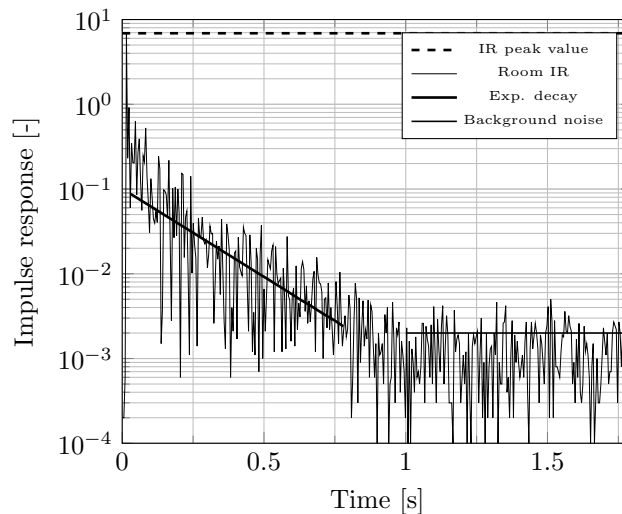


Figure 1.35: Example of a measured room impulse response, where the stationary background noise level is dominating after around 1 s.

1.4.6 What is the unit of an impulse response?

What is the correct unit of an impulse response? The answer depends on what we have defined as input signals and output signals, and whether we consider the impulse response to be the discrete-time version, $h(n)$, or the analog one, $h(t)$.

To start with a digital version; if we consider our "System-under-study" to include amplifiers and transducers, then we are exciting the system with a signal x which is an electric voltage. Also the response signal from the system, y , will be an electric voltage. The definition of the convolution *sum* will then lead to the fact that the IR, $h(n)$, has no unit. On the other hand, if we consider that the response signal, y , is really the sound pressure, with the unit pascal, then the IR will have the unit of Pa/V. Continuous-time impulse responses, $h(t)$, will always have an extra s^{-1} dimension, due to the definition of the convolution *integral*.

So, if we use $[f]$ to denote the unit of f :

$$[h(n)] = \frac{[y(n)]}{[x(n)]} \text{ for discrete-time signals}$$

$$[h(t)] = \frac{[y(t)]}{[x(t)]} \frac{1}{s} \text{ for continuous-time signals}$$

1.5 Appendices

1.5.1 Appendix - The impulse response, $h(n)$, vs its periodicized version, $h_{\text{periodic}}(n)$

If we use the *Discrete Fourier Transform* (DFT), then we assume, or rather enforce, time signals to be periodic, with a period time T , which corresponds to a period of $N = T \cdot f_s$ samples, where f_s is the sampling frequency. Since we can assume stationary conditions if signals are periodic, the convolution in Eq. (1.26) needs to be carried out only for one period, and we get the *circular or periodic convolution*, which is a special case of Eq. (1.26):

$$y_{\text{periodic}}(n) = x_{\text{periodic}}(n) \circledast h_{\text{periodic}}(n) = \sum_{k=0}^{N-1} h_{\text{periodic}}(k) x_{\text{periodic}}(n-k) \quad (1.46)$$

Here, the impulse response is denoted h_{periodic} but it should be noted that the impulse response will just *appear* to be periodic, even if it has a longer extension, because of this enforced periodicity. The periodicized impulse response, $h_{\text{periodic}}(n)$ will be constructed from cutting $h(n)$ into frames of length T seconds and add them on top of each other. The frames wrap around, so that $h_{\text{periodic}}(n)$ has length T second $\leftrightarrow N$ samples. $h_{\text{periodic}}(n)$ can be expressed as

$$h_{\text{periodic}}(n) = \sum_{m=0}^{\infty} h(n + m \cdot N) \quad (1.47)$$

where the sum needs to extend until the IR $h(n)$ has vanished completely. Note that if the IR has indeed died out within the period time T , i.e., if $h(n) = 0$ for $n \geq N$, then $h_{\text{periodic}}(n) = h(n)$. Fig. 1.36 illustrates an impulse response and its periodicized version.

Consequence of the above:

- When we use some DFT-based measurement method, or some other method based on periodic time signals, to find a system's impulse response, we will get $h_{\text{periodic}}(n)$ rather than $h(n)$ (which is probably what we actually would have liked to get).
- If we choose $N = n_{\text{DFT}}$ large enough, then
 - If $h(n)$ truly is a finite impulse response, then $h_{\text{periodic}}(n) = h(n)$.
 - If $h(n)$ is an infinite impulse response, but the wrapped-around components ($m > 0$ in Eq. (1.47)) are smaller than the background noise, then there is no detectable difference between $h_{\text{periodic}}(n)$ and $h(n)$.
- A note about the Discrete Fourier Transform vs. the Fourier Transform: The standard way to convert an impulse response to a frequency response is to use the DFT/FFT which, as mentioned above, gives

the frequency response's values at *discrete, regularly spaced frequencies*.

Assume that we have determined the impulse response of a loudspeaker with a sampling frequency of $f_S = 65536$ Hz (yes, this is a bit of an odd choice, but that is chosen to give a practical frequency resolution). If this impulse response has died out into the background noise of our measurement in, say, 20 ms, then we would need only $0.020 \cdot 65536 \approx 1311$ samples to describe $h(n)$ fully. Then we could use an FFT size of 2048 (or larger) to transform $h(n)$ to $H(\omega_k)$, for the frequencies $f_i = i \cdot 65536/2048 = i \cdot 32$ Hz. But, maybe we would really like to know how our loudspeaker reproduces the music note C_4 , which has a fundamental frequency of 261.626 Hz. We could get a higher frequency resolution with the DFT/FFT, by using zero-padding and use a much larger FFT size: 65536 would give us a frequency resolution of 1 Hz, but we would still not come any closer than 262 Hz. That's when the *Discrete-time Fourier transform (DTFT)*, rather than the DFT, could give us the answer since the DTFT will give us the value of $H(\omega)$ for any value of ω , rather than the discrete frequencies ω_k .

1.5.2 Appendix - Correlation functions

Auto- and cross-correlation functions are concepts that will be needed furtheron. In the same way as for the convolution, there is the general autocorrelation function for a discrete-time signal, which is a function of time shift, or lag, n samples:

$$r_{xx}(n) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{k=-N}^N x(k)x(k+n)$$

In contrast, the periodic autocorrelation function is

$$r_{xx,\text{periodic}}(n) = \frac{1}{N} \sum_{k=0}^{N-1} x_{\text{periodic}}(k)x_{\text{periodic}}(k+n) \quad (1.48)$$

The same versions exist for the cross- rather than auto-correlation functions, and the cross-correlation function that relates $y(n)$ to $x(n)$ is denoted $r_{xy}(n)$:

$$r_{xy}(n) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{k=-N}^N x(k)y(k+n)$$

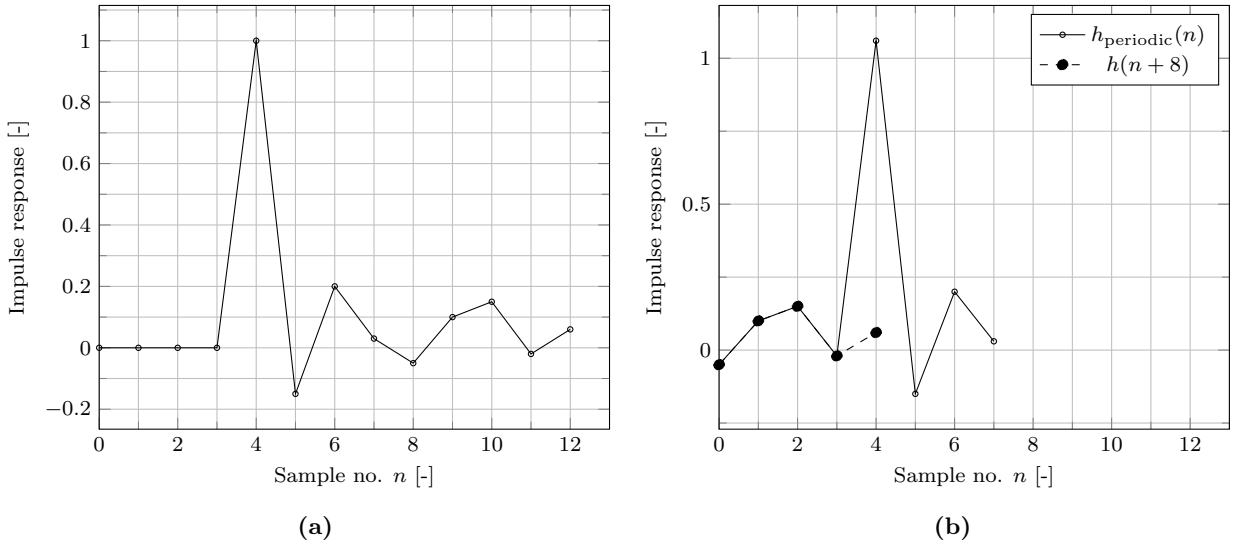


Figure 1.36: (a) An impulse response, $h(n)$, and (b) its periodicized version, $h_{\text{periodic}}(n)$, where samples no. 8-12 in $h(n)$ have been added to samples 0-4.

$$r_{xy,\text{periodic}}(n) = \frac{1}{N} \sum_{k=0}^{N-1} x_{\text{periodic}}(k) y_{\text{periodic}}(k+n) \quad (1.49)$$

These time-domain functions are related to the auto- and cross power spectral density functions,

$$R_{xx}(\omega) = \mathcal{F}[r_{xx}(n)], \quad R_{xx}(\omega_k) = \text{DFT}[r_{xx,\text{periodic}}(n)],$$

$$R_{xy}(\omega) = \mathcal{F}[r_{xy}(n)], \quad R_{xy}(\omega_k) = \text{DFT}[r_{xy,\text{periodic}}(n)],$$

The periodic correlation can, in the same way as for the circular convolution, be done much more efficiently using the DFT (or, preferably, the faster FFT version) (see details in Appendix A),

$$\begin{aligned} r_{xy,\text{periodic}}(n) &= \frac{1}{N} \text{DFT}^{-1} [\text{DFT}[x_{\text{periodic}}^{\text{mirrored}}(n)] \cdot \text{DFT}[y_{\text{periodic}}(n)]] \\ r_{xy,\text{periodic}}(n) &= \frac{1}{N} \text{DFT}^{-1} [(\text{DFT}[x_{\text{periodic}}(n)])^* \cdot \text{DFT}[y_{\text{periodic}}(n)]] \end{aligned} \quad (1.50)$$

where $(\cdot)^*$ represents the complex conjugate of $(\cdot)$. A mirroring in the time domain corresponds to taking the complex conjugate in the frequency domain.

A note on linear vs. circular convolution The linear convolution of signals x and h is

$$y(n) = x(n) * h(k) = \sum_{k=0}^{n_h-1} h(k)x(n-k)$$

and the output signal, y , will have the length $n_y = n_x + n_h - 1$ samples. Using Matlab, the function `conv` does exactly this:

$$\mathbf{y} = \text{conv}(\mathbf{x}, \mathbf{h});$$

The circular convolution assumes periodic signals, and Eq. (1.46) should be used instead.

The time-domain convolution is computationally expensive for long signals and/or impulse responses, so it is usually much faster to use the DFT-based circular convolution:

$$y_{\text{periodical}}(n) = \text{DFT}^{-1} [\text{DFT}[x_{\text{periodical}}(n)] \cdot \text{DFT}[h_{\text{periodical}}(n)]]$$

We should be aware that the DFT assumes periodical signals, and the output signal, $y_{\text{periodical}}(n)$, is one period of a periodic signal. This periodicity might then give us unexpected results through the wrap-around effect illustrated for the impulse response in Fig. 1.36: if the DFT size, n_{DFT} , is smaller than the known length of the output signal, $n_y = n_x + n_h - 1$ samples, then the samples of y after n_{DFT} will appear at the start of (the periodicized) y . However, by enforcing a DFT size which is at least as large as $n_x + n_h - 1$ samples (which means to zero-pad x and h), and by truncating the output signal y to the length $n_x + n_y - 1$ samples, then the result of the circular convolution is exactly the same as for the linear convolution. (Apart from “computation noise”). The Matlab function `cconv` does the circular convolution, and the DFT size can be enforced to be large enough and to get one of the efficient FFT-sizes like this:

$$\mathbf{y} = \text{cconv}(\mathbf{x}, \mathbf{h}, 2^{\text{nextpow2}(\text{length}(\mathbf{x}) + \text{length}(\mathbf{h}) - 1)});$$

$$\mathbf{y} = \mathbf{y}(1:\text{length}(\mathbf{x}) + \text{length}(\mathbf{h}) - 1);$$

1.5.3 Appendix - Cross correlation functions

The periodic correlation functions can also be expressed as circular convolutions with a mirrored function. Eq. (1.49) implies that

$$r_{xy,\text{periodic}}(-n) = \frac{1}{N} \sum_{k=0}^{N-1} x_{\text{periodic}}(k) y_{\text{periodic}}(k-n) \quad (1.5.1)$$

If we then introduce a mirrored function

$$y_{\text{periodic}}^{\text{mirrored}}(n) = y_{\text{periodic}}(-n)$$

Eq. (1.5.1) gets the form of a circular convolution, apart from the extra $1/N$ -scaling,

$$\begin{aligned} r_{xy,\text{periodic}}(-n) &= \frac{1}{N} \sum_{k=0}^{N-1} x_{\text{periodic}}(k) y_{\text{periodic}}^{\text{mirrored}}(n-k) \\ &= \frac{1}{N} x_{\text{periodic}}(n) \circledast y_{\text{periodic}}^{\text{mirrored}}(n) \end{aligned}$$

which implies that

$$r_{xy,\text{periodic}}(n) = \frac{1}{N} [x_{\text{periodic}}(n) \circledast y_{\text{periodic}}^{\text{mirrored}}(n)]^{\text{mirrored}} \quad (1.5.2)$$

But, the mirroring of the final result is equivalent with switching the mirroring to the x -function instead:

$$r_{xy,\text{periodic}}(n) = \frac{1}{N} x_{\text{periodic}}^{\text{mirrored}}(n) \circledast y_{\text{periodic}}(n) \quad (1.5.3)$$

Finally, the periodic correlation can, in the same way as for the circular convolution, be done much more efficiently using the DFT (or, preferably, the faster FFT version),

$$\begin{aligned} r_{xy,\text{periodic}}(n) &= \frac{1}{N} \text{DFT}^{-1} [\text{DFT}[x_{\text{periodic}}^{\text{mirrored}}(n)] \cdot \text{DFT}[y_{\text{periodic}}(n)]] \\ r_{xy,\text{periodic}}(n) &= \frac{1}{N} \text{DFT}^{-1} [(\text{DFT}[x_{\text{periodic}}(n)])^* \cdot \text{DFT}[y_{\text{periodic}}(n)]] \end{aligned} \quad (1.5.4)$$

where $(\cdot)^*$ represents the complex conjugate of $(\cdot)$. Thus, a mirroring in the time domain corresponds to taking the complex conjugate in the frequency domain.

1.5.4 Appendix - Acoustic quantities

Table 1.3: Acoustic quantities, part 1

Quantity	Symbol	Unit	Definition[1]	How is it measured +remarks
Sound pressure	p	Pa	difference between instantaneous total pressure and static pressure	With a microphone (in air) or a hydrophone (in water). p is a function of time, $p = p(t)$, but the symbol is often used without modification (e.g. without a subscript rms) for its r.m.s. value.
Sound particle displacement	$\delta(\xi, \eta, \zeta)$	m	instantaneous displacement of a particle in a medium from what would be its position in the absence of sound waves	Computed from measured values of sound particle velocity. δ is a function of time, $\delta = \delta(t)$, but symbol is often used without modification (e.g. without a subscript rms) for its r.m.s. value. (ξ, η, ζ) are the x -, y - and z -components of the vector quantity δ .
Sound particle velocity	$\mathbf{v}$ or $\mathbf{u}$ (u, v, w)	m/s	$\mathbf{v} = \frac{\partial \delta}{\partial t}$	Measured either directly with a special sensor, e.g., Microflown®, or indirectly by computing it from a sound pressure difference: $u = -\frac{1}{j\omega\rho} \frac{\partial p}{\partial x}$ $\mathbf{u}$ is a function of time, $\mathbf{u} = \mathbf{u}(t)$, but the symbol is often used without modification (e.g. without a subscript rms) for its r.m.s. value. (u, v, w) are the x -, y - and z - components of the vector quantity $\mathbf{v}$ or $\mathbf{u}$.
Sound volume velocity, sound volume flow rate	q	m ³ /s	surface integral of the normal component of the sound particle velocity over the cross-section (through which the sound propagates)	Computed from measured values of sound particle velocity. q is a function of time, $q = q(t)$, but the symbol is often used without modification (e.g. without a subscript rms) for its r.m.s. value.
Sound energy density	w	J/m ³	time-averaged sound energy in a given volume divided by that volume	
Sound power	W , P , P_a	W	through a surface, product of the sound pressure, p , and the component of the particle velocity, u_n , at a point on the surface in the direction normal to the surface, integrated over that surface	Computed from measured values of sound pressure, or sound intensity, depending on which of several methods is used.
Sound intensity	$\mathbf{i}$	W/m ²	$\mathbf{i} = p \cdot \mathbf{v}$	Intensity is sometimes known as energy flux density. $\mathbf{i}$ is a function of time, $\mathbf{i} = \mathbf{i}(t)$
Time-averaged sound intensity	$\mathbf{I}$	W/m ²	$\mathbf{I} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \mathbf{i}(t) dt$	Intensity is sometimes known as energy flux density.

Table 1.4: Acoustic quantities, part 2

Quantity	Symbol	Unit	Definition	How is it measured +remarks
Sound exposure	E	Pa^2/s	$E = \int_{t_1}^{t_2} p^2 dt$	p^2 is always understood to denote the square of a frequency-weighted and frequency-band-limited sound pressure. If specific frequency weightings, as specified in IEC 61672-1, are applied, this should be indicated by appropriate subscripts to the symbol E .
Characteristic impedance of a medium	Z_c	$\text{Pa} \cdot \text{s}/\text{m}$	At a point in a non-dissipative medium and for a plane progressive wave, the quotient of the sound pressure by the component of the sound particle velocity in the direction of the wave propagation.	$Z_c = \rho c$
Acoustic impedance	Z_a	$\text{Pa} \cdot \text{s}/\text{m}^3$	At a surface, the complex quotient of the average sound pressure over that surface by the sound volume flow rate through that surface. A more precise definition could have been: “At a surface, under sinusoidal conditions, (complex) quotient of the phasor representing the average sound pressure by the phasor representing the sound volume flow rate through that surface.”*	Computed from a measured sound pressure over a surface, and a measured volume velocity.
Specific acoustic impedance (not defined in ISO 80000-8)	Z_s	$\text{Pa} \cdot \text{s}/\text{m}$	At a point, under sinusoidal conditions, (complex) quotient of the phasor representing the sound pressure by the phasor representing the sound particle velocity, in a direction which must be specified.	With a microphone
Mechanical surface impedance	Z_m	$\text{N} \cdot \text{s}/\text{m}$	At a surface, the complex quotient of the total force on the surface by the component of the average sound particle velocity at the surface in the direction of the force. A more precise definition could have been: “At a surface, under sinusoidal conditions, (complex) quotient of the phasor representing the total force on the surface by the phasor representing the average sound particle velocity at the surface in the direction of the force.”	$Z_m = Z_a \cdot A^2$ where A is the area of the surface considered.

Table 1.5: Acoustic quantities, part 3

Quantity*	Symbol	Unit	Definition	How is it measured +remarks
Attenuation coefficient	α	Np · m ⁻¹	If a field quantity, $F(x)$, is a function of distance x given by $F(x) = A \cdot e^{-\alpha x} \cos[\beta(x - x_0)]$, then α is the attenuation coefficient and β is the phase coefficient.	$m = 2\alpha$ is called the power attenuation coefficient. When there is a risk of confusion with α for the absorption coefficient, $m/2$ is used instead of α for the attenuation coefficient.
Dissipation factor for sound power, dissipation	δ, ψ	1	At a surface, ratio of mean dissipated sound power and mean incident sound power.	
Reflection factor for sound power, reflectance	$r, (\rho)$	1	At a surface, ratio of mean reflected sound power and mean incident sound power.	
Transmission factor for sound power, transmittance	τ	1	At a surface, ratio of mean transmitted sound power and mean incident sound power.	$\delta + r + \tau = 1$
Absorption factor for sound power, absorbance	α	1	At a surface, ratio of mean dissipated and mean transmitted sound power to the mean incident sound power.	$\alpha = \delta + \tau$

A bit of wave fundamentals

In this chapter we will recapitulate some basic expressions for plane waves and spherical waves in acoustics. We will use application examples that are relevant for audio technology and for acoustic measurements. A slightly more detailed presentation than here is available in the course compendium "TFE4130 - Wave propagation, Sound waves, by Peter Svensson (NTNU 2022)".

2.1 The sound pressure - our primary quantity of interest

The primary quantity we are interested in is the *sound pressure*, both for studies of sound in the air and under water. The sound pressure is the small variation in pressure in the air or water, so we can view the static air pressure (the atmospheric air pressure) as a DC component, and the small variations around this DC component will be an AC component, and that is what we define as the sound pressure. Thus, in one specific point in space, the total air pressure is:

$$P_{\text{air,total}}(t) = P_{\text{atm.}} + p(t)$$

where $p(t)$ is the time-varying sound pressure signal. The sound pressure will also be a function of space, because sound pressure propagates in the form of waves.

A microphone, or a hydrophone for sound waves under water, is a transducer that gives an electrical signal which is proportional to the sound pressure at the transducer's sensing surface, e.g. the microphone membrane:

$$e_{\text{mic.}}(t) = C_{\text{sensitivity}} p_{\text{at mic.}} \quad (2.1.1)$$

where $C_{\text{sensitivity}}$ is a (hopefully frequency-independent) scale constant with the unit V/Pa. A measurement microphone's sensitivity is indeed expressed in datasheets in mV/Pa. The expression in Eq. (2.1.1) is valid for an *omnidirectional transducer*, but transducers typically are non-ideal, or directional by design, so that:

$$e_{\text{mic.}}(t) \propto D(\theta_{\text{in}}) p_{\text{in}} \quad (2.1.2)$$

where p_{in} is the sound pressure amplitude for an incident plane wave, the angle θ_{in} is the angle from which the incident plane wave is arriving, and $D(\theta_{\text{in}})$ is the *directivity function* of the microphone. Many microphones are rotationally symmetrical, and then it is enough to specify one incidence angle, θ_{in} . If there is no rotational symmetry, then we would need a directivity function which is $D(\theta_{\text{in}}, \varphi_{\text{in}})$. Our hearing has an eardrum which acts similar to a microphone membrane, with some additional aspects:

- We have two ears which facilitates directional hearing, as described in a later chapter.
- Our head, and the shape of the ear pinna, leads to a complicated frequency-dependent directivity function.

- The transmission of sound through air, leading to vibrations of the eardrum, which in turn leads to vibrations of the basilar membrane, is a remarkably linear process. But, when the conversion from vibrations to nerve signals happens, the processing will be non-linear.

As soon as a microphone has recorded a signal, the sound pressure can be viewed as a time signal, as in chapter 1. Below, we will look a bit further by viewing the sound pressure as a wave, that is, a time- and space-dependent quantity.

2.2 Sound waves

Sound originates from some sound source, which might be a vibrating surface which "pushes" on the surrounding air particles so the pressure increases near the surface. The pressure wants to return to the static air pressure so the overpressure "pushes" nearby air particles and a sound wave propagates, spherically. Fig. 2.1 illustrates how an impulsive sound might propagate away from a loudspeaker source.

Some observations can be made (if we had carried out such a physical experiment):

- The velocity of propagation, $c \approx 343$ m/s at 20 degrees C. This velocity depends on the temperature, around 0.6 m/s per increased degree C (near 20 degrees C).
- The amplitude of the sound pressure decreases as $1/r$ (starting at a short distance from the vibrating surface), which indicates that the sound wave radiates spherically.
- The shape of a pulse does not change (beyond a certain distance). This fact reflects that the sound propagation is *non-dispersive*.

Since we primarily use the $j\omega$ -method in our analysis, we can also inspect what would happen if the vibrating surface started vibrating sinusoidally at time 0, see Fig. 2.2.

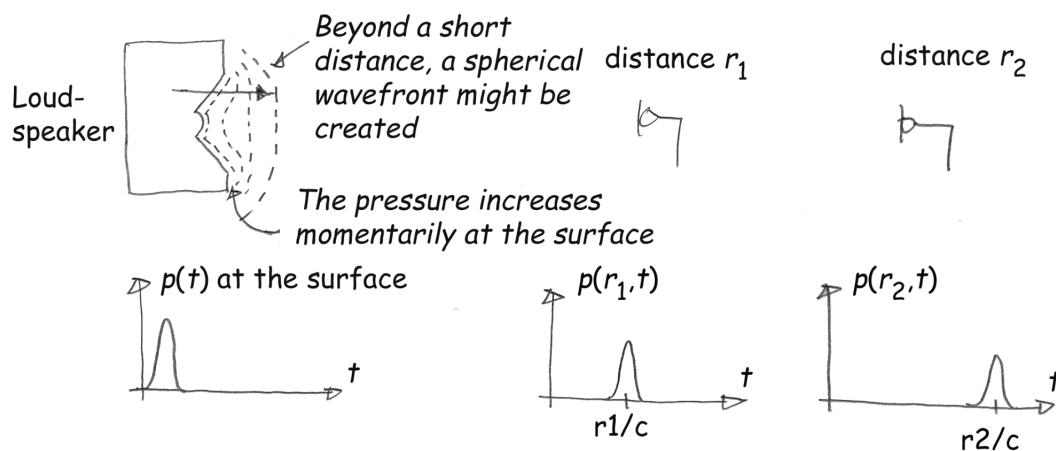


Figure 2.1: An illustration of a loudspeaker as a sound source, radiating an impulsive sound signal.

As exemplified in Figs. 2.1 and 2.2, sound waves typically spread spherically from a sound source, that is, the amplitude of the sound pressure varies as $1/r$ where r is the distance from a source to an observation point. However, for large distances from a sound source, as well as in situations like pipes (e.g., music instruments like flutes, trumpets etc, or the ear canal) and ventilation ducts, the amplitude of the sound wave will stay more or less constant and one gets *plane sound waves*.

Below, we will look a bit at those two special cases of sound waves: the plane wave and the spherical sound wave.

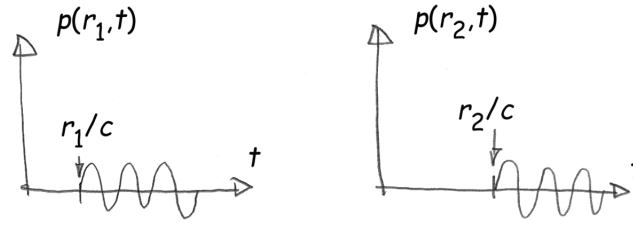


Figure 2.2: An illustration of a loudspeaker as a sound source, radiating a sinusoidal signal.

2.2.1 Plane sound waves

For the very useful special case of a plane sound wave, the sound pressure will be a function of just one space coordinate, and a general solution has two terms. These two terms describe two sound waves propagating in opposite directions:

$$p(x, t) = A_1 e^{j(\omega t - kx)} + A_2 e^{j(\omega t + kx)} = (A_1 e^{-jkx} + A_2 e^{jkx}) e^{j\omega t} \quad (2.2.3)$$

where the two (complex) amplitudes will be determined by the boundary conditions of a specific situation. We can acknowledge that the x -value affects only the phase, via $e^{\pm jkx}$, not the amplitude of each term.

An important application of plane wave modeling is to analyze the interference effect of two sound waves, for instance when an incident sound wave is reflected from a wall, as shown in Fig. 2.3. The plane wave model leads to the description of the sound field as (we leave out the time-harmonic factor $e^{j\omega t}$, so the quantity $p_{\text{tot}}(x)$ is the complex sound pressure amplitude, at position x , for a single-frequency sound field with angular frequency ω)

$$p_{\text{tot}}(x) = p_{\text{in}} (e^{-jkx} + R_p \cdot e^{jkx}) \quad (2.2.4)$$

Here the x -axis is oriented towards the right, the reflecting wall is placed at $x = 0$, and the wall has a *reflection factor* R_p . The sound pressure amplitude, for one frequency, as function of position x , will look like in Fig. 2.3(b). The sound pressure amplitude as function of frequency, for a fixed microphone position, will look like in Fig. 2.3(c): certain frequencies are attenuated; others are amplified. This phenomenon is called *destructive and constructive interference*, respectively.

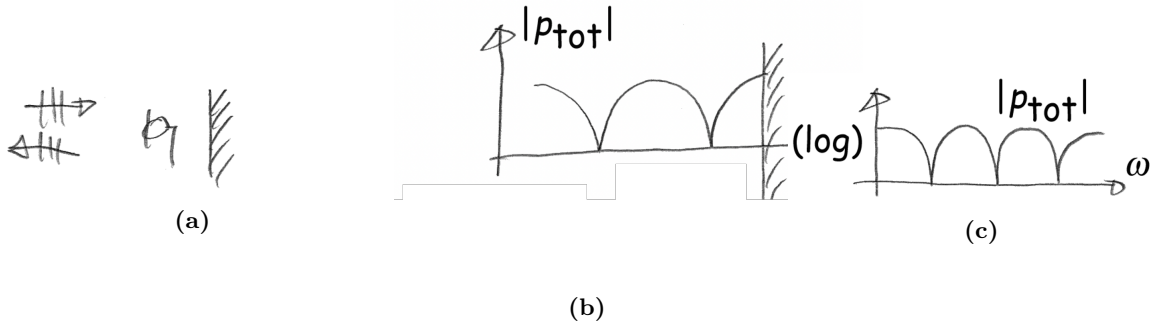


Figure 2.3: (a) A situation with a microphone near a reflecting wall. (b) The sound pressure amplitude, as function of distance from the wall, for a single frequency. (c) The sound pressure amplitude, as function of frequency, for one particular microphone position.

Comments:

- The microphone can be placed immediately at the surface. It will then experience twice as high $|p_{\text{tot}}(x)|$ as if the wall had not been there. This can be a good thing: the microphone gets a stronger signal and consequently a higher signal-to-noise ratio (SNR). This is the idea behind so-called *pressure-zone microphones*, or *boundary microphones*.
- If we put a microphone at a table, however, tables are finite in size, which limits this amplification effect at low frequencies.
- If we hold our head near a wall when we listen to music or speech we experience a change in timbre (a change in the subjective spectrum).
- If a speaking / singing person is close to a wall the same thing happens.

2.2.2 A plane sound wave in an arbitrary direction

For later studies of arrays, we need to find an expression for a plane sound wave which is propagating in an arbitrary direction, not only the $\pm x$ -directions. The compendium "TFE 4130 Wave propagation Sound waves" goes more into detail, but here we present the result. First, if a sound wave is a function of x and y (that is there is no z -dependence), then an incident plane wave coming from a direction which forms an angle θ with the x -axis can be described by the expression

$$p(x, y, t) = p_{\text{in}} e^{jk(x \cos \theta + y \sin \theta)} e^{j\omega t} \quad (2.2.5)$$

Please convince yourself that the two incident angles $\theta = 0$ and $\theta = \pi$ give the two terms in Eq. (2.2.3). Also convince yourself that the two incident angles $\theta = \pm\pi/2$ will give two forms that look like plane waves in the y -direction. A version of the expression in Eq. (2.2.5) is

$$p(x, y, t) = p_{\text{in}} e^{j(k_x x + k_y y)} e^{j\omega t} \quad (2.2.6)$$

where k_x, k_y are the x - and y -components, respectively, of the *wavenumber vector* $\mathbf{k} = (k_x, k_y) = (k \cos \theta, k \sin \theta)$. The wavenumber vector then contains information about the direction of the propagating plane wave, as well as information about the wavelength, $\lambda = 2\pi/k = 2\pi/\sqrt{k_x^2 + k_y^2} = 2\pi/||\mathbf{k}||$. The wavenumber vector form makes another, very compact, form possible:

$$p(x, y, t) = p_{\text{in}} e^{j\mathbf{k} \cdot \mathbf{x}} e^{j\omega t} \quad (2.2.7)$$

where $\mathbf{x} = (x, y)$ is the location of the microphone/sensor/observation point. This last form is particularly practical since it is valid also in three dimensions, with $\mathbf{k} = (k \sin \theta \cos \varphi, k \sin \theta \sin \varphi, k \cos \theta)$. Please note that in three dimensions, θ describes the angle relative to the z -axis, and φ describes the angle in the xy -plane relative to the x -axis.

2.2.3 Spherical waves

When we have a spherically symmetrical sound source, a monopole, see Fig. 2.4, the radiated sound pressure will be a function only of the distance r , not of the spherical angles θ or φ . The solution can then be written as

$$p(r, t) = \frac{A}{r} e^{j(\omega t - kr)} = \frac{A}{r} e^{-jkr} e^{j\omega t} \quad (2.2.8)$$

We see that the amplitude of the sound pressure, $|p| = |A|/r$, decreases as $1/r$. The constant A can be shown to be

$$A = \frac{\rho_0 S j \omega u_S}{4\pi}$$

where ρ_0 = the air density ($\approx 1.2 \text{ kg/m}^3$), u_S = the vibration velocity of the surface of the source, in $[\text{m/s}]$, S = source surface area, in $[\text{m}^2]$, and $k = \omega/c$ = wavenumber. This type of idealized sound source is called a

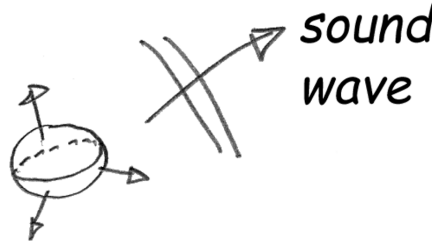


Figure 2.4: A spherically symmetrical sound source, a monopole, radiates an ideal spherical wave.

monopole source or point source. It is common to use a quantity called *source strength* $q_S = u_S \cdot S$, with the unit $[\text{m}^3/\text{s}]$. Then we write the sound pressure of the outgoing wave as

$$p(r, t) = \frac{\rho_0 j \omega q_S}{4\pi} \frac{e^{-jkr}}{r} e^{j\omega t} \quad (2.2.9)$$

The source strength is a *volume velocity*, and often, the symbol U is used for volume velocities. We also frequently use an expression for the rms-value of the sound pressure signal, squared, p_{rms}^2 ,

$$p_{\text{rms}}^2 = W \rho_0 c \frac{1}{4\pi r^2} \quad (2.2.10)$$

where W is the sound power of the source, which is expressed in watts. The expression for the complex sound pressure amplitude must give an equivalent result as the p^2 -expression and thus it must be true that

$$W = \frac{\rho_0 \omega^2 q_{S,\text{rms}}^2}{4\pi c} \quad (2.2.11)$$

Please note that in expression Eq. (2.2.9), both q_S and p are *complex amplitudes*. The *rms-value* will then be:

$$p_{\text{rms}} = \frac{\rho_0 \omega q_{S,\text{rms}}}{4\pi r} \quad (2.2.12)$$

Often, the rms-valued squared is used in various expressions, and the subscript "rms" is not always included, so for the monopole we might see

$$p^2 = \frac{\rho_0^2 \omega^2 q_S^2}{(4\pi)^2 r^2}$$

where p, q_S actually refer to the rms-values.

Transfer function/frequency response, and impulse response, of a monopole

The expression in Eq. (2.2.9) can be used to define a transfer function, or frequency response function, $H_S(\omega)$, for a monopole source,

$$H_S(\omega) = \frac{p}{q_S} = \frac{\rho_0 j \omega}{4\pi} \frac{e^{-jkr}}{r} \quad (2.2.13)$$

Comments:

1. A monopole/point source is a theoretical concept. Real sound sources can produce spherically symmetric sound fields at low frequencies (e.g. a small loudspeaker at 100 Hz) but rarely at high frequencies. However, real sound sources can often be described as a sum of point sources. Therefore the point source is important!

2. The expression in Eq. (2.2.9) describes an outgoing wave. In general, there could also be an incoming spherical wave, on the form $p(r, t) = B e^{i k r} e^{j \omega t}$. However, in practically all cases, we can claim that this incoming solution can not exist. Only if we were inside a perfect sphere with the source at the centre would we need to use this incoming wave.
3. We will also need to use the *impulse response* of a monopole source. The frequency-domain expression above corresponds to an impulse response, $h(t)$, such that the sound pressure time signal is a convolution between the impulse response and a source signal, $s(t)$:

$$p(t) = s(t) * h(t), \quad h(t) = \frac{1}{r} \delta\left(t - \frac{r}{c}\right) \quad (2.2.14)$$

where $s(t) = \rho_0 A_S(t)/(4\pi)$ and $A_S(t) = \frac{dq_S}{dt}$ is the *volume acceleration* of the monopole, expressed as a time signal.

4. We can investigate the relation between the spectrum of the sound pressure and the source strength; to jump a little bit ahead, this relation determines how large frequency range a single loudspeaker element can manage to reproduce. If the source has a volume velocity/ source strength, q_S as in Fig. 2.5 (left), the sound pressure becomes as in Fig. 2.5 (right). If we instead would desire a flat spectrum for $|p|$, as in Fig. 2.6 (right), the volume velocity of the source must have a frequency-dependence as in Fig. 2.6 (left). Ordinary (so-called direct radiating) loudspeakers happen to have $q_S \propto \omega^{-1}$ in the interesting frequency range $\Rightarrow$ flat freq. response for $|p|$. But! This means that the *displacement* of the loudspeaker must be proportional to $1/\omega^2$, which implies that the loudspeaker will need to give very large displacements at low frequencies. This $1/\omega^2$ -factor will strictly limit how loud a loudspeaker can play at low frequencies.*
5. We observe in Eqs. (2.2.13) and (2.2.14) that $|p|$ is proportional to $1/r$. This means that the amplitude of a spherical wave decreases with distance – if the distance doubles, the sound pressure amplitude halves. This corresponds to a 6 dB decrease in sound pressure level for each doubling of distance.
6. The form in Eq. (2.2.8) for an outgoing wave contains the factor $e^{j(\omega t - k r)}$. In physics and mathematics, an outgoing wave usually has a factor $e^{i(k r - \omega t)}$. It can be noted that the real parts of these two factors are identical, and different traditions in the fields of physics, mathematics and electrical engineering have lead to these two different choices.

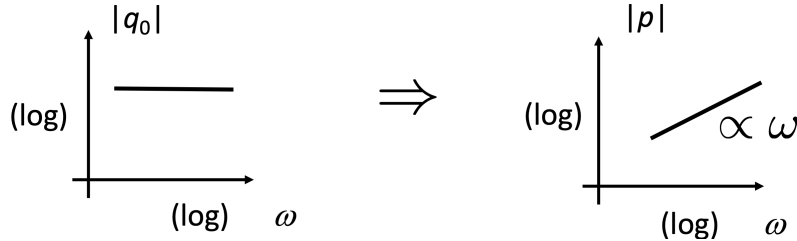


Figure 2.5: A monopole with a source signal spectrum as given in the left diagram will lead to a radiated sound pressure spectrum as in the right diagram.

The fact that the impulse response for the spherical radiation of a sound source is very simple: a scaled and delayed Dirac pulse is actually fantastically practical! This simple form means that we can generate/synthesize a room impulse response by adding a number of Dirac pulses that are delayed and scaled properly. The so-called direct sound will generate a first pulse, and then every reflection caused by the walls in a room will generate another pulse etc.

*Why must the displacement be proportional to $1/\omega^2$? Because the vibration velocity of a surface, $u_S(t)$, is the time-derivative of the momentary position = excursion of the surface, $\xi_S(t)$. For harmonic motion, the vibration and the excursion amplitudes are related as $u_S = j\omega \xi_S \Rightarrow \xi_S = u_S/(j\omega)$.

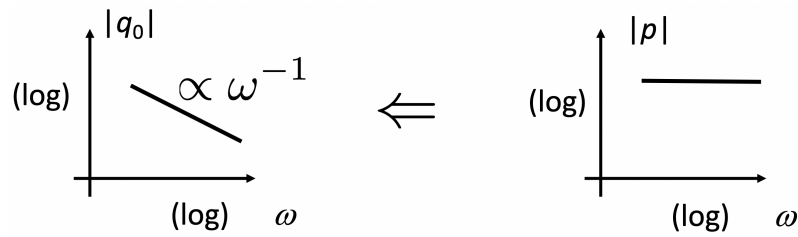


Figure 2.6: To get a flat sound pressure spectrum for the radiated sound from a monopole, as in the right diagram, the source signal spectrum must be as in the left diagram.

If a wall isn't perfectly rigid, it will not reflect the sound waves perfectly (that is, the reflection factor R_p will not be 1) - but the wall reflection pulse can then be replaced by a "wall impulse response" which is constructed to represent the frequency-dependent properties of the wall!

The idea presented above is the idea behind so-called *auralization*: an impulse response can be synthesized/constructed which represents a real room, and then some source signal (music, speech etc) can be convolved/filtered with the synthetic impulse response, and the output signal should be the sound pressure *which would have resulted* in the real situation!

2.3 Reflection of spherical sound wave, the image source method

In order to look into the reflection of a spherical sound wave from a wall, we will start with the case of an ideally rigid wall (e.g. a concrete wall, a hard floor, a water surface, a glass plate etc)

Fig. 2.7 shows a monopole with source strength Q near a rigid wall. Without going into the details, we claim that the direct sound wave from the source, and the reflection from the wall, can be generated exactly by the original source *plus an image source in free space, without the reflecting wall!*

The sound field must be the same in the two cases (the original situation and the one without a wall but with an image source) - at least to the right of the wall.

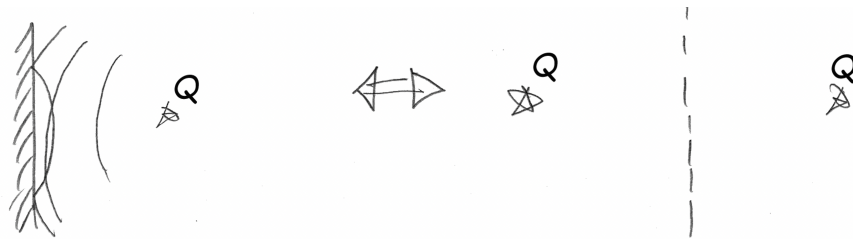


Figure 2.7: A monopole source next to a rigid wall can, according to the image source method, be replaced with two identical sources radiating in free space.

Loudspeakers

3.1 Introduction

Loudspeakers are transducers with an electrical input signal and the sound pressure of a radiated sound wave as output signal. Such a transducer can be viewed as a system, as illustrated in Fig. 3.1, with an electrical voltage as input signal and the sound pressure at some distance, in a particular direction, as output signal. If we express the signals in the frequency domain, the sound pressure at the receiver can be denoted $P_{1m}(\theta, \varphi, \omega)$. The *directivity* is a central characteristic of a loudspeaker, so the response signal can take this into account by introducing a directivity function. Thus, the sound pressure at the receiver is

$$P_{1m}(\theta, \varphi, \omega) = D(\theta, \varphi, \omega) \cdot P_{1m, \text{on-axis}}(\omega) \quad (3.1.1)$$

where $D(\theta, \omega)$ is the *Directivity function*, which gives the amplitude of the radiated sound pressure in direction θ , relative to the sound pressure in the reference direction, $\theta = 0, \varphi = 0$. The two angles are the standard ones in a spherical coordinate system, so that $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi]$.

The expression in Eq. (3.1.1) is based on the assumption that a directivity function $F(\theta, \omega)$ exists which is the same *regardless of the distance from the source*. This is true in the so-called *far-field* of a sound source, which is true beyond some critical transition distance.

In the far-field of a sound source, one can write the transfer function of the loudspeaker as

$$H(r, \theta, \omega) = D(\theta, \omega) \cdot H_{\text{on-axis}}(\omega) \frac{e^{-jk(r-r_{\text{ref}})}}{r} r_{\text{ref}} \quad (3.1.2)$$

where $r_{\text{ref}} = 1$ m (assuming that the distance of 1 m is indeed in the so-called far-field). The TF $H_{r_{\text{ref}} \text{ on-axis}}(\omega)$ will contain the sensitivity of the loudspeaker, which specifies what the sound pressure amplitude in Pa will be, at the distance r_{ref} , for an input voltage of 1V. The single-system transfer function of a loudspeaker, as illustrated in Fig. 3.1 often gets factorized into two, as in Fig. 3.2. In both of those system block diagrams, the spherical spreading factor, $e^{-jk(r-r_{\text{ref}})} r_{\text{ref}} r^{-1}$, in Eq. (3.1.2) is left out.

Quite different transducer principles are used in loudspeakers and other sound sources. Some are based on the *electromagnetic* equation $F = Bl \cdot i$ and the absolutely dominating transducer type for loudspeakers is the dynamic/electrodynamic transducer (often called the “moving coil” loudspeaker), both in loudspeakers and headphones. A cone-shaped membrane is constructed to be stiff and light so that it can move more or less like a piston, see Fig. 3.3.

Others are based on the *electrostatic*, or *piezoelectric* transduction principle. Other principles exist as well for special applications.

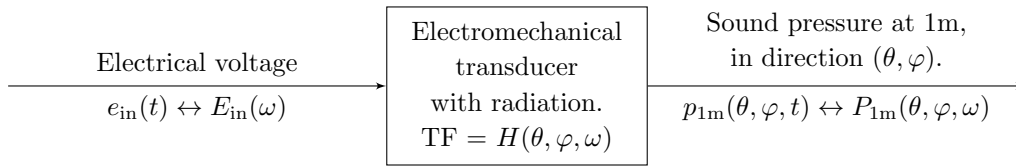


Figure 3.1: A single-system block diagram representing a loudspeaker

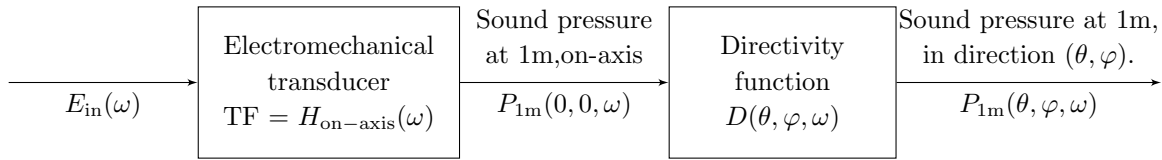


Figure 3.2: A two-part block diagram representing a loudspeaker, with the directivity function as a separate block.

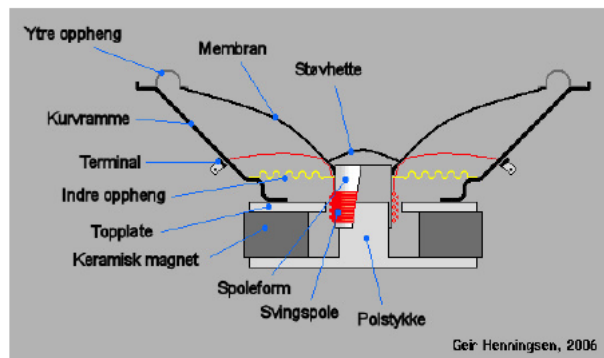


Figure 3.3: An electrodynamic, or moving-coil, loudspeaker element. Illustration from Wikimedia. *Svingspole* = voice coil. *Membran* = membrane. *Ytre oppheng* = outer suspension. *Indre oppheng* = inner suspension. *Støvhette* = dustcap. *Keramisk magnet* = ceramic magnet. *Kurvramme* = basket. *Polstykke* = pole piece. *Spoleform* = voice coil former.

3.2 Directivity functions, directivity factor, directivity index

To illustrate the relationships in the introduction, measurements were done on a small loudspeaker, a Meyer-sound MM-4XP[®], see Fig. 3.4 (a). Impulse responses were measured at a distance of 2.35 m in an anechoic room, for radiation angles in steps of 3 degrees, that is, for $\theta = 0, 3^\circ, 6^\circ, \dots, 180^\circ$, for the angle $\varphi = 0$. The impulse responses were converted to frequency responses, and they are all plotted on top of each other in Fig. 3.4 (b).

The frequency responses in Fig. 3.4 (b) tell us three things which are very typical for most loudspeakers:

1. The frequency response curves are very close to each other for low frequencies, which implies that the loudspeaker is *omnidirectional* for low frequencies.
2. The frequency response curves generally diverge more and more from each other as frequency increases, which implies that the loudspeaker has a *directivity which increases with frequency*.
3. The frequency response curves have a *high-pass filter characteristic*.

Items 1 and 2 will be explored further below, and item 3 will be discussed below, in section XX.

By dividing all the frequency response curves with the on-axis response, one can extract the directivity functions, $D(\theta, 0, \omega)$. Thus, Fig. 3.5 (a) shows the directivity functions for angles $\theta = 0^\circ, 30^\circ, 60^\circ, \dots$, plotted as function of frequency. Fig. 3.5 (a) also shows a curve marked as "Average", and that curve shows the average of $|D(\theta, 0, \omega)|^2$, that is, the curve is $10 \lg \langle |D(\theta, 0, \omega)|^2 \rangle_\theta$, where $\langle |D(\theta, \omega)|^2 \rangle_\theta$ is averaged over all θ -values*.

A final quantity, which might be the most common directivity descriptor, is the *directivity index*, which is the so-called *directivity factor*, expressed in dB. Both the directivity factor and the directivity index are computed from the $|D(\theta, \varphi, \omega)|^2$ -functions. The directivity factor is in some books denoted Γ , in other books Q , and still other times denoted DF , which is used here. Thus

$$DF(\theta, \varphi, \omega) = \frac{|D(\theta, \varphi, \omega)|^2}{\langle |D(\theta, \varphi, \omega)|^2 \rangle_{\theta, \varphi}}$$

Usually, the directivity factor is only presented for the main direction, "on-axis" ($\theta = 0, \varphi = 0$), and written as a function of frequency only:

$$DF(\omega) = \frac{|D(0, 0, \omega)|^2}{\langle |D(\theta, \varphi, \omega)|^2 \rangle_{\theta, \varphi}}$$

The directivity index, DI , is then

$$DI(\omega) = 10 \lg DF(\omega)$$

or more generally

$$DI(\theta, \varphi, \omega) = 10 \lg \frac{|D(\theta, \varphi, \omega)|^2}{\langle |D(\theta, \varphi, \omega)|^2 \rangle_{\theta, \varphi}}$$

The directivity index functions, for a few radiation angles, are shown in Fig. 3.5 (b). Also, in Fig. 3.6, the on-axis DI curve is plotted together with a simple approximation,

$$DI_{\text{approx.}} = 10 \lg(1 + ka)$$

where $k = 2\pi f/c$ and a is the (effective) radius of the loudspeaker element. In Fig. 3.6, the approximation uses the value $a = 40$ mm, which is close to the measured diameter of the loudspeaker element, divided by 2. The approximation illustrates that for low frequencies, when ka is very small, the DI is around 0 dB, which means that the sound source is omnidirectional (like in item 1 in the frame above). Also, for increasing frequency (increasing ka), the directivity index increases steadily (like in item 2 above).

3.3 The on-axis frequency response

As observed in Fig. 3.4 (b), the on-axis frequency response has a high-pass filter characteristic. Two of the more important design goals for a loudspeaker are

1. A flat frequency response (above the high-pass filter cut-off frequency).
2. A smoothly increasing directivity index curve.

Other design goals include the ability to play loud enough and to have low distortion. In some applications, such as sound reinforcement systems, a high directivity index is an important design goal.

*Strictly speaking, for a sound source radiating spherically, the average over radiation angles should take into account both the θ -angles and the φ -angles. What is presented here is a horizontal-plane-average.

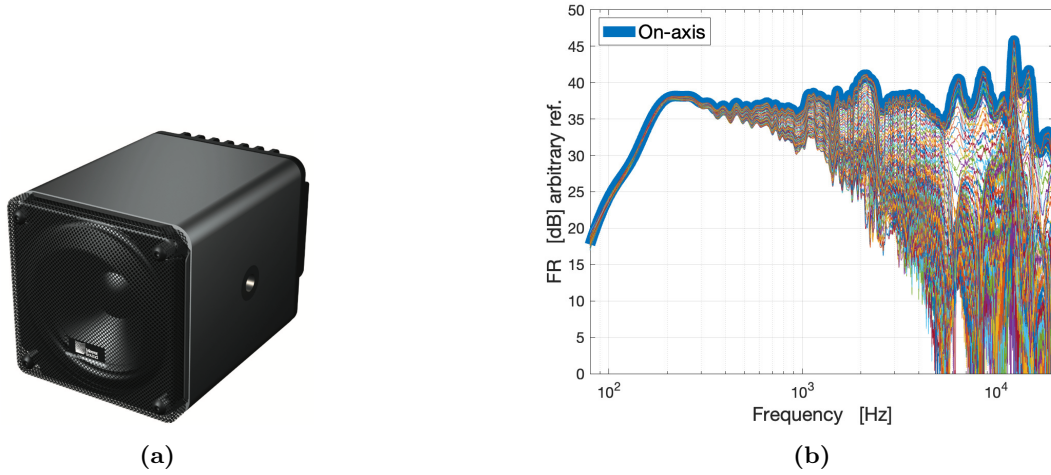


Figure 3.4: (a) A small loudspeaker, a Meyersound MM-4XP[®]. (b) Frequency responses for the loudspeaker in (a), measured in an anechoic room at a distance of 2.35 m, for radiation angles in steps of 3 degrees. The diagram shows $20 \lg |H(\theta, 0, \omega)|$ with an arbitrary reference for the dB value.

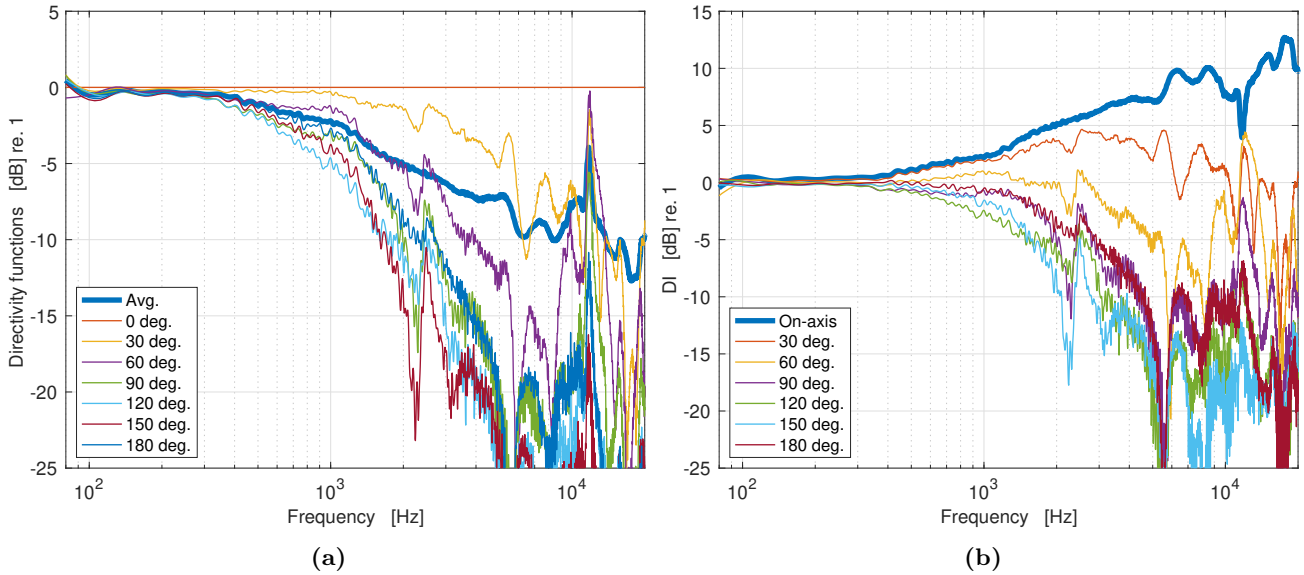


Figure 3.5: (a) Directivity functions for the loudspeaker in Fig. 3.4(a), based on the measured frequency responses in Fig. 3.4(b). The diagram shows $20 \lg |D(\theta, 0, \omega)|$. (b) Directivity index functions, for a few radiation angles.

3.3.1 Resonances and interferences

The on-axis frequency response in Fig. 3.4 (b) is flat-ish but has some clear peaks and dips. Such deviations from flatness are typically caused by two different phenomena:

- *Resonances* in the loudspeaker element/membrane, which implies that the vibration of the loudspeaker membrane gets stronger at certain frequencies. Resonance peaks are usually very similar for many radiation angles, so many of the curves in Fig. 3.4 (b) have the same peaks. The resonances are caused by imperfect vibration behavior of the loudspeaker membrane: ideally the membrane should vibrate as a *piston*, that is, like a perfectly rigid plate. But, the drawing in Fig. 3.3 shows that for the dynamic

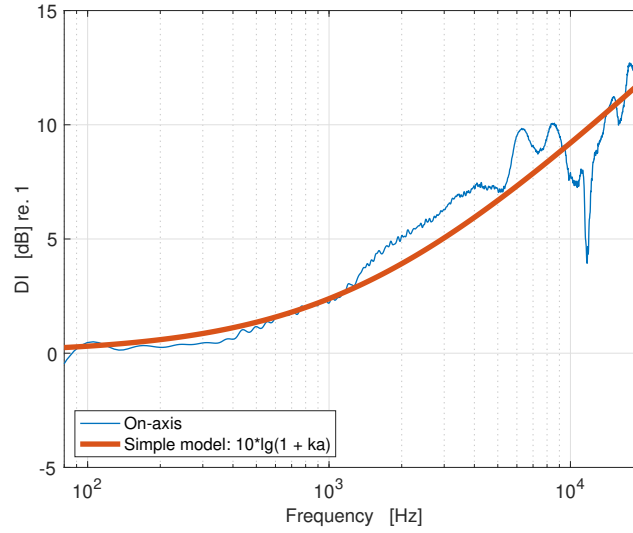


Figure 3.6: Directivity index on-axis for the loudspeaker in Fig. 3.4(a), based on the measured frequency responses in Fig. 3.4(b). Also plotted is an approximation, using the radius value $a = 40$ mm.

loudspeaker construction, the coil which is wound on the cylindrical "voice coil former" is attached to the cone-shaped membrane along a circle, close to the edge of the "dustcap". Thus, when the electromagnetic force is acting on the coil, the force is transmitted to the membrane along that circle. The membrane will, for low frequencies, move as a rigid (cone-shaped) plate but as frequency increases, the membrane can not move perfectly any more. Instead so-called bending waves are generated. They propagate in the membrane material, get reflected at the edge of the membrane, and build up resonances, with characteristic vibration patterns, for certain frequencies. The membrane material is constructed to be stiff and light, and various methods are used to suppress such resonances, more or less successfully.

- *Interference effects* are caused when the radiating sound wave generates reflections, e.g., from the edges of the loudspeaker enclosure, and interference results, between the sound wave radiated from the loudspeaker membrane, and those reflections. Positive and negative interference will create different peaks in different directions.

The directivity phenomenon is actually an interference effect, between the many sound waves generated by the different parts of the vibrating loudspeaker element.

3.3.2 The maximum excursion limit vs. the high-pass filter characteristic

At low frequencies, the loudspeaker apparently behaves like an omnidirectional monopole. Then we can look back at Eq. (2.2.12) which gave the sound pressure from a monopole. That expression used the source strength, q_S , but we replace it with the vibration velocity times the source area, and also clarify that the vibration velocity can be a function of ω : $q_S(\omega) = S_S \cdot u_S(\omega)$. Then Eq. (2.2.12) gives

$$p_{\text{rms}} = \frac{\rho_0 S_S}{4\pi r} \omega u_{S,\text{rms}}$$

We want to relate the vibration *velocity* to the vibration *excursion*, x_S . Since velocity is the time-derivative of the position/excursion, and we use the $j\omega$ -method, we get

$$u_S(t) = \frac{dx_S(t)}{dt} \rightarrow u_S(\omega) = j\omega x_S(\omega)$$

Thus,

$$p_{\text{rms}} = \frac{\rho_0 S_S}{4\pi r} \omega^2 x_{S,\text{rms}}$$

This expression implies some interesting facts:

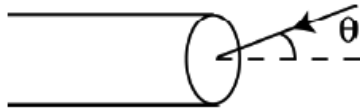
- In the flat part of the on-axis frequency response, where p_{rms} is more or less constant, the loudspeaker membrane will move with smaller and smaller amplitude as frequency increases.
- Since all loudspeaker constructions will have a physical excursion limit in terms of so and so many mm, $x_{S,\text{max}}$, the maximum sound pressure will decrease quickly as the frequency is decreased:

$$p_{\text{rms,max}} = \frac{\rho_0 S_S}{4\pi r} \omega^2 x_{S,\text{rms,max}}$$

Microphones

4.1 Introduction

Microphones are transducers which transform a sound pressure $p(t)$ into an electric voltage, $e_{out}(t)$, see Fig. 4.1. They have a directivity that can be described with a directivity function $D(\theta, \phi)$, and many microphones are rotationally symmetrical, which means that the directional response is solely a function of the angle θ , as indicated in Fig. 4.1.



$$e_{out}(t) \propto p(t) \cdot D(\theta)$$

Figure 4.1: A microphone is a transducer which gives an electrical output voltage, e_{out} , which is proportional to the sound pressure p , weighted with some directivity function, $D(\theta)$.

Microphones are obviously the opposite kind of transducer compared to the loudspeaker, but they share the directivity aspect and on-axis frequency response. Thus, equivalently to Eq. (3.1.2) for the loudspeaker, the transfer function of a microphone can be written

$$H_{mic.}(\theta, \varphi, \omega) = D(\theta, \varphi, \omega) H_{on-axis}(\omega) \quad (4.1.1)$$

where the transfer function $H_{on-axis}(\omega)$ includes a sensitivity factor, with the unit of V/Pa. To jump a little bit ahead, we can already acknowledge that microphones can have —em an order of magnitude flatter frequency responses than loudspeakers. Why is that? The primary reason is that microphones can be constructed much smaller than loudspeakers, and therefore both resonance and interference effects are pushed to higher frequencies, and can occur above the frequency range of interest.

Three large application areas exist for microphones, and the construction types and requirements vary quite much between these application areas.

1. *Measurement microphones* have the highest requirements for the flatness of the frequency response, simply because a measurement must have a known uncertainty as to what the sound pressure is *without the microphone*. As one example, ig. 4.2 shows the requirements for the frequency response of three different sizes of measurement microphones: WS1 has a diameter of approx. 1 inch, WS2 approx. 1/2 inch, and WS3 approx. 1/4 inch, [22]. For measurement microphones it is also of the utmost importance to *calibrate* the sensitivity of the microphone. Practically all measurement microphones are omnidirectional for most of the frequency range.

2. *Music recording microphones* have a more or less flat frequency response, but there are no formal requirements. It is rather the perceived sound quality of the recording engineer which determines the choice of microphone. On the other hand, most music recording microphones have some directivity such as cardioid, figure-of-8 etc, for much of the frequency range. The (frequency-dependent) balance of direct sound and reverberation will have a large impact on the perceived sound quality.
3. *Speech communication microphones* include all the billions of microphones produced for use in mobile phones. These microphones have a reasonably flat frequency response – but only in the frequency range supported by the specific application. In particular, the low-frequency response is not so important and might be shaped to suppress low frequencies.
4. *Microphones for arrays* can be used for speech or other sound signals. Such microphones usually need to be small in size as well as very similar in frequency response and sensitivity.

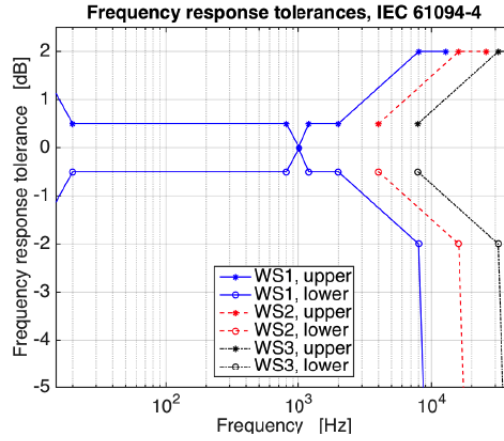


Figure 4.2: The frequency response tolerances for measurement microphones are specified in the standard IEC 61094-4, [22].

4.2 Directivity, first-order microphones

Equivalently to the loudspeakers, a microphone's directivity function, $D(\theta, \varphi, \omega)$, expresses the amplitude (and phase) response as function of a plane wave's incidence angle, and as also indicated in Fig. 4.1, microphones are often rotationally symmetrical, so that the directivity function is a function of θ only: $D(\theta)$. This directivity function contains a lot of details, and a much more practical quantity, which was also introduced for the loudspeaker, is the *directivity factor* DF , which expresses how much stronger the response squared is in the main direction (assumed to be $\theta = 0, \varphi = 0$) relative to the average of all directions:

$$DF(\omega) \equiv \frac{D^2(0, 0, \omega)}{\frac{1}{4\pi} \int \int D^2(\theta, \phi, \omega) \sin\theta d\theta d\phi}$$

and for rotational symmetry:

$$DF\omega \equiv \frac{D^2(0, \omega)}{\frac{1}{2} \int_0^\pi D^2(\theta, \omega) \sin\theta d\theta}$$

Furthermore it is more common to use the *directivity index*, DI , expressed in dB:

$$DI(\omega) = 10 \lg DF(\omega)$$

4.2.1 First-order microphones

As will be shown below, it is possible to construct microphones with a single microphone membrane but different mountings which lead to a directivity function, on this form (we leave out the dependence on ω for now)

$$D(\theta) = a + b \cos \theta \quad (4.2.2)$$

Microphones with directivities that can be described by this directivity function are called *first-order microphones* (because there is only the first term in a harmonic series $\cos n\theta$). Some very common directivity types are given in Table 4.1.

Table 4.1: Some first-order microphone types, their directivity functions, D , and their directivity indices, DI

Name	$D(\theta)$	DI [dB]
Omnidirectional	1	0
Figure-of-eight	$\cos \theta$	4.8
Cardioid	$0.5 + 0.5 \cos \theta$	4.8
Hypercardioid	$0.25 + 0.75 \cos \theta$	6.0

The directivity factor can be derived analytically for first-order microphones. We can look at the example of a so-called cardioid microphone,

$$\begin{aligned}
 D_{\text{cardioid}}(\theta) &= 0.5(1 + \cos \theta) \Rightarrow \text{DF}_{\text{cardioid}} = \frac{1}{\frac{1}{2} \int_0^\pi 0.25(1 + \cos \theta)^2 \sin \theta d\theta} \\
 &= \frac{8}{\int_0^\pi (\sin \theta + \sin 2\theta + \sin \theta \cos^2 \theta) d\theta} = \frac{8}{\left[-\cos \theta - \frac{1}{2} \cos 2\theta - \frac{1}{3} \cos^3 \theta\right]_0^\pi} \\
 &= \frac{8}{1 - (-1) + \frac{1}{2}(1 - 1) + \frac{1}{3}(1 - (-1))} = 3 \Rightarrow \text{DI}_{\text{cardioid}} = 10 \lg 3 \approx 4.8 \text{ dB}
 \end{aligned}$$

The directivity factor for any first-order microphone can be derived. As a first step in the derivation, the expression for the directivity function, in Eq. (4.2.2), is rewritten into a form which always gives the amplitude 1 for the angle 0, which is the main direction, such that

$$D(\theta) = 1 - b + b \cos \theta, \quad b \in [0, 1]$$

The resulting directivity factor then becomes (see the Appendix for derivation details)

$$\text{DF}_{\text{first-order}}(b) = \frac{3}{4b^2 - 6b + 3}, \quad b \in [0, 1]$$

and this is plotted, in the form of the directivity index, $\text{DI} = 10 \lg \text{DF}$, in Fig. 4.3. Interestingly, the cardioid and the figure-of-8 microphones both get exactly $\text{DF} = 3$, that is $\text{DI} \approx 4.8 \text{ dB}$. The highest directivity that can be achieved with a first-order microphone is $\text{DF} = 4$, that is, $\text{DI} \approx 6.0 \text{ dB}$, and that is how the hypercardioid microphone is defined.

4.2.2 The force acting on the microphone membrane

Most microphones use a membrane that vibrates due to the sound field, see Fig. 4.4*:

*Here we have ignored the static air pressure, but if the static air pressure is (slightly) different on the two sides of the membrane, which can happen if one side is enclosed, then there will be an additional *static* force, $F_{\text{m,static}} = P_{\text{static},1} - P_{\text{static},2}$. Such a static force would displace the membrane into a new resting position.

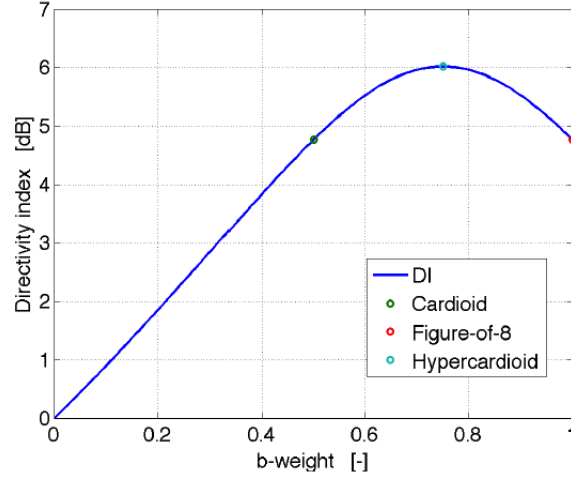
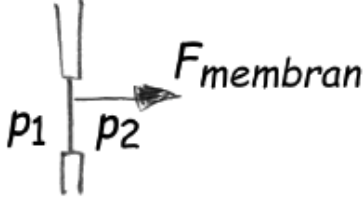


Figure 4.3: The directivity index for a first-order microphone, as function of the parameter b , when the directivity function is expressed as $D(\theta) = 1 - b + b \cos \theta$.



$$F_m = \int_{S_m} (p_1 - p_2) dS$$

Figure 4.4: A microphone membrane is exposed to a net force, F_m (shortened F_m), which is given by the net difference in sound pressures at the two sides.

One side or both sides of the membrane may be exposed to the sound field. If only one side is exposed, then we can assume that the sound pressure signal $p_2 = 0$ on the back of the membrane, and (remember that $p_2 = 0$ implies that the back side is just exposed to the static air pressure)

$$F_{m,\text{single-sided}} = \int p_1 dS \quad (4.2.3)$$

We will see below that this is the case for an omnidirectional microphone. For low frequencies (LF), the microphone is so small that the sound pressure is almost constant across the membrane, and then

$$\begin{cases} F_{m,\text{LF}} = (p_1 - p_2) \cdot S_m \\ F_{m,\text{single-sided},\text{LF}} = p_1 \cdot S_m \end{cases} \quad (4.2.4)$$

We realize that as the frequency increases, the sound pressure is not going to be constant across the membrane any more. For incidence from the main direction, however, we can still assume that the incident sound pressure is the same everywhere across the membrane, but for a sound wave arriving from a 90-degree angle of incidence, the sound pressure will, for high frequencies, display both positive and negative values across the membrane, and the integration will have a lower value than for incidence from the main direction. The varying results of this integration will lead to different vibration velocities of the membrane for different angles of incidence, and therefore different output voltages. The directivity function, $D(\theta)$, expresses this difference in output voltage from the microphone. It should be noted that the simple directivity functions in Eq. (4.2.2) are usually valid only for low enough frequencies, where the approximations in Eq. (4.2.4) are valid.

4.2.3 Omnidirectional microphones

A measurement microphone typically has a membrane placed at the end of a cylinder, and then only one side of the membrane is exposed to the sound field, as illustrated in Fig. 4.5. Such microphones become *omnidirectional* at low frequencies (LF),

$$F_{m,\text{single-sided,LF}} \approx p_1 S_m \Rightarrow D_{LF}(\theta) \approx 1$$

For higher frequencies (HF), the microphone starts to be directional, see Fig. 4.6.



$$F_m = \int p_1 dS$$

Figure 4.5: An omnidirectional microphone has one side of the membrane exposed to the sound field.



Figure 4.6: An omnidirectional microphone ideally has a polar directivity plot as on the left, but in reality, the directivity index will gradually increase from 0 dB with increasing frequency.

Some examples of omnidirectional microphones are shown in Fig. 4.7. It can be noted that omnidirectional microphones usually don't have any openings on the side of the microphone capsule.



(a) Miniature microphone from DPA® primarily for speech.



(b) Recording microphone from Audio Technica®.



(c) Measurement microphone from Earthworks®.

Figure 4.7: Examples of omnidirectional microphones.

4.2.4 Figure-of-eight microphones

Microphones can also be constructed with an open design, as illustrated in Fig. 4.8, such that the front and back sides of the membrane are exposed symmetrically. Then the difference in sound pressure can be approximated

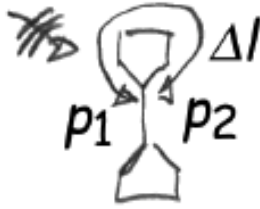
as*:

$$p_1 - p_2 \approx \Delta l \frac{\partial p}{\partial x} = \Delta l \cdot jk \cos \theta \cdot p_{in}$$

where θ is the angle relative to normal incidence. This microphone type is sometimes called a *gradient microphone*. So, for LF this type of microphone gets the directivity function

$$D_{LF}(\theta) = \cos \theta$$

which is often called a "figure-of-eight" directivity due to the polar plot of the directivity as shown in Fig. 4.9, to the left. As seen in Fig. 4.3, the figure-of-eight microphone gets the DI value 4.8 dB for LF, just like the cardioid microphone.



$$F_m = \int (p_1 - p_2) dS$$

Figure 4.8: A first-order gradient microphone of figure-of-eight type has both sides of the membrane exposed *symmetrically* to the sound field.

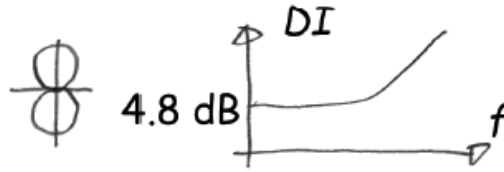


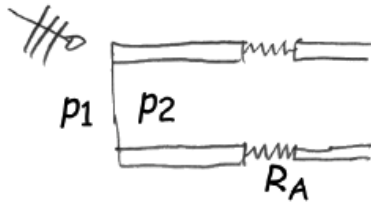
Figure 4.9: A figure-of-eight microphone ideally has a polar directivity plot as on the left, but in reality, the directivity index will gradually increase from 4.8 dB for increasing frequency.

4.2.5 Cardioid-type microphones

Microphones can be constructed as illustrated in Fig. 4.10, with openings that let the back side of the membrane be exposed to the sound field via carefully designed openings that act as acoustic resistances, R_A . By choosing the value of R_A , different kinds of cardioid-type microphones can result, with directivity functions (for LF) on the first-order microphone form in Eq. 4.2.2. As one example, a cardioid microphone has $D_{LF}(\theta) = 0.5 + 0.5 \cos \theta$, and this directivity function can be illustrated in a polar plot as in Fig. 4.11, to the left. For HF, also this type of microphone will get a directivity factor which increases with frequency, see 4.11, to the right. Examples of cardioid-type microphones, for music recording purposes, are shown in Fig. 4.12. Cardioid-type microphones can usually be identified by the openings with some mesh at the side of the microphone capsule, as seen in Fig. 4.12.

In [23], Example 6.1 demonstrates how one can, by choosing the value of an acoustic resistance, R_A , in the little openings on the side of the microphone, create a cardioid-type microphone. It is a challenge to find a material which has an acoustic resistance with a constant R_A value across a wide frequency range. As a consequence, it might be difficult to get the same directivity function for a wide frequency range.

*An incident plane wave, with angle θ to the x -axis, can be written on the form $p_{in}(x, y) = Ae^{jk \cos \theta \cdot x} e^{jk \sin \theta \cdot y}$



$$F_m = \int (p_1 - p_2) dS$$

Figure 4.10: A first-order gradient microphone of cardioid type has both sides of the membrane exposed to the sound field.

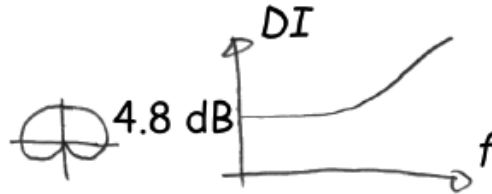


Figure 4.11: A cardioid microphone ideally has a polar directivity plot as on the left, but in reality, the directivity index will gradually increase from 4.8 dB for increasing frequency.



(a) Hypercardioid recording microphone from Neumann®.



(b) A stereo pair of cardioid recording microphones from Rode®.

Figure 4.12: Examples of cardioid-type microphones.

Bass amplification

Microphones with their back exposed (symmetrically or asymmetrically) will produce a bass amplification if placed too close to a source (which is radiating spherical waves). The reason is that the output signal is proportional to the sound pressure gradient (see "Figure-of-eight microphones" above), and for normal incidence we get a spectrum modification for a close spherical source but not for a plane wave:

$$\begin{cases} \text{In a plane wave} & \frac{\partial p_{in}}{\partial x} = jk p_{in} \Rightarrow e_{out} \propto \omega p_{in} \\ \text{In a spherical wave} & \frac{\partial p_{in}}{\partial r} = -\left(jk + \frac{1}{r}\right) p_{in} \Rightarrow e_{out} \propto \left(jk + \frac{1}{r}\right) p_{in} \end{cases}$$

The relationship for the spherical wave implies that the frequency response will be different in the near-field compared to the far-field.

4.2.6 Microphones with switchable directivity

There are also double-membrane constructions, where the directivity can be changed with a switch. It is a change of the polarization voltage that changes the sensitivity of the two membranes, and that changes the directivity of the synthesized microphone that results from adding the two signals. An example is shown in Fig. 4.13.



Figure 4.13: A double-membrane microphone from AKG[®], with a directivity-switch.

4.2.7 Synthesizing first- and higher-order microphones

In the sections above, the cardioid-type, and figure-of-eight microphones, were all constructions that use a single membrane, and a physical construction that is tailored to give a specific, constant directivity.

In a later chapter, on "Microphone array signal processing" it will be shown that *any* first-order microphone can be *synthesized* from two small omnidirectional microphones, or from a pair of cardioid-type of microphones. With four such microphones it is, in addition, possible to control in which direction the synthesized/virtual microphone should have its maximum sensitivity. This opens up for many interesting applications, as will be shown in that later chapter.

This "synthetic microphone" idea can be taken even further: by including more and more microphone elements, it is possible to synthesize so-called *higher-order* microphones, which can be more directional than first-order microphones. Stay tuned to the chapter, on "Microphone array signal processing"!

4.2.8 High directivity microphones

Different possibilities exist to obtain higher directivity than the 5-6 dB that can be achieved by a simple one-element microphone (1st-order gradient). Some examples of these are shown in Fig. 4.14

4.3 Microphone types

Microphones can be constructed based on different conversion technologies, similarly to loudspeakers. There is, however, an interesting difference: the electrostatic principle is most common for microphones. Thus, the various



(a) A parabola microphone with an extra little disc for amplifying the microphone signal.



(b) A shotgun microphone from Sony®.



(c) AA microphone array with 128 microphones from Norsonic®.

Figure 4.14: Examples of directional microphones.

microphone (or hydrophone, for recording in water) technologies are:

- The *condenser microphone* uses the electrostatic principle: a moveable stretched thin foil membrane is placed at a very small distance (a few 10s to a 100 μm) from a fixed electrode. The sound pressure gives a forces which acts on the membrane and moves it. Then the capacitance changes, and this change in capacitance can be measured very accurately. This electrostatic principle can be implemented with a DC-voltage applied (48V for studio microphones, 200 V for measurement microphones), or with an electret material which can hold a charge over very lon periods of times. Electret microphones don't need an external DC-voltage.
- The *dynamic microphone* has a moving voice coil attached to the membrane. They are popular to use as voice recording microphones for live music performances.
- *MEMS microphones*. MEMS stands for Micro-ElectroMechanical Systems, and the term is used for sensors and other devices that are made from silicon, with techniques from the microelectronics industry. It is possible to make very small microphones, and with very constant properties, and cheaply, with these techniques, but the performance has been inferior to condenser microphones and electret microphones.

It is claimed, [24], that a big step forward for the MEMS microphones happened in 2009, when the smartphone iPhone 4 started to use MEMS microphones instead of electret microphones. That ref. also gives numbers for how many MEMS microphones are sold globally:

2009: 433 million

2012: 2060 million

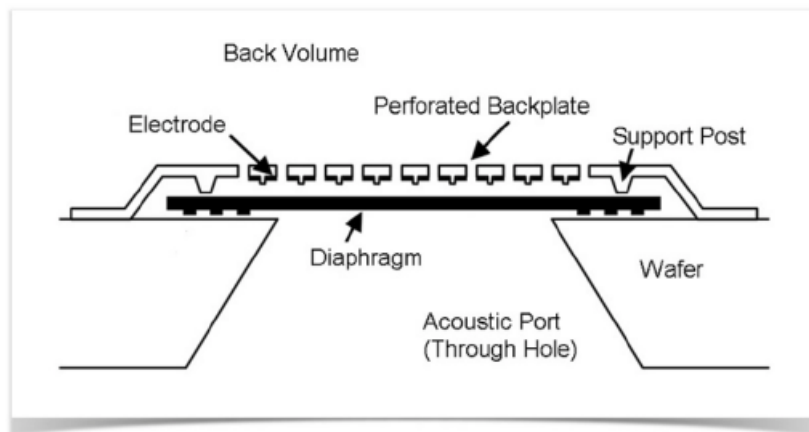
2016: (predicted in 2013) 4650 million units, to a total value of 1 billion \$. So, the prices per microphone then seems to be around $0.24\$ \approx 2\text{NOK}$.

An illustration of a MEMS microphone is given in Fig. 4.15, from *.

*<https://www.comsol.com/blogs/mems-microphone-model-presented-asa-166-san-francisco/>

This microphone becomes very similar to a condenser microphone: a DC voltage is connected across the condenser which consists of "Electrode" and "Diaphragm". A vibration in the membrane ("diaphragm") will give a change in capacitance, which generates an AC output voltage. Since the diaphragm/membrane is moving but is attached at the edge, a mass/spring system effect will apply here too. In the same way as for the condenser microphone, the resonance frequency should be above the frequency range of interest.

- An "optical" microphone exists which uses an optical system to convert the membrane vibration to an electric voltage. The company sensiBel in Oslo (www.sensibeltech.com), has commercialized a principle that was developed at SINTEF and Oslo university, where an optical reading of the membrane vibrations is used, instead of a capacitive. The illustration in Fig. 4.16 is taken from [25].



Sketch of the MEMS microphone motor (not to scale). The diaphragm has a thickness of $1\text{ }\mu\text{m}$, the gap between the backplate and the diaphragm is $4\text{ }\mu\text{m}$, the diameter of the perforations in the backplate is $10\text{ }\mu\text{m}$, and the thickness of the backplate is $2\text{ }\mu\text{m}$. The distance across the motor from support post to support post is $590\text{ }\mu\text{m}$. Sketch courtesy of Knowles Electronics.

Figure 4.15

When each microphone element can be made cheaply, it is possible to construct microphone arrays with large numbers of microphones. The company Squarehead Technology in Oslo (www.sqhead.com) constructed an array with 300 microphones in a circular plate with 2 m diameter, see Fig. 4.17 [26]. A recently developed type of MEMS microphone called "Vesper"[®], uses a piezoelectric rather than capacitive sensor principle, and that leads to a higher signal-to-noise ratio. The illustration in Fig. 4.18 is from Ref. *.

Among the biggest advantages for MEMS microphones, compared to traditional electret microphones, is that the variation from element to element gets reduced significantly. This gives much better possibilities to construct arrays with many microphones, because the amplitude and phase variations from element to element limits the performance of arrays.

Finally, it is possible to construct microphones with a larger bandwidth than traditional audio-range microphones. Then a microphone could also handle ultrasonic signals. The company Elliptic Labs in Oslo (www.ellipticlabs.com) has developed a technology where the loudspeaker of a smartphone emits ultrasound pulses, and the microphone analyses the response and detects movements in front of the smartphone (a hand in front of the phone gives a reflection).

*http://www.eetimes.com/document.asp?doc_id=1324827

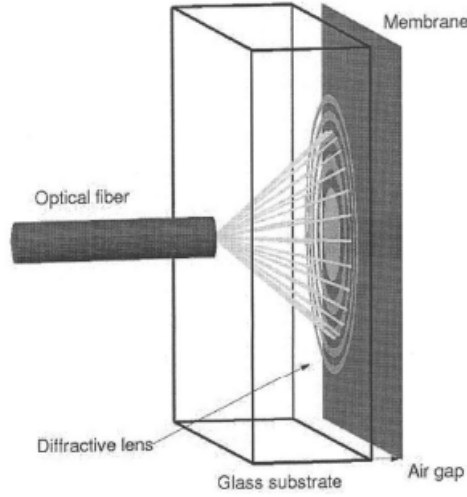


Fig. 1. Simplified geometry of the diffractive microphone. For compact and/or low-cost applications, the optical fiber can be replaced by an LED source and photodiode detector attached to the substrate, sacrificing some of the immunity toward electromagnetic noise.

Figure 4.16

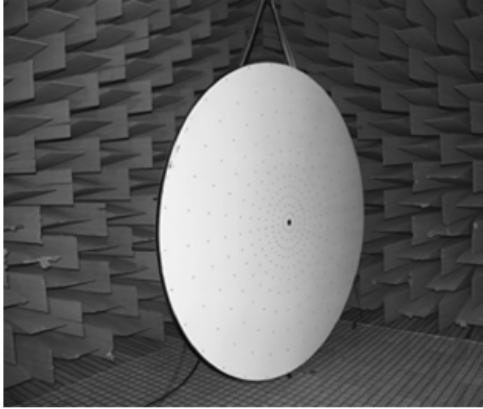


Fig. 10. Array in anechoic chamber.

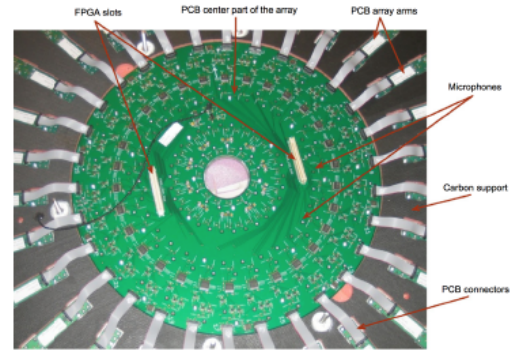


Fig. 6. Inside of the array. Printed circuit board with microphones and the FPGA connectors.

Figure 4.17: Large array of microphones, from Squarehead AS.

4.4 Appendix

4.4.1 The directivity factor for a first-order microphone

In order to derive the directivity factor for *any* first-order microphone, we can choose to scale the directivity function such that the response amplitude is always 1 for the angle $0 =$ the main direction. Then we have a single parameter, b , and the directivity function is

$$D(\theta) = 1 - b + b \cos \theta, \quad b \in [0, 1]$$

Then the directivity factor becomes

$$\text{DF}_{\text{first-order}} = \frac{2}{\int_0^\pi (1 - b + b \cos \theta)^2 \sin \theta d\theta}$$

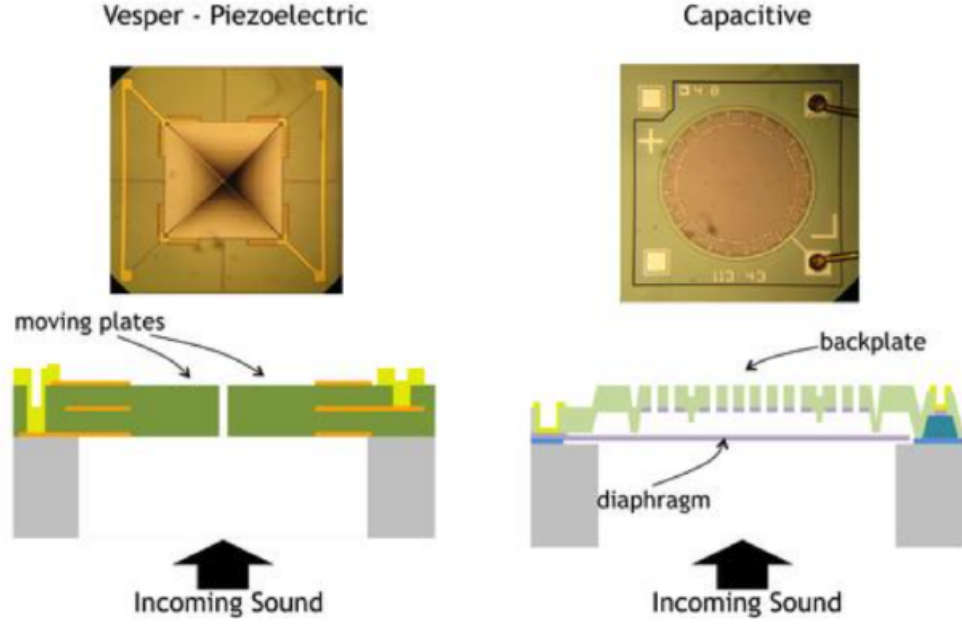


Figure 4.18: Vesper's piezoelectric MEMS microphone (left) gains a 5dB advantage over everybody else's capacitive MEMS microphones (right), giving it a signal-tonoise ratio of 70 – 80 dB, the highest in the industry. (Image: Vesper)

$$\begin{aligned}
 &= \frac{2}{\int_0^\pi [(1-b)^2 \sin \theta + b(1-b) \cdot 2 \sin \theta \cos \theta + b^2 \sin \theta \cos^2 \theta] d\theta} \\
 &= \frac{2}{(1-b)^2 [-\cos \theta]_0^\pi + b(1-b) \left[-\frac{1}{2} \cos 2\theta\right]_0^\pi + b^2 \left[-\frac{1}{3} \cos^3 \theta\right]_0^\pi} \\
 &= \frac{2}{2(1-b)^2 + \frac{2b^2}{3}} = \frac{3}{3(1-b)^2 + b^2} = \frac{3}{4b^2 - 6b + 3}
 \end{aligned}$$

To find which b -value gives the highest DF-value, we can look for the minimum of the denominator. So, we find the derivative of the denominator, with respect to b :

$$f = \frac{d}{db}(4b^2 - 6b + 3) = 8b - 6$$

and set this derivative to zero

$$f = 0 \Rightarrow 8b - 6 = 0 \Rightarrow b = \frac{3}{4}$$

Inserting this b -value in the expression for DF, we get

$$\max(\text{DF}_{\text{first-order}}) = 4 \Rightarrow \max(\text{DI}_{\text{first-order}}) = 10 \lg 4 = 6.0 \text{ dB}$$

Hearing

5.1 Hearing - The basic structure

A very simplified view of the basic structure of the outer, middle, and inner ear (cochlea) is shown in Fig. 5.1. We can follow a sound signal inwards.

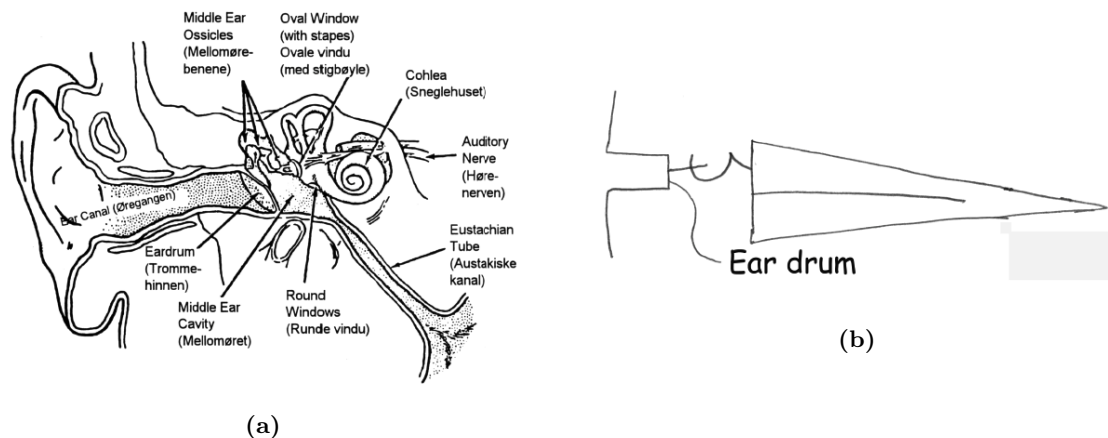


Figure 5.1: (a) Illustration of the outer and middle ear, from [27]. (b) A simplified drawing of the outer ear, middle ear, and cochlea, unwound.

1. *The sound pressure at the eardrum, caused by an incoming sound wave, gives vibrations of the eardrum*

A net force excites the eardrum, $F_{\text{net}} = p_{\text{eardrum}} \cdot S_{\text{eardrum}}$ since the static air pressure is the same on the inside and the outside. The eardrum is like a tensioned membrane (which has a first damped resonance around 1.5 kHz), so the net force will cause vibrations of the eardrum, see Fig. 5.2. Compare this situation with the different types of microphones in the chapter on microphones. The eardrum is thus like a microphone with one side of the membrane exposed to the sound field.

2. *The eardrum vibrations are transmitted to the cochlea*

The vibrations of the eardrum are transmitted via the *ossicles* ("malleus, incus, and stapes") to the *oval window*. The ossicles act as a transformer: a weak force but a large excursion at the eardrum is transformed to a larger force but smaller excursion at the oval window. When the oval window is vibrating, a pressure wave is created in the liquid-filled cochlea. See Fig. 5.3 for simplified illustrations.

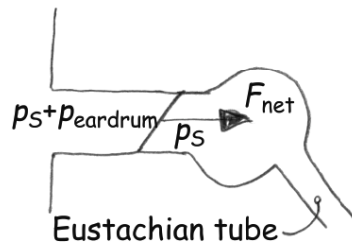


Figure 5.2: A simplified drawing of the eardrum, with the sound pressures on the outer and inner sides.

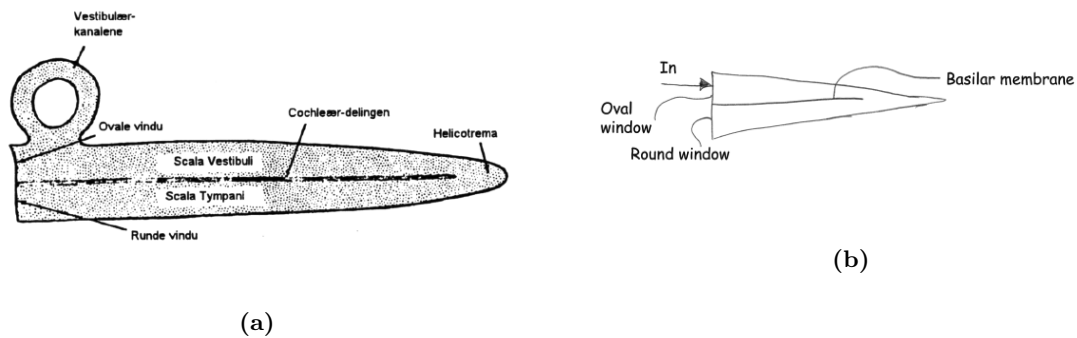


Figure 5.3: (a) Illustration of the cochlea with two of the three liquid-filled ducts illustrated. (b) Simplified drawing.

3. The basilar membrane $\approx$ a soft floor in the cochlea

The vibrations of the oval window cause a pressure wave into the liquid inside the cochlea. The liquid is kept inside a duct which surrounds the *basilar membrane* (and the two connected ducts have got different names: "scala vestibuli" and "scala tympani"). Whenever a pressure wave is excited inside the cochlea, each point on the basilar membrane will be exposed to slightly different pressure values (because of the different travel times to the upper and lower side), see Fig. 5.4, and this pressure difference will deform the basilar membrane. This deformation, in turn, leads to that small hairs, that are connected to the so-called inner haircells, are bent. The bending of these hairs cause a nerve signal to be emitted from the hair cells. This transduction occurs in the so-called "Organ of Corti".

There are approx. 3500 inner haircells placed along the basilar membrane, and this lengthwise distribution of haircells corresponds to a frequency analysis: haircells at different locations along the basilar membrane are sensitive to different frequencies. This is the so-called *tonotopic organization*.

- The basilar membrane performs a mechanical filtering, so that different haircells get a large excursion for different frequencies. That is why we can view the hearing as a frequency analyzer with a number of overlapping bandpass filters.
- There are also so-called outer hair cells; several rows of hair cells that are parallel to the inner haircells. These outer haircells act as "motors" - they actually make the membrane vibrate even more and thus, they amplify the vibrations so that the inner hair cells can pick up weak sounds.

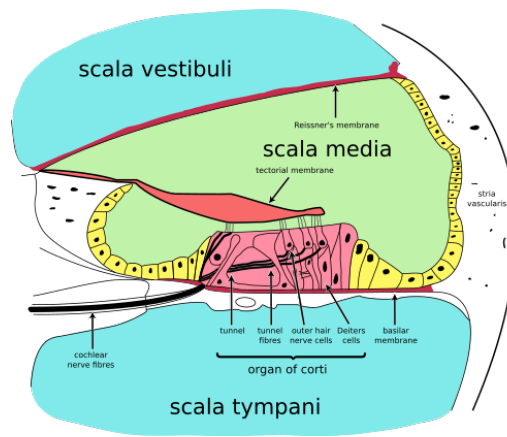


Figure 5.4: A cross-section of the cochlea, illustrating the three different ducts, and the organ of Corti, and basilar membrane between two of them, from Wikimedia Commons.

5.2 Masking

For a sinusoidal input signal (sound wave), a sinusoidal vibration results on the basilar membrane. The amplitude for these vibrations will schematically look like in Fig. 5.5, where x is the position along the basilar membrane, and there is one point, $x_{\max}$, where the membrane vibrations get a maximum.

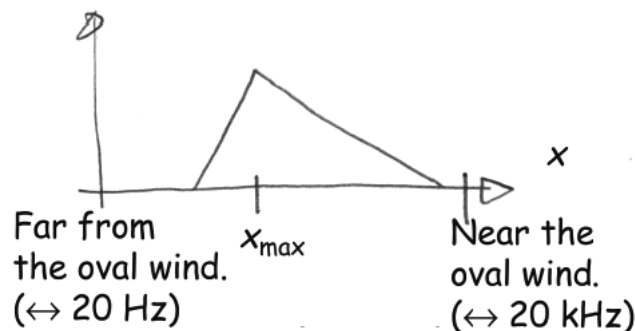


Figure 5.5: Simplified drawing of the excitation amplitude along the basilar membrane for a sinusoidal input.

The excitation is asymmetric as function of frequency, and this behavior can be imitated by assigning a different bandpass filter, to each point along the basilar membrane. One such example is illustrated schematically in Fig. 5.6.

Note! A sinusoidal signal leads to a stronger excitation for x -values above $x_{\max}$ than below $x_{\max}$, see Fig. 5.5, and these higher x -values correspond to higher frequencies. This effect can be imitated by a filter that "lets through" more of the frequencies below the max.sensitivity frequency! That is why the bandpass filter must be a kind of mirror image of the excitation pattern.

The excitation pattern shown in Fig. 5.5 has an important consequence which is called *masking*. If we hear a strong sinusoidal, which gives a primary excitation pattern like the one in Fig. 5.5, and a second sinusoidal gives another excitation pattern *which stays below the primary excitation pattern*, then we can not tell if the secondary tone is there or not!

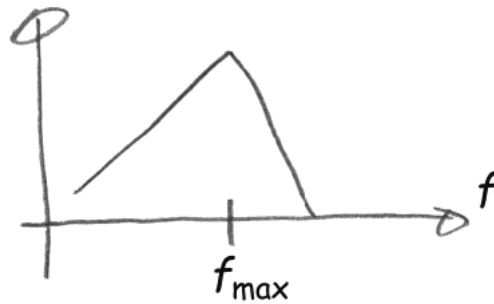


Figure 5.6: Simplified drawing of the frequency response of a single haircell, i.e., one "analysis channel" out of 3500.

The primary, stronger sound is called *the masker*.

The secondary, weaker sound is called *the masked sound*.

The masking phenomenon is the basis for all perceptually based audio compression techniques! See more below to understand how this is done.

This masking effect has been investigated a lot through perception (listening) experiments. In such experiments, a primary, louder, masking sound is played, and secondary, weaker sound is played. The listening person is then asked if the secondary sound can be heard or not, and a threshold of audibility is established. One example of results from such experiments is shown in Fig. 5.7 for sinusoids as masking and masked sounds.

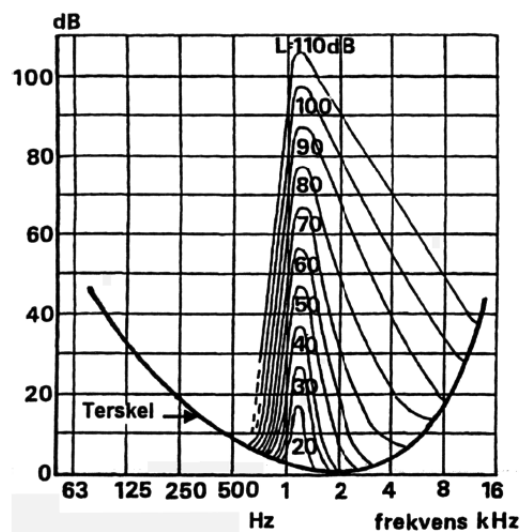


Figure 5.7: Results of masking experiments: the curves show the threshold of hearing, when a sinusoidal tone of 1.2 kHz is present. (From [27]).

As mentioned in the box above, the masking phenomenon is the basis for all perceptually based audio compression techniques. Parts of a sound signal that can not be heard can safely be discarded. But, how can one discard parts of a signal? By letting the so-called *quantization noise* be as high as possible, without being audible. The quantization noise will then "ruin" those parts of the sound signal that are not audible. The quantization noise is given a desired level simply by choosing the number of bits resolution in each frequency

band. The number of bits resolution can be made different for different frequency bands either by dividing the signal into a number of frequency bands - subbands - or by using a frequency transform of a block of the time signal.

5.3 The auditory filter bandwidth and the frequency axis of the hearing

A filter that mimics the hearing displays asymmetry, as illustrated qualitatively in Fig. 5.6, and a clear level dependence, as one can understand from Fig. 5.7. A level dependent frequency response means a non-linear filter, which is more complex to work with ordinary LTI-filters. A set of such filters is illustrated in Fig. 5.8.

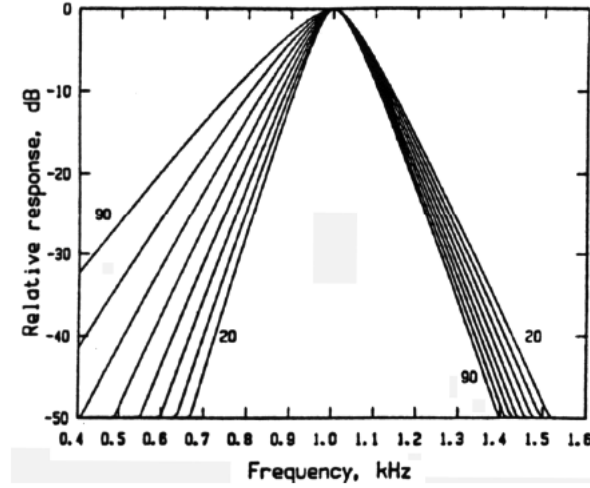


Figure 5.8: Auditory filter shapes, derived from perception experiment results as in Fig. 5.7 (From [27]).

The inner hair cells, the "sensors" in our hearing, are basically linearly distributed along the basilar membrane, and therefore a filter model of our hearing should preferably have filters distributed along a "warped" frequency axis that follows the maximum-sensitivity frequency distribution of our basilar membrane. A frequency scale that follows our hearing is the so-called Bark-scale that mimics the hearing's function (the Bark-value follows the x -value along the basilar membrane). The Bark-scale is very close to a logarithmic frequency axis above approx. 700 Hz, and close to a linear frequency axis below 700 Hz. The distance between, e.g., 7 Bark and 8 Bark, is the width of a so-called *critical band*, which is a term for the bandpass filters of the hearing [28]. The bandwidths of these critical bands are plotted in Fig. 5.9, together with 1/4-octave bands and the so-called ERB filters, which will be introduced below.

We see that the 1/4 octave bands fit quite well to the Bark bands. Later research has shown that the hearing's filters are even a bit more narrow than the Bark bands, and that at low frequencies, the hearing's filters are actually more like the logarithmic distribution than the old Bark bands. The newer filters are often called *ERB (Equivalent Rectangular Bandwidth) filters* [29]. Apparently, 1/6-octave band filters fit very well to the ERBs above 1000 Hz. The diagram in Fig. 5.10, from [29], illustrates some measured data.

The Bark value (which was determined through perception experiments) can be approximated with a mathematical function [28]:

$$B \approx 13 \arctan(0.76 f_{\text{kHz}}) + 3.5 \arctan\left(\frac{f_{\text{kHz}}}{7.5}\right)^2 \quad (5.3.1)$$

where B is the Bark value and f_{kHz} is the frequency in kHz. It can be noted that there are other mathematical functions which approximate the *empirically established* Bark scale. The widths of the Bark bands (which are plotted in Fig. 5.9) can be calculated numerically from Eq. (5.3.1). The value of the ERB as function of

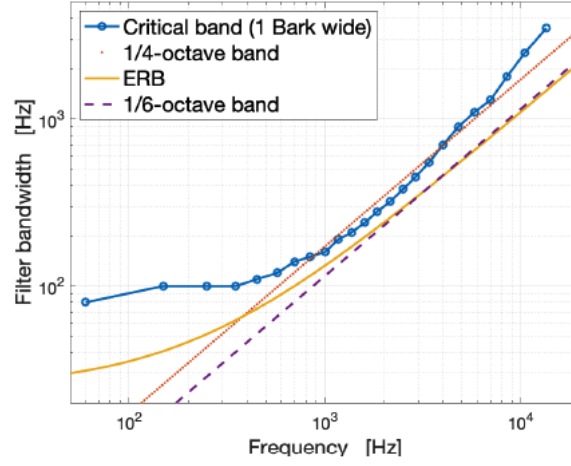


Figure 5.9: Bandwidths of critical bands, ERB-bands, and two different fractional-octave-band filters.

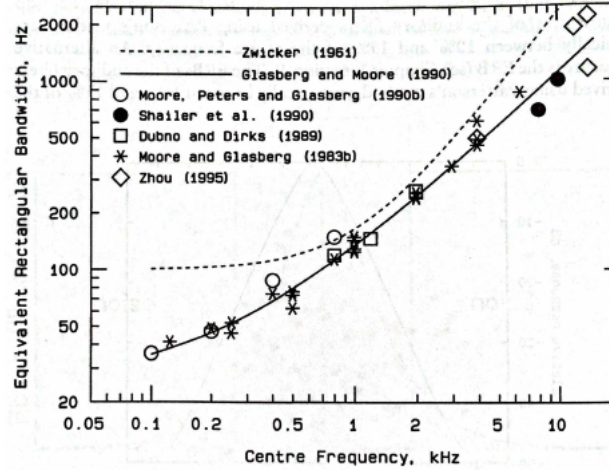


Figure 5.10: The dashed curve shows the traditional value of the critical band width as a function of frequency (Zwicker, 1961). The solid curve was combined from the results of several experiments. Diagram from [29].

frequency can be represented by the relation

$$\text{ERB} = 24.7 (4.37 f_{\text{kHz}} + 1) \quad (5.3.2)$$

A scale closely related to the Bark-scale is the so-called mel-scale, which has been established through perception experiments where listeners have judged the perceived distance in pitch between pairs of tones. So, the mel is the unit for a quantification of the sensation of *pitch*. The mel-value has been defined to be 1000 for a sinusoidal tone of 1000 Hz, so the mel-value is close to 140 times higher than the Bark value. One, of several, mathematical functions that approximate the empirically found mel-scale is

$$m = 2595 \cdot \lg \left(1 + \frac{f_{\text{kHz}}}{0.7} \right) \quad (5.3.3)$$

Fig. 5.11 shows a comparison of the two hearing-related frequency scales: the Bark-scale and the mel-scale. The mel-scale has become very popular to use for converting sounds into a compressed hearing-related scale, which can be effective for machine-learning purposes. The conversion from a linear-frequency-axis spectrum,

which results from an ordinary FFT of a time signal, to this mel-scale is done by constructing a small set of band-pass filters with simple triangular shapes in the linear-frequency-axis domain. The spectrum that results from an FFT can then be multiplied with such band-pass filters and give a total energy value, which represents one mel-value. The set of amplitudes that results from such a set of bandpass filters applies to an FFT-based spectrum is often called a *mel-spectrum*.

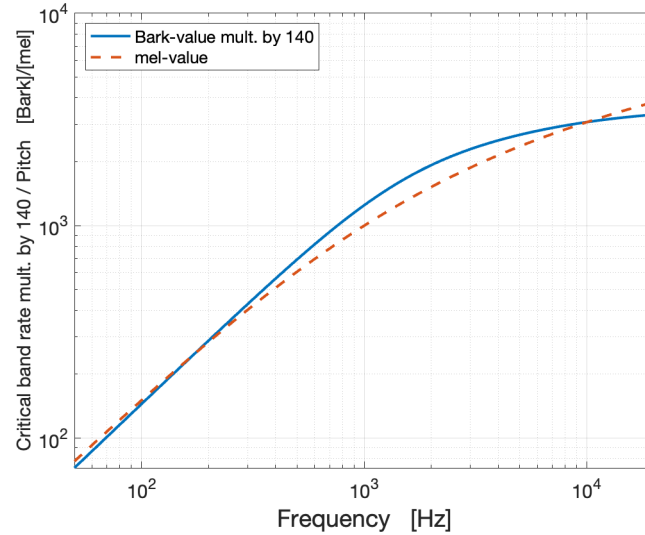


Figure 5.11: Plots of 140 times the critical band rate value, $140 \cdot B$, where B is calculated according to Eq. (5.3.1). Also plotted is the mel-value, according to Eq. (5.3.3).

The two hearing-related frequency scales can be better understood from the diagram in Fig. 5.12, which is from [30]. The diagram illustrates several parallel scales: the position along the basilar membrane (0-32 mm); a difference limen (DL) scale, where 1 unit indicates a just audible difference; the so-called mel-scale; the Bark scale as shown in the diagrams above; and finally the frequency scale. It can be noted that according to this reference, the mel-value is simply 100 times the Bark-value.

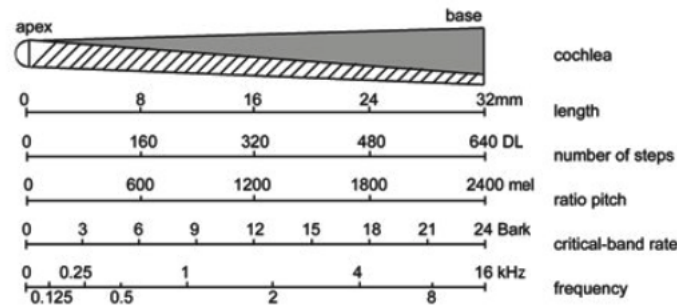


Figure 5.12: A number of linear scales that are related to the position along the basilar membrane, and the (almost) logarithmic frequency scale, from [30].

The length along the basilar membrane is relevant, because the ≈ 3500 inner hair cells are linearly distributed along the ≈ 32 mm length of the basilar membranw. The mel-scale is a scale of pitch with the unit of mel. Apparantly the mel value is quite similar to the number of hair cells. Comparing the mel scale and the DL scale, we see that approx. 4 mel is the change in pitch that is just possible to detect.

5.4 The auditory filter shape

The sections above looked at the bandwidths of the auditory filters, as well as their distribution along the frequency axis. We need more information, however: the exact shape of the auditory filter response. The first diagram in the section "The auditory filter bandwidth" had one diagram which showed some different filter responses, at different levels. So, the hearing must be non-linear (because the filter shape changes with input signal amplitude), but moderately so (because the filter shapes are quite similar to each other).

A number of different filter shapes have been proposed in the literature, and an overview is given in [31].

A set of auditory filter as described above can be very attractive for processing a sound signal in the same way as the hearing does. Any model (neural network, speech recognition, etc) might be more efficient if it is fed with a signal which is pre-processed in a similar way as the hearing system does.

5.5 Directional hearing - the three mechanisms

With two ears we can localize which direction a sound wave is coming from, [32]. Three mechanisms are used by the brain to figure out where a source could be:

- (1) Time differences between the signals in the two ears; the so-called interaural time difference, ITD;
- (2) Level differences between the signals in the two ears, interaural level difference, ILD;
- (3) Spectral content of the signals in the two ears.

The first two of these are called "The duplex theory of sound localization", which was suggested by Lord Rayleigh in the early 1900s. All these three aspects are included in the so-called *Head-Related Transfer Functions (HRTF)* which are the transfer functions of the sound pressure at the two ears, for a given angle of incidence. Nevertheless, it's still common to treat the time and level differences separately.

5.6 Interaural time difference, ITD

For a frontal sound source, a 1° difference in azimuth is audible under favorable conditions (anechoic room), see Fig. 5.13(a). This so-called Minimum Audible Angle corresponds to a $10 \mu s$ time difference between the two ear signals, which is the smallest time difference between the two ear signals that we can perceive. To obtain an audible time difference between the ear signals, when the sound comes from the side, see Fig. 5.13(b), a 10° change in azimuth is required. The diagram in Fig. 5.14 shows the so-called "Minimum audible angle" from a study in 1958 [33]. The "azimuth" in the figure caption is the horizontal incidence direction angle.

Comments

:

- The angular resolution is much better when we are facing the sound source. It is a natural reaction to turn towards a sound to get a more accurate localization. Of course, we might also be able to see the cause of the sound when we face the sound.
- The maximum time difference between the two ears is about $700 \mu s$. This is obtained for a sound incidence straight from the side (see a formula for the calculation of the time difference further down).
- Up to about 1 - 2kHz the phase information that the hearing extracts is correct; the nerve impulses follow the (half-wave rectified) waveform. The two ear signals may then look like in Fig. 5.15(a) (before conversion to nerve impulses):
- Over 1 - 2kHz our hearing follows the envelope, i.e. the nerve impulses can't keep up with the waveform variations, see Fig. 5.15(b). Only transients in the signal are then detected.

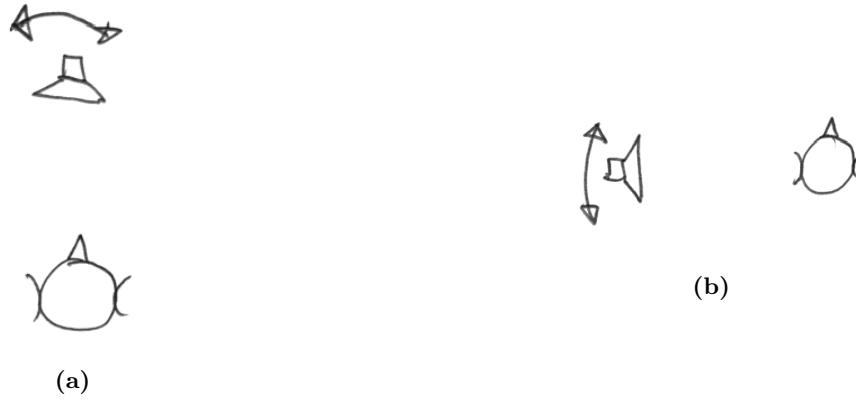


Figure 5.13: The smallest horizontal/lateral movement that a listener can detect is called the Minimum Audible Angle, the value of which differs substantially for (a) frontal incidence, and (b) incidence from a 90° angle.

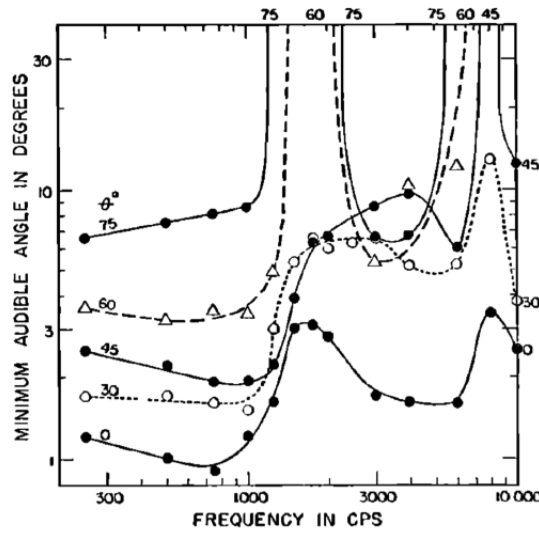


Figure 5.14: Average minimum audible angle as a function of the stimulus frequency. The parameter (θ) is the azimuth (horizontal incidence angle) of the reference tone pulse. From [33].

5.6.1 ITD for the spherical head model

We can easily calculate Δt , the ITD, for an ideal *spherical head model*, the so-called "Woodworth model" via a geometric study of the distance from the source to the two ears [34]. Please note that the extra delay caused by the propagation through the ear canal adds equally to both sides, so it can be left out for finding the interaural time difference. See Fig. 5.16.

So, the interaural time difference is, [34],

$$\Delta t = \frac{\Delta l}{c} = \frac{l_1 + l_2}{c}$$

$$l_1 = a \cdot \varphi, \quad l_2 = a \cdot \sin \varphi \quad \Rightarrow \quad \Delta t = \frac{a}{c}(\varphi + \sin \varphi) \quad (5.6.4)$$

where a = the radius of a sphere which is equivalent to the head, typically chosen to be 9 cm* This expression

* A human's head isn't really spherical, but the sphere model still fits very well with measured ITD. There have been studies to find the best "equivalent sphere size" from measurements of a person's head measures, [35].

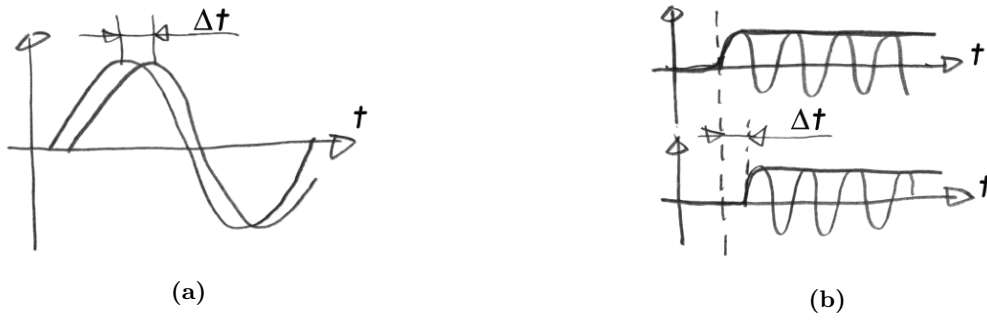


Figure 5.15: (a) At VLF (≈ 100 Hz) a typical Δt between the two ear signals results in a very small phase difference which is hard to detect. (b) At high frequencies (above 1-2 kHz), the nerve impulses can not follow the detailed waveform but only the envelope of the signal.

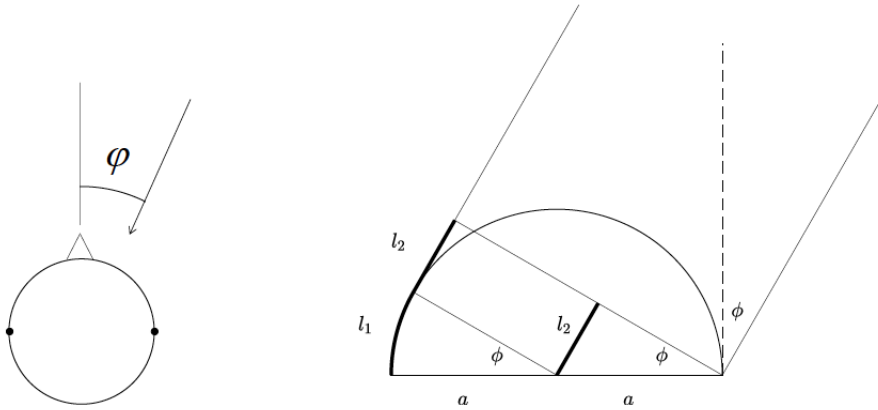


Figure 5.16: A spherical head model, illustrating that the sound wave has a longer path, of length $l_1 + l_2$ to the shadow side of the ear compared to the "sunny" side of the ear.

is valid for $|\varphi| \in [0, \pi/2]$. Beyond the angle $\pi/2$, we need to modify:

$$\Delta t = \frac{a}{c}(\pi - \varphi + \sin \varphi), \text{ if } |\varphi| > \pi/2$$

We can also calculate the time difference for two microphones placed in free air, and then

$$\Delta t_{\text{free air}} = \frac{a}{c}(\sin \varphi + \sin \varphi) = \frac{2a}{c} \sin \varphi \quad (5.6.5)$$

See the diagram in Fig. 5.17 for a comparison of ITD calculated with these two expressions.

Comments

:

- Apparently, the head leads to longer propagation paths to the "shadow-side" ear (than for microphones placed in free-field), but this seems noticeable only for angles beyond 40 degrees or so.
- The hearing system has a problem after it has detected an ITD value: there are two angles that fit each ITD value, because of symmetry. There is actually a cone of incidence angles, around the axis through the two ears, for which the ITD will be the same. This cone is called the "cone of confusion" because it

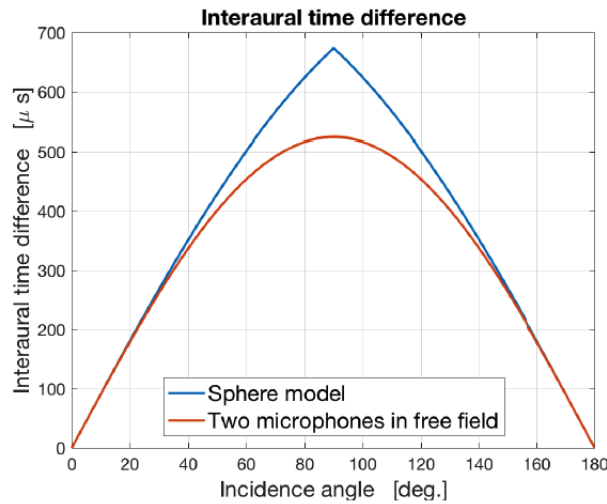


Figure 5.17: The ITD for a spherical head model, Eq. (5.6.4), and for two microphones in free air, Eq. (5.6.5).

is very difficult for the hearing to discriminate between all the possible incidence angles along this cone. The hearing will have to use extra information: frequency-dependent level differences might differ because of the not completely symmetrical head shape.

- It turns out that ear positions around 100 degrees, rather than 90 degrees, fit a bit better with reality. The expressions for estimating the ITD become somewhat more complicated then [32].
- At low frequencies (LF), it turns out that the maximum ITD becomes higher than this geometrically established value. The reason is that sound waves diffract along many different paths towards the "shadow-side" ear, and quite surprisingly, the LF asymptotic value has a very simple expression,

$$\Delta t_{LF} = \frac{3a \sin \varphi}{c}$$

5.6.2 Aliasing

Our hearing can translate a detected ITD to an incidence angle (with the complication caused by the "cone of confusion", see above). For stationary sinusoids, there is, however, the ambiguity that the brain (the hearing system) can not tell if the right ear signal is 0.1 periods behind the left ear signal, as in Fig. 5.18(a), or if the right ear signal is 0.9 periods ahead of the left ear signal, as in Fig. 5.18(b).

However, for the frequency 200 Hz, which has a period time of 5 ms, the case in Fig. 5.18(b) implies that the right ear signal is 4.5 ms ahead of the left ear. Such a large ITD can not occur (Fig. 5.16 shows that the ITD must be below approximately 700 μs), which the brain has learnt through the constant analysis of ear signals. Thus, the brain is absolutely certain that for 200 Hz, the case in Fig. 5.18(a) is the correct one, and the incidence angle is around 60 degrees to the left.

The first ambiguity happens for the frequency 700 Hz, with a period time of ≈ 1.4 ms: both sound incidence from the left and from the right will give ear signals which are exactly opposite in phase, see Fig. 5.19. Both the blue alternative (the left ear signal is 0.7 ms ahead of the right ear signal) and the red alternative (the right ear is 0.7 ms ahead of the left ear signal) could be possible, so the hearing can not use this time difference to detect if the sound wave comes from 90 degrees to the left or to the right. Furthermore, the level difference is quite small at 700 Hz, so the hearing doesn't get much help from trying to analyze the level difference.

We look at a last example, illustrated in Fig. 5.20. For the frequency 1.4 kHz, the hearing would have to decide if the left ear signal is 0.18 ms ahead of the right ear (blue arrow, which corresponds to sound incidence 20 degrees to the left), or if the right ear is 0.54 ms ahead of the left signal (red arrow, which corresponds to

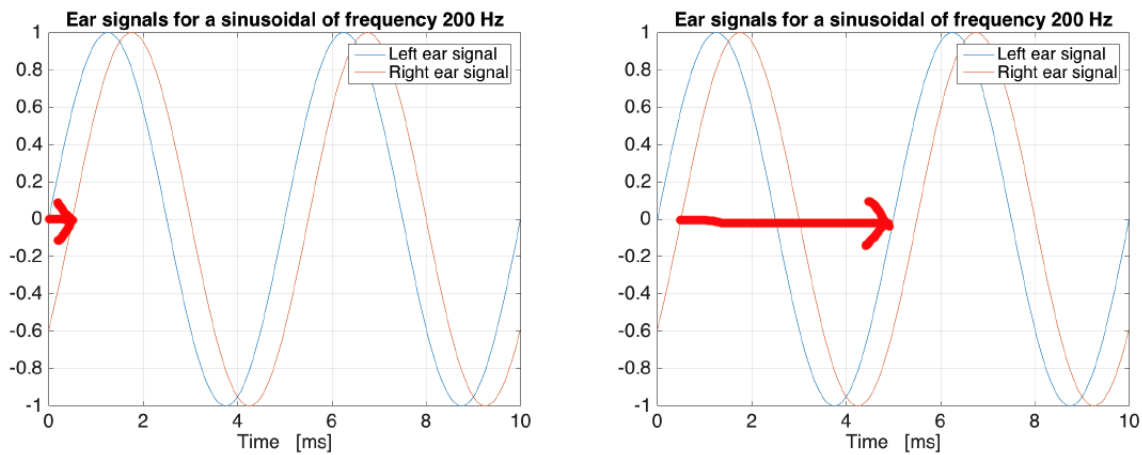


Figure 5.18: Illustration of the challenge for the hearing to decide which ear receives the signal first, when the signal is a stationary sinusoidal.

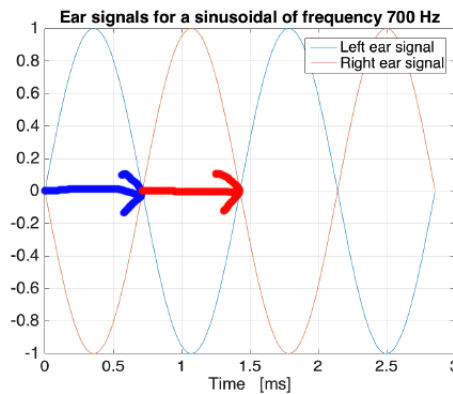


Figure 5.19: When a sinusoidal signal hits the ears from a 90-degree angle, for some frequencies, like the illustrated 700 Hz, it is impossible for the hearing to say which one comes first.

65 degrees to the right). Both of these time delays can happen in reality and therefore the hearing can not tell the difference, based on this time difference alone.

Does this mean that our hearing can not say for sure where a sound comes from, if the frequency is above 700 Hz ? Our hearing has several techniques that help:

- The brain can solve the ambiguity problem by analyzing any level differences in addition, see the next section.
- The brain can decide to rotate the head and this can resolve the ambiguities, because the correct solution would stay in place whereas the ambiguous alternative would move.
- Any sound signal which contains more than a single frequency will be less ambiguous, since two frequencies will have identical alternatives for one of the two possible solutions, but different angle alternative for the other possible solution. Then the hearing can draw the conclusion that the angle alternative which is shared between two frequencies will be the correct one.

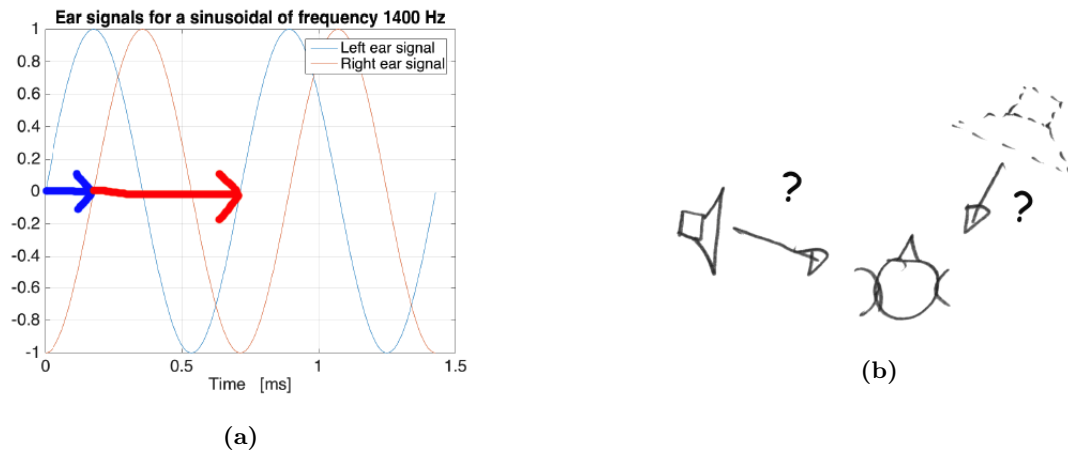


Figure 5.20: (a) When the two ears are hit by a sinusoidal of 1400 Hz, both the illustrated possible incidence angles give plausible interaural time delays (less than 0.7 ms). (b) Two possible incidence angles.

5.7 Interaural level difference, ILD

At LF the sound pressure is roughly the same all over the head (since the head's diameter $\ll \lambda$: The level differences between the left and right ears are small below approx. 500-700 Hz. At high frequencies the screening / shielding effect of the head is strong and may cause level differences of 10-20 dB.

Figure 5.21 shows the magnitude of two HRTFs, *Head-Related Transfer Functions*, to the two ears for a certain angle of incidence. We see that the magnitude is fairly similar at low frequencies but for higher frequencies the difference increases. Hence level differences can be used (by the brain) to determine the angle of incidence for high frequencies.

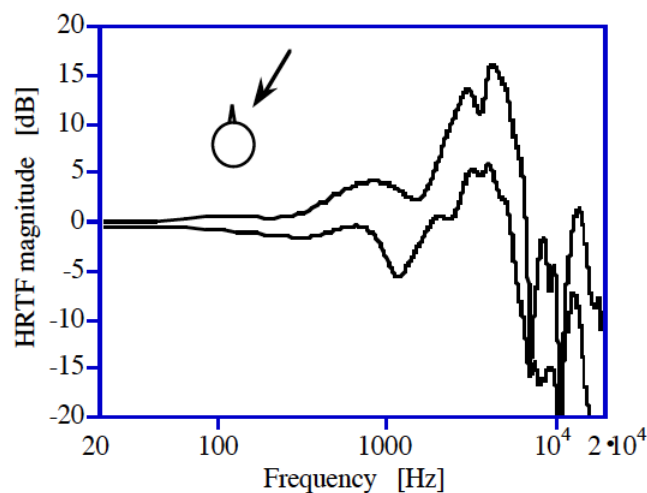


Figure 5.21: An example of a pair of Head-Related Transfer Functions, from the MIT KEMAR database.

The HRTF example in Fig. 5.21 is from the openly available measurement database for the KEMAR dummy head: <http://sound.media.mit.edu/resources/KEMAR.html>. The diagram in Fig. 5.21 illustrates that there isn't really such a thing as a constant level difference (across frequency) - the level difference varies quite much.

The rigid-sphere model can be used as a first attempt to model the level differences (as well as the time differences, see above), and the diagram in Fig. 5.22 shows the frequency responses for microphones placed on a sphere, and sound incidence from many different angles. Please note that the angle 0 degrees in the diagram corresponds to sound hitting the "sunny-side" ear from 90 degrees, that is, the angle in the diagram is relative to the ear position [36].

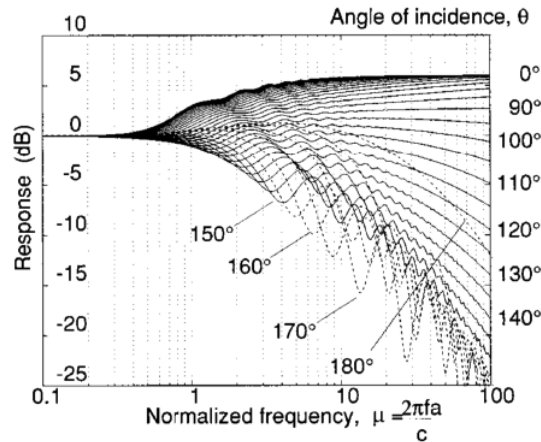


Figure 5.22: Frequency response functions for a rigid sphere, from [36].

Magnitude response for an infinitely distant source. Roughly speaking, the response is flat when the angle of incidence is around 100 degrees, exhibits a 6-dB boost at high frequencies near the front of the sphere, and - except for the bright spot at the very back - falls off with frequency around the back of the sphere. Interference effects caused by waves propagating in various directions around the sphere introduce ripples in the response that are quite prominent on the shadowed side. We can note that the frequency responses for this sphere model are much smoother than those measured for a real head. The real-head measurements do, however, include the ear canal resonances which affect the responses substantially. We can note that the frequency responses for this sphere model are much smoother than those measured for a real head. The real-head measurements do, however, include the ear canal resonances which affect the responses substantially. The normalized frequency 1 corresponds to ≈ 607 Hz for a 9 cm head radius.

5.8 Spectral content of the ear signals

When a sound source is in the so-called median plane, see Fig. 5.23, there won't be any time- or level differences between the two ear signals, and it will be difficult for the hearing to determine where the sound comes from. When our two ears receive identical signals, we have learnt (through our repeated exposure to sounds with visual feedback telling us the true direction to sound sources) that sound coming from a specific direction will amplify certain frequency ranges (caused by the shape of the head and the ears).

The frequency ranges marked above the diagram in Fig. 5.24 are called *Directional bands*. Those bands mean that if we hear a signal in the median plane, with a strong level in such a band, then we have a tendency to guess that the sound comes from a specific direction. When we detect that there are no time- or level- differences between the two ears signals, we also have the possibility to turn the head and analyze the differences that result (from [32])

5.9 HRTF

The time- and level differences, and spectral variations, that are mentioned above are described completely by so-called Head Related Transfer Functions (HRTF).

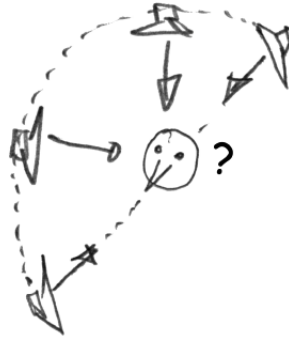


Figure 5.23: For incidence in the so-called median plane, there are practically no time or level differences for the binaural hearing to analyze.

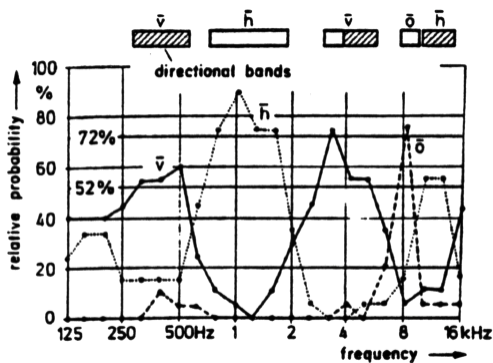


Figure 5.24: Results of listening experiments which illustrates the concept of "directional bands", from [32].

A HRTF is defined by [32]:

$$\text{HRTF}(\Phi, \theta) = \frac{p_{\text{at ear}}(\Phi, \theta)}{p_{\text{ref}}} = \frac{p_{\text{at ear}}(\Phi, \theta)}{p_{\text{without the head}}}$$

where (Φ, θ) are the angles that describe the direction of the incident sound wave, see Fig. 5.25. They are usually called the azimuth ($\Phi, \in [-180^\circ, +180^\circ]$) and elevation ($\theta, [-90^\circ, +90^\circ]$) angles. In the common spherical coordinate system, the θ angle would usually refer to the angle relative to the vertical direction, and $\theta \in [0, 180^\circ]$.

There is one HRTF for each ear and they might look like in Fig. 5.26, as HRTFs or HIRs (Head Impulse Responses) (example for one person from [37]):

Comments to HRTF/HIR (Head Impulse Response)

- We can observe that the level differences increase with frequency. For LF, the level is around 0 dB, which means that the sound is diffracted around the head, leading to the same amplitude everywhere around the head. In addition, the level is almost the same as if the head hadn't been there.
- From approx. 1-2 kHz the level starts falling on the shadow side while it is magnified quite significantly on the "sunny side" of the head. There are also clear interference phenomena (peaks and valleys) and these are caused by the fact that the sound can travel around the head and interfere with the direct sound. Furthermore, reflections off the shoulders can also cause interference effects.
- In the time domain, the arrival time differences can clearly be seen. This is caused by the different path lengths to the two ears for the direct sound as discussed earlier (ITD).

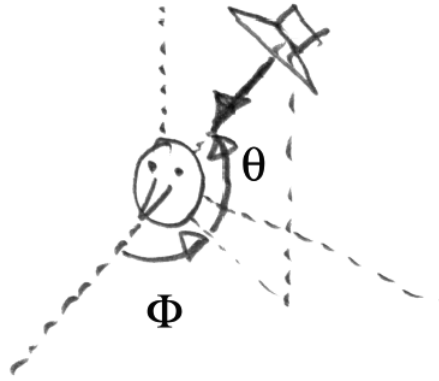


Figure 5.25: Illustration of the spherical coordinate system used for Head-Related Transfer Functions.

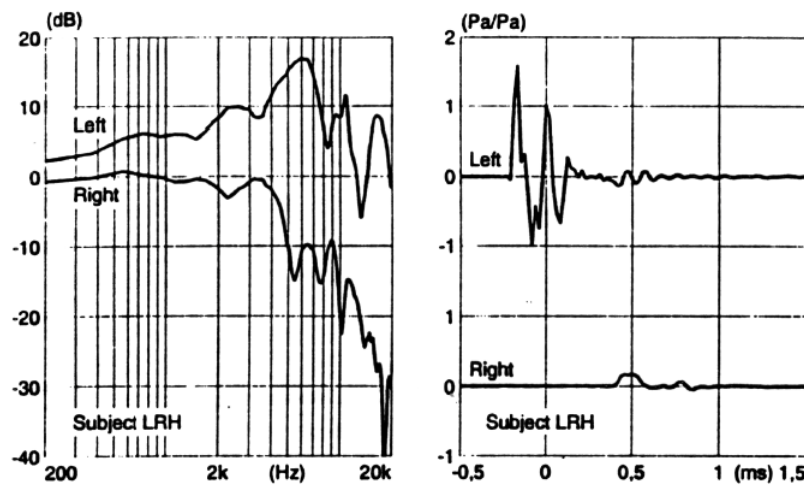


Figure 5.26: Example of HRTFs (left) and HIRs (right), from [37].

- HRTFs (often in the form of HIRs) are *the basis for so-called binaural synthesis*.
- There are large individual differences between such HRTFs. The size of the head, and to a slightly smaller extent the shape of the head, will affect the HRTFs. For higher frequencies, the detailed shape of the pinna (the visible part of the ear) will determine the details of the HRTFs.
- There will be a data acquisition/storage issue: even if we could use a single set of HRTFs (measured on a single person, or on an artificial head), we would have to measure and store hundreds of HRTFs for a complete set of angles of incidence (θ, Φ) [38].
- Each time-domain HRTF (HIR) can usually fit within approx. 2 ms (see Fig. 5.26). With a sampling frequency of 48 kHz, approx. 100 FIR filter coefficients will be needed to describe each HIR.
- Quite much research has been aimed at handling HRTFs/HIRs more efficiently. Examples:
 - By using a spherical head model, the HRTFs/HIRs can be calculated quite quickly for an arbitrary angle of incidence. An individual fitting (i.e., by choosing a head radius a which fits an individual person) will make such mathematical HRTFs more similar to a person's own HRTFs [35].

- It is possible to describe all the hundreds of HRTFs as weighted sums of a small (5-10) number of basis functions, which leads to an efficient storage and use of such a large library/set of HRTFs. Principal components analysis (PCA) is one technique for finding such basis functions.
- By carefully choosing the microphone position in the ear for measuring the sound pressure $p_{\text{at ear}}$, it is possible to minimize the individual differences substantially, as shown in Fig. 5.27 [37].

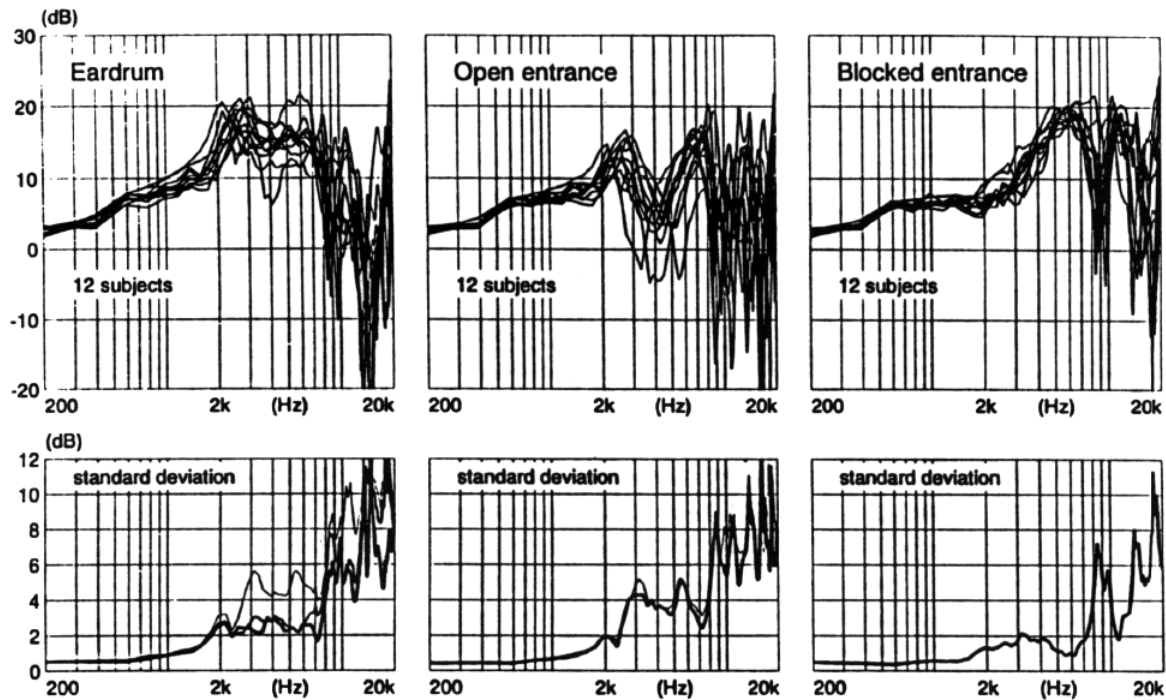


Figure 5.27: From [37].

- The idea to measure the HRTFs right at the entrance of a blocked ear canal is that the transmission through the ear canal is a plane wave transmission which *does not depend on the angle of incidence of the sound wave*.
- It has been shown that binaural recordings, or binaural synthesis give the clearly best localization effect if a person's own HRTFs are used. If another person's HRTFs are used, larger error will result in localization experiments. Using HRTFs from an artificial head results in even worse localization [37].
- When HRTFs are used in binaural synthesis, there is a factor which will be discussed again in the section on headphone presentation of binaural content: whether headtracking is used or not. In real-life listening, we turn our heads deliberately as well as involuntarily, and we have learnt how the binaural sound changes when we do. If fixed HRTFs are used, our hearing will receive wrong signals when we are turning our heads during the listening. By tracking the head movements, and by dynamically update the HRTFs, this defect can be corrected! With the development of head-mounted displays, head-tracking has become a necessary technology for updating the displays, and the head tracking can also be employed for the sound.

5.10 Multiple sources

When two stereo loudspeakers emit the same signal, apart from a small time- or level difference, we experience a *virtual source*, see Fig. 5.28(a). This is the basis for usual two-channel stereo, and also for some multi-channel

systems that use two or three loudspeakers to simulate virtual sources in-between each loudspeaker pair or triplet. It is possible to create time and/or level differences at the recording or mixing stage, for creating a stereo signal. The amplitude panning technique is the most common (and it is used in all mixing consoles), and it leads to an amplitude difference only between the two stereo channels (for each individual sound source).

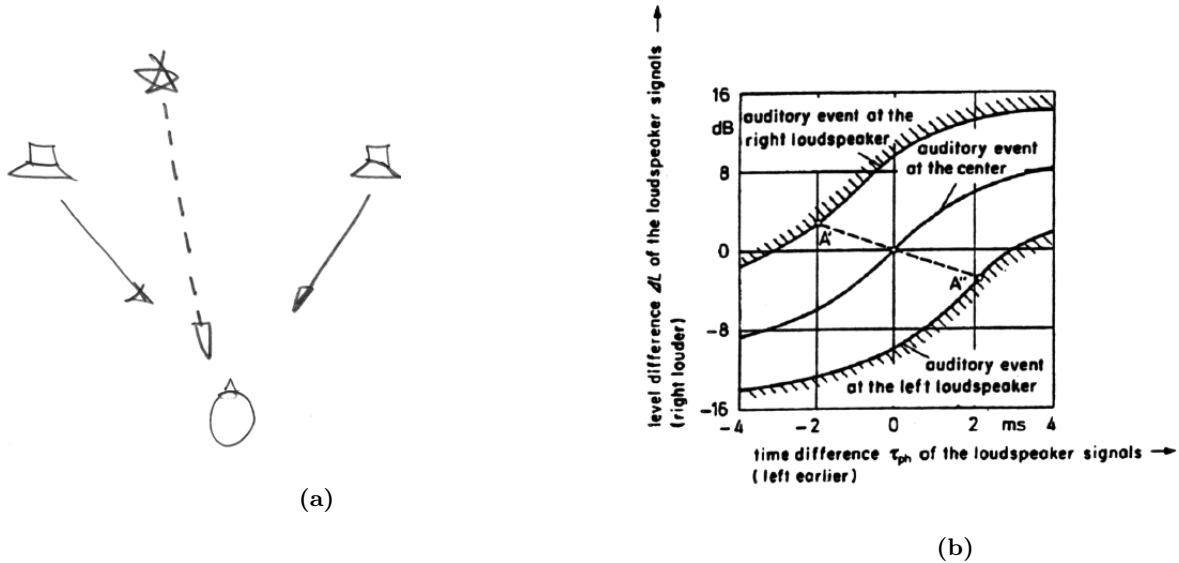


Figure 5.28: (a) When two symmetrically placed loudspeakers emit the same signals, apart from a time and/or intensity difference, a phantom source somewhere between the speakers will be perceived. (b) Various combinations of time and level differences can lead to the same perceived location (from [27]).

Microphone arrays

6.1 General

A microphone array is a construction with two or more microphone elements positioned in a systematic way, and with the output signals of the various microphone elements combined to give a desired effect. Two common applications are

1. Create a directional "virtual" microphone.
2. Find the direction to a sound source.

Application 2 can use a directional microphone, but can also use other techniques. We start by looking at the simplest possible array, with two microphone elements.

6.2 Directional microphone array with two elements

In the chapter on microphones, we encountered directional microphones - cardioid microphones and figure-of-8 microphones - which can be constructed with a single microphone membrane, and a microphone enclosure that acts as an acoustic filter, which in turn can lead to various directivity patterns. Such microphones have a fixed directivity, but in this chapter, we will see that it is possible to create directional microphones where the directivity can be controlled through processing of two or more microphone signals.

6.2.1 First-order microphones

From the microphone chapter, we recall the term first-order microphone, which is a rotationally symmetrical microphone with a directivity function that can be written*

$$D(\theta) = a + b \cdot \cos \theta \quad (6.2.1)$$

It is very common to normalize this directivity function so that it has a value 1 in the main direction, which is usually $\theta = 0$, and this normalization was also introduced in the microphone chapter, giving

$$D(\theta) = (1 - b) + b \cdot \cos \theta, \quad b \in [0, 1] \quad (6.2.2)$$

Thus the first-order microphone can be a function of a single parameter, b . Now we have a single parameter b and can study how the directivity changes. First some examples of first-order directivity plots (using the Matlab code by <http://2inarow.org/wordpress/>) in Fig. 6.1.

*To jump a little bit ahead, a higher-order microphone also has directivity terms like $\cos^2 \theta, \cos^3 \theta$ etc. Also, we will see first-order microphones that are functions of the two angles (θ, φ) in a spherical coordinate system, see section 6.3.2.

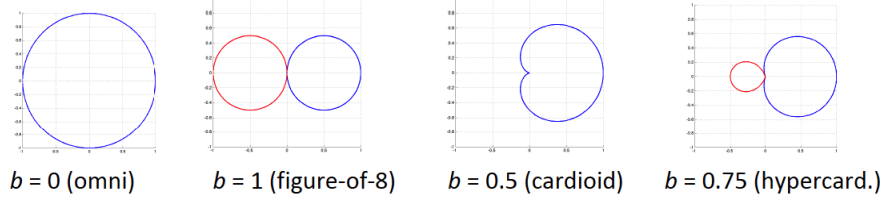


Figure 6.1: Four different first-order microphone directivities, according to Eq. (6.2.1), with the additional normalization of Eq. (6.2.2).

In the microphone chapter, the directivity factor DF as function of b , for the first-order microphone, given by Eqs. (6.2.1) - (6.2.2), was shown to be

$$DF = \frac{3}{4b^2 - 6b + 3} \quad (6.2.3)$$

We also expressed this in the form of the directivity index, DI, in dB,

$$DI = 10 \cdot \lg DF \quad (6.2.4)$$

which is plotted in Fig. 6.2, as function of the parameter b . We acknowledged already in the chapter on micro-

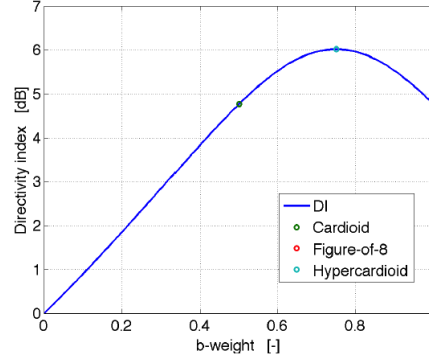


Figure 6.2: The directivity index for a first-order microphone, according to Eqs. (6.2.3) and (6.2.4).

phones that the highest directivity index that can be achieved with a first-order microphone is $DF=4$, that is, $DI \approx 6.0$ dB, and that is exactly how the hypercardioid microphone is defined. We can also notice that there will be a (rotationally symmetrical) zero-direction which will vary between 180 degrees (for the cardioid, $b = 0.5$) and 90 degrees (for the figure-of-8, $b = 1$). A zero can not be achieved for angles smaller than 90 degrees if the maximum sensitivity should be in the 0 degree-direction.

Below, we will see that a first-order microphone can be implemented in a few different ways:

1. By summing the output signals of one omnidirectional and one figure-of-eight microphone, that are placed very close to each other, using scale factors a and b , see section 6.2.2.
2. By summing the output signals of two cardioid-type microphones pointing in opposite directions, using proper scale factors, see section 6.2.4.
3. By using two omnidirectional microphone elements at a small distance, and combine their two output signals to simulate a virtual figure-of-eight or cardioid-type microphone, and subsequently create any first-order microphone with method 1 or 2 above, see section 6.2.3.

6.2.2 Synthesizing a first-order directivity with an omni mic. and a figure-of-8 mic.

Equation (6.2.1) tells us that a first-order microphone can result if we have one omnidirectional microphone and one figure-of-eight microphone, placed (almost) in the same location, their outputs are added with weight/scale factors a and b , as illustrated in Fig. 6.3. Adding the two microphone signals such, we get the output signal of a virtual/synthesized microphone with a directivity function on the form in Eq. (6.2.1).

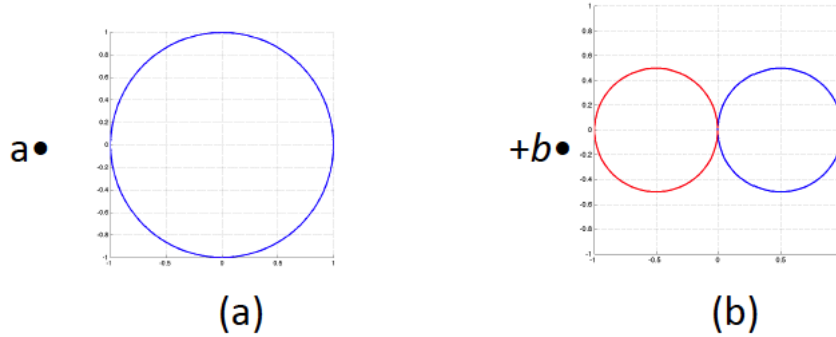


Figure 6.3: Illustration of the directivities of (a) an omnidirectional microphone, and (b) a figure-of-eight microphone. The " $a \bullet$ " and " $+b \bullet$ " indicate that the outputs from the two microphones are mixed together, that is, scaled and summed.

6.2.3 Synthesizing an omni microphone and a figure-of-8 microphone from two omni microphones

Let's assume that we have two omnidirectional microphones, the output signals of which are filtered and then summed, as illustrated in Fig. 6.4.

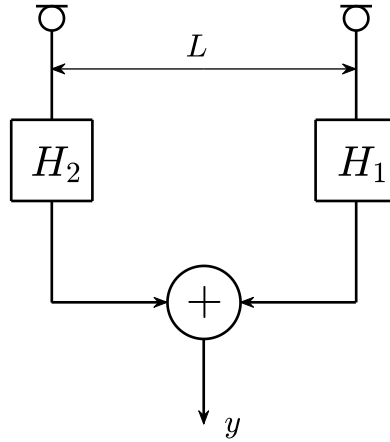


Figure 6.4: An array consisting of microphone elements at a distance of L . The microphone output signals are filtered and summed into a "virtual microphone" signal y .

We want to know how the output signal of such a two-element array depends on the incidence angle of a plane wave. Therefore we start by introducing the general description of an incident plane wave in an (x, y) -coordinate system, as illustrated in Fig. 6.5. The sound pressure for such a plane wave can be written as (with the time-harmonic factor $e^{j\omega t}$ left out).

$$p(x, y) = p_{\text{in}} e^{j(k \cos \theta)x} e^{j(k \sin \theta)y}$$

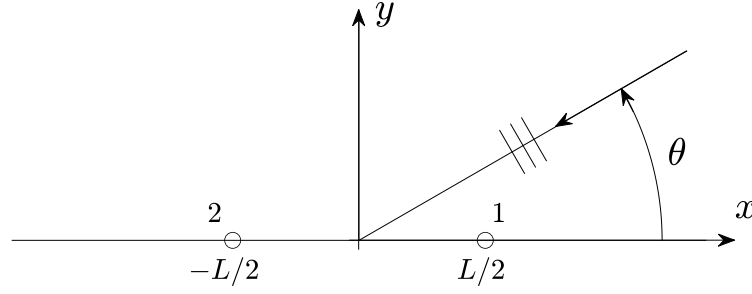


Figure 6.5: An incident plane wave with an angle θ relative to the positive x -axis. Two microphones are placed symmetrically along the x -axis in positions $(\pm L/2, 0)$.

Therefore, the sound pressure at the two microphones will be

$$\begin{cases} p_1 = p_{\text{in}} e^{jk \frac{L}{2} \cos \theta} \\ p_2 = p_{\text{in}} e^{-jk \frac{L}{2} \cos \theta} \end{cases} \quad (6.2.5)$$

Now we add the two microphone signals, after multiplying them with factors H_1 and H_2 , which could be simple scale constants, or filters. Thus we create an output signal, y :

$$y = H_1 p_1 + H_2 p_2 \quad (6.2.6)$$

and the question is how we can choose H_1 and H_2 to simulate a virtual microphone with a desired directivity. That is, we expect the output signal y to be:

$$y = p_{\text{in}} D_{\text{virtual mic.}}(\theta)$$

This is a generalized analysis of the derivation in the microphone chapter for a figure-of-8 microphone. We insert the expressions in Eqs. (6.2.5) in Eq. (6.2.6) and get

$$\begin{aligned} y &= p_{\text{in}} \left[H_1 e^{jk \frac{L}{2} \cos \theta} + H_2 e^{-jk \frac{L}{2} \cos \theta} \right] \\ \Rightarrow D_{\text{virtual mic.}}(\theta) &= H_1 e^{jk \frac{L}{2} \cos \theta} + H_2 e^{-jk \frac{L}{2} \cos \theta} \end{aligned} \quad (6.2.7)$$

Low-frequency (LF) analysis of the directivity

It is challenging to analyze the directivity from the expression in Eq. (6.2.7), but it is straightforward to study the low-frequency behaviour, by using the Taylor series expansion for the exponential function. For small x

$$e^x \approx 1 + x$$

So, the expression in Eq. (6.2.7) is, for LF,

$$\begin{aligned} D_{\text{virtual mic., LF}}(\theta) &\approx H_1 \left(1 + jk \frac{L}{2} \cos \theta \right) + H_2 \left(1 - jk \frac{L}{2} \cos \theta \right) \\ &= (H_1 + H_2) + (H_1 - H_2) jk \frac{L}{2} \cos \theta \end{aligned} \quad (6.2.8)$$

Here we realize that we can create an omnidirectional microphone, with $D_{\text{LF}} \approx 1$, if

$$\begin{cases} H_1 + H_2 = 1 \\ H_1 - H_2 = 0 \end{cases}$$

which is fulfilled if

$$H_1 = 0.5, \quad H_2 = 0.5$$

Also, a figure-of-8 microphone, with $D_{LF} \approx \cos \theta$, results if

$$\begin{cases} H_1 + H_2 = 0 \\ (H_1 - H_2)jk\frac{L}{2} = 1 \end{cases}$$

which is fulfilled if

$$H_1 = \frac{1}{jkL}, \quad H_2 = -\frac{1}{jkL}$$

By constructing those two virtual microphone types: an omnidirectional and a figure-of-8, from the same pair of (spaced omnidirectional) microphone elements, we can add the signals from those two virtual microphones and get any kind of first-order microphone, exactly like in section 6.2.2. We can, however, also analyze the directivities for higher frequencies.

High-frequency analysis of the directivity

We can analyze the directivity also for higher frequencies by computing the DF, and DI, for the expression in Eq. (6.2.7) numerically. We recall that the expression for DF uses the directivity function, $D(\theta)$, of the microphone response, so DF is:

$$DF = \frac{|D(\theta_{\text{main}})|^2}{0.5 \int_0^\pi |D(\theta)|^2 \sin \theta d\theta} \quad (6.2.9)$$

The virtual omnidirectional microphone

For the virtual omnidirectional alternative in the previous section, there is no main direction for low frequencies, but as the frequency increases, the main direction is going to be $\theta = 90$ degrees, and there,

$$D_{\text{virtual omni}}(\theta_{\text{main}}) = D_{\text{virtual omni}}(\pi/2) = 1 \quad (6.2.10)$$

A microphone array which has its main direction at 90 degrees is called a *broadside array*. Furthermore, we find an expression for the directivity function for the entire frequency range, for the virtual omni mic., if we insert $H_1 = H_2 = 0.5$ in Eq. (6.2.7):

$$D_{\text{virtual omni}}(\theta) = \cos \left(\frac{kL}{2} \cos \theta \right) \quad (6.2.11)$$

which can be inserted in Eq. (6.2.9) and give

$$DF_{\text{virtual omni}} = \frac{1}{0.5 \int_0^\pi \cos^2 \left(\frac{kL}{2} \cos \theta \right) \sin \theta d\theta} \quad (6.2.12)$$

The virtual figure-of-8 microphone

For the figure-of-8 alternative, on the other hand, the main direction for low and high frequencies is $\theta = 0$ so,

$$\begin{aligned} D_{\text{virtual fig.-of-8}}(\theta_{\text{main}}) &= D_{\text{virtual fig.-of-8}}(0) = \frac{1}{jkL} \left[e^{jkL/2} - e^{-jkL/2} \right] \\ &= \frac{2}{kL} \sin \frac{kL}{2} \end{aligned} \quad (6.2.13)$$

A linear microphone array with the main direction in the length direction is called an *endfire array*. The directivity function for the entire frequency range is found by inserting $H_1 = 1/(jkL)$ and $H_2 = -1/(jkL)$ in Eq. (6.2.7), which gives

$$D_{\text{virtual fig. of 8}}(\theta) = \frac{2}{kL} \sin \left(\frac{kL}{2} \cos \theta \right) \quad (6.2.14)$$

which can be inserted in Eq. (6.2.9) and give

$$DF_{\text{virtual fig. of 8}} = \frac{\sin^2 \frac{kL}{2}}{0.5 \int_0^\pi \sin^2 \left(\frac{kL}{2} \cos \theta \right) \sin \theta d\theta} \quad (6.2.15)$$

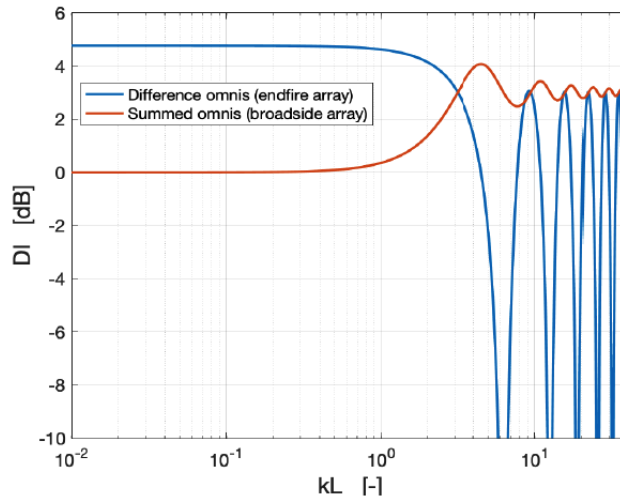


Figure 6.6: The directivity index for two different two-microphone arrays, numerically computed by Eqs. (6.2.12) and (6.2.15), and expressed as DI.

These integrals, in Eqs. (6.2.12) and (6.2.15), will have to be computed numerically and the results are presented in Fig. 6.6, which shows the directivity index, $DI = 10 \lg(DF)$, for those two alternatives, as function of frequency (expressed as kL).

For low frequencies ($kL < 0.5$), we do indeed see the +4.8 dB and 0 dB values that were observed in the microphone chapter. For higher frequencies, the difference approach (virtual figure-of-8 microphone) apparently leads to a very low DI for regularly spaced frequencies, and the reason is that the difference approach gives zero output signal in the main direction whenever the distance between the microphones is an odd number of half wavelengths.

We can see that the broadside array (virtual omni mic.) has a directivity index around 3 dB for high frequencies. Without looking into the analysis, we can mention that if we increase the number of microphones in a linear array, and use the simple summing principle of the virtual omni mic., then we get *a 3 dB increase in directivity index for each time we double the number of microphone elements*. That is, 4 elements would give $DI \approx 6$ dB at high frequencies, 8 elements would give $DI \approx 9$ dB etc.

6.2.4 Synthesizing any first-order microphone from two cardioid-type microphones

It turns out that we could also combine the output signals of two cardioid microphones, and get the same flexible possibilities as with one omnidirectional and one figure-of-8 microphone, or with two omnidirectional elements - but with one important advantage over the latter: we will not need a filter of the type $1/(j\omega)$!

A previous exam question demonstrates this:

Question: We have two hypercardioid microphones, both with the directivity function $D = 0.25 + 0.75 \cos \theta$. We want to use these two to synthesize a virtual omnidirectional microphone and a virtual figure-of-eight microphone.

- If we put the microphones back to back (so that $\theta_{\text{mic.2}} = \pi - \theta_{\text{mic.1}}$, how should the two microphone signals be mixed together to give a virtual omnidirectional microphone?
- If we put the microphones back to back (so that $\theta_{\text{mic.2}} = \pi - \theta_{\text{mic.1}}$, how should the two microphone signals be mixed together to give a virtual figure-of-8 microphone, with $D_{\text{virtual fig.8}} = \cos \theta$?

Answer:

- a) Since $\theta_{\text{mic.2}} = \pi - \theta_{\text{mic.1}}$, we could write the directivity function of hypercardioid number 2:

$$D_2 = 0.25 + 0.75 \cos(\pi - \theta) = 0.25 - 0.75 \cos \theta$$

Now, we want to mix together the two signals, that is, the output signal of the microphone e_{total} is determined by two scale factors/gain factors, H_1 and H_2 and we want to find their values:

$$\begin{aligned} y &= H_1 y_{\text{mic.1}} + H_2 y_{\text{mic.2}} \Rightarrow D_{\text{total}} = H_1 D_1(\theta) + H_2 D_2(\theta) \\ &= 0.25 (H_1 + H_2) + 0.75 (H_1 - H_2) \cos \theta \end{aligned}$$

We wish that $D_{\text{total}} = 1$ (omnidirectional), which is fulfilled if

$$H_1 = 2, \quad H_2 = 2$$

- b) Now we want $D_{\text{total}} = \cos \theta$ (figure-of-eight), which is fulfilled if

$$H_1 = \frac{2}{3}, \quad H_2 = -\frac{2}{3}$$

6.2.5 Signal-to-noise ratio

Now we will look into the signal-to-noise ratio for the two-microphone arrays in section 6.2.3 that use two omnidirectional microphones, the output signals of which are filtered and then summed.

For a noise analysis we assume that in addition to the sound pressure signals, p_1 and p_2 , the two microphones also pick up two independent random noise signals, n_1 and n_2 . These noise signals could be electronic noise generated in the microphones and their respective preamplifiers. We describe it such that the electronic output signal of each microphone, y_i , is the sum of the sound pressure and the noise*

$$y_i = p_i + n_i \quad (6.2.16)$$

When we deal with random noise signals, we want to compute the *expected value*, $E[\cdot]$, of the output signal power, for the two hypothetical cases:

1. "no noise, only signal": that is the "signal" in the numerator in Eq. (6.2.17).
2. "no signal, only noise": that is the "noise" in the denominator in Eq. (6.2.17).

$$\text{SNR} = \frac{E[|y_{n=0}|^2]}{E[|y_{p=0}|^2]} \quad (6.2.17)$$

We assume that the signal-to-noise ratio, SNR, which each individual microphone experiences is the same for both microphones:

$$\text{SNR}_{\text{one mic.}} = \frac{E[|p_i|^2]}{E[|n_i|^2]} = \frac{p_{2,\text{r m s}}^2}{n_{2,\text{r m s}}^2} = \frac{p_{1,\text{r m s}}^2}{n_{1,\text{r m s}}^2}$$

Using the model in Eq. (6.2.16), for the two-microphone array, the output signal of the array, y , is

$$\begin{aligned} y &= H_1 y_1 + H_2 y_2 = H_1 (p_1 + n_1) + H_2 (p_2 + n_2) \\ &= [H_1 p_1 + H_2 p_2] + [H_1 n_1 + H_2 n_2] \end{aligned} \quad (6.2.18)$$

* It is a bit inelegant to use p to represent the electrical voltage which is proportional to the sound pressure, but we hope that is not confusing.

Now we analyze the signal-to-noise ratios for the two processing approaches that we used in the previous section: virtual omni, and virtual figure-of-8 microphone. The signal strength will depend on the direction of the incident sound wave, but we are primarily interested in the signal strength for the main direction.

Virtual omni mic.: $H_1 = H_2 = 0.5$

For the main direction $\theta = 90$ degrees, we can find the noise-free output signal $y_{n=0}$ as $p_{\text{in}} \cdot D(\theta_{\text{main}})$, with $D(\theta_{\text{main}})$ from Eq. (6.2.10):

$$y_{n=0} = p_{\text{in}} \Rightarrow \text{E} [|y_{n=0}|^2] = p_{\text{in,rms}}^2$$

The signal-free output signal is

$$y_{p=0} = 0.5n_1 + 0.5n_2 = 0.5 \cdot \sqrt{2}n \Rightarrow \text{E} [|y_{p=0}|^2] = \frac{1}{2}n_{\text{rms}}^2$$

where n is a random noise signal with the same statistical properties as n_1 and n_2^* . So, the SNR is

$$\text{SNR}_{\text{virtual omni}} = \frac{\text{E} [|y_{n=0}|^2]}{\text{E} [|y_{p=0}|^2]} = 2 \frac{p_{\text{in,rms}}^2}{n_{\text{rms}}^2} = 2 \cdot \text{SNR}_{\text{one mic}} \quad (6.2.19)$$

Virtual figure-of-8 microphone: $H_1 = 1/(jkL), H_2 = -1/(jkL)$

For the main direction $\theta = 0$ degrees, we can find the noise-free output signal $y_{n=0}$ as $p_{\text{in}} \cdot D(\theta_{\text{main}})$, with $D(\theta_{\text{main}})$ from Eq. (6.2.13):

$$y_{n=0} = p_{\text{in}} \frac{2}{kL} \sin \frac{kL}{2} \Rightarrow \text{E} [|y_{n=0}|^2] = p_{\text{in,rms}}^2 \frac{4}{(kL)^2} \sin^2 \frac{kL}{2}$$

The signal-free output signal is

$$y_{p=0} = \frac{1}{jkL} [n_1 - n_2] = \frac{1}{jkL} \sqrt{2}n \Rightarrow \text{E} [|y_{p=0}|^2] = \frac{2}{(kL)^2} n_{\text{rms}}^2$$

where, again, n is a random noise signal with the same statistical properties as n_1 and n_2 . Using these two expressions, the SNR for the virtual figure-of-8 microphone is

$$\text{SNR}_{\text{virtual fig.-of-8}} = \frac{\text{E} [|y_{n=0}|^2]}{\text{E} [|y_{p=0}|^2]} = 2 \sin^2 \left(\frac{kL}{2} \right) \frac{p_{\text{in,rms}}^2}{n_{\text{rms}}^2} = 2 \sin^2 \left(\frac{kL}{2} \right) \text{SNR}_{\text{one mic}} \quad (6.2.20)$$

The SNR values of Eqs. (6.2.19) and (6.2.20) are plotted in Fig. 6.7. Here it becomes apparent that the higher directivity of the "endfire array" (for low frequencies), which we saw in Fig. 6.6 comes at the price of a much worse signal-to-noise ratio. The virtual omni mic. (broadside array) apparently gives an SNR which is +3 dB across the frequency range. In a similar way as for the directivity, we state without looking into the details that if increase the number of microphone elements in a linear array, *the SNR will increase by 3 dB for each doubling in number of microphone elements.*

6.2.6 Steering the main direction of the virtual omnidirectional microphone array

The analysis above showed that the simple summing of two omnidirectional microphone elements gives a maximum signal strength for the incidence angle 90 degrees, "broadside". But what happens if we delay the signal of microphone 1 with the amount Δt ? Let's analyse it. The two filters will be[†]

$$H_1 = 0.5 \cdot e^{-j\omega\Delta t}, \quad H_2 = 0.5$$

*Note that when we add two independent random signals with the same rms-value, the sum will have an rms-value which is $\sqrt{2}$ higher than each of the two noise signals.

[†]We are using a frequency-domain representation of a delay.

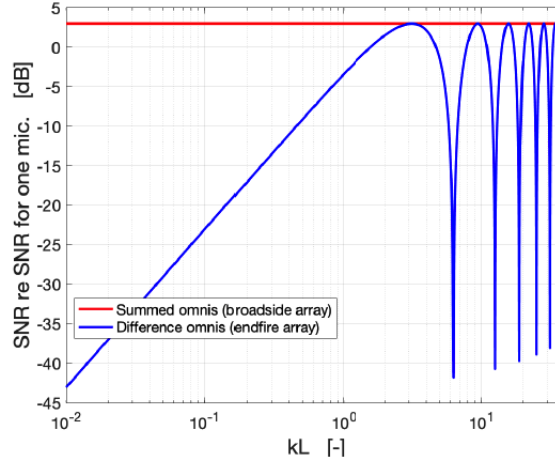


Figure 6.7: The SNR for two different two-microphone arrays, according to Eqs. (6.2.19) and (6.2.20).

We insert those expressions in Eq. (6.2.7) and get the directivity function for this new type of "delay array":

$$\begin{aligned}
 D_{\text{delay array}}(\theta) &= 0.5 \cdot e^{-j\omega\Delta t} e^{jk\frac{L}{2}\cos\theta} + 0.5e^{-jk\frac{L}{2}\cos\theta} \\
 &= 0.5 \cdot e^{-j\omega\Delta t/2} \left[e^{-j\omega\Delta t/2} e^{jk\frac{L}{2}\cos\theta} + e^{j\omega\Delta t/2} e^{-jk\frac{L}{2}\cos\theta} \right] \\
 &= 0.5 \cdot e^{-j\omega\Delta t/2} \left[e^{j\frac{kL}{2}(\cos\theta - \frac{c\Delta t}{L})} + e^{-j\frac{kL}{2}(\cos\theta - \frac{c\Delta t}{L})} \right]
 \end{aligned} \tag{6.2.21}$$

We learnt above that when we don't use any delay Δt , we get the virtual omnidirectional microphone array, which has its maximum signal strength when $\cos\theta = 0$. Now, with a delay Δt , as in Eq. (6.2.21) above, we get the maximum signal strength when

$$\left(\cos\theta_{\text{main}} - \frac{c\Delta t}{L} \right) = 0 \Rightarrow \Delta t = \frac{L}{c} \cos\theta_{\text{main}} \tag{6.2.22}$$

So, we can apparently steer the main direction away from 90 degrees by delaying one of the signals! This is the idea behind the very common "Delay-and-sum beamforming" technique, which can easily be expanded to more array elements, see next chapter. We might realize that when the incoming sound wave is coming a little bit earlier to microphone 1, we could delay signal 1 by exactly that delay amount before we sum it with the signal from microphone 2. Thus, the two signals will be aligned in time and we get maximum positive interference!

6.3 Microphone arrays with more than two elements

When we introduce more than two elements we obviously get more possibilities to do interesting things. A general term for creating directional microphones with an array of microphone elements is *beamforming*, which can be done in different ways.

6.3.1 Delay-and-sum beamforming

A classical technique for beamforming is the *Delay-and-Sum* beamformer. It can be used for any open array of sensors, and it is based on summing delayed versions of the microphone signals so that the delayed versions

are aligned in time for the wanted direction of incidence, see section 6.2.6 for a two-element version. Figure 6.8 illustrates that for one specific direction of (plane wave) incidence, the projection of the microphone positions on the vector of the incidence direction will give positions that reveal how much the incident signal is delayed to the different microphones. This arrival time for a plane wave is straightforward to compute: if a microphone is placed at the coordinates $\mathbf{x}_{\text{mic}} = [x_{\text{mic}}, y_{\text{mic}}, z_{\text{mic}}]$, and we want to have a main direction $\mathbf{x}_{\text{main}}$, then the projected position of the microphone position onto the vector from the origin towards $\mathbf{x}_{\text{main}}$ is

$$d_{\text{projected}} = \frac{\mathbf{x}_{\text{main}} \cdot \mathbf{x}_{\text{mic}}}{|\mathbf{x}_{\text{main}}|} \quad (6.3.23)$$

where $\mathbf{x} \cdot \mathbf{y}$ is the dot product of two vectors, and this d -value can be positive or negative. The distance is easily converted to arrival time via $t = d/c$. Thereby, by delaying the signals by these delays, with the opposite signs, the delayed signals will all be lined up in time and the sum will be an amplified version of the incident sound pressure signal. The expression in Eq. (6.3.23) is equivalent with the one in Eq. (6.2.22), which we can understand if we use the cosine formula for the dot product:

$$\mathbf{x}_{\text{main}} \cdot \mathbf{x}_{\text{mic}} = |\mathbf{x}_{\text{main}}| |\mathbf{x}_{\text{mic}}| \cos \theta$$

where θ is the angle between two vectors. For our two-microphone array, $|\mathbf{x}_{\text{mic}}| = L$, and we can choose to set $|\mathbf{x}_{\text{main}}| = 1$ since $\mathbf{x}_{\text{main}}$ is just used to indicate a direction, not a length or distance. Thus, if we use this in Eq. (6.3.23), together with $\delta t = d_{\text{projected}}/c$, we get

$$\Delta t = \frac{d_{\text{projected}}}{c} = \frac{\mathbf{x}_{\text{main}} \cdot \mathbf{x}_{\text{mic}}}{c} = \frac{L \cos \theta}{c}$$

which is the same as in Eq. (6.2.22). For all other directions of incidence than θ_{main} , the addition will give positive and negative interference effects, and the sum will be amplified less than for the main direction. A common array geometry is a linear array of equidistant sensors.

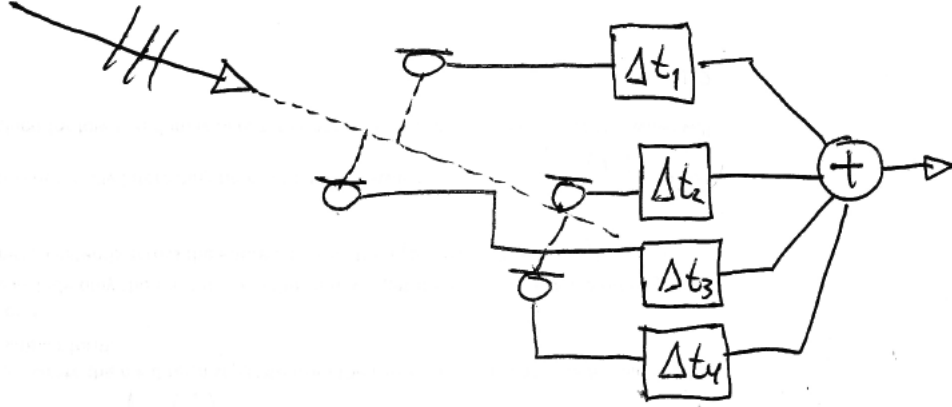


Figure 6.8: The signals from 4 microphones are delayed so that the outputs for one specific plane wave incidence get time-aligned.

For N microphone signals, x_i , this processing could be formulated in the continuous-time domain as

$$y(t) = \sum_{i=1}^N x_i(t - \Delta t_i)$$

and in the sampled, discrete-time (DT) domain

$$y(n) = \sum_{i=1}^N x_i(n - n_i)$$

where $n_i = \Delta t_1 f_s$. In the DT domain, one is restricted to integer values for n_i , which might severely restrict the possibilities one has. Various techniques exist for increasing the temporal resolution. Oversampling the signals x_i is a straightforward technique, which might be computationally expensive. Other techniques involve fractional delay algorithms [39]. Two different geometries are shown in Fig. 6.9.

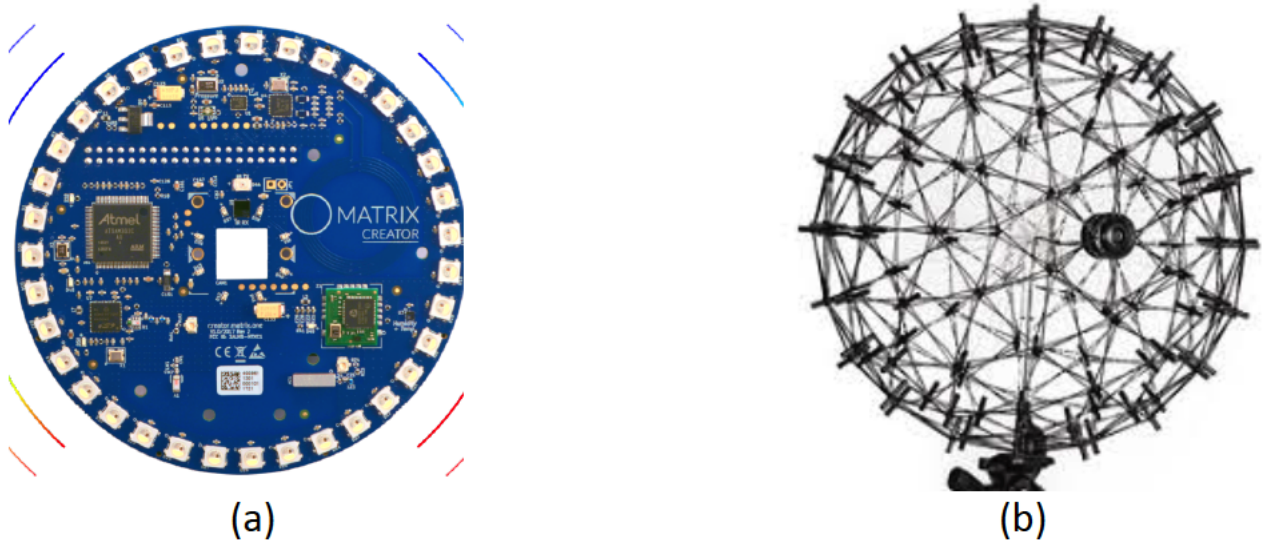


Figure 6.9: Two examples of microphone arrays (a) Circular flat array, "Matrix Creator[®]" and (b) Spherical, open array, "Acoustic Camera[®]"

6.3.2 Soundfield microphone and the B-format

In section 6.2.2, we saw that the combination of an omnidirectional microphone and a figure-of-eight microphone made it possible to create a microphone with a directivity that can be chosen by mixing together the two microphone signals. It is possible to introduce two more figure-of-eight microphones, and arrange them perpendicularly to each other, to be arranged along the x -, y - and z -axes. Then we get four microphone signals which can be mixed together as illustrated in Fig. 6.10.

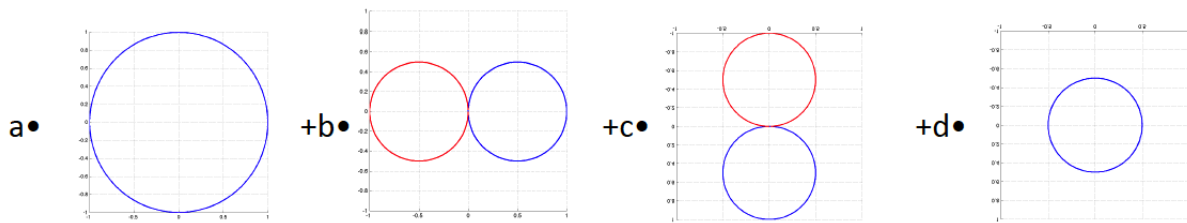


Figure 6.10: Illustration of the directivities of (a) an omnidirectional microphone, and figure-of-eight microphones arranged along the (b) x -axis, (c) y -axis, and the (d) z -axis. The "a •", "+b •", "+c •" and "+d •" indicate that the outputs from the four microphones are scaled and summed.

This is exactly the idea behind the so-called *Soundfield microphone* which will be described below, and the *first-order Ambisonics recording format* (which will be described further in the chapter on sound reproduction formats). The Ambisonics so-called *B-format* contains those four microphone signals, and they are denoted W ,

X, Y, Z , where W is the omnidirectional signal scaled down by 3 dB. If p_{in} is the incoming sound signal:

$$\begin{cases} W = \frac{1}{\sqrt{2}} \cdot p_{in} \\ X = \sin \theta \cos \varphi \cdot p_{in} \\ Y = \sin \theta \sin \varphi \cdot p_{in} \\ Z = \cos \theta \cdot p_{in} \end{cases} \quad (6.3.24)$$

where θ is the angle that runs from 0 ("Northpole") to π ("Southpole"), while φ is the angle that describes the latitude around the sphere, from 0 to 2π , see Fig. 6.11.

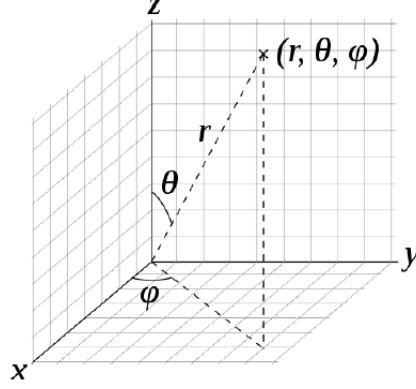


Figure 6.11: Illustration of the spherical coordinate system used here, with θ running from 0 ("Northpole") to π ("Southpole"), and φ running from 0 to 2π ,

Consequently, with this choice of coordinate system, a rotationally symmetrical microphone should point in the z -direction. By weighting those four microphone signals, with weights A, B, C, D ,

$$\text{Output} = A \cdot W + B \cdot X + C \cdot Y + D \cdot Z \quad (6.3.25)$$

we synthesize a microphone with the directivity pattern

$$D(\theta, \varphi) = \frac{A}{\sqrt{2}} + B \sin \theta \cos \varphi + C \sin \theta \sin \varphi + D \cos \theta \quad (6.3.26)$$

This gets much more complex to analyze than the case in section 6.2.2, but we realize that if we want to implement the directivities from section 6.2.2, then we set $B = C = 0$, $A = \sqrt{2}(1 - D)$, and D will be equal to b in section 1.1. We will see below that if we want the main direction to be something else than $\theta = 0$, then we can "rotate the incoming soundfield", and then apply the microphone which has a main direction $\theta = 0$!

We can note a very important aspect of Eq. (6.3.25): the mixing/adding together of the four B-format signals uses frequency-independent scale factors. Thus, the mixing can be done very cheaply, directly on the time-domain signals, or in the frequency domain, after a DFT. Also, the mixing can be done in the analog domain (which would have to be time-domain signals) or in the digital domain.

If the four time-domain B-format signals, of length N samples, are stored in a matrix B (of size $[4, N]$), then Eq. (6.3.25) can be expressed as a matrix multiplication which gives the output signal as a vector of size $[1, N]$:

$$\text{Output} = \mathbf{W}B \quad (6.3.27)$$

where $\mathbf{W}$ is a vector with the weighting values stored as $\mathbf{W} = [A, B, C, D]$, of size $[1, 4]$.

This B-format is very flexible. If one records the four channels W, X, Y, Z , then it is possible afterwards, through postprocessing, to do several things:

1. Synthesize a single microphone with any first-order directivity, pointing in any direction, as indicated above.
2. (As an extension of 1) Synthesize two stereo microphones, L, R , of arbitrary directivity and pointing directions.
3. Implement a simple adaptive filter which synthesizes a microphone which finds the direction to a single incoming sound wave by synthesizing a cardioid microphone, which has a sharp minimum. By letting the adaptive filter minimize the output signal from the cardioid signal, the cardioid microphone will point away from the sound source.

Before we look into how to do point 1 above, we will take a look at various Soundfield microphone constructions, see Fig. 6.12. A Soundfield microphone has 4 microphone signals that are called the "A-format" in the Ambisonics terminology. These 4 are usually never handled by the user; instead they are mixed together and give the so-called "B-format" which also has 4 channels, denoted W, X, Y, Z , as described above.

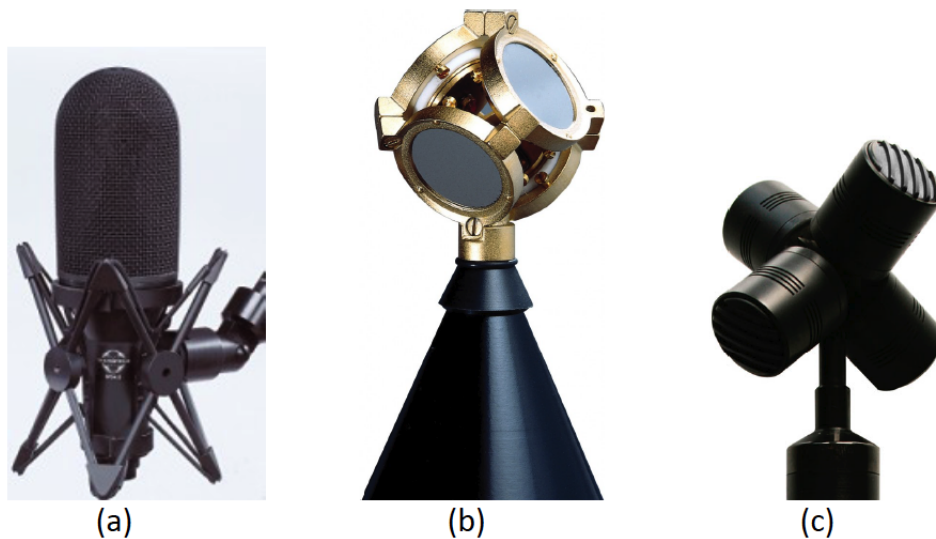


Figure 6.12: (a) The Soundfield microphone, which contains an arrangement of four microphone capsules illustrated in (b). A close-up of the four individual microphone elements of the Soundfield microphone. (c) Another construction which shows four cardioid microphones that form a Soundfield microphone.

6.3.3 Constructing a Soundfield microphone

We can note that even if X, Y and Z should in principle be the output signals of figure-of-8 microphones, most Soundfield microphone synthesize the W, X, Y, Z from four omnidirectional or cardioid microphones. In section 6.2.3, it was shown how two cardioid microphones could be mixed into a virtual figure-of-eight microphone. So, one possible construction could be to use six cardioid microphone elements, and place them as pairs, symmetrically around the origin, in the x -, y -, and z -axes. By applying the expressions in section 6.2.3, it is possible to simulate figure-of-8 microphones. However, this approach requires 6 microphones, but it is possible to use only four microphone elements, as indicated in Figs. 6.12 (b) and (c). Four cardioid microphones, pointing in the directions of a four-sided pyramid, can be mixed into three perpendicular figure-of-eight microphones, using a similar approach as in section 6.2.3, but we will not present the derivation here. For some constructions, the conversion from the A-format (the microphone signals) to the B-format, can be done with simple frequency-independent scaling:

$$\mathbf{B} = \mathbf{F}\mathbf{A}$$

where $\mathbf{A}$ contains N samples of four microphone signals (that is, $\mathbf{A}$ is a matrix of size $[4, N]$), and $\mathbf{F}$ is a $[4, 4]$ matrix which represents the mixing together of the four microphone signals. An interesting open-hardware project for constructing your own Soundfield microphone (=first-order Ambisonics microphone) as well as a second-order Ambisonics microphone is described in [40]. See also <https://cmgitlab.stanford.edu/ambisonics/SpHEAR/>.

6.3.4 Rotating the B-format signals

The B-format offers a practical way to point a synthesized directivity in another direction, by using rotation matrices. Thus, if we want to rotate the pattern by an angle α_x around the x -axis, we should modify the m_x, m_y , and m_z scale factors by multiplying by a rotation matrix (the m_w factor is not affected):

$$\begin{bmatrix} m'_x \\ m'_y \\ m'_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha_x & -\sin \alpha_x \\ 0 & \sin \alpha_x & \cos \alpha_x \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = \mathbf{R}_x \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} \quad (6.3.28)$$

Similarly, rotation matrices for rotating around the y -axis and around the z -axis are:

$$\mathbf{R}_y = \begin{bmatrix} \cos \alpha_y & 0 & \sin \alpha_y \\ 0 & 1 & 0 \\ -\sin \alpha_y & 0 & \cos \alpha_y \end{bmatrix}, \mathbf{R}_z = \begin{bmatrix} \cos \alpha_z & -\sin \alpha_z & 0 \\ \sin \alpha_z & \cos \alpha_z & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (6.3.29)$$

By making rotations in two steps, it is possible to point in any direction. Thus, for microphone signals stored on the B-format in a matrix $\mathbf{B}$, a rotated version $\mathbf{B}'$ is

$$\mathbf{B}' = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0}^T & \mathbf{R}_2 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0}^T & \mathbf{R}_1 \end{bmatrix} \mathbf{B} \quad (6.3.30)$$

where $\mathbf{R}_{1,2}$ is a $[3,3]$ -sized rotation matrix like in Eqs. ((6.3.28)-(6.3.29) and $\mathbf{0} = [0, 0, 0]$. The $[4,4]$ -sized rotation matrices in Eq. (6.3.30) imply that the W -component of the B-format does not get changed by the rotation, which seems to make sense.

6.3.5 More general microphone arrays

One of the most common aims for a microphone (or other sensor) array is to have a higher directivity than a single sensor, and in addition, to have the possibility to steer the main directivity to a desired direction. This process is called beamforming, and it is a large research field in itself. Important application areas include medical ultrasound, remote sensing, "acoustic cameras" etc. Different geometries exist as shown in Fig. 6.13.

Later, we will look at another common task for a microphone array: to find the direction of a sound source. More advanced algorithm exist, some of which are for certain array geometries. For the spherical array, for instance, a family of algorithms based on the *spherical harmonics transform* exist. The derivations that were done by hand at the start of this chapter, for so-called first-order microphones, can be generalized to higher-order spherical harmonics functions, which can be employed when an array has more microphones than the four needed for first-order spherical harmonics transform in 3D.

In [41], Rafaely compared the Delay-and-Sum beamformer to one using spherical harmonics processing, called "Phase-mode" processing. The diagrams in Fig. 6.14, from that paper, illustrate that for fourth-order spherical harmonics processing (which requires $(4+1)^2 = 25$ microphones), one can achieve $\text{DI} \approx 14$ dB for the "phase-mode" processing, but the signal-to-noise ratio, quantified as "White noise gain (WNG)", is much worse for the phase-mode processing. These examples are just generalizations of our results from above, for the

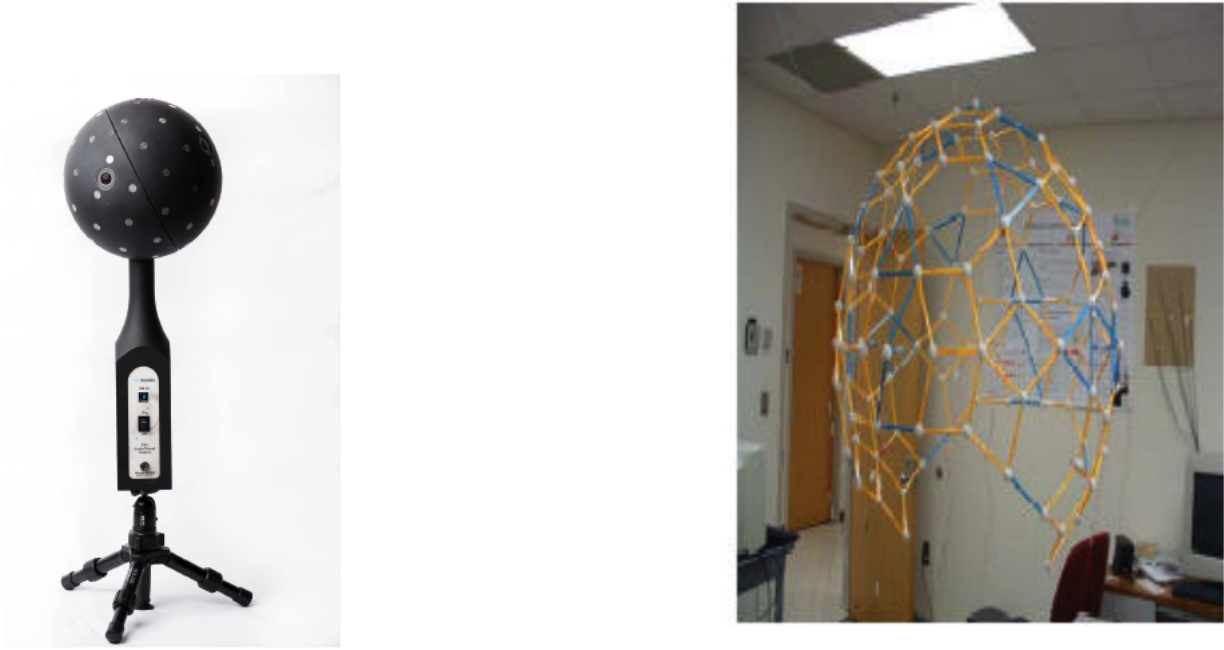


Figure 6.13: Examples of various multi-microphone arrays (a) Spherical, closed array, "Visisonics"®, and (b) Randomly positioned microphones in an open array [Raykar & Duraiswami, ICASSP 2004].

"two-microphone-array". Note the linear frequency axis in Fig. 6.14, rather than the logarithmic one we used in Section 6.2.4. Also, here, the frequency axis does not extend to the high frequencies that we illustrated in Figs. 6.6-6.7.

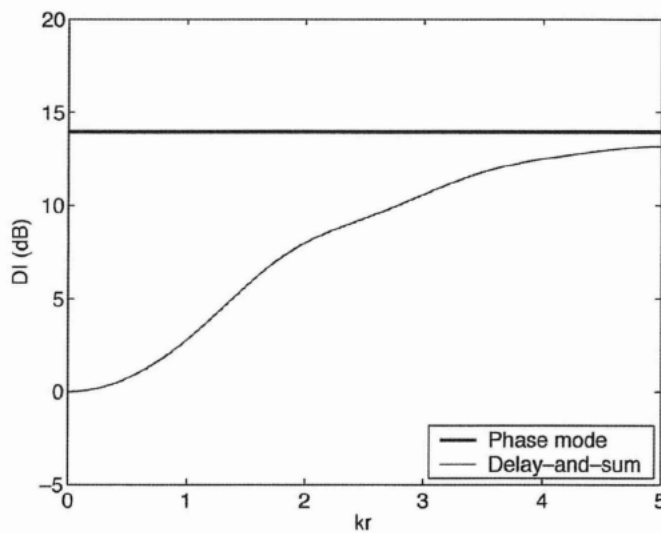


Fig. 3. DI of the phase-mode processing array and the delay-and-sum array, both of order $N = 4$.

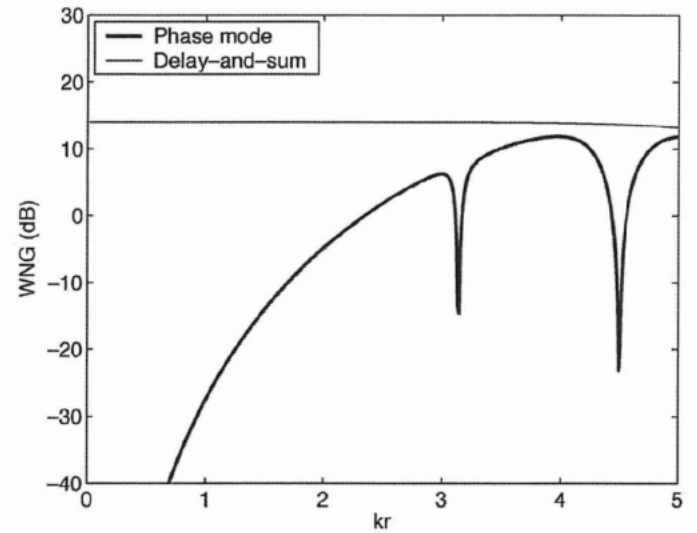


Fig. 4. WNG of the phase-mode processing array and the delay-and-sum array, both of order $N = 4$.

Figure 6.14: Results of simulations of the performance of a spherical array of microphones, from [41].

6.4 Application example: Find the direction to a sound source

A very common application for microphone arrays is to find the direction of arrival (DOA) of a sound source. Existing methods to estimate the DOA by using an array of acoustic sensors can be classified as methods based on one of two categories:

- The time difference of arrival (TDOA) estimation
- The steered response power, with two subcategories
 - Spectral estimation methods.
 - Eigenvector based techniques

6.4.1 Time difference of arrival (TDOA) estimation

The use of **TDOA methods** and compact arrays is attractive because of the simplicity of the signal processing algorithm, the low number of microphones required and the relatively small spatial extent of the final array. We analyze a simple array with three microphones, positioned to form an equilateral triangle, with sidelength d , as in Fig. 6.15.

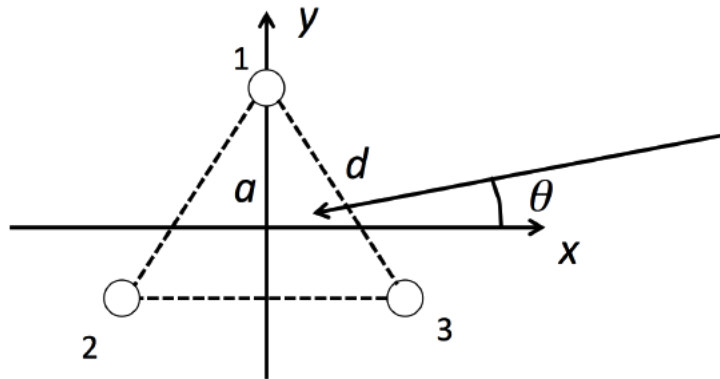


Figure 6.15: A triangular array of three microphones, and an incoming plane wave, at angle θ .

An incident plane sound wave, from the direction θ will give different arrival times to the three microphones, in the same way as illustrated for the delay-and-sum beamformer above. The difference in arrival times to two microphone can be found using crosscorrelation.

Cross-correlation

If we have two discrete-time signals, $x(n)$ and $y(n)$, the cross-correlation function $r_{xy}(l)$ is defined as

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n+l)y(n), \quad l = 0, \pm 1, \pm 2, \pm 3, \dots \quad (6.4.31)$$

This function gives a value, r_{xy} , for each value of time difference, or lag, l , expressed in number of samples. If the cross correlation function has a high value, it means that the two signals are very similar. So, if two signals are identical, apart from a time difference of l samples, then $r_{xy}(l)$ will give a high value.

This fact indicates the basic idea for finding the time difference between two signals: calculate $r_{xy}(l)$ for a number of l -values, and find which l gives the highest $|r_{xy}(l)|$. The time difference is then given by $\Delta t = l/f_s$ where f_s is the sampling frequency.

In Fig. 6.16 (a), we see an example of two identical signals, with a small displacement of exactly three samples. The sampling frequency is 4000 Hz, which corresponds to a time delay of 0.25 ms between each sample. Fig. 6.16 (b) below shows the cross correlation function, and Fig. 6.16 (c) shows a magnified version of the cross correlation function. In the last figure, 6.16 (c), we can indeed see a peak for the delay 0.75 ms = 3 samples.

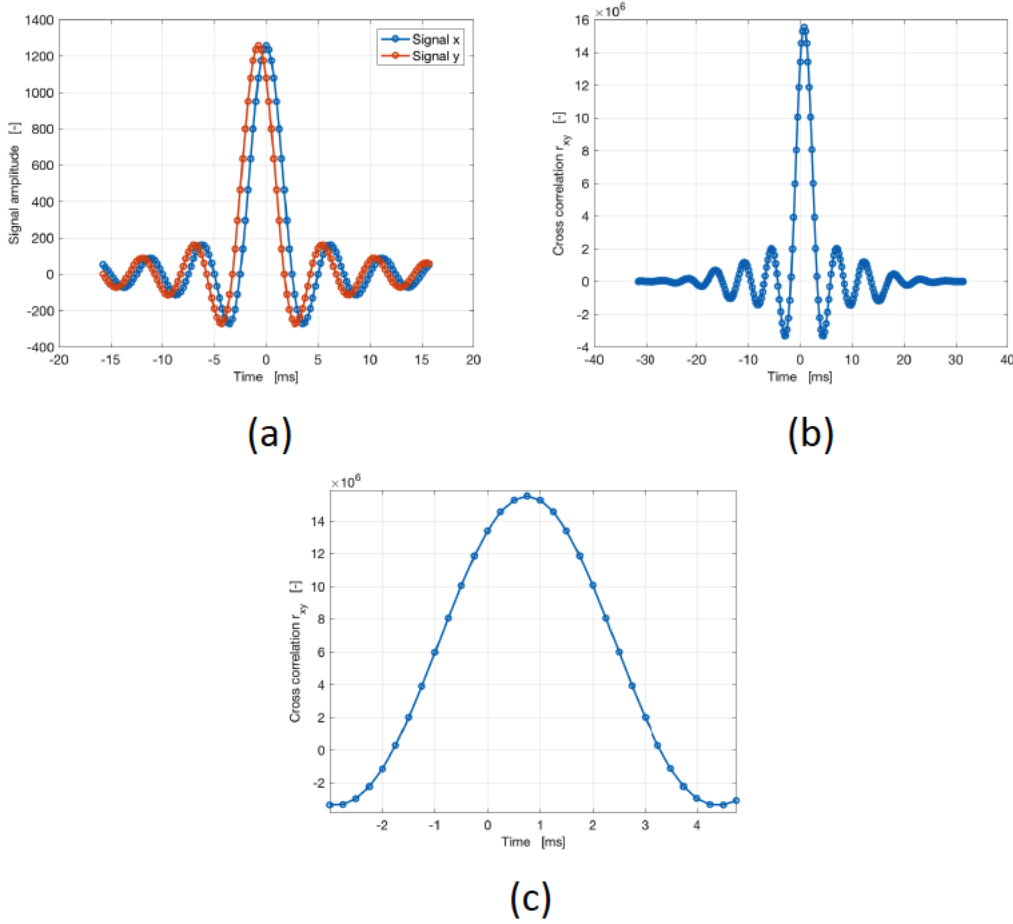


Figure 6.16: Two identical signals, apart from a displacement by three samples. (b) The cross correlation function for the two signals in (a). (c) A magnified version of the diagram in (b).

Now we construct one more example, where the delay between the two signals is 0.25 samples, that is 62.5 μ s. Fig. 6.17 shows in (a) the magnified version of the cross correlation function. We realize that when we look for the peak, our error in the estimate of the time delay will then be up to ± 0.5 samples = $\pm 125 \mu$ s! It is, however, possible to upsample the signal, and Fig. 6.17 (b) shows that with a 16-times upsampling, to $f_s = 64$ kHz, we can find the peak with an error of ± 0.5 samples = 7.8μ s! We can also study the example above (a three samples delay), but with quite much gaussian noise added to both signals (independent noise sequences to the two signals), see Fig. 6.18 (a). Fig. 6.18 (b) shows the cross-correlation function, and the noise seems to influence the cross correlation function less than it affects the signals itself (in (a)). Zooming in around the peak of the cross correlation function, it becomes clear that if the noise had been even stronger, the peak could

have tipped over to 2 or 4 samples. It might help to band-pass filter the signals before carrying out the cross correlation, but only if some frequency-range has a higher signal-to-noise ratio than the entire broadband range.

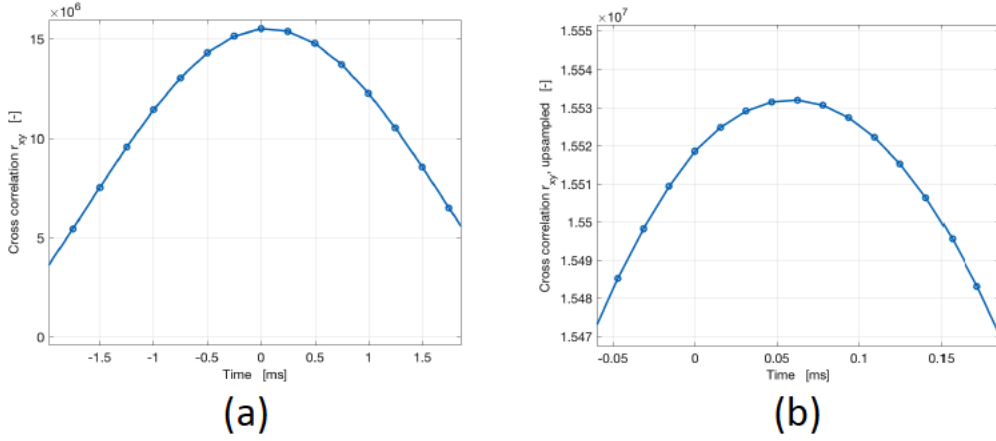


Figure 6.17: (a) The cross correlation between two signals that are identical apart from a delay of 0.25 samples at an $f_{s_s} = 4\text{kHz}$. (b) An upsampled version of the function in (a), with Matlab's interp-function for a 16-times upsampling to $f_s = 64\text{kHz}$.

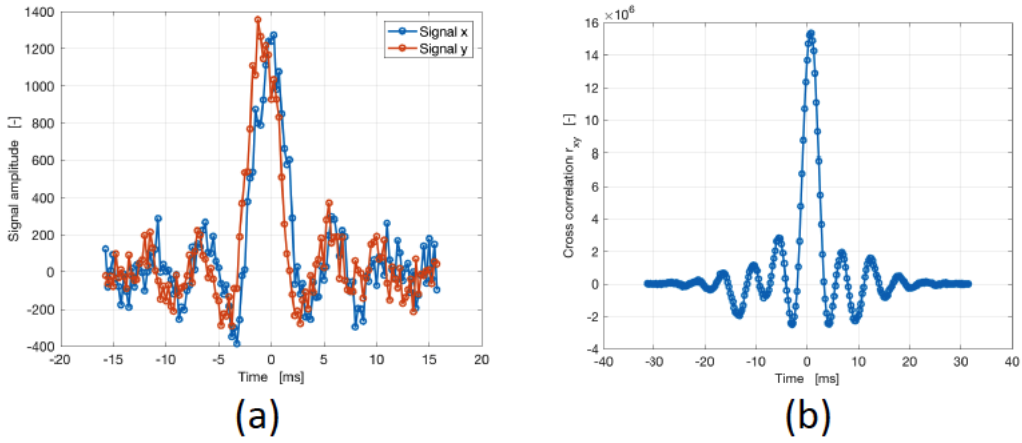


Figure 6.18: (a) Two identical transient signals, apart from a three-sample delay, with gaussian noise added. (b) The cross correlation function for the two signals in (a).

Generalized cross-correlation

The cross-correlation in Eq. (6.4.31) can also be computed via the frequency-domain:

$$r_{xy}(l) = \text{DFT}^{-1}[G_{xy}(\omega)]$$

where $G_{xy}(\omega)$ is the cross-spectrum of the two signals,

$$G_{xy}(\omega) = E[X(\omega)Y^*(\omega)]$$

As before, $E[\cdot]$ indicates the expected value, which has to be found with averaging samples of the signals. The asterisk indicates the complex conjugate, which represents the fact that a cross correlation is identical to a convolution, where one of the signals has been timereversed - which corresponds to taking the complex conjugate in the frequency-domain. This frequency-domain formulation also makes it possible to introduce various weighting functions, $\psi(\omega)$,

$$r_{xy}(l) = \text{DFT}^{-1} [\psi(\omega) G_{xy}(\omega)] \quad (6.4.32)$$

and two common weighting functions are the so-called PHAT (Phase-transform) weighting, and the maximum-likelihood weighting functions [42]. We will look into the PHAT version, which often gives very good results in *sharpening the cross-correlation function* [43]. So, PHAT effectively removes the magnitude function before taking the IDFT, see Fig. 6.19. Apparently, the PHAT version creates a very sharp peak, since the spectral content, the magnitude function, is reflected in the sharpness of the time-domain equivalence, like an impulse response.

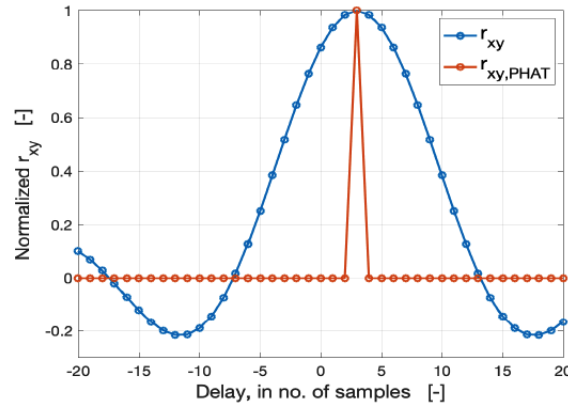


Figure 6.19: The cross correlation function plotted in Fig. 6.16(c), together with the cross correlation which uses the PHAT weighting function, according to Eq. (6.4.32).

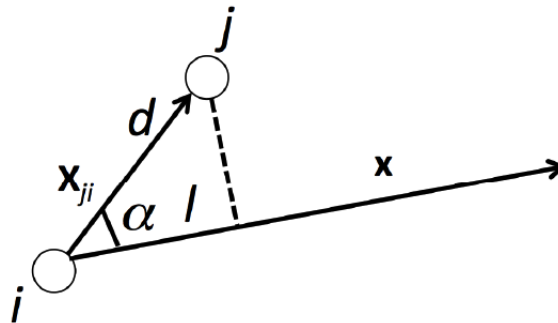


Figure 6.20: One pair of microphones, and an incident sound wave which is propagating in the direction - $\mathbf{x}$ (which is unknown). The vector $\mathbf{x}_{ji}$ goes from microphone i to microphone j .

Finding the direction of incidence from a set of time differences – 2D

We can assume that we have found, using cross-correlation as described above, three time differences, between each pair of microphones. How can this give us a direction of incidence? We inspect one particular pair, as in Fig. 6.20. Two vectors are introduced:

$\mathbf{x}$ = the unknown vector which describes the direction of incidence (with the direction of propagation being $-\mathbf{x}$).

$\mathbf{x}_{ji}$ = the vector from microphone i to microphone j ; this vector is known and has the length d .

The length l corresponds to the delay from sensor i to sensor j , which is known from the cross-correlation:

$$\tau_{ji} = -\frac{l}{c} \quad (6.4.33)$$

There is a minus-sign because the wave is incident in the direction of $-\mathbf{x}$, and then τ_{ji} (the time from microphone i to microphone j) must be negative. l is related to the (unknown) angle α :

$$l = d \cos \alpha = |\mathbf{x}_{ji}| \cos \alpha \quad (6.4.34)$$

We could have stopped already here, and found α from $\alpha = \cos^{-1} \left[-\frac{c\tau_{ji}}{d} \right]$. However, in order to generalize to more than two sensors, we will use an approach that at first seems more complicated. We can find $\cos \alpha$ from the dot-product of the two vectors:

$$\mathbf{x}_{ji} \cdot \mathbf{x} = |\mathbf{x}_{ji}| |\mathbf{x}| \cos \alpha \Rightarrow |\mathbf{x}_{ji}| \cos \alpha = \frac{\mathbf{x}_{ji} \cdot \mathbf{x}}{|\mathbf{x}|} \quad (6.4.35)$$

Note that these relationships were used also in the section for the delay-and-sum beamforming technique. Here, we actually don't want to use α , because that angle changes for each new sensor pair we analyze - it's more practical to let all sensor pairs try to find the same incidence vector $\mathbf{x}$. We get rid of α by inserting the expressions from eqs. (6.4.33) and (6.4.34) in eq. (6.4.35):

$$|\mathbf{x}_{ji}| \cos \alpha = l = -c\tau_{ji} = \frac{\mathbf{x}_{ji} \cdot \mathbf{x}}{|\mathbf{x}|}$$

Again, we know τ_{ji} (from the cross-correlation), and we can choose to define $\mathbf{x}$ such that $|\mathbf{x}| = 1$ *

: $\mathbf{x} = [\cos \theta, \sin \theta]$ where θ is an angle in the xy -coordinate system and thus independent of where the sensors are placed (see Fig. 6.15). Then

$$\mathbf{x}_{ji} \cdot \mathbf{x} + c\tau_{ji} = 0 \quad (6.4.36)$$

The dot product can alternatively be written out explicitly (with $\mathbf{x} = [\cos \theta, \sin \theta]$), such that

$$\mathbf{x}_{ji} \cos \theta + \mathbf{y}_{ji} \sin \theta + c\tau_{ji} = 0$$

Now, with three microphones/sensors, we get three equations like in Eq. (6.4.36):

$$\begin{aligned} \mathbf{x}_{12} \cdot \mathbf{x} + c\tau_{12} &= 0 \\ \mathbf{x}_{13} \cdot \mathbf{x} + c\tau_{13} &= 0 \\ \mathbf{x}_{23} \cdot \mathbf{x} + c\tau_{23} &= 0 \end{aligned} \quad (6.4.37)$$

* We should preferably have defined that $|\mathbf{x}| = 1$ m, rather than just the number 1. Out of lazyness, we don't write out this meter unit - but as a consequence, we will get inconsistent units in some of our equations!

where the unknown is $\mathbf{x} = [x, y]$, a vector of size $(1, 2)$, and the known vectors $\mathbf{x}_{ij} = [x_{ij}, y_{ij}]$ are also vectors of size $(1, 2)$, which describe where the sensors are related to each other. There are two approaches to solve the set of equations in (6.4.37).

Approach 1 – matrix equation, solved by pseudo-inverse

The three equations, (6.4.37), can be formulated as a single matrix equation,

$$\mathbf{X}_{\text{array}} \mathbf{x}^T = -c\mathbf{T} \quad (6.4.38)$$

where

$$\mathbf{T} = [\tau_{12}, \tau_{13}, \tau_{23}]^T, \text{ a column vector of size } (3, 1)$$

$$\mathbf{X}_{\text{array}} = [\mathbf{x}_{12}^T, \mathbf{x}_{13}^T, \mathbf{x}_{23}^T]^T, \text{ a matrix of size } (3, 2)$$

The matrix equation (6.4.36) represents an overdetermined system: two unknowns, three equations. An overdetermined system can be solved by the least-squares method, which can be carried out using the so-called pseudo-inverse* of the matrix $\mathbf{X}_{\text{array}}$, which can be denoted $\mathbf{X}_{\text{array}}^+$, where $\mathbf{X}_{\text{array}}^+$ fulfills $\mathbf{X}_{\text{array}} \mathbf{X}_{\text{array}}^+ \mathbf{X}_{\text{array}} = \mathbf{X}_{\text{array}}$, and then

$$\mathbf{x}^T = -c\mathbf{X}_{\text{array}}^+ \mathbf{T} \quad (6.4.39)$$

In Matlab, the pseudo-inverse is implemented as the pinv function.

Approach 2 – derive the least mean squares solution explicitly

We can also derive an explicit solution to the set of equations (6.4.37), with the least-mean squares approach (which leads to the same solution as with the pseudo-inverse above). The method is the same as is used to find the best straight-line fit to a set of data points. We realize that the two values we look for, x and y , can not fulfil the three equations (6.4.37). That means that the right-hand sides of the equations will not be 0, but some (hopefully) small value. So, we can define an error measure, and we will then look for values of x and y that minimize this error value. We choose as error measure *the sum of the squared values of the three equations*:

$$\varepsilon^2 = \sum_{ji} (\mathbf{x}_{ji} \cdot \mathbf{x} + c\tau_{ji})^2 = \sum_{ji} (x_{ji}x + y_{ji}y + c\tau_{ji})^2 \quad (6.4.40)$$

We find the values of x and y by setting the derivative of $\varepsilon^2 = 0$, both with respect to x and to y . The values x and y that fulfill those equalities will give a minimum (it could have been a maximum, but we know that the squared value ε^2 is always > 0 , and is a squared function of x and y , so there will be one extreme value, which must be a minimum):

$$\begin{cases} \frac{\partial \varepsilon^2}{\partial x} = \frac{\partial}{\partial x} \sum_{ji} (x_{ji}x + y_{ji}y + c\tau_{ji})^2 = \sum_{ji} 2(x_{ji}x + y_{ji}y + c\tau_{ji}) x_{ji} = 0 \\ \frac{\partial \varepsilon^2}{\partial y} = \frac{\partial}{\partial y} \sum_{ji} (x_{ji}x + y_{ji}y + c\tau_{ji})^2 = \sum_{ji} 2(x_{ji}x + y_{ji}y + c\tau_{ji}) y_{ji} = 0 \end{cases}$$

$$\begin{cases} x \sum_{ji} 2(x_{ji}^2) + y \sum_{ji} 2(x_{ji}y_{ji}) + \sum_{ji} 2(c\tau_{ji}x_{ji}) = 0 \\ x \sum_{ji} 2(x_{ji}y_{ji}) + y \sum_{ji} 2(y_{ji}^2) + \sum_{ji} 2(c\tau_{ji}y_{ji}) = 0 \end{cases} \Rightarrow \begin{cases} xA + yB + C = 0 \\ xB + yD + E = 0 \end{cases}$$

This can be solved:

$$x = \frac{CD + BE}{AD - B^2}, \quad y = -\frac{BC + AE}{AD - B^2} \quad (6.4.41)$$

*The ordinary matrix inversion applies only to square matrices.

If we place the sensors symmetrically around the origin, like in Fig. 6.15, then the value $B = 0$ because $y_{23} = 0$ (the vector from sensor 3 to sensor 2 has a zero y -component), and $x_{12} = -x_{13}$, while $y_{12} = y_{13}$

$$x_{12}y_{12} + x_{13}y_{13} + x_{23}y_{23} = 0$$

So, eq. (6.4.41) is simplified:

$$x = \frac{C}{A}, \quad y = -\frac{E}{D} \quad (6.4.42)$$

We inspect Fig. 6.15 again and write down the positions for the three sensors:

$$\begin{aligned} \mathbf{x}_{\text{sensor } 1} &= [0, 1]a \\ \mathbf{x}_{\text{sensor } 2} &= [-\sqrt{3}/2, -0.5]a \\ \mathbf{x}_{\text{sensor } 3} &= [\sqrt{3}/2, -0.5]a \end{aligned}$$

so, the vectors from sensor to sensor are:

$$\begin{aligned} \mathbf{x}_{21} &= \mathbf{x}_{\text{sensor } 2} - \mathbf{x}_{\text{sensor } 1} = [-\sqrt{3}/2, -1.5]a \\ \mathbf{x}_{31} &= \mathbf{x}_{\text{sensor } 3} - \mathbf{x}_{\text{sensor } 1} = [\sqrt{3}/2, -1.5]a \\ \mathbf{x}_{32} &= \mathbf{x}_{\text{sensor } 3} - \mathbf{x}_{\text{sensor } 2} = [\sqrt{3}, 0]a \end{aligned}$$

Then

$$\begin{aligned} A &= 2 \sum (x_{ji})^2 = 9a^2 \\ D &= 2 \sum (y_{ji})^2 = 9a^2 \\ C &= 2c \sum x_{ji}\tau_{ji} = 2ca \left(-\frac{\sqrt{3}}{2}\tau_{21} + \frac{\sqrt{3}}{2}\tau_{31} + \sqrt{3}\tau_{32} \right) \\ &= \sqrt{3}ca (-\tau_{21} + \tau_{31} + 2\tau_{32}) \\ E &= 2c \sum y_{ji}\tau_{ji} = 2ca \left(-\frac{3}{2}\tau_{21} - \frac{3}{2}\tau_{31} \right) = -3ca (\tau_{21} + \tau_{31}) \end{aligned}$$

Inserting these values in Eq. (6.4.40) gives us:

$$\begin{aligned} x &= \frac{\sqrt{3}ca (-\tau_{21} + \tau_{31} + 2\tau_{32})}{9a^2} = \frac{c}{3\sqrt{3}a} (-\tau_{21} + \tau_{31} + 2\tau_{32}) \\ y &= -\frac{-3ca (\tau_{21} + \tau_{31})}{9a^2} = \frac{c}{3a} (\tau_{21} + \tau_{31}) \end{aligned}$$

Now we can convert these two values in to a suprisingly simple expression for the angle, θ :

$$\theta = \arctan \frac{y}{x} = \arctan \left(\sqrt{3} \frac{\tau_{21} + \tau_{31}}{-\tau_{21} + \tau_{31} + 2\tau_{32}} \right) \quad (6.4.43)$$

As stated earlier, the estimates of x and y above will be the same as via the pseudoinverse in Approach 1.

Finding the direction of incidence from a set of time differences – 3D

With the matrix equation in Eq. (6.4.38), it becomes quite clear that we could easily expand this to 3D, by introducing a fourth microphone sensor [44]. The unknown, $\mathbf{x}$, will now have the size $(1, 3)$. The matrix of known array vectors contains 6 three-element vectors (x, y and z),

$$\mathbf{X}_{\text{array}} = [\mathbf{x}_{12}^T, \mathbf{x}_{13}^T, \mathbf{x}_{14}^T, \mathbf{x}_{23}^T, \mathbf{x}_{24}^T, \mathbf{x}_{34}^T]^T, \text{ a matrix of size } (6, 3)$$

and the time-delay vector, $\mathbf{T}$, has size $(6, 1)$:

$$\mathbf{T} = [\tau_{12}, \tau_{13}, \tau_{14}, \tau_{23}, \tau_{24}, \tau_{34}]^T$$

Again, the matrix equation (6.4.37) will be overdetermined but the least-squares solution can be found using the pseudo-inverse of the matrix $\mathbf{X}_{\text{array}}$, like in Eq. (6.4.39).

6.4.2 Spectral estimation methods

Spectral estimation methods calculate the power of the output of a beamformer as a function of the "look direction" and takes the argument of the highest peak(s) in the power output as the DOA of the source(s) [45]. One example is to use a delay and sum beamformer. Another example is a so-called minimum variance distortionless beamformer, [46], resulting in enhanced resolution but deteriorated performance in the presence of correlated sources (e.g. ground reflections).

6.4.3 Eigenvector based methods

The most popular eigenvector based techniques are the MUSIC, [47], and the ESPRIT, [48], algorithms which take benefit of the eigenstructure of the correlation matrix. Both algorithms are high resolution techniques. The performance of both algorithms decreases in the presence of correlated sources because such sources reduce the rank of the correlation matrix, but spatial smoothing techniques exist to overcome this problem.

Auralization - virtual acoustics

7.1 Introduction

A very common computational task to solve, in many different fields of acoustics, is to predict the sound pressure at a listener's position, given:

- the properties of a sound source: source power, or source signal, $s(t)$, plus the source directivity,
- the propagation properties of the environment: speed of sound (which is usually assumed constant indoors and non-constant for long-distance outdoor cases) and frequency-dependent air absorption,
- the reflection properties of the environment: the geometry and reflection properties of a room's surfaces, or of an outdoors case.

and we want to know what the sound pressure, $p(t)$, is in a specific receiver/listener position. This general problem is illustrated in Fig. 8.7, with an indoor and outdoor example.

In this course, we are particularly interested in how to create an *auralization*, that is, to be able to listen to the computer simulated sound pressure signal $p(t)$, with the important temporal, spectral and spatial aspects of the sound faithfully reproduced [49], [50]. Auralization requires various signal processing steps in order to take a computer simulation result and make it listenable.

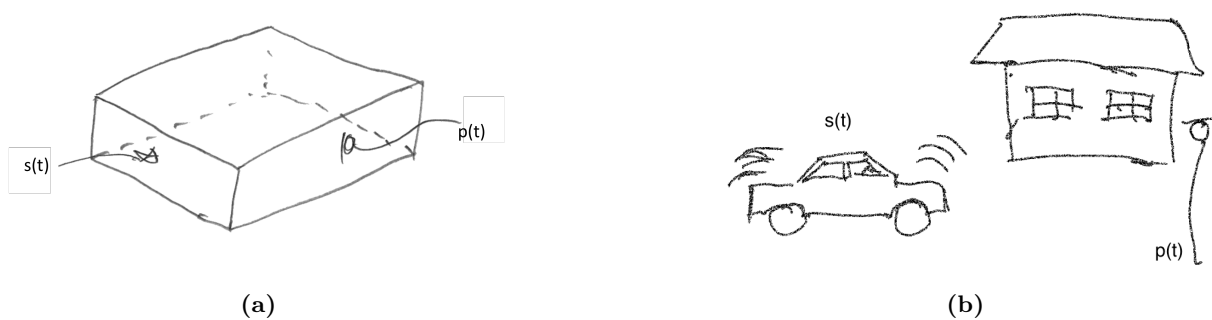


Figure 7.1: Illustration of acoustical prediction problems: if a source signal, s , is known, together with information about the environment; what is the sound pressure, p , in a receiver position? This is a common task both for (a) indoor cases and (b) outdoor cases.

Examples where auralization is very useful can include:

- Listen to various designs of a new loudspeaker construction, in a specific listening room
- Listen to various designs of a new concert hall
- Listen to various designs of a new car cabin, and listen to the sound system in the car
- Listen to various solutions of the sound system for a large concert arena
- Listen to how the building of a new highway close to a residential area will affect the noise level, and investigate various noise barrier solutions
- etc

Depending on the complexity of the problem, the accuracy demands, and the frequency range of interest, different calculation methods are used. Three common categories are:

1. Diffuse-field methods are very useful and popular for simplified studies of indoor cases. See the appendix for a few notes on this.
2. Geometrical acoustics (GA) – which is the topic of this lecture, see further below.
3. Wave-theoretical methods are very accurate, but can be used only for some specific cases. See the appendix for a few notes on this.

Here we will focus on category 2, geometrical acoustics, because GA-based methods easily give the possibility for auralization.

7.2 Geometrical acoustics (GA)

Geometrical acoustics (GA) is based on some assumptions:

- GA assumes that the sound field can be viewed as a sum of direct sound and reflections, *and that each such reflection can be computed separately*

$$p = p_{\text{direct}} + \sum_{i=1}^{\infty} p_{\text{refl.},i} \quad (7.2.1)$$

- GA further assumes that sound travels along straight lines and thus does not diffract/propagate around corners ^a.
- GA also assumes that reflections are either specular reflections, which can be represented by image sources, or diffuse reflections, the computation of which are described further below.

^aIn outdoor sound and under water, a non-constant speed of sound implies that sound travels along curved paths.

There are two approximations in the model of each reflection. Both approximations are most in error for low frequencies, and can be solved to some degree by extensions to geometrical acoustics:

1. As demonstrated in the chapter on reflections and diffraction, GA creates truncated wavefronts from each finite-sized reflector, and that is clearly wrong. It also means that GA predicts that sound can not travel around corners, which is also wrong. For rigid surfaces, diffraction waves can be added and reduce this error substantially, [51], as also demonstrated in the chapter on reflections and diffraction.

2. GA uses the model of a specular reflection, with the plane-wave reflection factor, which is inaccurate when a surface is not ideally hard and the source and receiver are both close to the reflecting surface. Individual reflections, such as the ground reflection outdoors, could be calculated by more sophisticated methods, and reduce the error caused by standard GA.

So, standard GA works poorly at LF but accurately at medium and high frequencies.

Numerical implementations use the image source method, ray tracing or radiosity, and they are described in an overview article, [52], and briefly below. Many commercial programs for auditorium acoustics, sound reinforcement systems etc, use these methods. Examples of such software are CATT-Acoustic (www.catt.se), Odeon (www.odeon.dk), Ease (<http://ease.afmg.eu>) and others. So, the geometrical acoustics approach looks at a sound field in a room in terms of wall reflections, as illustrated in Fig. 7.2.

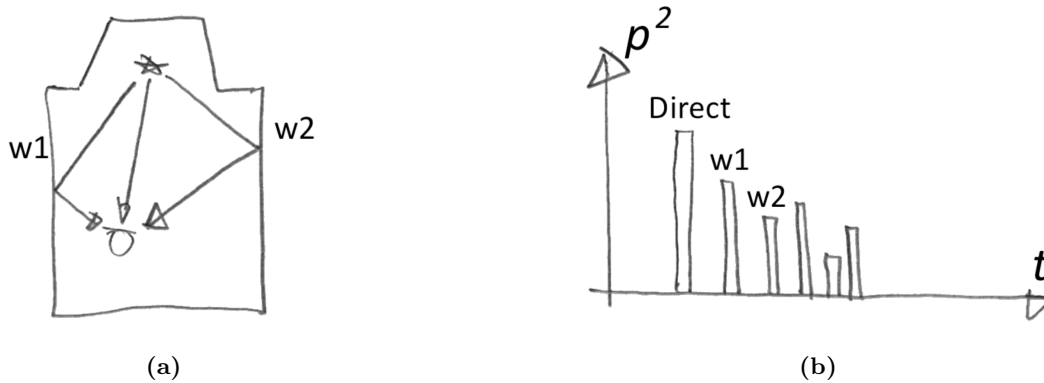


Figure 7.2: (a) Illustration of a room with a direct sound and two specular wall reflections marked, and (b) the corresponding echogram.

The calculation of the sound pressure, p , in Eq. (7.2.1), with GA-based methods is done in two steps:

1. Find all possible wall reflections, that is, sound paths, from one source position to one receiver position. This is a purely geometrical task, and can be done with techniques similar to those used in computer graphics.
2. Compute each reflection's contribution, $p_{\text{refl},i}$, in Eq. (7.2.1).

7.2.1 Step 1 - the geometrical task

Wall reflections are either *specular*, obeying Snell's reflection law, or *diffuse*. A surface's smoothness and flatness determines whether a reflection is pre-dominantly specular or diffuse (there will practically always be some degree of both). These two types of reflections are easy to handle for a computerized algorithm:

- Specular reflections can be studied accurately by the *image source method*, approximately with the *ray tracing* method, or accurately with the *beam-tracing method* (a special type of ray tracing which can handle only specular reflections).
- Diffuse reflections can be studied with the *ray tracing* method, or by the *radiosity* method.

The image source method

The image source method is quite straightforward to implement: the algorithm has to construct (i.e., compute the coordinates of) *image sources* for all reflecting surfaces in a CAD model of a room. Then, for a receiver to be reached by a specular reflection from a certain wall, the line from the image source to the receiver must pass

through the finite reflecting wall. One example for a reflection from the side wall on a stage is shown in Fig. 7.3. The image source concept is valid only for specular reflections, therefore the image source method can not handle diffuse reflections.

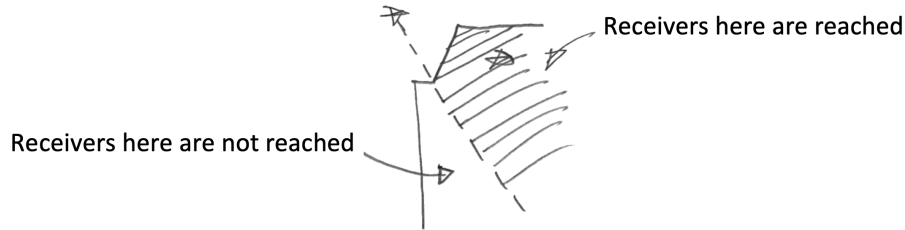


Figure 7.3: The image source method assumes that a wall reflection will reach a receiver if the receiver can "see" the image source through the finite reflecting plane - exactly as a mirror reflects light.

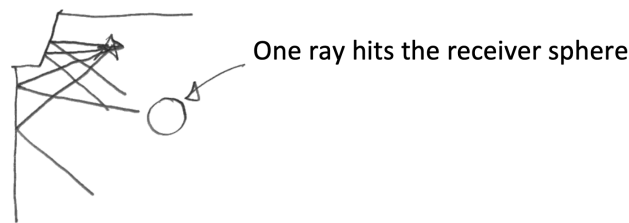


Figure 7.4: Illustration of the ray tracing method.

The image source method can be extended to higher-order reflections, and will manage to find all valid specular sound paths - but the computational cost grows exponentially with reflection order. The so-called beam tracing method is much more efficient for higher reflection orders, [53], [54].

Ray tracing

The ray tracing method is also quite straightforward: a large numbers of rays (lines) are sent out from a source. Each ray is checked against each of the walls, and whenever a wall is hit, the ray is reflected back into the room, either specularly or diffusively. When a ray hits a receiver, then a "valid reflection" is marked for that receiver (the first ray tracing algorithm for room acoustics was presented by A. Krokstad et al from NTH in 1968, [55]), see Fig. 7.4.

7.2.2 Before step 2 - computing the sound pressure, or the impulse response?

Before discussing how to calculate the contribution by each reflection, $p_{\text{refl},i}$, in Eq. (7.2.1), we can contemplate if we should calculate the *sound pressure*, or the corresponding *impulse response*. We are usually interested in what the sound pressure is at a receiver point. Obviously, the sound at a receiver point depends on the source signal, so if we want to calculate/predict the sound at a receiver point for different source signals, we have to recalculate for each new type of source signal that we want to study.

A very powerful technique is to calculate the system's impulse response rather than the sound pressure. Then one can generate/synthesize the sound pressure signal for any source signal, by convolving the source signal with the computed impulse response. This is one of the underlying ideas for auralization.

It is a very small difference between computing the output sound pressure and the impulse response – the latter results when we use a Dirac pulse as a source signal, $s(t)$, in Fig. 1. The basic block diagram for a linear,

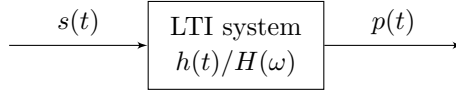


Figure 7.5: A room, or environment, can be represented as a linear, time-invariant (LTI) system, which can be represented by its impulse response, $h(t)$, or its frequency response function, $H(\omega)$.

time-invariant system (LTI) in Fig. 7.5 reminds us about the roles of the source signal, $s(t)$, the output sound pressure, $p(t)$, and the impulse response, $h(t)$.

But, what exactly is the source signal? Two alternatives are common:

1. For theoretical derivations, it is common to use a source signal, $s_{\text{source}}(\omega)$, which is, in the frequency-domain and time-domain, respectively:

$$s_{\text{source}}(\omega) = \frac{j\omega\rho_0 U_{\text{source}}(\omega)}{4\pi} = \frac{\rho_0 A_{\text{source}}(\omega)}{4\pi}$$

$$s_{\text{source}}(t) = \frac{\rho_0 A_{\text{source}}(t)}{4\pi}$$

where U_{source} is the volume velocity of the source, A_{source} is the volume acceleration of the source, and ρ_0 is the density of air. Then the sound pressure, the output signal of the system, is in the frequency domain, $p(\omega)$, given as the multiplication of $s_{\text{source}}(\omega)$ with the transfer function/frequency response, $H(\omega)$. For a monopole source in free-space radiation, $H(\omega)$ is easy:

$$H_{\text{free-field}}(\omega) = \frac{e^{-jkr}}{r}$$

where r is the distance from source to receiver. In the time domain, the sound pressure at the receiver, $p(t)$, is the convolution of the source signal, $s(t)$, with the impulse response for a monopole source in free-space radiation:

$$h_{\text{free-field}}(t) = \frac{\delta(t - r/c)}{r}$$

2. For practical purposes, it turns out to be much more useful to define the sound pressure at 1 m distance as the source signal:

$$s_{\text{source}}(t) = p_{\text{free-field}, 1\text{m}}(t)$$

and then the impulse response will be

$$h_{\text{free-field}}(t) = \frac{\delta[t - (r - 1\text{m})/c]}{(r/1\text{m})} \quad (7.2.2)$$

If we want to be able to listen to a calculated/predicted sound pressure, we must have it in the form of a time-domain signal, $p(t)$. But, thanks to the linearity (that we are assuming), it works perfectly fine to make all calculations in the frequency-domain - and convert the final sound pressure, $p(\omega)$, to a time-domain signal.

7.2.3 Step 2 - computing the contribution by each reflection

The expressions above for the impulse response for the free-space radiation makes it straightforward to derive what the transfer function/impulse response is for a *reflection*, that is, the transfer function of one image source:

$$H_{\text{refl}, i}(\omega) = \frac{e^{-jkr_i}}{r_i} D_S(\theta_i, \varphi_i, \omega) \Pi_{j=1}^{n_{\text{refl}, \text{order}, i}} R_{p, j}(\omega) \quad (7.2.3)$$

where r_i is the total travel distance from the image source number i , via $n_{\text{refl.order},i}$ reflections, to the receiver. D_S is the directivity function of the source in the radiation direction (θ_i, φ_i) for the angular frequency ω , and $R_{p,j}(\omega)$ is the frequency-dependent reflection factor for reflection number j along the reflection sequence.

In the time domain, the multiplications of frequency-dependent factors becomes a convolution of corresponding time-domain functions instead.

Numerical example

We look at a numerical example of an impulse which represents the situation in Fig. 7.6, which is simple enough that we can derive the impulse response by hand. If the wall is perfectly rigid, so that $R_p = 1$, and there is no attenuation in the air, then both the direct sound and the wall reflection will be delayed and scaled Dirac pulses. We choose the sound pressure in free-field, 1m from the source, as the source signal which defines our impulse response according to Eq. (7.2.2). Fig. 7.7 shows the impulse response and the corresponding frequency response.

7.2.4 Sound pressure, or sound pressure squared?

It turns out that the product of reflection factors in Eq. (2) is very difficult to calculate in practice – because we rarely have access to the complex impedances that are needed to calculate R_p . Remember: for the reflection of a plane wave, the reflection factor for perpendicular incidence is

$$R_p = \frac{Z_{\text{wall}} - Z_{\text{air}}}{Z_{\text{wall}} + Z_{\text{air}}}$$

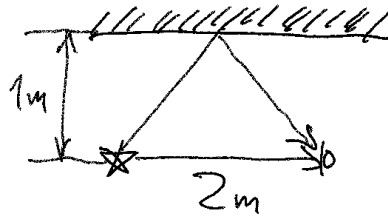


Figure 7.6: The geometry for a numerical example.

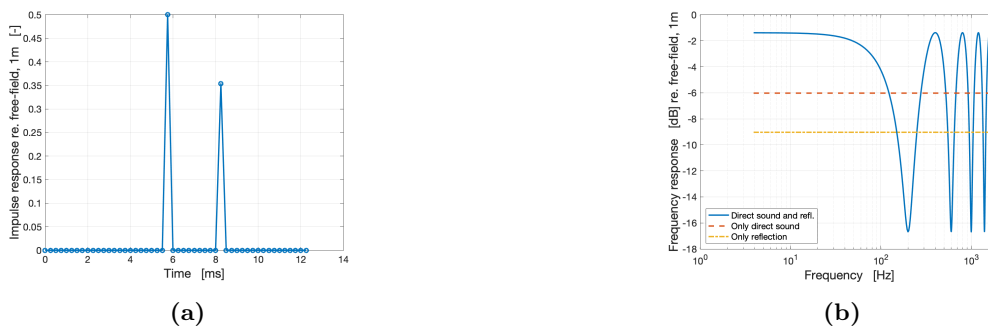


Figure 7.7: (a) The impulse response for the geometry in Fig. 7.6, with the sound pressure at 1m in free-field defined as source signal (however, the time delay is relative to the position of the sound source!), and (b) the corresponding frequency response.

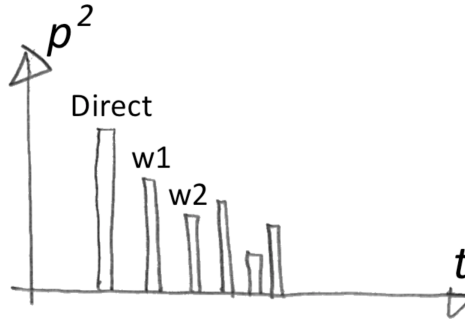


Figure 7.8: An echogram, that is, a sequence of pulses; one for each sound wave (caused by one combination of wall reflections) which reaches a receiver. The amplitude of each pulse gives p^2 for that sound wave, for one frequency band (typically an octave band or third-octave band).

So, a pragmatic choice is to calculate what is called the *echogram* rather than the impulse response. Then each reflection gives a contribution to the *squared magnitude* of the frequency response function:

$$\langle |H_{\text{refl.},i}|^2 \rangle = \frac{1}{r_i^2} \text{DF}_S(\theta_i, \varphi_i) e^{-\mu r_i} \prod_{j=1}^{n_{\text{refl.order},i}} (1 - \alpha_j)$$

where DF_S is the directivity *factor* rather than the directivity *function* ($p \propto D$, $p^2 \propto \text{DF}$), μ is the dissipation coefficient of air (which is strongly frequency dependent), α is the absorption factor ($p \propto R_p$, $p^2 \propto (1 - \alpha)$), and $\langle \rangle$ indicates averaging across a frequency range, typically an octave band. By choosing $\langle |H_{\text{refl.},i}|^2 \rangle$ instead of $H_{\text{refl.},i}(\omega)$, one loses information about phase shifts that happen at each reflection, and any possible phase shifts of the directivity function. Again, this loss of information is because we don't usually have the information needed to compute the phase shifts correctly!

One irritating challenge is that even if we have only enough information to calculate $\langle |H_{\text{refl.},i}|^2 \rangle$, for a small number of octave bands, we still need an impulse response in order to do auralization!

So, what to do? Well, we can try to recreate/synthesize an impulse response, which can be done as follows*

7.2.5 Synthesis/construction of an impulse response from p^2 -information

The $|H|^2$ -values are calculated for every octave band or 1/3-octave band. A plot of these $|H|^2$, or p^2 -values, as function of time, for one frequency band, is often called an *echogram*, as shown in Figs. 7.2(b) and 7.8.

An impulse response can be synthesized, based on such an echogram. This synthesis technique can be described as follows:

For each reflection in the echogram, a short FIR filter is constructed (see below for details) based on the p^2 -, or $|H|^2$ -values for octave bands (which might be just 6 values):

1. A DFT-size, n_{DFT} , is chosen that should be used for the transformation from the frequency domain to the time domain. This value gives a frequency resolution, $\Delta f = f_s/n_{\text{DFT}}$.
2. The few magnitude values are interpolated and extrapolated into a regularly sampled magnitude function from 0 to $f_s/2$, with the resolution Δf . It will give $n_{\text{DFT}}/2 + 1$ values.

*It must be realized that it is impossible to recreate an exact impulse response from an echogram, because a lot of information has been thrown away (or not been used) for the calculation of the echogram. But the idea is that a lot of inaudible details can safely be neglected.

3. A phase function is constructed, based on a linear phase or minimum phase assumption.
4. The set of complex amplitudes from 0 to $f_S/2$, is mirrored to the frequency range $f_S/2$ to f_S (using complex conjugate!), so that one gets a reflection transfer function, $H_{\text{one reflection}}(\omega)$, sampled with n_{DFT} points from 0 to $f_S - \Delta f$.
5. A short FIR filter is calculated using DFT:

$$h_{\text{singlereff.}}(t) = \text{DFT}^{-1} \{H_{\text{one reflection}}(\omega)\}$$

6. This short FIR filter is added to the total room impulse response, at the delay given by the propagation distance r , for that reflection combination.

Such a synthesized impulse response would then ideally be audibly similar to the impulse response that we would measure in a room.

To construct an impulse response that represents a single wall reflection implies that we would like to construct an impulse response with a sampling frequency of, e.g., 44.1 kHz, which would correspond to a measurement of a wall reflection impulse response. A fictive example is shown in Fig. 7.9, with its corresponding frequency response.

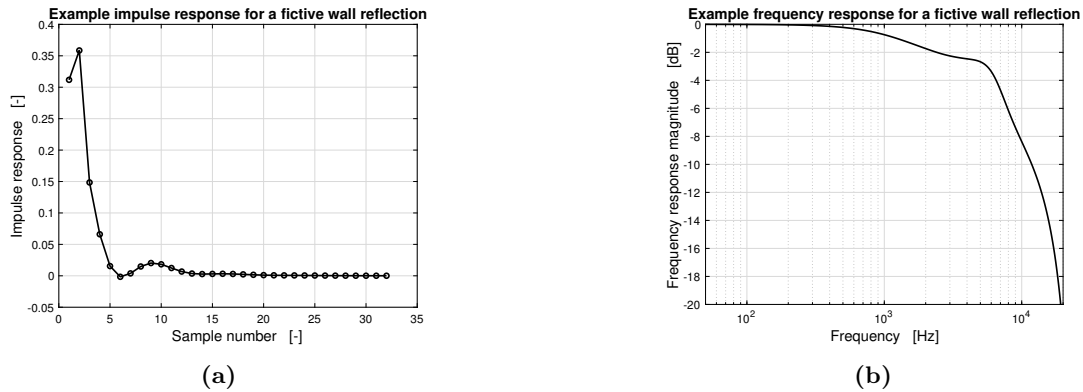


Figure 7.9: (a) The impulse response and (b) the corresponding frequency response function for a fictive wall reflection.

Now, the tricky situation is that this synthesis must be based on the knowledge of only six values of $|H|^2$: the averages for the 125 Hz octave band up to the 4 kHz octave band (because the computer prediction software most probably has tables only of absorption coefficients for those six octave bands). Fig. 7.10 shows the frequency response from Fig. 7.9(b) again, with six average values plotted as well.

So, the process of synthesizing an impulse response has to do this "backwards": start from those six average values, in Fig. 7.10, then interpolate to get a detailed frequency response curve, plus extrapolate below 125 Hz down to 0 Hz, as well as above 4 kHz up to half the sampling frequency. Once that interpolation has given a more detailed, smooth frequency response, (and one has "guessed" a phase response) it is possible to use the inverse Discrete Fourier Transform to arrive at an impulse response, as in Fig. 7.9 (a).

This process is repeated for each reflection path that has been found (which can be tens of thousands!), and each little wall reflection impulse response is added with the extra propagation delay that is given by the path length, and the result might look like in Fig. 7.11. The impulse response illustrated in Fig. 7.11 is a bit extreme since it corresponds to an ideal diffuse field, which means that no distinct specular reflections can be seen, but a huge number of weaker scattering wall reflections.

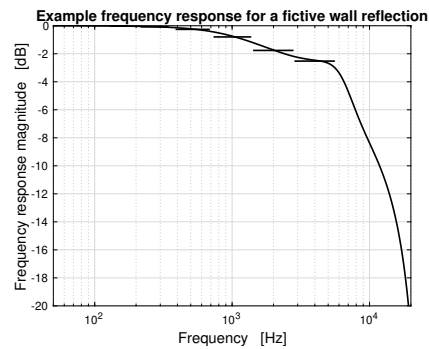


Figure 7.10: The frequency response of Fig. ??, with six octave-band-averaged levels marked.

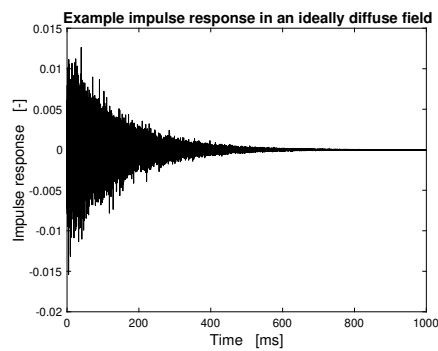


Figure 7.11: A synthetic room impulse response which represents a diffuse sound field. No distinct wall reflections can be seen, since each individual wall reflection might look like in Fig. 7.12 (b).

7.2.6 Specular reflection vs. diffuse reflection

We can make a demonstration to illustrate why a pulse (which could represent a specular reflection), and a noise sequence (which could represent a diffuse reflection) are not as different as they might look! A very simplified model of a flat wall reflection could be as shown in Fig. 7.12 (a), whereas a very rough wall section could give a response as in Fig. 7.12 (b).

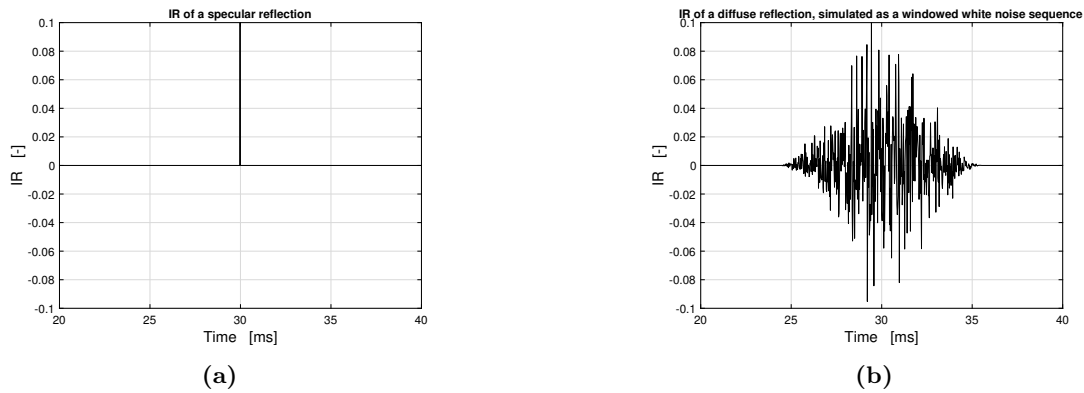


Figure 7.12: The impulse response of (a) a specular wall reflection and (b) a diffuse wall reflection, represented approximately as a windowed noise sequence.

These two time signals are scaled so that they have the same energy, i.e.

$$\sum_{n=0}^{N-1} h^2(n) = \text{constant}$$

If we convolve the two reflections with a music signal, we will realize that the signal does not sound like noise – there is just a bit of colouration difference. Note, however, that they will sound approximately equally loud.

Sound examples:

- Music convolved with a specular vs. diffuse reflection
- Diffuse-field impulse response

The purpose of demonstration 1 was to demonstrate that the details of each reflection might not be critical – but it turns out to be very important that the correct level (average amplitude squared across frequency, or time) is computed.

7.2.7 Adding the directional hearing effect

A crucial component of the auralization technique is to add the head-related transfer functions, or head-related impulse responses! For each reflection, the computer software can find out the incidence angle towards the receiver. Then, the short wall reflection impulse response as in Fig. 7.9(a), or as in Fig. 7.12 can be convolved with the head-related impulse responses, for the left and right ear, for that specific incidence angle towards the listener. A two channel impulse response results, and this can be convolved with the source signal, and the result can be listened to over headphones. Then, if the head-related impulse responses are those of the listener's head, the resulting two ear signals should be similar to the ear signals when listening in the real sound field! See Fig. 7.13 for an example with a wall reflection impulse response which is a Dirac pulse, that is, the wall is perfectly rigid.

This is how the audio part of virtual reality can be generated!

We will see a bit more about this in later lectures on sound reproduction formats, but we take the chance to show the block diagram for an auralization system already now:

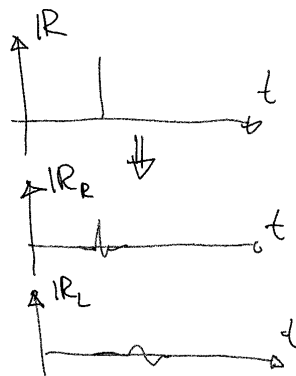


Figure 7.13: Illustration of how the wall reflection impulse response in Fig. 7.7(b) (that is, without the direct sound), can be convolved with the Head-Related Impulse Responses for the right and left ear, respectively, for the specific incidence angle that the wall reflection hits the listener with. Note that the right-ear impulse response starts a bit earlier than the wall reflection at the top, because the right ear is a bit closer to the wall.

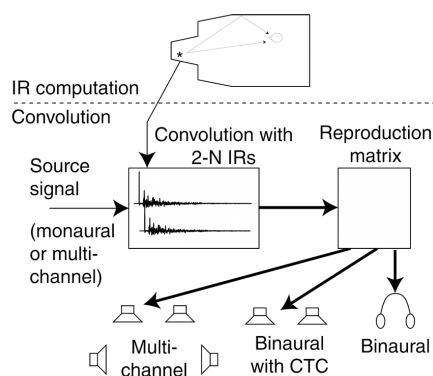


Figure 7.14: Block diagram which illustrates the auralization technique

7.3 Appendices

7.3.1 Appendix - Diffuse-field, or statistical, methods

These methods assume a so-called *diffuse sound field*, i.e. many enough reflections from all directions (i.e., isotropic) to make the sound level the same in all positions in the room (i.e., homogeneous). This means that individual reflections are ignored – it is the (expected value of the) sum of p^2 for all these reflections that is studied. Diffuse-field methods provide few details – only the sound level (as function of distance from the source) and reverberation time. They are valid *above* a certain frequency (the so-called Schroeder frequency), that is, the method is inaccurate at low frequencies. A few classical equations are well-known and very often used:

The sound pressure level, L_p , at a distance, r , from a source with source power level L_W and directivity factor DF_S , in a room with absorption area A is, according to classical diffuse-field theory:

$$L_p = L_W + 10 \lg \left(\frac{DF_S}{4\pi r^2} + \frac{4}{A} \right)$$

A more modern version is Barron's revised formula, which predicts that the reverberation level is also somewhat distance-dependent:

$$L_p = L_W + 10 \lg \left(\frac{DF_S}{4\pi r^2} + \frac{4}{A} e^{-2\delta r/c} \right)$$

where $\delta = 3 \ln 10 / T_{60} \approx 6.91 / T_{60}$ and c is the speed of sound. Furthermore, the absorption area, A , can be found from Sabine's equation, which involves the room volume, V , and the reverberation time T_{60} :

$$A = \frac{24 \ln 10}{c} \cdot \frac{V}{T_{60}} \approx \frac{0.161V}{T_{60}}$$

The Schroeder frequency, above which the diffuse-field methods can be expected to work quite well, is

$$f_{\text{Schr}} = 2000 \sqrt{\frac{T_{60}}{V}}$$

Barrons' revised formula can be used to create a diffuse-field impulse response (which uses the free-field sound pressure at 1 m distance as input signal) with these two parts:

1. A direct sound pulse, at the sample number which corresponds to the distance r . The direct sound pulse should have the amplitude $1/r$.
2. A reverberation part which is a Gaussian/white noise signal, which is multiplied by a time function $e^{-\delta t}$ and $\delta = 3 \ln 10 / T_{60}$. This exponentially decreasing noise sequence should, as a first step, start from time zero. It should be scaled so that $\text{sum}(ir_{\text{reverberation}}^2)$ gets the value $16\pi/(DF_S \cdot A)$. As a next step, the noise sequence should be truncated at the start so that it starts right after the direct sound pulse.

With this scaling, the resulting impulse response will give correct relative sound levels at different distances, according to the Barron's revised formula.

Application examples in audio signal processing

In the previous chapters, there was quite much signal processing for microphone arrays. Here, we will look at two other applications for audio signal processing: inverse filtering for loudspeakers, and adaptive signal processing for active noise control.

8.1 Inverse filtering

The concept of an inverse filter is simple: we might want to construct a filter $G(\omega)$ such that the cascading (series connection) of the filter $G(\omega)$ and a measured transfer function $H(\omega)$ becomes 1, see Fig. 8.1. As an example, a perfect loudspeaker would have the transfer function 1 - all frequencies would be reproduced exactly equally. (This is at least our first, "naive" view of the perfect loudspeaker). Since we would, in principle, like that $G(\omega) \cdot H(\omega) = 1$, we simply get $G(\omega) = 1/H(\omega)$.

Such an inverse filter can be useful in many applications: improve the quality of a loudspeaker for sound reproduction, remove the influence of a measurement loudspeaker, etc.

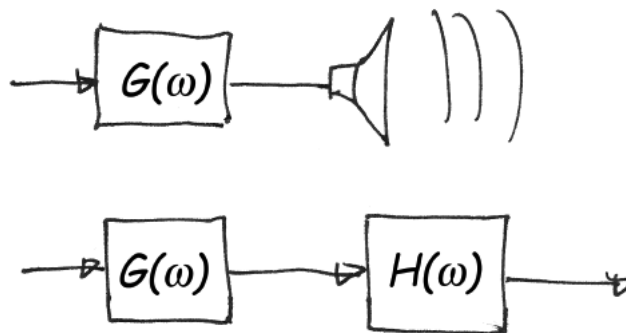


Figure 8.1: A loudspeaker can be represented by its measured frequency response function, $H(\omega)$. An inverse filter is then a filter, $G(\omega)$, which is such that $G(\omega) \cdot H(\omega) = 1$, in some sense.

8.1.1 The straightforward/naive approach

Let's try with a measured loudspeaker response. On the MIT web page with measured HRTFs for the KEMAR dummy head, we can find a measured loudspeaker IR: <http://sound.media.mit.edu/resources/KEMAR/headphones+spkr.zip> contains wave-files with measured impulse responses. Try the one called Optimus.wav which simply is a measured IR for a small loudspeaker (in an anechoic room). Our first straightforward/naive attempt could be via this MATLAB code:

```
File = 'Optimus.wav';
[ir,fs] = audioread(File);
nfft = 16384;
nir = length(ir);

%-----
% First we look at the IR and TF

tvec = [0:nir-1]/fs;
figure(1)
plot(tvec*1e3,ir)
axis([0 100 -0.4 0.3])
xlabel(['Time [ms]'])
ylabel(['IR [-]'])
title(['Measured IR of loudspeaker'])
grid

fvec = fs/nfft*[0:nfft/2-1];
H = fft(ir,nfft);

figure(2)
semilogx(fvec,20*log10(abs(H(1:nfft/2))))
xlabel(['Frequency [Hz]'])
ylabel(['TF magnitude [dB]'])
title(['Measured TF of loudspeaker'])
grid
```

The impulse response and frequency response are shown in Fig. 8.2.

Note that the IR in the file is actually close to 400 ms long but only the first 100 ms are plotted in Fig. 8.2. Now we try an inversion with this Matlab code (which is an extension of the Matlab code above):

```
G = 1./H;
g = real(ifft(G));

figure(3)
plot(tvec*1e3,g)
xlabel(['Time [ms]'])
ylabel(['IR [-]'])
title(['IR of loudspeaker inverse filter'])

figure(4)
semilogx(fvec,20*log10(abs(G(1:nfft/2))))
xlabel(['Frequency [Hz]'])
ylabel(['TF magnitude [dB]'])
title(['TF of loudspeaker inverse filter'])
grid
```

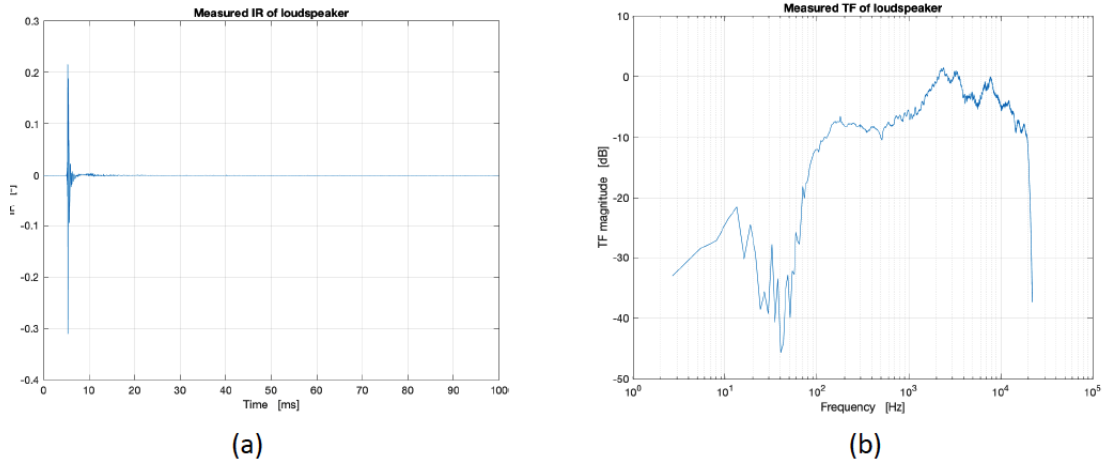


Figure 8.2: (a) The impulse response and (b) the corresponding frequency response for a loudspeaker, "Optimus", as found in the dataset with HRTFs for the Kemar dummy head, from MIT.

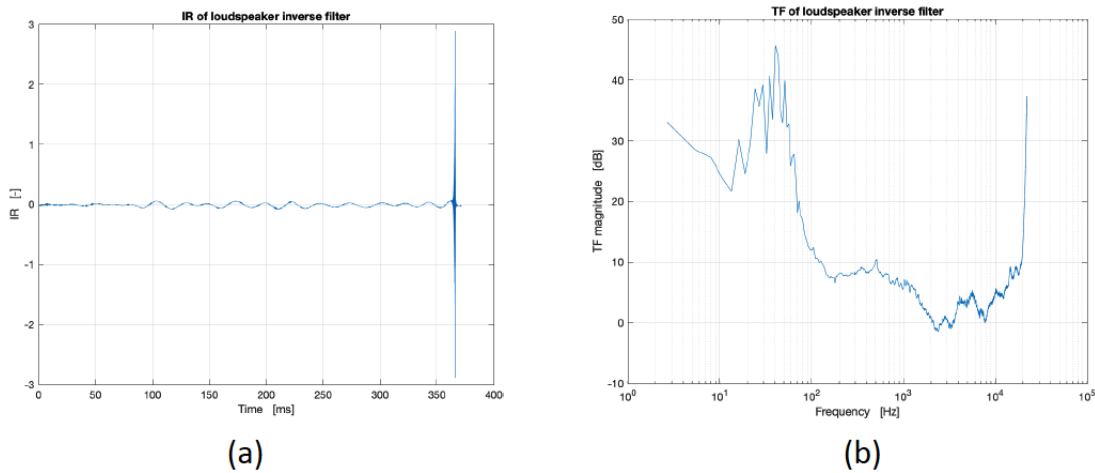


Figure 8.3: (a) The inverse impulse response and (b) the corresponding frequency response for the loudspeaker with responses as in Fig. 8.2. The inverse filter was done by a naive $G(\omega) = 1/H(\omega)$.

We use `g= real(ifft(G));` because `g` should in principle be a real-valued function but round-off errors will introduce a very small imaginary part in `g`.

By looking at the inverse IR in Fig. 3(a), we will observe two things that we don't like:

1. An impulse response should preferably die out when we move away from the main peak. Here it doesn't happen - there is some "ringing" that is almost constant.
2. An impulse response should preferably have a main peak in the beginning (to give a short FIR filter), or a symmetrical peak in the middle (to give a linear-phase FIR filter). Here, the main peak is at the end of the plot.

We can understand point 1 by looking at the inverse TF in Fig. 8.3 (b): the inverse filter's frequency response, $|G(\omega)|$ has a very high amplitude below approx. 80 Hz and above 20 kHz. We can call this "noise

blow-up” - the measurement has a limited signal-to-noise ratio (SNR), so the measurement contains only noise below approx. 40 Hz, and above 20 kHz. So, the inversion will give a high amplitude for these noise components. The obvious solution is to *band-pass filter* the inverse filter.

Point 2 can be addressed easily by introducing a *modelling delay*. With the naive approach we claim that we want the equalized loudspeaker to have a response ”1” (in the frequency domain), which corresponds to that the loudspeaker’s IR when convolved with the equalizer’s IR should give a Dirac pulse at time zero. Since the loudspeaker IR has a certain delay (caused by the distance from the loudspeaker to the microphone), we force the equalizer IR to be *non-causal*!

These two observations lead to the following: we should not have a perfect 1 as a *target response*, but a bandpass filter, properly delayed!

8.1.2 Use a non-perfect target response

Now we try to construct an inverse filter using a delayed bandpass filter as a target response.

1. We try with an FIR bandpass filter, with e.g., 70 Hz and 20kHz as cutoff frequencies, and 4096 filter coefficients, and use the Matlab function `fir1` to design such an FIR filter.
2. We introduce a delay of 1000-2000 samples in the target response (the bandpass filter) (If you choose a 4096 sample linear-phase FIR filter, then you automatically get a delay of 2048 samples and don’t need to add any extra. If you choose to try an IIR filter, then you might want to add initial zeros). We construct the FIR filter but convert it to the frequency domain using the same FFT size as for the measured IR:

```
Bbp = fir1(.....);
Target = fft(Bbp,nfft);
G = Target./H;
```

We can measure our success by the following criteria:

- The inverse filter IR should have a clear peak, and die out fast as we move away from the peak (in both directions).
- The convolution of the inverse filter and the loudspeaker IR should give a TF which is very flat between the frequencies that we used in the target response. We can tweak this approach a bit:
 - Truncate the measured IR to exclude the parts that are only noise (hint: look at `semilogy(abs(ir))` to see when the IR contains noise only).
 - Window the inverse filter around its main peak with a rectangular or other window to give it a shorter length.

8.1.3 Other methods - 1: Minimum-phase inverse IIR filter

Another popular approach is to *equalize the magnitude only*, using an IIR equalizer. The MATLAB function `yulewalk` can be used, and for that function one needs to specify a set of frequency points and a corresponding set of magnitudes. An important little detail is that the specification must include the frequency 0 Hz and half the sampling frequency, and these values might have to be extrapolated from neighboring frequency values.

8.1.4 Other methods - 2: Optimum Wiener inverse FIR filter

It is possible to find the optimum inverse FIR filter with N coefficients using the Wiener filter approach. We won’t discuss it further here but the Wiener filter will minimize the error (squared magnitude difference) between a model FIR and the real IR. This method can be extended by introducing a frequency-range weighting, so that the error is made smaller in some frequency range, [20]. Fig. 8.4 shows a block diagram from [20].

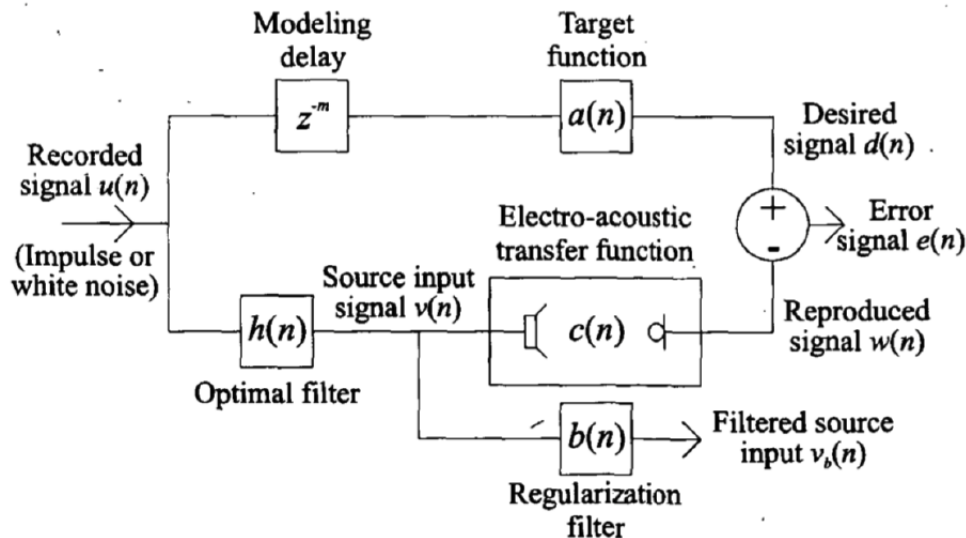


Figure 8.4: A block diagram for the estimation of an optimal filter, which minimizes and error signal $e(n)$. From [20].

8.1.5 Other methods - 3: Adaptive LMS filter

The LMS method is an adaptive method to derive an FIR filter, e.g., an inverse filter. It will lead to the final optimum Wiener FIR filter. Often the "filtered-X" version, X-LMS, needs to be used. We will see more about the LMS filter further on.

8.1.6 Details in the measured response

An FIR inverse filter will try to correct, as well as it manages, with N FIR filter coefficients, all the deviations from a flat response. It is clear, however, that many of the small deviations in the measured response are irrelevant because:

- The deviations could be caused by noise.
- The deviations could be caused by various spurious reflections. Such spurious reflections can typically be caused by other objects than the loudspeaker: the stand, other reflecting objects in the room. The later the reflections arrive, the less relevant they are for the equalization. We should realize that late reflections will change (in amplitude, spectrum, arrival time) if we move the microphone a little distance or move the loudspeaker to another room.
- A sharp peak in the frequency response is less audible/disturbing than a broad peak of the same height. This might seem counterintuitive.
- All of this leads to that one should really check, after constructing an inverse filter, that the response will be improved also if we measure in a slightly different position.

8.2 Adaptive filters - The LMS algorithm

The LMS algorithm, [56], derives an optimal FIR filter in an iterating way. That is, the algorithm tries to find an FIR filter which is as similar as possible to some unknown system. The illustration in Fig. 8.5 is from the Wikipedia article about LMS.

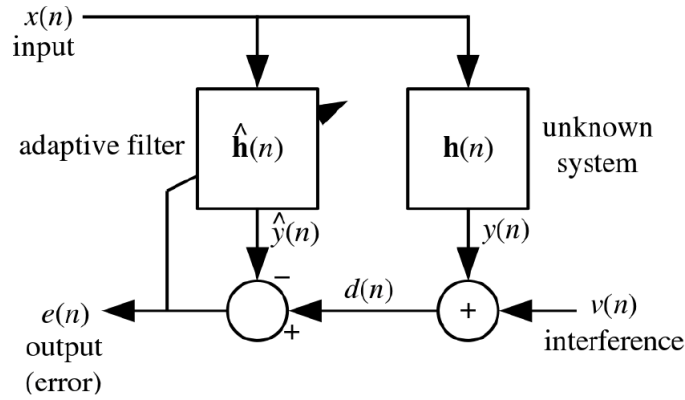


Figure 8.5: A block diagram for the LMS algorithm, from the Wikipedia article about LMS.

This situation is a bit different from the inverse filtering in the previous section, where we knew the impulse response of a loudspeaker. We will, however, see below that this LMS approach can be used also for finding inverse filters to loudspeakers.

The diagram in Fig. 8.5 illustrates a situation where we have access to an input signal, $x(n)$, (we might or might not be able to choose this ourselves), and the output signal $d(n)$, which is the output signal of the system, $y(n)$ + some unknown interference/disturbance signal $v(n)$. We want to find out what the unknown impulse response $\mathbf{h}$ is (the impulse response is denoted as a vector $\mathbf{h}$ rather than $h(n)$). **We also realize/allow that $\mathbf{h}$ might change, from time sample to time sample!** Therefore, $\mathbf{h}(n)$ refer to the values of N FIR filter coefficients at time instant n .

The LMS algorithm is based on that one starts by guessing a first estimate of the filter coefficients, $\hat{\mathbf{h}}(\text{old})$, and then one can compare the output of the real system with the output of our guess, or model. The difference between these is the error, $e(n)$, which is one value for each time sample. The LMS algorithm tries to minimize $e^2(n)$ by adjusting the values of $\hat{\mathbf{h}}$ according to this formula:

$$\hat{\mathbf{h}}(\text{new}) = \hat{\mathbf{h}}(\text{old}) + \alpha \cdot e(n) \cdot \mathbf{x}_{\text{mirrored}} \quad (8.2.1)$$

where

$\hat{\mathbf{h}}(\text{new})$ are the N "old" FIR filter coeffs,

$\hat{\mathbf{h}}(\text{new})$ are the N "new" FIR filter coeffs,

α is a so-called convergence coefficient,

$\mathbf{x}_{\text{mirrored}}$ are the N *last* samples of the input signal $x(n)$, but reversed in time.

The LMS algorithm will lead to the optimal Wiener filter solution if the convergence coefficient is not set too high. A low value of α leads to a very slow adaptation, a too high value of α leads to a fast but non-converging algorithm. In the next chapter, on active noise control, a Matlab demonstration will illustrate how we can find a model of our measured IR $h_P(n)$. The demonstration will also show that the filter model adapts to changes in the system - it is an adaptive filter.

One very useful property of the LMS algorithm is that the impulse response model, $\hat{\mathbf{h}}$, can be adapted **while it is being used!** Also, the LMS is a "cheap" algorithm since the formula in Eq. (8.2.1) is very simple.

We will try to understand the LMS algorithm by looking at an example system with only two FIR filter coefficients:

$$\mathbf{h}_{\text{true}} = [h_0, h_1]$$

We call our estimate

$$\hat{\mathbf{h}} = [\hat{h}_0, \hat{h}_1]$$

Inspecting Fig. 8.5, we realize that the error signal can be written as follows, where we write out the convolution sums involved,

$$\begin{aligned} e(n) &= y(n) - \hat{y}(n) = x(n)h_0 + x(n-1)h_1 - [x(n)\hat{h}_0 + x(n-1)\hat{h}_1] \\ &= x(n) [h_0 - \hat{h}_0] + x(n-1) [h_1 - \hat{h}_1] \end{aligned} \quad (8.2.2)$$

Obviously, if $\hat{h}_0 = h_0$ and $\hat{h}_1 = h_1$, the error $e(n)$ is zero. Now, eq. (8.2.1) expresses how the two coefficients should be updated:

$$\begin{cases} \hat{h}_{0,\text{new}} = \hat{h}_{0,\text{old}} + \alpha \cdot e(n) \cdot x(n) \\ \hat{h}_{1,\text{new}} = \hat{h}_{1,\text{old}} + \alpha \cdot e(n) \cdot x(n-1) \end{cases}$$

Inserting the expression for $e(n)$ from Eq. (8.2.2) gives

$$\begin{cases} \hat{h}_{0,\text{new}} = \hat{h}_{0,\text{old}} + \alpha \cdot x^2(n) [h_0 - \hat{h}_{0,\text{old}}] + \alpha \cdot x(n)x(n-1) [h_1 - \hat{h}_{1,\text{old}}] \\ \hat{h}_{1,\text{new}} = \hat{h}_{1,\text{old}} + \alpha \cdot x(n)x(n-1) [h_0 - \hat{h}_{0,\text{old}}] + \alpha \cdot x^2(n-1) [h_1 - \hat{h}_{1,\text{old}}] \end{cases} \quad (8.2.3)$$

We can imagine two types of input signals, $x(n)$: those without a DC-component and those with a DC-component. Without a DC-component, the expected value of $x(n)x(n-1)$ is zero while the expected values of $x^2(n)$ and $x^2(n-1)$ are larger than zero. Then any difference $[h_1 - \hat{h}_{1,\text{old}}]$ will, in the long run, not affect the updating of the value of $\hat{h}_0$, and any difference $[h_0 - \hat{h}_{0,\text{old}}]$ will, in the long run, not affect the updating of the value of $\hat{h}_1$.

If the x -signal does have a DC-component, then the expected value of $x(n)x(n-1)$ will not be zero but the expected value of $x^2(n)$ and $x^2(n-1)$ will be higher. So, the fact that the expected value of $x(n)x(n-1)$ is not zero just slows down the convergence somewhat.

Furthermore, the first equation in (8.2.3) shows that if $\hat{h}_{0,\text{old}}$ is smaller than h_0 , then $\hat{h}_{0,\text{new}}$ will be larger than $\hat{h}_{0,\text{old}}$. If, on the other hand, $\hat{h}_{0,\text{old}}$ is larger than h_0 then $\hat{h}_{0,\text{new}}$ will be smaller than $\hat{h}_{0,\text{old}}$. Thus, the $\hat{h}$ -coefficients will tend closer and closer to the true h -coefficients.

The filtered-X LMS version

Two versions of a slightly different type of system than the one analyzed above are shown in Fig. 8.6. A third block has been added, with a supposed-to-be-known impulse response $c(n)$. In both of these systems, the LMS algorithm will work towards minimizing $e(n)$ which in the end has these results:

$$\begin{cases} (a) : \hat{f}(n) * c(n) \rightarrow h(n) \\ (b) : c(n) * \hat{f}(n) \rightarrow h(n) \end{cases}$$

These two results are identical, but the implementation of the LMS algorithm becomes slightly different. Remember that for the LMS algorithm, the two signals $x(n)$ and $y(n)$ are known, and here we also know the impulse response $c(n)$. Then:

For the ordinary LMS algorithm:

1. Calculate the latest value, for sample n , of $\hat{y}(n)$ by convolving the N last samples of $x(n)$ with $\hat{\mathbf{f}}$.
2. Calculate the latest value of the error signal $e(n) = y(n) - \hat{y}(n)$
3. Update the values of $\hat{\mathbf{f}}$ using $e(n)$ and the N last values of $x(n)$, and the value α .

For the system in Fig. 8.6(a):

1. Calculate the latest value, for sample n , of $\hat{y}(n)$ by convolving the N last samples of $x(n)$ with $\hat{\mathbf{f}}$. We also need the N last samples of previously calculated $\hat{y}(n)$.
2. Calculate the latest value, for sample n , of $\hat{y}'(n)$ by convolving the N last samples of $\hat{y}(n)$ with $c(n)$.
3. Calculate the latest value of the error signal $e(n) = y(n) - \hat{y}'(n)$
4. Update the values of $\hat{\mathbf{f}}$ using $e(n)$ and the N last values of $x(n)$, and the value α .

For the system in Fig. 8.6(b) (which is called the filtered-X LMS algorithm):

1. Calculate a new input signal, $x'(n)$, by convolving the input signal $x(n)$ with $c(n)$.
2. Calculate the latest value, for sample n , of $\hat{y}(n)$ by convolving the N last samples of $x'(n)$ with $\hat{\mathbf{f}}$.
3. Calculate the latest value of the error signal $e(n) = y(n) - \hat{y}(n)$
4. Update the values of $\hat{\mathbf{f}}$ using $e(n)$ and the N last values of $x'(n)$, and the value α .

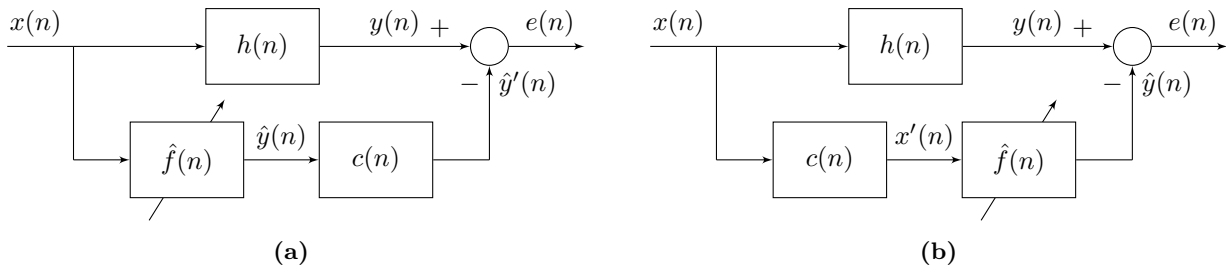


Figure 8.6: Two equivalent versions of a slightly more elaborate system than the ordinary LMS model.

Comparing the system in Fig. 8.6(b) with the block diagram in Fig. 8.4, we might realize that this "filtered-X LMS" algorithm can be used to find an inverse filter for, e.g., a loudspeaker via these steps:

1. The known impulse response (IR) $c(n)$ in Fig. 8.6(b) is the measured loudspeaker IR.
2. The IR $h(n)$ is a target IR, which could be delayed Dirac pulse, with a delay larger than the delay of the measured loudspeaker IR.
3. The iteratively updated FIR filter, $\hat{f}(n)$, will become an inverse filter.

It should be noted that the output signal $y(n)$ will not contain any energy at the frequencies outside the loudspeaker's passband, so the inverse filter will not try to correct the response outside the passband. Thus, there is no problem with the "noise explosion" which happens with a DFT-based inversion of the response, as in Fig. 8.3.

8.3 Active noise control

We will take a brief look at active noise control (ANC) systems, using many examples from [19]. ANC systems are categorized into two distinct types:

1. Feedforward systems, and
2. Feedback systems

ANC systems might use adaptive filters that can be dynamically adapting to a changing acoustic environment, or they can use static filters that are derived once. We will start by looking at the static situation.

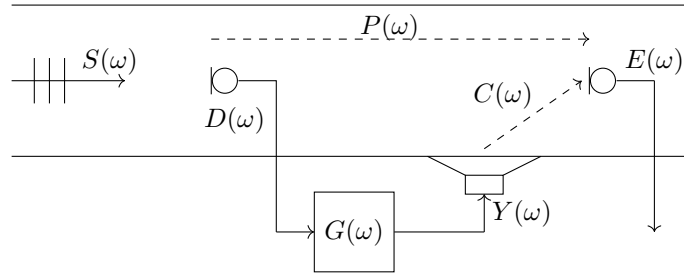


Figure 8.7: Illustration of a feedforward active noise control system in a ventilation duct. A primary noise source at the left end of the duct generates a sound wave with the primary source sound pressure $S(\omega)$ at the detection microphone. The signal picked up by the detection microphone is $D(\omega)$, which is fed to the controller, which is a filter with a frequency response $G(\omega)$. A signal, $Y(\omega)$, is fed to the secondary source, the loudspeaker, which generates a secondary sound pressure at the error microphone, which has the output signal $E(\omega)$.

8.3.1 Feedforward ANC systems

A prototypical application for a feedforward ANC system is a ventilation duct, where a fan blows air through some duct distribution system. The fan, and its motor, also generates noise, which propagates through the canal. This noise is often called the primary sound signal, as the primary noise source causes it. Fig. 8.7 illustrates such a situation, and $S(\omega)$ is the primary sound pressure, expressed in the frequency domain. A *secondary source*, a loudspeaker, is introduced, and the ANC system uses a *detection microphone*, which picks up the signal $D(\omega)$. The *controller* is a filter that takes the signal from the detection microphone, filters it, and feeds it to the secondary source. The secondary source radiates sound so that a second microphone, the error microphone, picks up the two sound waves. The goal is that the signal from the error microphone, $E(\omega)$ should be minimized, because then also the sound wave that propagates "downstream," past the right end of the duct, will be reduced.

Simple analysis of the feedforward ANC system

The system in Fig. 8.7 can be represented by a block diagram, as in Fig. 8.8. Note that extraneous noise contributions, N_1 and N_2 , have been added to the two microphones. Those noise contributions are caused by turbulence noise in the airflow, noise in the microphone amplifiers, etc.

If we, to begin with, assume a noise-free situation, then $N_1 = N_2 = 0$, and the output signal is

$$E(\omega) = S(\omega) [P(\omega) + G(\omega)C(\omega)]$$

and we realize that the output signal, the signal at the error microphone, can be zero if

$$P(\omega) + G(\omega)C(\omega) = 0 \Rightarrow G(\omega) = -\frac{P(\omega)}{C(\omega)} \quad (8.3.4)$$

What kind of filter would this transfer function represent? We can use the simplest possible model of the sound propagation in the duct - a plane wave model, which is valid up to a certain frequency, where *cross-modes* can exist. For a plane-wave model,

$$P(\omega) = e^{-jkL}$$

where L is the distance between the two microphones. For the transfer function $C(\omega)$, we assume for now that the loudspeaker has a flat frequency response for this particular application, and we assume that the distance, D , from the secondary source loudspeaker to the error microphone is large enough that plane waves will radiate*.

*Close to the secondary source loudspeaker, the sound waves will not be perfectly plane, but beyond some distance, the radiated waves will be plane, up to the cross-mode cutoff frequency.

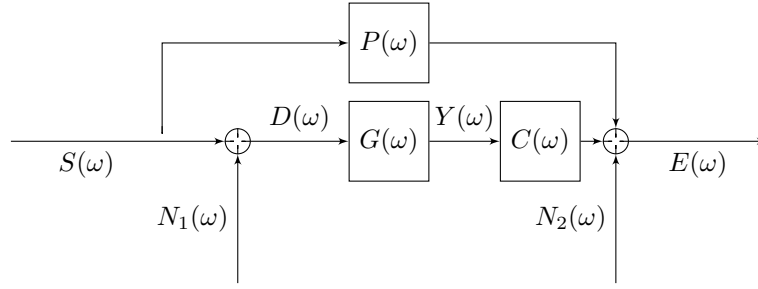


Figure 8.8: Block diagram for the ANC feedforward system in Fig. 8.7, with extraneous noise contributions N_1 and N_2 at the two microphones.

The transduction factor from the electrical signal Y to the sound pressure at the error microphone can be represented by a constant A (with the unit Pa/V).*, so:

$$C(\omega) = Ae^{-jkD}$$

and then the controller should have the filter frequency response

$$G(\omega) = -\frac{e^{-jkL}}{Ae^{-jkD}} = -\frac{1}{A}e^{-jk(L-D)} \quad (8.3.5)$$

We can easily transform this to an impulse response:

$$g(\tau) = -\frac{1}{A}\delta\left(\tau - \frac{L-D}{c}\right) \quad (8.3.6)$$

So, the controller filter simply needs to be a single delay, with a negative polarity, and a delay which is determined by the distance difference $L - D$! That is very easy to implement, and hopefully it seems intuitive. It also tells us that if $D > L$, that is, if the secondary source loudspeaker is placed *to the left* of the detection microphone, then the system can not work, because then the impulse response in Eq. (8.3.6) would need to be *non-causal*: the Dirac pulse would need to arrive before the time zero, which is impossible to implement. That might also seem intuitively right: if the detection microphone picks up a primary sound wave which is a pulse, and the secondary source wants to send out a similar pulse with the opposite polarity, then the secondary sound wave will be sent out too late.

With a reference to the section on the adaptive LMS FIR filter, it is easy to let an LMS algorithm iteratively find the optimal form of the impulse response $g(\tau)$. Here we can make the observation that if we can accept an ANC system which does not adapt to changes in the acoustic environment, e.g. due to changes in the speed of sound (which in turn can be due to changes in temperature), then the error microphone is needed only when the system is installed, and not during the actual use of the system! The error microphone can be installed at the system setup, and the LMS algorithm might be used to develop the optimal filter impulse response, g . Then this filter could be permanently installed, and the sound from the primary source will be canceled at the "virtual error microphone", and therefore also downstream of the error microphone position. Having said that, an adaptive system that can change dynamically might be necessary in some situations.

In the next subsection, we will see that there are some aspects that makes ANC systems in practice a bit more complex than shown above.

A more complex model of the feedforward ANC system - taking feedback into account

The simple model in Fig. 8.8 omitted a very important factor: there will be *feedback* from the secondary source loudspeaker to the detection microphone, via a feedback transfer function $F(\omega)$, as indicated in Fig. 8.9 and in

*Here, we treat $E(\omega)$ as if it is a sound pressure, rather than an electrical voltage, but there should be a direct proportionality between the sound pressure at the microphone and the electrical voltage from the microphone.

the block diagram in Fig. 8.10.

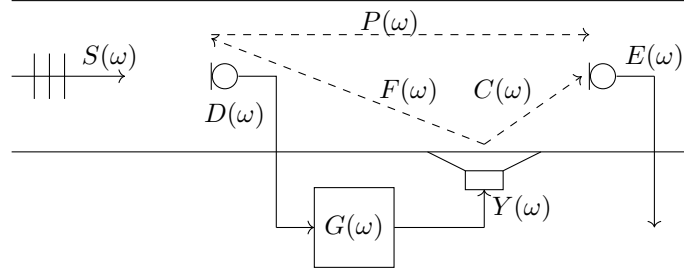


Figure 8.9: Illustration of a feedforward active noise control system in a ventilation duct. A primary noise source at the left end of the duct generates a sound wave with the primary source sound pressure $S(\omega)$ at the detection microphone. The signal picked up by the detection microphone is $D(\omega)$, which is fed to the controller, which is a filter with a frequency response $G(\omega)$. A signal, $Y(\omega)$, is fed to the secondary source, the loudspeaker, which generates a secondary sound pressure at the error microphone, which has the output signal $E(\omega)$.

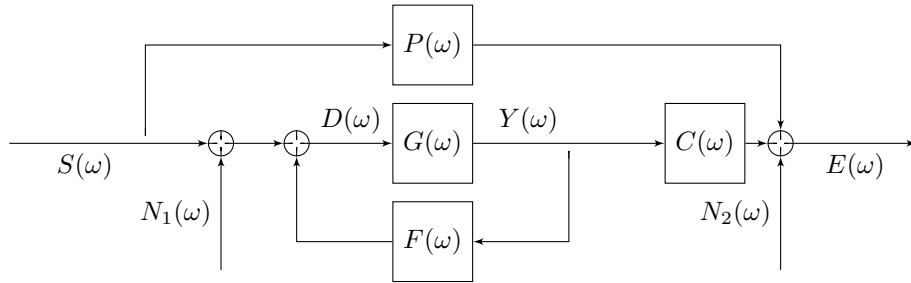


Figure 8.10: Block diagram for the ANC feedforward system in Fig. 8.9, including the feedback path, $F(\omega)$. The two summations for the detection microphone is equivalent with a single summation.

To analyze this more complex block diagram, we can again start by studying a noise-free situation, such that $N_1 = N_2 = 0$. Also, we can identify a positive feedback loop formed by the two transfer functions/frequency responses $G(\omega)$ and $F(\omega)$. Such a feedback loop can be replaced by a *closed-loop transfer function*,

$$H_{\text{closed-loop}}(\omega) = \frac{G(\omega)}{1 - F(\omega)G(\omega)} \quad (8.3.7)$$

as illustrated in Fig. 8.11. Now we might realize that we get the same block diagram as in Fig. 8.8, if we replace the block $G(\omega)$ in Fig. 8.8 with $H_{\text{closed-loop}}(\omega)$. As a consequence, the correct value for $H_{\text{closed-loop}}(\omega)$ in order to achieve that $E(\omega) = 0$ is, according to Eq. (8.3.4),

$$H_{\text{closed-loop}}(\omega) = -\frac{P(\omega)}{C(\omega)} \quad (8.3.8)$$

Since it is $G(\omega)$ that we can adjust (it is an electronic filter), and not $H_{\text{closed-loop}}(\omega)$, we get an expression for $G(\omega)$ by combining Eqs. (8.3.7) and (8.3.8):

$$\begin{aligned} G(\omega) &= \frac{H_{\text{closed-loop}}(\omega)}{1 + F(\omega)H_{\text{closed-loop}}(\omega)} = \frac{-\frac{P(\omega)}{C(\omega)}}{1 - F(\omega)\frac{P(\omega)}{C(\omega)}} \\ &= \frac{-P(\omega)}{C(\omega) - F(\omega)P(\omega)} \end{aligned} \quad (8.3.9)$$

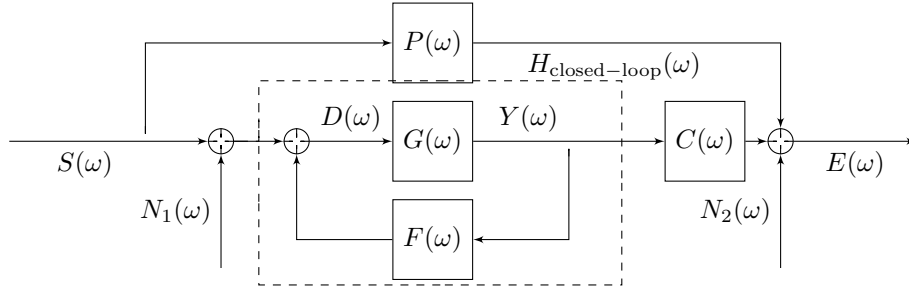


Figure 8.11: The same block diagram as in Fig. 8.10, with the dashed line indicating that everything inside the dashed-line box can be replaced by a single-input, single-output transfer function/frequency response, $H_{\text{closed-loop}}(\omega)$ given in Eq. (8.3.7)

So, if we manage to construct a controller with this filter frequency response $G(\omega)$, then the error microphone signal $E(\omega) = 0$ (in an ideal noise-free situation). Can we understand what kind of filter this will be? Yes, if we use the same plane-wave transfer functions as in the simpler analysis*, using the fact that the distance from the secondary source to the detection microphone must be $L - D$:

$$P(\omega) = e^{-jkL}, \quad C(\omega) = Ae^{-jkD}, \quad F(\omega) = Ae^{-jk(L-D)}$$

Then $G(\omega)$ is, according to Eq. (8.3.9),

$$\begin{aligned} G(\omega) &= \frac{-e^{-jkL}}{Ae^{-jkD} - Ae^{-jk(L-D)}e^{-jkL}} = \frac{-e^{-jkL}}{Ae^{-jkD}} \frac{1}{1 - e^{-jk2L}e^{jk2D}} \\ &= \frac{-1}{A} \frac{e^{-jk(L-D)}}{1 - e^{-jk2(L-D)}} \end{aligned} \quad (8.3.10)$$

The second fraction in this expression indicates that for the frequencies that give $e^{-jk2(L-D)} = 1$ will cause infinite amplitude for the filter $G(\omega)$, which seems to be problematic. First, we denote the distance $L - D = M$. Then, the dangerous frequencies are when

$$k2M = n \cdot 2\pi \Rightarrow f = n \frac{c}{2M}$$

or

$$\lambda = \frac{2M}{n} \Rightarrow M = n \frac{\lambda}{2}$$

So, when the distance from the secondary source loudspeaker to the detection microphone, M , is a number of half wavelengths, then the filter $G(\omega)$ requires an infinite amplitude, but we don't give up so easily. First, we realize that the filter defined by the frequency response in Eq. (8.3.10), with $M = L - D$, can be implemented with a closed-loop filter construction like in Fig. 8.12. We will shortly see this structure again when we try to figure out what kind of impulse response such a filter corresponds to.

As a second step, we use the infinite series expansion

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

and get the infinite series expression for Eq. (8.3.10)[†]

$$G(\omega) = \frac{-e^{-jkM}}{A} [1 + e^{-jk2M} + e^{-jk4M} + e^{-jk6M} + \dots]$$

*Please note that we are assuming that these transfer functions contain direct sound only, that is, no reflections from either end of the duct!

[†]We happily ignore, for now, that this infinite series doesn't converge when e^{-jk2M} .

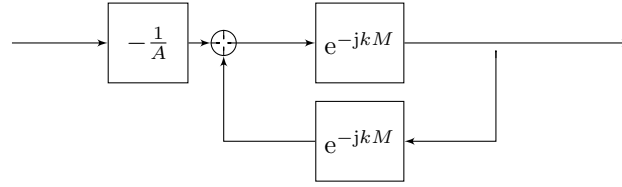


Figure 8.12: A filter structure for $G(\omega)$ which is equivalent to the filter given by Eq. (8.3.10), using $M = L - D$.

Terms of this type are easy to transform to the time domain since $e^{-jkx} \Leftrightarrow \delta(t - x/c)$, so

$$\begin{aligned} g(\tau) &= -\frac{1}{A} \delta\left(t - \frac{M}{c}\right) * \left[\delta(t) + \delta\left(t - \frac{2M}{c}\right) + \delta\left(t - \frac{4M}{c}\right) + \dots \right] \\ &= -\frac{1}{A} \left[\delta\left(t - \frac{M}{c}\right) + \delta\left(t - \frac{3M}{c}\right) + \delta\left(t - \frac{5M}{c}\right) + \dots \right] \end{aligned} \quad (8.3.11)$$

Apparently, the impulse response, $g(\tau)$, of the controller should be an infinite sequence of pulses with a constant time delay between the pulses. Even if an "infinite sequence of pulses" also sounds a bit problematic, it (possibly) doesn't sound as dramatic as the "infinite amplitude of the frequency response $G(\omega)$ ". This is especially true when we realize that the infinite sequence of pulses is very easy to generate with an *infinite impulse response (IIR)* filter as in Fig. 8.13, with $N = f_S M/c$.

Something quite unexpected will result from the impulse response structure shown in Fig. 8.13 and Eq. (8.3.11). Apparently, the controller filter, $G(\omega)$, or $g(\tau)$, will emit an infinite sequence of pulses every time the primary source generates some sound pulse that reaches the detection microphone. Nevertheless, the signal $Y(\omega)$, or rather its corresponding time signal, $y(t)$, will have a single negative pulse, which will exactly cancel the pulse from the primary source that reaches the error microphone!

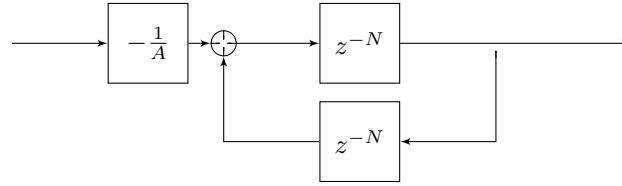


Figure 8.13: An IIR filter structure which is the digital filter equivalence of the filter structure in Fig. 8.12.

More practical aspects of the feedforward ANC system

The models used in the previous subsections indicate that the feedforward ANC system could achieve perfect noise canceling, since the signal $E(\omega)$ could be made zero. In practice, that will not happen for these reasons:

- There will be extraneous noise at the detection sensor, that is, $N_1(\omega) \neq 0$. The detection sensor can not separate the signal from the primary source, $S(\omega)$, and the extraneous noise, and thus the secondary source will try to cancel the extraneous noise - but the extraneous noise will never reach the error microphone! As a result, the secondary source will generate sound that isn't canceling anything, but rather generates more sound.
- The secondary source loudspeaker can not generate all frequencies, so the frequency-independent scale factor A is instead $A(\omega)$, and for very low and very high frequencies, the amplitude of A is low. When that happens, the secondary source loudspeaker can not cancel the primary sound field.

- The transfer functions, P , C , and F , will be less ideal than in the derivations above. There will be reflections in all three of them, and as a result, $G(\omega)$ will need to be more complicated than a single Dirac pulse, or a sequence of equi-distant Dirac pulses.

We also have the plane-wave limitation: we assume that the sound wave is completely described by a single microphone: a plane wave has the same sound pressure across the duct and then we can measure anywhere, even with a microphone mounted in the wall. We remind ourselves of the general expression for sound wave propagation in a duct, which includes cross-modes. We let z be the coordinate along the duct, and then we get (we leave out the time-harmonic factor $e^{j\omega t}$ which should be multiplied to all terms), for a duct with rigid side walls, and the cross dimensions l_x by l_y :

$$p(x, y, z) = A_{00}e^{-jkz} + \sum_{\substack{n_x, n_y=0 \\ n_x+n_y>0}}^{\infty} A_{n_x, n_y} \cos \frac{n_x \pi x}{l_x} \cos \frac{n_y \pi y}{l_y} e^{-jk_z z} \quad (8.3.12)$$

where the wavenumber $k = \omega/c$ and

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2}, \quad k_x = \frac{n_x \pi}{l_x}, \quad k_y = \frac{n_y \pi}{l_y}$$

The expression in Eq. (8.3.12) has the plane-wave component, the zero-th mode with amplitude A_{00} , outside the sum, and the entire sum is ignored when we use a plane-wave model. For a low enough frequency, the wavenumber in the z -direction will become complex as soon as n_x or n_y is larger than zero. In such a case, only plane waves can propagate*.

The amplitudes $A_{00}, A_{10}, A_{01}, A_{11}, \dots$ tell how much each mode n_x, n_y contributes to the total sound pressure, and this set of mode amplitudes depend on the source position. Therefore the primary source and the secondary source will excite/generate different mode amplitudes values. As a consequence, the secondary source can cancel only of these amplitudes, which will have to be the plane wave amplitude. We would have needed one secondary source for each mode we would like to cancel!†

A number of applications are possible for feedforward ANC systems, and particularly promising are situations where low-frequency noise is a problem. In propeller aircrafts, systems have been used where the detection sensor is a sensor placed at the engine, and there are (many) secondary source loudspeakers mounted inside the passenger compartment of the aircraft. The detection sensor just needs to be any sensor which can pick up a (preferably noise-free) signal that is correlated with the primary sound wave that reaches the error microphone.

Some noise canceling headsets have a microphone on the outside of the headphone. That indicates a feedforward ANC structure, so the microphone on the outside is a detection microphone. Typically there is no error microphone on the inside, so the system is designed to minimize the sound at a virtual error microphone (which was a real microphone during the development of the headset).

8.3.2 Feedback ANC systems

A so-called feedback ANC system works in the way that one perhaps would guess that an ANC system should look like: it uses just a single (error) microphone, a secondary source loudspeaker, and a controller (a filter) which tries to minimize the signal at the error microphone. Fig. 8.14 illustrates such a situation. We construct a block diagram for this situation, see Fig. 8.15.

The block diagram in Fig. 8.15, leads to a closed loop transfer function, from the primary/incident sound pressure signal $S(\omega)$ to the error microphone signal, $E(\omega)$, which is

$$\frac{E(\omega)}{S(\omega)} = \frac{1}{1 - C(\omega)G(\omega)} \quad (8.3.13)$$

*Higher-order modes can still exist in the neighborhood of the source: imaginary wavenumbers in the z -direction implies that waves die out rapidly as z increases.

†In addition, all the secondary sources must be placed in different locations across the duct, to generate independent sets of mode amplitudes!

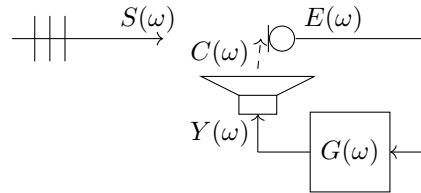


Figure 8.14: Illustration of a feedback active noise control system. A primary/incident sound pressure has the value $S(\omega)$ at the error microphone. The signal picked up by the error microphone is $E(\omega)$, which is fed to the controller, which is a filter with a frequency response $G(\omega)$. A signal, $Y(\omega)$, is fed to the secondary source, the loudspeaker, which generates a secondary sound pressure at the error microphone, via the transfer function $C(\omega)$.

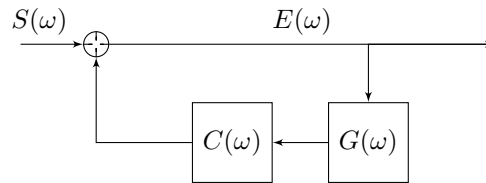


Figure 8.15: A block diagram for the situation in Fig. 8.14.

This expression implies that the controller filter, $G(\omega)$, should be such that the value $1 - C(\omega)G(\omega)$ should be as large as possible, that is, the product $C(\omega)G(\omega)$ should be as far away as possible from the value $+1$ in the complex number plane (remember that we are using the $j\omega$ -method, that is, we are assuming a single frequency, and a sound signal is expressed with a complex amplitude, for that particular frequency). So, it seems like the controller could perhaps just be an amplifier, with a very high gain, and a negative polarity?

There is, however, a *stability issue* lurking in the expression in Eq. (8.3.13). According to the so-called *Nyquist stability criterion*, if the product $C(\omega)G(\omega)$ is plotted as function of frequency in the complex number plane, the curve *is not allowed to pass to the right of the point $+1$* !. This turns out to be a severely limiting criterion which in practice can be fulfilled for a limited frequency range.

An interesting connection back to the chapter on digital audio can be made: if the controller is a digital filter which introduces some extra processing delay (caused by, e.g., delays in AD- and DA-conversions), then the stability situation is made much worse! Therefore, feedback ANC systems with digital filters need to carefully design the system to minimized any delays.

A noise canceling headphone might be constructed like this, with an error microphone on the inside of the headset, and no microphone on the outside.

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